# Removal of the spatial cutoff for weakly coupled $P(\varphi)_2$: norm convergence of the ground-state net on the quasi-local algebra

Part I — unified manuscript

Alexander Stottmeister

Institut für Theoretische Physik, Leibniz Universität Hannover,
Appelstraße 2, 30167 Hannover, Germany

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**Repair status (16 August 2026; superseded on 31 October 2026, see below).** This is the canonical Part I manuscript. The proof below is an implication from the imports in §2.4; it is not a proof of those imports. In particular, the exact slab-uniform moment estimate $U_{\mathrm{GJS}}$ is now stated explicitly in Assumption A8(vi). The printed GJS text gives strong evidence for that reading, but Appendix C is a source-mechanism audit and a proof of analytic prerequisites, not a reconstruction of the GJS cluster expansion. Consequently Theorem 12.2 is conditional on $U_{\mathrm{GJS}}$ and the other enumerated imports. The localization omission found in the CE2-to-gap passage and the ultraviolet-mollifier specification are repaired below.

**Scope amendment (5 September 2026; superseded on 31 October 2026, see below).** The status of $U_{\mathrm{GJS}}$ is re-based. It is no longer described as a quantifier strengthened beyond the source: it is the cutoff-uniform quantifier that [7] itself states (p. 629) and consumes in its printed proofs of Theorems 1.1.7 and 1.1.1 (pp. 594–597). Assumption A8(vii′) quotes the passages verbatim and the Scope paragraph of Assumption A8 records the reading. $U_{\mathrm{GJS}}$ remains a named hypothesis of Theorem 12.2: it is imported at the source's own scope, not re-derived here. No proof in the manuscript changed.

**Re-sourcing of A8(vi) (31 October 2026; superseded on 7 November 2026, see below).** Theorem 12.2 is proved from the literal imports A1–A8. Its moment input A8(vi) is GJS Theorem 1.1.11, (1.1.16), at $B = I$, $n = 0$ and any $0 < T < 1$ (p. 596), the case GJS name on p. 629 and specialize to $T = 0$ in their proof of Theorem 1.1.8 (p. 600); the record does not re-derive the cluster expansion. The two paragraphs above are kept for the record: the reading of 5 September 2026 used the same pages as evidence for Theorem 1.1.8 and is the same mathematics with different bookkeeping. No proof in the manuscript changed. The planned derivation of the local moment from Theorem 1.1.7 alone, and the one case of Theorem 1.1.7 that its printed proof does not reach (Scope paragraph of Assumption A8), are recorded in `ugjs/PLAN.md`.

**Integration of Appendix D (7 November 2026).** Theorem 12.2 is proved from printed statements only, with one GJS theorem imported: Theorem 1.1.7, which is also used for sharp-time observables across strips transverse to the time-zero line, a case its printed statement covers and its printed proof does not reach (listed separately as A8(iv)-T). The uniform local moment is a theorem of the record (Appendix D, Theorem D.4; Theorem D.3 supplies the moment bounds behind the uniform gap; both are derived from Theorem 1.1.7 and Appendix C); A8(vi) is retired. In the same pass the three unimported inputs of Appendix C found by its audit of 2 November

2026 were repaired (the Wick powers of the time-zero and the Euclidean field are now constructed in Assumptions A1 and A3, the kernel facts derived, every constant explicit), the count $N_I$ was put on the columns of the source's squares, and the constants of Theorem 5.1, Corollary 5.2, Lemmas 7.1 and 7.6 changed accordingly; the constant $M$ now contains the finite, unevaluated constant of Theorem C.5. The human readings of audit item C5 (items 1–4 and 10–12, with item 9(c)) remain open. The record of the derivation (council, drafts, four audit rounds) is `ugjs/`; the gate record of the integration is `part_i/README.md`.

**Abstract**

Let $H(g)$ be the spatially cutoff $P(\varphi)_2$ Hamiltonian, $\Omega_g$ its ground state, and $\omega_g$ the induced state on the time-zero quasi-local algebra $\mathfrak{A}$. Glimm and Jaffe (1970) constructed an infinite-volume vacuum as a $w^*$ *limit point* of space-averaged $\omega_g$; the cited construction does not prove convergence of the net itself. Guerra–Rosen–Simon (1975) formulated analogous Euclidean local-convergence questions as their Problems 1–2. We prove: in the Glimm–Jaffe–Spencer weak-coupling region $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$, the *full net* $(\omega_g)$ — no subnets, no averaging — converges in norm on every local algebra,

$$\lim_{g\in\mathcal{G}} \big\| (\omega_g - \omega_\infty)|_{\mathfrak{A}(O)} \big\| = 0,$$

with a super-polynomial rate in $\operatorname{dist}(O, \{g < 1\})$, explicit up to the finite constant of Theorem C.5; the limit is locally normal, translation invariant, $\mathbb{Z}_2$-invariant for even $\mathscr{P}$, and is a ground state of the Glimm–Jaffe dynamics $(\mathfrak{A}, \alpha)$ in the spectral sense. The proof is Hamiltonian: a local-perturbations-perturb-locally estimate built on an exact spectral filter and the exact light cone makes the net Cauchy; its two hypotheses — a cutoff-uniform gap $\operatorname{spec} H(g) \subseteq \{0\} \cup [m_1, \infty)$ and a uniform local interaction moment — are derived from one printed theorem of Glimm–Jaffe–Spencer (Ann. of Math. 100 (1974)), the exponential-clustering Theorem 1.1.7, imported as printed and not re-proved; the uniform local moment is derived from it by comparing a cutoff with its restriction to a box of the source's lattice squares (Appendix D), which uses the printed statement also for sharp-time observables across strips transverse to the time-zero line, a case its printed proof does not reach. The proof also imports, and explicitly identifies in §2, the standard cutoff-construction results on self-adjointness and the simple positive ground state, Feynman–Kac–Nelson transfer, finite-volume stability and hypercontractivity, domain smoothing and a common core, finite propagation and patching, and covariance. No Euclidean uniqueness theorem, global Markov property for the infinite-volume measure, or reverse Osterwalder–Schrader reconstruction is used. We do not claim uniqueness of $\omega_\infty$ among all locally normal ground states, a sharp exponential rate, or anything outside the weak-coupling region.

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# 1 Introduction

## 1.1 The problem

The $\lambda\varphi_2^4$ (more generally $P(\varphi)_2$) model was constructed by Glimm and Jaffe in the series I–IV [1, 2, 3, 4]. Paper III builds the physical vacuum as follows: the finite-volume ground states $\Omega_g$ of the cutoff Hamiltonians $H(g)$ induce states $\omega_g$ on the quasi-local algebra $\mathfrak{A}$; after a space-translation averaging, a $w^*$-compactness argument produces a *limit point* $\omega$, which is shown to be locally Fock [3, Thms. 2.1–2.3]. Whether the net $(\omega_g)$ *converges* — so that the vacuum is canonical rather than selected by a limit procedure — was not proved there; the Expositions volume states plainly that the $g \to 1$ limit of the vacuum expectation values "has not been proved" [6, §4]. Guerra, Rosen and Simon list analogous *Euclidean* local $L^p$ and finite-volume Gibbs-state convergence statements as Problems 1–2 of [10, pp. 125–126]; their note added on p. 128 records Newman's solution of the small-coupling $p = 1$ versions. (Some later surveys state the Hamiltonian cutoff limit as a convergence; the cited primary construction itself is unambiguously a limit-point construction.)

On the Euclidean side, convergence of *Schwinger functions* at weak coupling follows from the cluster expansion of Glimm–Jaffe–Spencer [7]. Albeverio–Høegh-Krohn–Zegarlinski prove Euclidean Gibbs uniqueness under their stated hypotheses [13]; Bauerschmidt–Dagallier–Weber prove uniqueness of the invariant measure for the $\Phi_2^4$ stochastic-quantization equation above the critical temperature [14]. Neither statement is, by itself, a theorem about norm convergence of this Hamiltonian ground-state cutoff net. The cited primary construction does not prove that local-norm convergence, and no such theorem is identified in the references reviewed here. The Hamiltonian-renormalisation programme of Rodriguez Zarate–Thiemann cited here is explicitly restricted to finite volume [15].

## 1.2 The result

> *In the GJS weak-coupling region the full net $(\omega_g)_{g\in\mathcal{G}}$ converges in norm on every local algebra; the limit is locally normal, translation and (for even $\mathscr{P}$) $\mathbb{Z}_2$ invariant, and is a ground state of the Glimm–Jaffe dynamics. A rate holds, explicit up to the finite constant of Theorem C.5: $\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| \le \Psi_{m_1}(d)$ for $g \equiv 1$ on $O_d$, with $\Psi_{m_1}$ decaying faster than every inverse power.*

See Theorem 12.2. The logical structure is deliberately Hamiltonian and compactness-free:

| step | content | where proved |
|---|---|---|
| (i) | GJS CE2 (Thm 1.1.7), printed, incl. A8(iv)-T | Assumption A8 (§2) |
| (i′) | CE2 ⇒ cutoff-uniform moment bounds | Appendices C, D |
| (ii) | uniform local moment ($M_2$) | §5 |
| (iii) | uniform gap (G): $\operatorname{spec} H(g) \subseteq \{0\} \cup [m_1, \infty)$ | §7 |
| (iv) | LPPL/Cauchy: (G)+($M_2$) ⇒ full-net convergence + rate | §§8–11 |
| (v) | the limit is an $\alpha$-ground state; invariances | §12 |

Steps (ii)–(iii) convert Euclidean input to Hamiltonian statements through slab Feynman–Kac–Nelson transfers in which every limit is taken at fixed $g$; no uniform gap is presupposed anywhere in their proofs (the gap of step (iii) is an output). Step (iv) replaces the compactness of [3] by a Cauchy argument: the same gap that yields clustering also yields a local-perturbations-perturb-locally estimate, which bounds $\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\|$ whenever both cutoffs equal 1 on $O_d$. Step (v) is an exact spectral identity: the ground-state property of the limit costs an identity, not an estimate.

The dependency graph, as established by the independent audit of the predecessor manuscript (Appendix B) and updated on 7 November 2026 for Appendices C and D, is:

$$\begin{array}{c}\text{cutoff self-adjointness, simple positive ground state, FKN, domains} \\ + \quad \text{GJS CE2 (Thm 1.1.7, incl. A8(iv)-T), scaling} \\ + \quad \text{stability (App. C), box comparison (App. D)}\end{array} \Longrightarrow \begin{array}{c}(\mathrm{M}_2)\text{ uniform local moment} \\ (\mathrm{G})\text{ uniform cutoff gap}\end{array}$$

$$(\mathrm{M}_2) + (\mathrm{G}) + \text{finite propagation} + \text{patching} \\ + \text{affiliation} + \text{path regularity} \Longrightarrow \text{LPPL estimate} \Longrightarrow \begin{array}{c}\text{full-net local-norm convergence} \\ \text{local normality}\end{array}$$

$$\text{local-norm convergence} + \text{patching} + \text{covariance} \Longrightarrow \begin{array}{c}\text{spectral ground state} \\ \text{translation and parity invariance.}\end{array}$$

### 1.3 What is not claimed

(a) *Uniqueness*: Theorem 12.2 identifies the limit of *this* cutoff family; it does not show that every locally normal ground state of $(\mathfrak{A}, \alpha)$ equals $\omega_\infty$. (b) *Sharp rate*: in the exactly solvable quadratic model the true rate is $\mathrm{poly}(d)\, e^{-2m_1 d}$ (companion note `m2/`); $\Psi_{m_1}$ is super-polynomial but not exponential, and this is intrinsic to the filter method used. (c) *Beyond weak coupling*: under the explicit phase-classification hypotheses stated in Hypothesis 13.1, a cutoff-uniform gap is incompatible with that two-phase structure (Theorem 13.2). Thus an extension to such a broken phase would have to abandon the uniform-gap input; no claim is made outside $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. (d) *Wavelet uniformity*: uniformity in a lattice/wavelet resolution is a separate problem (Part II of this programme) and is not addressed here.

### 1.4 The proof strategy in operator-algebraic terms

*The topology, and why it is the right one.* Each $\omega_g$ is the vector state of the simple positive ground vector $\Omega_g$ of $H(g)$, restricted to the quasi-local algebra $\mathfrak{A}$; the cutoffs form a directed set under inclusion of plateaus. The classical construction [3] produced a vacuum as a $w^*$ *limit point* of space-averaged $\omega_g$: compactness selects a state but proves neither convergence nor canonicity. Theorem 12.2 asserts convergence of the full, unaveraged net in the norm of every local dual $\mathfrak{A}(O)^*$ — the strongest locally uniform statement available — and Corollary 12.3 then makes the historical construction subsequence- and averaging-independent. Norm convergence on local duals also transports normality: the limit lies in the local Fock folia (Proposition 11.2).

*Why standard tools fail.* The difference $V(g') - V(g)$ has infinite norm, so the quasi-adiabatic/automorphic-equivalence machinery for gapped lattice systems [16, 17] does not start; ergodicity-type arguments give extremality of limit points, not uniqueness; and Euclidean convergence of Schwinger functions lives in the wrong topology — there is no soft passage from moments to norms on local von Neumann algebras. The architecture here is instead a *Cauchy* argument built from a local-perturbations-perturb-locally estimate: if $g \equiv g' \equiv 1$ on $O_d$, then $\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \le \Psi_{m_1}(d)$ with $\Psi_{m_1}$ super-polynomial (Proposition 10.4, Lemma 10.5). Plateaus are cofinal and local duals are complete, so the net converges (Theorem 11.1) — no compactness anywhere.

*Pillar 1: two cutoff-uniform Hamiltonian estimates from Euclidean input.* The Cauchy estimate consumes exactly two nonstandard inputs, both uniform over the whole Hamiltonian

cutoff class $\mathcal{C}_{\mathrm{H}}$: the gap (G) of Theorem 7.8, and the local moment ($\mathrm{M}_2$) of Theorem 5.1. Both are transferred from one cluster-expansion estimate of [7] for the interacting Euclidean measures $dq_h$, the exponential clustering of Source Fact 2.2: the moment bounds that the transfer needs are derived from it in Appendix D, by comparing the cutoff with its restriction to a box of the source's lattice squares and bounding the box term with the normalization and stability results of Appendix C (Theorem D.3 at fixed cutoff, across strips parallel to the time-zero line, for the gap; Theorem D.4 uniformly in the cutoff, across transverse strips as well, for the moment). Appendix C also records the source mechanism; neither appendix replaces the source's polymer expansion. The transfer device is the Feynman–Kac–Nelson slab formula: $\psi_T = e^{-TH(g)}\Omega_0$ normalizes to the ground state, and sharp-time Euclidean moments become matrix elements of $e^{-tH(g)}$. One structural point deserves emphasis: $\hat{\psi}_T \to \Omega_g$ uses *simplicity only* (Lemma 4.4) — no gap is presupposed, and the gap of Theorem 7.8 is an output, so there is no circularity. Converting semigroup decay on centered polynomial vectors into a spectral gap requires that these vectors be total; this is the density theorem (Theorem 7.7), proved from the factorial moments supplied by Theorem D.3 (Lemma 7.1) via exponential integrability, strip analyticity of characteristic functionals, Fourier uniqueness, and martingale convergence.

*Pillar 2: the derivative estimate.* Along the affine interpolation $g_s = (1-s)g + sg'$, first-order perturbation theory for the simple ground state gives $\dot{\Omega}_s = -H_s^{-1}Q_s V(v)\Omega_s$ (Proposition 8.2). Three moves control this, each with an operator-algebraic flavor. *(i) An exact inverse-energy filter* (Lemma 9.1) writes $H^{-1}(1-P_0) = \int W_\gamma(t)\, e^{itH}(1-P_0)\, dt$, converting the reduced resolvent into dynamics. The kernel $W_\gamma$ decays faster than every inverse power but *cannot* decay exponentially: a kernel with exponential tails has a transform analytic in a strip, which cannot vanish on a real neighbourhood of 0 — as $\widehat{W}_\gamma$ must — without vanishing identically. This is why the final rate is super-polynomial rather than exponential; it is intrinsic to the method (cf. §14). *(ii) The exact light cone*: $e^{itH_s}$ turns $A \in \mathfrak{A}(O)$ into $\beta_t^{g_s}(A) \in \mathfrak{A}(O_{|t|})$ with speed $\leq 1$ exactly (Proposition 2.7); no Lieb–Robinson theory is needed. *(iii) Clustering with one unbounded local argument* (Lemma 9.3): the correlation between a far perturbation cell $V_j$ and the localized $\beta_t(A)$ is bounded by $2\|V_j\Omega_s\|\|A\|e^{-\gamma R/2}$. The mechanism is a quantitative substitute for the strip-analyticity arguments familiar from positive-energy representations: the two boundary functions have spectral supports in $[\gamma,\infty)$ and $(-\infty,-\gamma]$, locality makes them agree on the interval $|t| < R$, and an optimized Gaussian separator (Lemma 9.2) bounds the value at $t = 0$. Note what the moment ($\mathrm{M}_2$) is doing: the perturbation is unbounded, but only $V_j$ *evaluated in the ground state* enters — the uniform moment is the operator-algebraic softening of the unboundedness. A unit-scale partition of the perturbation, the two-regime split of each time integral, and the count of at most eight cells per distance shell assemble the estimate (Lemma 10.1 through Proposition 10.4).

*Assembly and identification.* Cauchy-ness in each $\mathfrak{A}(O)^*$ gives compatible limit functionals and the state $\omega_\infty$; local normality follows by a monotone-net argument (Proposition 11.2). The ground-state property costs an identity, not an estimate: for each cutoff and $|t| < d(g,O)$, patching (P) makes $\omega_g(A\alpha_t(B))$ literally the Fourier–Stieltjes transform of a measure supported in $[0,\infty)$, and the positive-energy condition passes to the limit by locally uniform convergence (Proposition 12.1). Translation invariance needs no averaging: $g \mapsto \tau_a g$ is an order isomorphism of the cutoff net with cofinal range, so covariance (T) plus net convergence suffices; parity likewise.

*Boundaries.* All of this sits in the weak-coupling region through the explicit import ledger of §2.4 and the coordinate convention of Remark 3.3. The theorem does not identify $\omega_\infty$ among *all* locally normal ground states of $(\mathfrak{A},\alpha)$, does not reach a sharp exponential rate (the filter obstruction above; the solvable quadratic model has $\mathrm{poly}(d)\, e^{-2m_1 d}$), and claims nothing beyond weak coupling; there, Theorem 13.2 records the expected mechanism — for

even $\mathscr{P}$, a cutoff-uniform gap is incompatible with the fully classified two-phase structure of Hypothesis 13.1. In one sentence: *the same uniform gap that gives exponential clustering also gives a local-perturbation stability estimate through an exact inverse-energy filter and the exact light cone; the printed clustering theorem of [7], through the box comparison of Appendix D, supplies the gap and one local moment uniformly in the cutoff; and Cauchy-ness in the local duals replaces every compactness step of the classical construction.*

### Provenance and relation to the frozen sources

This document unifies, at full explicitness, the following frozen components of the research record (file pointers in Appendix B): the audited manuscript `m6/main.tex` (v2), the two structured proof reconstructions `audit_2026/lamport_euclidean_to_gap.tex` and `audit_2026/lamport_gap_to_main.tex`, the source-scope and constant gates `m4/r1-check.tex` and `m4/r2-check.tex`, the spectral-passage note `m0/`, the dichotomy note `m3/`, and, since 7 November 2026, the comparison note `ugjs/comparison.tex` (Appendix D). Every repair identified by the independent audit of 9 August 2026 (register in Appendix B) is incorporated. External inputs are the printed theorems of [7] and the explicitly cited construction results from [1, 2, 5, 11, 12, 10, 6]; they are enumerated once, as Assumptions A1–A8, and no other external mathematical input is used (Corollary 12.3 additionally quotes the averaged-state formula [3, (2.3)–(2.5)] as the definition of the averaged states it identifies).

## 2 Setting, notation, and standing imports

### 2.1 The model

$\mathcal{H} = L^2(\mathcal{S}'(\mathbb{R}), d\phi_0)$ is $Q$-space over the time-zero free field of mass $m_0 > 0$: $d\phi_0$ is Gaussian with covariance $C = (2\mu)^{-1}$, $\mu = (-\partial_x^2 + m_0^2)^{1/2}$. Let $\pi_0$ be the time-zero momentum field, the sources' $\pi(x,0)$ carried to $Q$-space by the Fock space–$Q$-space dictionary of the Scope paragraph of Assumption A1. With the Weyl operators $W(f,h) = \exp i(\varphi_0(f) + \pi_0(h))$, the local net is

$$\mathfrak{A}(O) = \{W(f,0),\, W(0,h) : f,h \in C_c^\infty(O) \text{ real}\}'', \qquad \mathfrak{A} = \overline{\textstyle\bigcup_O \mathfrak{A}(O)}^{\,\|\cdot\|},$$

$O$ ranging over bounded open intervals. Equivalently, $\mathfrak{A}(O)$ is the von Neumann algebra generated by the bounded Borel functions of the time-zero fields $\varphi_0(f)$ and $\pi_0(f)$, $f \in C_c^\infty(O)$ real: $W(tf,0) = e^{it\varphi_0(f)}$ and $W(0,tf) = e^{it\pi_0(f)}$, and the von Neumann algebra generated by the unitary group $e^{itT}$, $t \in \mathbb{R}$, of a self-adjoint $T$ contains every bounded Borel function of $T$ (an operator commuting with every $e^{itT}$ commutes with the spectral projections of $T$). These are the local algebras of the sources (Assumption A6(iii)); until 7 November 2026 the generators were written as all $W(f,h)$, $f,h \in C_c^\infty(O)$, and no proof below uses a generator other than $W(f,0)$ and $W(0,h)$. Let $\mathscr{P}(\xi) = \sum_{n=0}^{2p'} a_n \xi^n$, $a_{2p'} > 0$, $p' \geq 1$, and for real $h \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ of compact support

$$V(h) := \lambda \int h(x) :\!\mathscr{P}(\varphi_0(x))\!:_C dx := \lambda \sum_{n=0}^{2p'} a_n :\!\varphi_0^n(h)\!:_C,$$

where $:\!\varphi_0^n(h)\!:_C$ is the Wick power of the time-zero field smeared with $h$, the element of the $n$-th chaos of $L^2(d\phi_0)$ fixed by its pairings with the Wick monomials (Assumption A1, paragraph *Wick powers*; the $n = 0$ term is the constant $\lambda a_0 \int h$). Thus $V(h)$ is a real element of $L^2(d\phi_0)$, hence a self-adjoint multiplication operator in $Q$-space; it is affiliated with $\mathfrak{A}(U)$ for every bounded open interval $U \supseteq \operatorname{supp} h$, and with $\mathfrak{A}(O')'$ for every bounded

open interval $O'$ with $\operatorname{dist}(\operatorname{supp} h, O') > 0$ (*Locality* below), and finitely additive on common domains ($V(\sum_j h_j)\xi = \sum_j V(h_j)\xi$ whenever all terms are defined), since $h \mapsto :\varphi_0^n(h):_C$ is linear. Until 7 November 2026 this manuscript wrote $V(h)$ as the formal integral only and stated no construction of its Wick powers (Appendix B, corrections history). For bounded $h$ of compact support and $n \le 2p'$, Lemma N.23′(iii) of Appendix D, with Lemma N.20(iii) by linearity and scaling, proves that $:\varphi_0^n(h):_C$ is the $L^2(d\phi_0)$-limit of the mollified Wick powers $W_{n,\kappa}(h) := \int h(x) :\varphi_\kappa(x)^n: dx$, with $\varphi_\kappa(x) := \varphi_0(\rho_\kappa(x - \cdot))$ as defined before Lemma N.20; so $V(h)$ is also an explicit limit.

*Locality.* Let $U \supseteq \operatorname{supp} h$ be a bounded open set, not necessarily an interval, and $\Sigma_U$ the $\sigma$-algebra generated by the $\varphi_0(f)$, $f \in C_c^\infty(U)$ real. The constant term ($n = 0$) is $\Sigma_U$-measurable, and so is every $:\varphi_0^n(h):_C$, $1 \le n \le 2p'$. For bounded $h$: split $h$ into finitely many pieces supported in intervals of length 2, scale each to $\|\cdot\|_\infty \le 1$, and use the limit just stated; for $\kappa > 1/\operatorname{dist}(\operatorname{supp} h, \mathbb{R} \setminus U)$ every test function $\rho_\kappa(x - \cdot)$, $x \in \operatorname{supp} h$, is supported in $U$, so $W_{n,\kappa}(h)$, an $L^2$-valued integral of $\Sigma_U$-measurable variables (Lemma N.23′(i)), lies in the closed subspace $L^2(\Sigma_U)$, and so does its limit. For unbounded $h$: the truncations $h_N := h\mathbf{1}_{\{|h| \le N\}}$ are bounded, supported in $\operatorname{supp} h$, and $\|:\varphi_0^n(h):_C - :\varphi_0^n(h_N):_C\|_2 \le c\,\|h - h_N\|_2 \to 0$ by (V.6b) with $p = 2$, $\varepsilon = 1$ (Assumption A1(d)) and dominated convergence; this case uses the reading of Theorem V.2(b) recorded in the Scope paragraph of Assumption A1, and every $h$ whose locality is used below is bounded. Hence $V(h)$ is $\Sigma_U$-measurable. The von Neumann algebra $\mathcal{M}_U$ generated by the unitaries $W(f, 0) = e^{i\varphi_0(f)}$, $f \in C_c^\infty(U)$ real, contains every bounded $\Sigma_U$-measurable multiplication operator: it contains every bounded Borel function of finitely many of the commuting self-adjoint operators $\varphi_0(f_1), \dots, \varphi_0(f_k)$, whose unitary groups $W(tf_j, 0)$ lie in it (joint functional calculus); the sets $E \in \Sigma_U$ with $\mathbf{1}_E \in \mathcal{M}_U$ form a monotone class (strong limits of monotone sequences of projections in $\mathcal{M}_U$ stay in $\mathcal{M}_U$) containing the algebra of cylinder sets, hence all of $\Sigma_U$. The spectral projections of the real $\Sigma_U$-measurable multiplication operator $V(h)$ are multiplications by indicators of sets in $\Sigma_U$, so they lie in $\mathcal{M}_U$, and $V(h)$ is affiliated with $\mathcal{M}_U$. If $U$ is an interval, $\mathcal{M}_U \subseteq \mathfrak{A}(U)$, so $V(h)$ is affiliated with $\mathfrak{A}(U)$. For a general $U$: the connected components of $U$ are disjoint open intervals, so only finitely many of them, $I_1, \dots, I_m$, meet the compact support of a real $f \in C_c^\infty(U)$, and $\operatorname{supp} f \cap I_k = \operatorname{supp} f \setminus \bigcup_{l \ne k} I_l$ is a compact subset of $I_k$; hence $f_k := f\mathbf{1}_{I_k} \in C_c^\infty(I_k)$, $f = \sum_k f_k$, and $W(f, 0) = \prod_k W(f_k, 0)$ (a product of multiplication operators, $\varphi_0$ being linear), with $W(f_k, 0) \in \mathfrak{A}(I_k)$. Therefore $\mathcal{M}_U \subseteq \mathfrak{A}(O')'$ for every bounded open interval $O'$ disjoint from $U$, by the locality of the net, $\mathfrak{A}(I_k) \subseteq \mathfrak{A}(O')'$ (Assumption A6(iii)). Given a bounded open interval $O'$ with $d := \operatorname{dist}(\operatorname{supp} h, O') > 0$, the open set $U := \{x : \operatorname{dist}(x, \operatorname{supp} h) < d/2\}$ is bounded, contains $\operatorname{supp} h$ and is disjoint from $O'$; so $V(h)$ is affiliated with $\mathfrak{A}(O')'$. (Until 7 November 2026 the affiliation was asserted here without an argument.)

For a cutoff $g$ as below,

$$K(g) := H_0 + V(g), \quad E(g) := \inf \operatorname{spec} K(g), \quad H(g) := K(g) - E(g),$$

with normalized ground state $\Omega_g$ (Assumption A1), $\omega_g := \langle \Omega_g, \cdot\, \Omega_g \rangle$, and $\beta_t^g = \operatorname{Ad} e^{itH(g)}$.

## 2.2 The three cutoff classes

**Definition 2.1** (cutoff classes)**.**

$$\begin{aligned}
\mathcal{C}_{\mathrm{E}} &:= \{h : \mathbb{R}^2 \to [0,1] \text{ measurable, } \operatorname{supp} h \text{ compact}\} && \text{(Euclidean class)},\\
\mathcal{C}_{\mathrm{H}} &:= \{g : \mathbb{R} \to [0,1] \text{ measurable, } \operatorname{supp} g \text{ compact}\} && \text{(Hamiltonian class)},\\
\mathcal{G} &:= \{g \in C_c^\infty(\mathbb{R}) : 0 \le g \le 1,\ g \equiv 1 \text{ on a nonempty open interval}\} && \text{(plateau net class)}.
\end{aligned}$$

The first two are the printed classes of [7, p. 589] and [7, (1.6), p. 588], respectively. $\mathcal{G}$ is directed by $g \preccurlyeq g' \iff \{g = 1\}^\circ \subseteq \{g' = 1\}^\circ$. For a bounded interval $O$ and $r \geq 0$ set

$$O_0 := O, \qquad O_r := \{x \in \mathbb{R} : \operatorname{dist}(x, O) < r\} \quad (r > 0), \qquad d(g, O) := \operatorname{dist}\big(O, \mathbb{R}\backslash\{g = 1\}^\circ\big).$$

The roles are fixed once and for all: the Euclidean estimates of Assumption A8 quantify over $\mathcal{C}_{\mathrm{E}}$; the uniform moment and gap of §§5, 7 quantify over $\mathcal{C}_{\mathrm{H}}$; the convergence theorem quantifies over the subnet class $\mathcal{G} \subset \mathcal{C}_{\mathrm{H}}$. The elementary compatibility statements (memberships and affine-path stability) are Lemma 3.1.

## 2.3 Notation freeze

All later sections conform to the following table; in particular every symbol imported from a frozen source is converted to this notation at transcription time (see `part_i/README.md` for the conversion log).

| symbol | meaning / convention |
|---|---|
| $\mathscr{P}$, $2p'$ | interaction polynomial; $\deg \mathscr{P} = 2p'$, $a_{2p'} > 0$, $p' \geq 1$ |
| $\langle \cdot, \cdot \rangle$ | inner product, conjugate-linear in the *first* argument |
| $C$, $\mu$ | time-zero covariance $C = (2\mu)^{-1}$, $\mu = (-\partial_x^2 + m_0^2)^{1/2}$ |
| $\mathcal{C}_{\mathrm{E}}$, $\mathcal{C}_{\mathrm{H}}$, $\mathcal{G}$ | cutoff classes of Definition 2.1, in the roles fixed there |
| $K(g), E(g), H(g), \Omega_g, \omega_g, \beta^g$ | cutoff Hamiltonian data (§2.1) |
| $\alpha$, $\tau_a$, $U_a$ | patched dynamics (Assumption A6); space translations and their $Q$-space unitaries |
| $\Theta$, $\theta$ | parity unitary $\Theta = \Gamma(-1)$; parity automorphism $\theta = \operatorname{Ad}\Theta$ |
| $K_1, K_2, C_7, m_1$ | GJS constants (Assumption A8); normalization $K_2 \geq 1$ |
| $\|\cdot\|_{\mathrm{GJS}}$ | printed observable norm (1.1.7)–(1.1.9) of [7] |
| $N_I$ | number of columns $[k - \frac{1}{2}, k + \frac{1}{2})$, $k \in \mathbb{Z}$, of the squares of [7, p. 593] meeting the interval $I$ in positive measure (Appendix D, Definition N.1) |
| $M$ | uniform local moment constant of ($\mathrm{M}_2$) (§5); $M = M_B$ of Theorem D.4 |
| $S_2(B)$, $c_U$, $A$, $B'$ | constants of Appendix D ($S_2$ finite, not evaluated) |
| $\gamma$ vs. $m_1$ | $\gamma$ = abstract gap parameter in §§8–10; instantiated $\gamma = m_1$ in §11 |
| $\mathfrak{m}$ | total-variation bound in the spectral-separation lemma (§9) |
| $h_{T,g}$, $\psi_s$, $\hat{\psi}_s$, $c_g$ | slab objects (Assumption A2; §4) |
| $\Psi_\gamma$ | rate function of the one-path estimate (§10) |
| $d_0$ | LPPL threshold $d_0 = 4 + 2/\gamma$ |

Two deliberate renamings relative to the frozen sources: the separation-lemma variation bound is $\mathfrak{m}$ (it was $M_0$ in `lamport_gap_to_main.tex`, colliding with the moment constant $M$), and $\Theta = \Gamma(-1)$ is reserved for the parity unitary throughout.

## 2.4 Standing imports

Each import below states (a) the exact statement used, (b) the pinpoint source, (c) a scope remark verifying that the used statement is no stronger than the printed one, (d) the sections that consume it, and (e) the page images documenting it in the audit dossier (`audit_2026/evidence/`, index in `audit_2026/citation_screenshot_dossier.tex`).

**Assumption A1** (cutoff construction)**.** For every $g \in \mathcal{C}_{\mathrm{H}}$: $K(g) = H_0 + V(g)$ is essentially self-adjoint and bounded below; its closure (again denoted $K(g)$) has $E(g) = \inf \operatorname{spec} K(g)$ as a *simple* eigenvalue, and the normalized eigenvector $\Omega_g$ may be chosen strictly positive

$d\phi_0$-a.e. in $Q$-space. For every real $h \in L^1 \cap L^2$ of compact support, $V(h)$ is self-adjoint on its maximal multiplication domain.

*Wick powers of the time-zero field* (added 7 November 2026; the construction behind $V(h)$ of §2.1). Let $\mathcal{H}_C$ be the real Hilbert space obtained by completing the real-valued $\mathcal{S}(\mathbb{R})$ in the inner product $2\langle f, Cg\rangle$ ($C$ is bounded and strictly positive, so this is an inner product, and $\mathcal{S}(\mathbb{R})$ is a dense continuously imbedded subspace, $2\langle f, Cf\rangle \le \|f\|_2^2/m_0$). $Q$-space with $d\phi_0$ and $\varphi_0$ is the Gaussian random process indexed by $\mathcal{H}_C$ in the source's normalisation, "(c) $\langle\phi(v)\phi(w)\rangle = \frac{1}{2}\langle v, w\rangle$, the inner product on $\mathcal{H}$" [12, Definition, p. 15]: $(\varphi_0(f))_f$ is a centred Gaussian family with $\mathbb{E}[\varphi_0(f)\varphi_0(g)] = \langle f, Cg\rangle$; such a process is determined by its index space up to isomorphism [12, Thm. I.6, p. 15]. The source's definitions are used as printed. *(a)* For Gaussian random variables $f_1, \dots, f_k$, "The Wick product $:f_1^{n_1} \cdots f_k^{n_k}:$ is defined recursively in $n \equiv n_1 + \cdots + n_k$ by $:f_1^0 \cdots f_k^0: = 1$, $\langle :f_1^{n_1} \cdots f_k^{n_k}:\rangle = 0\,(n \neq 0)$ and $\frac{\partial}{\partial f_i}:f_1^{n_1} \cdots f_k^{n_k}: = n_i :f_1^{n_1} \cdots f_i^{n_i-1} \cdots f_k^{n_k}:$" [12, p. 12]; for one variable, $:f^n:$ is the coefficient of $\alpha^n/n!$ in $:\exp(\alpha f): = \exp(\alpha f - \frac{1}{2}\alpha^2\langle f^2\rangle)$ [12, (I.15), (I.18a), p. 10], explicitly $:f^n: = \sum_{m=0}^{[n/2]} \frac{n!}{m!\,(n-2m)!}\, f^{n-2m}\big(-\frac{1}{2}\langle f^2\rangle\big)^m$ [12, (I.18b), p. 10]. *(b)* $\langle :f^n: :g^m:\rangle = \delta_{nm}\, n!\, \langle fg\rangle^n$ for jointly Gaussian random variables $f, g$ [12, Thm. I.3, (I.21), p. 11] (its proof uses that $\alpha f + \beta g$ is Gaussian; every use below is within one Gaussian process), and $\langle :f_1 \cdots f_n: :g_1 \cdots g_m:\rangle = 0$ if $n \neq m$ [12, Cor. I.4, (I.23a), p. 12]. *(c)* $\Gamma_n(\mathcal{H})$ is "the closed subspace[ ] generated by $\{:\phi(f_1) \cdots \phi(f_n): \mid f_i \in \mathcal{H}\}$" [12, p. 16] (p. 17 says "spanned by"), with $\Gamma(\mathcal{H}) = L^2(d\phi_0)$ for $\mathcal{H} = \mathcal{H}_C$; by (b) the spaces $\Gamma_n$, $n \ge 0$, are mutually orthogonal, $\Gamma_0$ being the constants. *(d)* For $n \ge 1$ and $g \in L^{1+\varepsilon}(\mathbb{R})$, $0 < \varepsilon \le 1$, the Wick power $:\varphi_0^n(g):_C$ is "the unique random variable in $\Gamma_m(\mathcal{H})$ with $\langle :\phi(h_1) \cdots \phi(h_m):, :\phi^m(g):\rangle = m! \int [\prod_{j=1}^m h(x_j) S(x_j, y)] g(y)\, dx\, dy$" [12, (V.3), p. 136] (printed with $h(x_j)$ where $h_j(x_j)$ is meant [sic]), taken with $m = n$, $\mathcal{H} = \mathcal{H}_C$ and the kernel $S$ of the source's (V.1), "$\frac{1}{2}\langle f, g\rangle_{\mathcal{H}} = S(f \otimes g)$" [12, p. 134], which for the inner product $2\langle f, Cg\rangle$ of $\mathcal{H}_C$ is $S(x, y) = C(x - y)$, the kernel of $\langle \cdot, C\,\cdot\rangle$ (Lemma N.19); that is,

$$\big\langle :\varphi_0(f_1) \cdots \varphi_0(f_n):,\ :\varphi_0^n(g):_C \big\rangle = n! \int \Big[ \prod_{j=1}^{n} \int f_j(x_j)\, C(x_j - y)\, dx_j \Big] g(y)\, dy \qquad (f_j \in \mathcal{S}(\mathbb{R})). \tag{W.1}$$

It exists for this class: "$:\phi_F^n:$ exists for any $n$ as a random process for $L^{1+\varepsilon}(R)$ for each $\varepsilon \,\epsilon\, (0, 1]$. Moreover, for any such $\varepsilon$ and any $p \ge 2$, $\|:\phi_F^n(g):\|_p \le c(p-1)^{n/2}\|g\|_{1+\varepsilon}$ (V.6b)" [12, Thm. V.2(b), p. 138], with $\|:\phi^m(g):\|^2 = m! \int g(x) g(y) S(x, y)^m\, dx\, dy$ [12, p. 137]. The source prints (V.3) for $g \in \mathcal{S}$ and says "If the map $g \mapsto :\phi^m(g):$ from $\mathcal{S}$ to $\Gamma_m(\mathcal{H})$ extends to a larger test function space, $X$, than $\mathcal{S}$ we say $:\phi^m:$ exists for $X$" [12, p. 136]: for $g \in L^{1+\varepsilon}$, $:\varphi_0^n(g):_C$ is that extension, continuous for $\|\cdot\|_{1+\varepsilon}$ by (V.6b) at $p = 2$, and (W.1) persists for it, both sides being continuous in $g$ (the right side by Hölder's inequality, since $\prod_j \int f_j(x) C(x - \cdot)\, dx$ is bounded and integrable for $f_j \in \mathcal{S}(\mathbb{R})$, $C \in L^1(\mathbb{R})$); an element of $\Gamma_n(\mathcal{H}_C)$ is determined by its pairings with the Wick monomials with $f_j \in \mathcal{S}(\mathbb{R})$, as the source's "unique" in (V.3) asserts. For $n = 0$ and $g \in L^1(\mathbb{R})$, (W.1) with the empty product determines the constant $:\varphi_0^0(g):_C := \int g$; it is not continuous for $\|\cdot\|_{1+\varepsilon}$, and (V.6b) is used only for $n \ge 1$. Every real $h \in L^1 \cap L^2$ of compact support lies in $L^{1+\varepsilon}(\mathbb{R})$, so $V(h) = \lambda \sum_n a_n :\varphi_0^n(h):_C$ of §2.1 is defined, real, and linear in $h$.

*Sources.* [7, p. 588]; existence and uniqueness of the cutoff vacuum: [2, Thms. 2.2.1, 2.3.1]; lower bounds and positivity: [5]; the semigroup construction: [11, Thm. 5.3]; the Wick powers: [12, pp. 10–12, 15–17, 134–138], read on the page images on 7 November 2026.

*Scope.* Strict positivity of $\Omega_g$ is the standard positivity-improving semigroup argument of the cited construction; simplicity is used below only through the two consequences $c_g := \langle \Omega_g, \Omega_0\rangle > 0$ and the absence of a second ground vector. *Isolation of $E(g)$ (a gap) is*

*not imported*; the gap of §7 is an output. Of the Wick powers: the facts (a)–(c) are printed in Chapter I for the Gaussian process indexed by any real Hilbert space, and the definition (V.3) for a real Hilbert space $\mathcal{H}$ containing $\mathcal{S}_R(R^n)$ as a dense continuously imbedded subspace, with the kernel $S$ of (V.1) locally integrable [12, pp. 134–135]; $\mathcal{H}_C$ satisfies these hypotheses (above), and $C$ is locally integrable (Lemma N.19). The normalisation of the source's "time zero Minkowski field" $\phi_F$ of Theorem V.2(b) was not re-checked; its existence statement and (V.6b) are used for $\phi_F$ read as the process indexed by $\mathcal{H}_C$. For bounded $h$ of compact support, the only $h$ for which a proof below uses $V(h)$ (the general statements of this assumption, of Assumption A5(i) and of Proposition 2.6 are applied only to such $h$), the same element is constructed in this manuscript as an explicit $L^2(d\phi_0)$-limit (Lemma N.23′(iii) with Lemma N.20(iii)), so nothing below rests on that reading alone. The sources of the cutoff construction define $V(g)$ by normal ordering in Fock space; that their $V(g)$ is the object defined here is the Fock space–$Q$-space dictionary of [12, Ch. I], a reading of the kind recorded in Assumption A8, Scope (d).

*Used in.* §2.1 (the construction and locality of $V(h)$), §4 (slab limits), §5 (Step 1; weak-graph passage), §7 (density, spectral projections), §8 (simple Perron root), §12 and §13 (covariance and parity via Lemma 2.8), Appendix C (Lemma C.2 uses item (c)), Appendix D (Lemmas N.20 and N.23′).
*Dossier.* `gj2-ground-existence-08`, `gjii-ground-12`, `gj71-lower-bound-02`, `gj71-theorem2-04`, `rosen-semigroup-22`, `rosen-selfadjoint-23`, `gjs-source-05/06` (E04–E06, E10–E13, E20); Wick powers (added 7 November 2026): `simon-wickgen-032`, `simon-wickpair-033`, `simon-wickprod-034`, `simon-gaussproc-037`, `simon-gamman-038/039`, `simon-wickdef-156/157/158`, `simon-wickthm-159/160`.

**Assumption A2** (Feynman–Kac–Nelson transfer)**.** Let $d\mu_0$ be the free Euclidean Gaussian measure with covariance $(-\Delta_{\mathbb{R}^2}+m_0^2)^{-1}$, $J_t$ the sharp-time embeddings, and, for $h\in\mathcal{C}_{\mathrm{E}}$, $\mathcal{U}_h:=\lambda\int h(z)\!:\!\mathscr{P}(\Phi(z))\!:dz$. For all bounded time-zero functions $f,h'$, $T>0$, and $g\in\mathcal{C}_{\mathrm{H}}$,

$$\langle f, e^{-TK(g)}h'\rangle = \int \overline{(J_T f)}\,(J_0 h')\, e^{-\mathcal{U}_{\mathbf{1}_{[0,T]}\otimes g}}\, d\mu_0,$$

with the interaction over exactly the slab $[0,T]$ carrying the two embeddings, together with the analogous multi-time formulas on finite slabs (used in §4 in the two- and one-time forms, where the symmetric slab cutoffs $h_{T,g}:=\mathbf{1}_{[-T,T]}\otimes g\in\mathcal{C}_{\mathrm{E}}$ enter through the *normalized* measures $dq_{h_{T,g}}$).

*Sources.* Free-field FKN: [12, Thm. III.6]; interacting form: [12, Cor. V.14, Thm. V.15]; equivalently [10, Thms. II.14, II.16].

*Scope.* [12, Thm. III.12] is a projection-product estimate, *not* an FKN theorem, and is nowhere used (audit finding S1). The identification of sharp-time Wick functionals with $Q$-space multiplication operators is the conventions lock, Remark 2.5.

*Used in.* §4 (slab identities), §5 (Step 1: the sharp-time embedding $J_0$), §8 (path formula), Appendix C (through Lemma N.25(i)), Appendix D.
*Dossier.* `simon-free-fkn-111`, `simon-fkn-184/185`, `grs-fkn-39/40`, `simon-wrong-iii12-119` (E27–E43 range).

**Assumption A3** (finite-volume stability)**.** For every $h\in\mathcal{C}_{\mathrm{E}}$: $\mathcal{U}_h\in L^p(d\mu_0)$ for every $p<\infty$, and $e^{-\varrho\mathcal{U}_h}\in L^1(d\mu_0)$ for every finite $\varrho>0$. In particular $0<Z(h):=\int e^{-\mathcal{U}_h}d\mu_0<\infty$, and the interacting measures

$$dq_h := Z(h)^{-1}e^{-\mathcal{U}_h}\,d\mu_0, \qquad h\in\mathcal{C}_{\mathrm{E}},$$

are well-defined probability measures.

*Construction of $\mathcal{U}_h$* (added 7 November 2026; repair R8 of the V3 audit, Appendix B). Let $\mathcal{H}_G$ be the real Hilbert space obtained by completing the real-valued $\mathcal{S}(\mathbb{R}^2)$ in the inner product $2\langle f, (-\Delta_{\mathbb{R}^2} + m_0^2)^{-1} g\rangle$; $(\mathcal{S}'(\mathbb{R}^2), d\mu_0, \Phi)$ of Assumption A2 is the Gaussian random process indexed by $\mathcal{H}_G$ in the source's normalisation (as for $\mathcal{H}_C$ in Assumption A1), so that the kernel of (V.1) is $G(z - z')$ of Lemma N.19(v), which is locally integrable. The Wick products, the pairing facts and the chaoses $\Gamma_n(\mathcal{H}_G) \subseteq L^2(d\mu_0)$ are those of Assumption A1, items (a)–(c), for $\mathcal{H} = \mathcal{H}_G$; for $n \geq 1$ and $g \in L^{1+\varepsilon}(\mathbb{R}^2)$, $0 < \varepsilon \leq 1$, the Wick power $:\Phi^n(g):$ is the unique element of $\Gamma_n(\mathcal{H}_G)$ with

$$\big\langle :\Phi(f_1)\cdots\Phi(f_n):,\, :\Phi^n(g):\big\rangle = n! \int \Big[\prod_{j=1}^{n} \int f_j(z_j)\, G(z_j - z)\, dz_j\Big] g(z)\, dz \qquad (f_j \in \mathcal{S}(\mathbb{R}^2)), \tag{W.2}$$

[12, (V.3), p. 136] (for $g \notin \mathcal{S}$ the continuous extension, as in Assumption A1(d)); it exists for this class, "$:\phi^n:$ exists for any $n$ as a random process for $L^{1+\varepsilon}(R^2)$ for each $\varepsilon\,\epsilon\,(0,1]$" [12, Thm. V.2(a), p. 137], with

$$\|:\Phi^n(g):\|^2_{L^2(d\mu_0)} = n! \iint g(z)\, g(z')\, G(z - z')^n\, dz\, dz' \tag{W.3}$$

[12, p. 137], and, for every $p \geq 2$, $\|:\Phi^n(g):\|_p \leq c\,(p-1)^{n/2}\|g\|_{1+\varepsilon}$ with $c$ depending on $\varepsilon$ and $n$ only [12, (V.6a), p. 137; Remark 1, p. 138]. For $n = 0$ and $g \in L^1(\mathbb{R}^2)$, (W.2) with the empty product gives the constant $:\Phi^0(g): := \int g$, as in Assumption A1(d); the bound above is used only for $n \geq 1$. Every $h \in \mathcal{C}_{\mathrm{E}}$ is bounded with compact support, hence in $L^1 \cap L^{1+\varepsilon}(\mathbb{R}^2)$, and

$$\mathcal{U}_h := \lambda \sum_{n=0}^{2p'} a_n\, :\Phi^n(h): \in \bigoplus_{n=0}^{2p'} \Gamma_n(\mathcal{H}_G), \qquad :\Phi^0(h): = \int h\, dz,$$

which is linear in $h$ and lies in every $L^p(d\mu_0)$ by (V.6a); this makes the first clause of the assumption a consequence of the construction, and the clause on $e^{-\varrho\mathcal{U}_h}$ is the import proper. Appendix C shows that $\mathcal{U}_h$ is the $L^2(d\mu_0)$-limit of its mollified versions (Lemma C.4 with Lemma N.23′(iii)). The second quantization of a contraction $c$ on $\mathcal{H}_G$ acts on the Wick monomials by $\Gamma(c)\,:\Phi(f_1)\cdots\Phi(f_n): = :\Phi(cf_1)\cdots\Phi(cf_n):$ [12, (I.32), p. 25]; in particular $\Gamma(e^{-t}\mathbf{1})$ is multiplication by $e^{-nt}$ on $\Gamma_n(\mathcal{H}_G)$.

*Source.* The first clause: [12, Thm. V.2(a), (V.6a), p. 137]; the second, with $P := \varrho\lambda\mathscr{P}$ (semibounded, since $\varrho\lambda \geq 0$ and $a_{2p'} > 0$) and $g := h \geq 0$: [12, Thm. V.7(a), (V.28a), p. 152] ("Let $P$ be a fixed semibounded polynomial. Then (a) If $g \geq 0$, $g\,\epsilon\,L^1 \cap L^{1+\epsilon}(R^2)$, then $\exp(-\int g(x){:}P(\phi(x)){:}d^2x)\,\epsilon\bigcap_{p<\infty} L^p(Q_N, d\mu_{0,N})$", dossier image `simon-exp-174`) and its chapter context; the construction: [12, pp. 135–137], read on the page images on 7 November 2026 (until then the assumption cited Theorem V.7 for both clauses and stated no construction; V3 audit, findings M-3 and m-9).

*Scope.* Only finite slabs and compactly supported $h$ occur below; no infinite-volume stability is imported. A self-contained quantitative lower bound on $Z(h)$ and the $h$-uniform $L^p$ bound on $e^{-\mathcal{U}_h}$ are proved in Appendix C from the construction above and Assumption A4. That the source's $\int g\,{:}P(\phi){:}$ in Theorem V.7 is the $\mathcal{U}_h$ constructed from its own (V.3) is read off the chapter (both are built on the same Wick powers); that it is the $V(h)$ of [7, (1.8), p. 589], which prints "Wick order with respect to the measure $d\Phi$" and no construction, is a reading of the kind recorded in Assumption A8, Scope (d).

*Used in.* §3 (Lemma 3.1), §4, §8 (Hölder/Vitali chains), Appendix C, Appendix D.
*Dossier.* `simon-exp-174`, `simon-transfer-178/179/180` (E33–E43 range); construction of $\mathcal{U}_h$ (added 7 November 2026): `simon-wickdef-156/157/158`, `simon-wickthm-159`, `simon-secondq-047`.

**Assumption A4** (free hypercontractivity)**.** The free time-zero Markov semigroup $\mathsf{P}_T$ satisfies

$$\|\mathsf{P}_T F\|_q \le \|F\|_p \qquad \text{whenever} \qquad \frac{q-1}{p-1} = e^{2m_0 T}, \quad 1 < p \le q < \infty.$$

More generally (the abstract form of the same source): for the second quantization $\Gamma(c)$ of a contraction $c$ on the one-particle space of any Gaussian $L^2$-space, $\|\Gamma(c)F\|_q \le \|F\|_p$ whenever $\|c\|^2 \le (p-1)/(q-1)$; the display is the case $c = e^{-T\mu}$, and the Mehler case $c = e^{-t}\mathbf{1}$ on $L^2(d\mu_0)$ is the form used in Appendix C.

*Sources.* [12, Thm. I.17] (the abstract form and the display); the time-region form is [12, Thm. III.17]. The moment comparison on the chaoses that Appendix C derives from the abstract form is also printed as [12, Thm. I.22, (I.47), p. 38]: "if $\psi \in \bigoplus_{n=0}^{m} \Gamma_n(\mathcal{H})$, then $\|\psi\|_p \le (p-1)^{m/2}\|\psi\|_2$ for all $p > 2$" (read on the page image on 7 November 2026).

*Scope.* [12, Thms. V.10, V.12] are, respectively, a transfer-matrix application and an essential-self-adjointness theorem; neither is the hypercontractivity input and neither is cited for it (audit finding S2). The abstract form is printed for $p, q \in (1, \infty)$, $p \le q$ [12, Thm. I.17, p. 34]; every use below has $1 < p \le q < \infty$ (§8, Step 1: $p = 1 + e^{-m_0 T}$; Appendices C and D: $p = 2$). How $\Gamma(c)$ acts on the chaoses, (I.32), is recorded in Assumption A3.

*Used in.* §8 (the pairing bound), Appendix C (tail estimate; the comparison $\|Y\|_q \le (q-1)^{p'}\|Y\|_2$), Appendix D (the same comparison at $q = 4$).

*Dossier.* `simon-hyper-056`, `simon-timeslice-125/126`; Theorem I.22 (added 7 November 2026): `simon-thm122-060`.

**Assumption A5** (domain smoothing; common core)**.** *(i)* For every $g \in \mathcal{C}_{\mathrm{H}}$ and $t > 0$,

$$\operatorname{Ran} e^{-tK(g)} \subseteq D(H_0) \cap D(V')$$

for every polynomial multiplication operator $V'$ of the time-zero field with coefficients in $L^1 \cap L^2$ of compact support (in particular for every $V(h)$ as in §2.1). *(ii)* For every affine path $g_s = (1-s)g + sg'$, $g, g' \in \mathcal{G}$, there is a common core $\mathscr{C}$ for all $K(g_s)$, $s \in [0,1]$, on which $K(g_s)\xi = K(g_{s_0})\xi + (s - s_0)V(g' - g)\xi$.

*Sources.* (i) is the model-specific smoothing theorem [6, Ch. 7, Thm. 7.4]. (ii) is part of the cutoff construction; essential self-adjointness pinpoints, in their roles per the audit: [12, Thm. V.12], [11, Thm. 5.3], [2].

*Scope.* [6, Thm. 7.3] alone assumes a potential bounded below and is therefore *not* sufficient here; the citation is to Theorem 7.4 (audit finding S3).

*Used in.* Proposition 2.6, §4 (dominators), §8 (eigenvalue differentiation).

*Dossier.* `gjexp-domain-165/166/167` (E17–E19).

**Assumption A6** (free propagation; patching; the dynamics $\alpha$)**.** *(i)* The free dynamics $\operatorname{Ad} e^{itH_0}$ has propagation speed one (Klein–Gordon finite propagation), and the Trotter product formula for $e^{itK(g)}$ converges strongly. *(ii)* There is a one-parameter automorphism group $\alpha$ of $\mathfrak{A}$, independent of $g$, with

$$\alpha_t(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|}), \qquad \alpha_t(A) = \beta_t^g(A) \ \text{ if } g \equiv 1 \text{ on } O_{|t|} \ (A \in \mathfrak{A}(O)). \tag{P}$$

*(iii)* The net is local: $\mathfrak{A}(O) \subseteq \mathfrak{A}(O')'$ for disjoint bounded open intervals $O$, $O'$ (listed on 7 November 2026; used before without being listed).

*Sources.* [1, §III]; [2, §§3.5–3.6]; [6, §4.1 and Ch. 4]; (iii): [2, p. 397, proof of Thm. 3.5.1], read on the page image on 7 November 2026.

*Scope.* (P) is the Glimm–Jaffe patching construction verbatim; the $g$-uniform propagation bound for the *cutoff* dynamics is not imported but derived (Proposition 2.7). Item (iii) is

printed for the algebra the source defines, for an open set in space, "as the weak closure of the bounded functions of the $t = 0$ fields" $\int \varphi(x,0) f(x)\,dx$ and $\int \pi(x,0) f(x)\,dx$ "as $f$ runs over the $C^\infty$ functions with support in" that set: for two "disjoint open sets", "elements of" the two algebras "commute". The algebra so defined for an interval $O$ is the $\mathfrak{A}(O)$ of §2.1, which is generated by the bounded Borel functions of $\varphi_0(f)$ and $\pi_0(f)$, $f \in C_c^\infty(O)$ real, once the source's Fock-space fields $\varphi(x,0)$, $\pi(x,0)$ are read as the $Q$-space fields $\varphi_0$, $\pi_0$: the Fock space–$Q$-space dictionary of the Scope paragraph of Assumption A1, a reading of the source. Items (i) and (ii) are imported under the same reading. (Until 7 November 2026 §2.1 wrote the generators of $\mathfrak{A}(O)$ as all $W(f,h)$; matching that generating set to the source's would need the Weyl relations, which this manuscript does not state and does not use for the local algebras; the remark on the coordinate convention in §3 presupposes them for its scaling automorphism, of which only the action on $W(f,0)$ and $W(0,h)$ enters $\sigma_s(\mathfrak{A}(O)) = \mathfrak{A}(sO)$.)

*Used in.* Proposition 2.7, §9 (locality identity), §10, §12; item (iii): §2.1 (*Locality*, hence Proposition 2.7 and the two-regime bound of §10) and §13 (Step 1 of the proof of the dichotomy theorem).

*Dossier.* `gj1-locality-06`, `gjexp-054/055/056`, `gj2-cutoff-independence-38`, `gj2-local-algebra-40` (E03, E06, E14–E16); item (iii) (added 7 November 2026): `gj2-locality-37`.

**Assumption A7** (covariance data)**.** Let $U_a$ implement the space translation $\tau_a$ on $Q$-space and let $\Theta := \Gamma(-1)$, so that $(\Theta\psi)(q) = \psi(-q)$ in $Q$-space. Then, for every $g \in \mathcal{C}_{\mathrm{H}}$,

$$U_a K(g) U_a^* = K(\tau_a g), \qquad \Theta H_0 \Theta = H_0, \qquad \Theta V(g) \Theta = V(g) \ \text{ for even } \mathscr{P}.$$

*Sources.* [2, p. 398] (cutoff covariance); the transformation of Wick powers under $\Gamma(-1)$ is the parity of the polynomial.

*Scope.* Only the displayed operator identities are imported; the state-level consequences are derived in Lemma 2.8.

*Used in.* Lemma 2.8; thence §12, §13.

*Dossier.* `gj2-cutoff-independence-38` (E06).

**Assumption A8** (the GJS package)**.** The Euclidean input is [7], used exactly as printed, as follows.

*(i) Measures.* For $h \in \mathcal{C}_{\mathrm{E}}$, $dq_h = Z(h)^{-1} e^{-\mathcal{U}_h} d\mu_0$ as in Assumption A3; the Hamiltonian cutoff class of [7, (1.6), p. 588] is $\mathcal{C}_{\mathrm{H}}$.

*(ii) Observable class.* The class (1.1.5) of [7] consists of sharp-time monomials $Q(x^0) = \int d\vec{x}\, w(\vec{x}) \prod_i {:}\Phi(x_i)^{m_i}{:}$ with $L^2$ kernels; equal times are allowed, and products of individually Wick-ordered powers are admissible without reordering. Equation (1.1.6) of [7], $Q = \int dx^0\, Q(x^0)$, defines the time-integrated monomials with kernels in $L^2$ of space-time; "localized" is defined for both forms by support in a product of unit lattice squares "centered at the lattice side [sic] $j_i \in \mathbb{Z}^2$" (p. 593), i.e. squares centred at integer points. The spatial cells used in §§5–6 are the columns $[k - \frac{1}{2}, k + \frac{1}{2})$, $k \in \mathbb{Z}$, of these squares (Definition N.1 fixes the convention on the lattice lines); the count bounds used below (at most three columns for a support of length 2, Theorem 5.1, Step 2; $N_I \le \lfloor L \rfloor + 2$, Lemma 6.1) are proved on them. (Until 7 November 2026 these sections counted in the cells $[j, j+1)$, a lattice shifted by one half unit, on which the displayed norm bounds of Proposition 6.2 can fail for the norm of p. 594; Appendix B, corrections history.) GJS bound the degree of each *individual* Wick factor: $n_i < \bar{n}$ for a fixed finite $\bar{n}$, any choice $\bar{n} \ge \deg \mathscr{P}$ being sufficient; the printed text adds "Since there is no restriction on $r$, $Q$ still may have an arbitrarily large degree" (p. 593). Footnote 9 permits $\bar{n} = \infty$ at the price $N(\Delta)! \to (N(\Delta)!)^3$ in (1.1.7), (1.1.8), (2.4.5). Every observable used below has individual factor degrees $\le \max(2p', 1)$, so the fixed finite choice

$\bar{n} = 2p' + 1$ applies and the printed first-power weights are used as displayed; the *total* degrees of the observables of §7 are unbounded, exactly as the printed sentence licenses.

*(iii) Printed norms.*

$$\begin{aligned} (1.1.7)\quad & \|Q\|_{\mathrm{GJS}} = K_1\Big[\prod_{\Delta} K_2^{2N(\Delta)} N(\Delta)!\Big]\|w\|_2, \\ (1.1.8)\quad & \|Q(x_0)\|_{\mathrm{GJS}} = K_1\Big[\prod_{\Delta} K_2^{2N(\Delta)} N(\Delta)!\Big]\|w(x_0,\cdot)\|_2, \end{aligned} \tag{1}$$

where $N(\Delta)$ is the total degree localized in the unit square $\Delta$, and (1.1.9): a nonlocalized polynomial is normed by summing its localized pieces. Replacing $K_2$ by $\max\{1, K_2\}$ only enlarges right-hand sides; *we fix* $K_2 \geq 1$.

*(iv) CE2.*

**Source Fact 2.2** (CE2 = [7, Thm. 1.1.7], p. 594)**.** For localized monomials $Q_1, Q_2$ with disjoint localization squares separated by a strip of width $d$ bounded by two parallel lines (any orientation),

$$\Big|\int Q_1 Q_2\, dq_h - \int Q_1 dq_h \int Q_2 dq_h\Big| \leq C_7 e^{-m_1 d}\|Q_1\|_{\mathrm{GJS}}\|Q_2\|_{\mathrm{GJS}},$$

with $m_1 > 0$ and $C_7$ *independent of* $h$, $Q_1$, $Q_2$ (so stated in the source).

*(iv-T) The transverse case, listed separately (A8(iv)-T; added 7 November 2026).* GJS Theorem 1.1.7 is printed for localized monomials separated by a strip of any orientation (p. 594). Its printed proof rotates the strip into the time direction (p. 597) and so reaches sharp-time monomials only across strips parallel to the time-zero line. Lemma 7.4 uses only that case; Appendix D also uses the transverse case, listed separately as A8(iv)-T. Namely, *A8(iv)-T* is Source Fact 2.2 applied to a sharp-time monomial $Q_1$ of type (1.1.5) and a localized monomial $Q_2$ across a strip that is not parallel to the time-zero line (transverse in the sense of Definition N.6), at every $h \in \mathcal{C}_{\mathrm{E}}$: a case of the printed statement that its printed proof does not reach (Scope, point (b)). In every use that a proof of this manuscript makes (Theorem D.4), $Q_2$ is of type (1.1.6) and the strip is bounded by two lines of constant $x^1$. It is listed for transparency, not as a different import standard: it is the printed statement at its printed scope, exactly as items (i)–(iv). It is not a narrower form of the retired item (vi): that was a moment bound on the slab class and stated no clustering, and neither statement is known to imply the other. Used in Theorem D.4 (through Lemma D.2), hence in Theorem 5.1 and in every statement that consumes its constant $M$ (Lemma 10.2 onward); not in §7, whose Theorem D.3 uses parallel strips only.

*(v) CE1 (provenance; not used since 7 November 2026).*

**Source Fact 2.3** (CE1 = [7, Thm. 1.1.8], p. 595)**.** The printed estimate is

$$\Big|\int Q(x^0)\, dq_h\Big| \leq C_8\|Q(x^0)\|_{\mathrm{GJS}},$$

stated uniformly as the source cutoff tends to one. This item records the literal one-sentence statement of Theorem 1.1.8. The cutoff-uniform form that §§5 and 7 consumed until 7 November 2026 was item (vi), read off the printed Theorem 1.1.11 (p. 596) in the specialization the source itself prints as its proof of Theorem 1.1.8 (p. 600); since that date the moment bounds are Theorems D.3 and D.4 of Appendix D, proved from item (iv) with items (i)–(iii), Assumptions A2–A4 and Appendix C, and neither this item nor (vi) is consumed by any proof.

*(vi) The cutoff-uniform moment estimate $U_{\mathrm{GJS}}$ (GJS Theorem 1.1.11 at $B = I$, $n = 0$; imported as printed from 31 October 2026; **retired on 7 November 2026**: no proof of this manuscript uses it; the item, its tag and its label are kept for provenance).* For

$$\mathcal{C}_{\mathrm{E}}^{\mathrm{slab}} := \{h_{T,g} = \mathbf{1}_{[-T,T]} \otimes g : T > 0,\ g \in \mathcal{C}_{\mathrm{H}}\} \subset \mathcal{C}_{\mathrm{E}},$$

one and the same constant $C_8$ satisfies

$$\left| \int Q(x^0)\, dq_{h_{T,g}} \right| \le C_8 \|Q(x^0)\|_{\mathrm{GJS}} \quad (T > 0,\ g \in \mathcal{C}_{\mathrm{H}}). \tag{$U_{\mathrm{GJS}}$}$$

The quantifiers include every affine cutoff $g_s = (1-s)g + sg'$ used later, because $g_s \in \mathcal{C}_{\mathrm{H}}$. Equation ($U_{\mathrm{GJS}}$) was the CE1 uniformity needed in §§5 and 7 until 7 November 2026 (Theorems D.3 and D.4 now supply what those sections need, from item (iv) and the inputs listed in Appendix D); from 31 October 2026 it was read off one printed theorem of the source, as follows.

**Source Fact** ([7, pp. 596, 600 and 629])**.** *(a) p. 596.* "We let $A$ and $B$ denote monomials of type (1.1.5) of degree $|A|$ and $|B|$ and with kernels $w_A$ and $w_B$ respectively. We assume that $A$ is localized at times $x_{i0} \le 0$, and that $B$ is localized at times $x_{i0} \ge 0$. Let $B_T$ denote the translate of $B$ at time $T > 0$, so that $B_T$ is localized at times $\ge T$." "In the next two results, we take all times $x_{1,0} = t = \cdots = x_{r,0}$ equal in $Q(x_0)$. To indicate this restriction, we write $Q(t)$ in place of $Q(x_0)$." [...] "THEOREM 1.1.11. Let $n$ and $\varepsilon > 0$ be given and let $\gamma = (n+1)m_0(1-\varepsilon)$. For any $A$, we may choose $n^{\text{th}}$ degree polynomials $Q_n(t_j)$ of the form (1.1.5) with $r = 1$, so that

$$\sum\nolimits_{t_j > 0} e^{\gamma t_j} \|Q_n(t_j)\| \le O(1)\|A\| \tag{1.1.14}$$

and so that

$$Q = \sum\nolimits_{0 < t_j < T} Q_n(t_j) \tag{1.1.15}$$

satisfies

$$\left| \int (A - Q) B_T\, dq_h \right| \le O(1)\|A\|\, \|B\|\, e^{-\gamma T}, \tag{1.1.16}$$

uniformly in $h$. We shall choose $t_j = j$, $j \in \mathbb{Z}$." *(b) p. 600.* "Proof of Theorem 1.1.8, assuming Theorem 1.1.10 [sic]. Choose $T = 0$ and $B = I$ in Theorem 1.1.11. Then $Q = 0$, so Theorem 1.1.8 is a special case of Theorem 1.1.10 [sic]." *(c) p. 629.* "Theorem 1.1.11 follows from this inequality, while Theorem 1.1.8 is the special case $B = I$, $n = 0$."

*Reading (vi) off the source fact.* Take $n = 0$ and $B = I$ in Theorem 1.1.11, as the source does on p. 600 ($B = I$, $T = 0$) and names on p. 629 ($B = I$, $n = 0$). For $0 < T < 1$ the sum (1.1.15) over the integers $0 < t_j < T$ is empty, so $Q = 0$ and $I_T = I$, and (1.1.16) reads $|\int A\, dq_h| \le O(1)\|A\|\, \|I\|\, e^{-\gamma T} \le O(1)\|A\|\, \|I\|$, uniformly in $h$, for every monomial $A$ of type (1.1.5) localized at times $\le 0$; the source's own choice $T = 0$ (p. 600) gives the same bound, and $0 < T < 1$ is used here because the source introduces $B_T$ for $T > 0$. By (1.1.7)–(1.1.8), $\|I\| = K_1$ (no Wick factors: the product is empty and the kernel is the constant 1), so ($U_{\mathrm{GJS}}$) holds with

$$C_8 := O(1)\, K_1,$$

$O(1)$ the constant of Theorem 1.1.11 at $n = 0$ and a fixed $\varepsilon \in (0,1)$; it holds for every $h \in \mathcal{C}_{\mathrm{E}}$, in particular on $\mathcal{C}_{\mathrm{E}}^{\mathrm{slab}}$, and $C_8$ depends on the printed thresholds only. Three remarks make the reading exact. *(1)* A sharp-time monomial at an arbitrary time $x^0$ is carried to a time in $(-1, 0]$ by an integer time translation, which maps the lattice squares of p. 593

onto themselves, preserves the norms (1.1.7)–(1.1.8), the class $\mathcal{C}_{\mathrm{E}}$ and the measure $d\mu_0$, and carries $dq_h$ to $dq_{h'}$ with $h'$ the translated cutoff; the bound at $h'$ is the bound at $h$. *(2)* A general $Q(x^0)$ with kernel of bounded support is a finite sum of localized monomials, and $\|Q(x^0)\|_{\mathrm{GJS}}$ is the sum of their norms by (1.1.9); the bound is applied to each piece, whose times are all equal, as the source requires for Theorems 1.1.10–1.1.11 (p. 596). *(3)* The smallness hypotheses under which Theorem 1.1.11 holds are not printed in its statement; the source prints them with Theorem 4.1, from which Theorems 1.1.8 and 1.1.11 are proved on pp. 628–629 ("Let $m_0$, $K_2$ and $K_1$ be sufficiently large, and let $\lambda$ be sufficiently small"), and on p. 600 ("convergence of our expansions requires both sufficiently small $\lambda$ and sufficiently large $m_0$"); §2.6 fixes the working constants accordingly. The admissibility of the degree-zero $B = I$ rests on the source's own use of it (pp. 597, 600, 629) and on p. 593's "there is no restriction on $r$".

*(vii) Printed corroboration of (vi).*

**Source Fact 2.4** ([7, p. 629, §4])**.** "It is important to work with a translation invariant Hamiltonian, in order that time translation preserve subspaces of bounded momentum. The estimates of Theorems 1.1.8, 1.1.11, 3.1 and 4.1 are uniform in the space cutoff $h$, and the path space integrals converge as $h \to 1$. Thus these theorems hold in the infinite volume limit, and we proceed to work in the limit $h = 1$."

*(vii′) Printed use of the quantifier by the source (corroboration).* The following passages show the source stating the cutoff-uniform quantifier of ($U_{\mathrm{GJS}}$) and consuming it for arbitrary cutoff functions of its class, not only in the limit $h \to 1$. The source's $V(g)$ is the interaction $\mathcal{U}_g$ of item (i); its $\langle F\rangle_h$ is $\int F\, dq_h$.

**Source Fact** ([7, pp. 594–597 and 629])**.** *(a) Proof of Theorem 1.1.1, pp. 594–595.* "For test functions $f_i \in C_0^\infty(\mathcal{O})$ and for cutoff functions $h_1$ and $h_2$, we assert that

$$\Big|\int \prod_i \Phi(f_i)\, dq_{h_1} - \int \prod_i \Phi(f_i)\, dq_{h_2}\Big| \le 0(1)e^{-m_1 d/2} \tag{1.1.11}$$

[sic: p. 594 prints the prefactor of (1.1.11) with an upright digit zero, "0(1)", where the $O(1)$ of (1.1.10) on the same page is meant; transcribed as printed since 7 November 2026] where $d = \operatorname{dist}(\mathcal{O}, \operatorname{suppt} h_1 - h_2)$, and then Theorem 1.1.1 follows. To prove (1.1.11) let $g = h_2 - h_1$, and define $g_i = g\chi_{\Delta_i}$, where $\chi_{\Delta_i}$ is the characteristic function of $\Delta_i$. Also we use the notation $\langle F\rangle_h = \int F\, dq_h$, and set $Q = \prod_i \Phi(f_i)$. Then

$$\begin{aligned}
\langle Q\rangle_{h_2} - \langle Q\rangle_{h_1} &= \int_0^1 d\alpha\, \frac{d}{d\alpha}\langle Q\rangle_{h_1+\alpha g} \\
&= -\int_0^1 d\alpha \big[\langle QV(g)\rangle_{h_1+\alpha g} - \langle Q\rangle_{h_1+\alpha g}\langle V(g)\rangle_{h_1+\alpha g}\big] \\
&= -\int_0^1 d\alpha \sum_{i\in\mathbb{Z}^2} \big[\langle QV(g_i)\rangle_{h_1+\alpha g} - \langle Q\rangle_{h_1+\alpha g}\langle V(g_i)\rangle_{h_1+\alpha g}\big].
\end{aligned} \tag{1.1.12}$$

We apply Theorem 1.1.7 to each term in the sum to obtain

$$|\langle Q\rangle_{h_2} - \langle Q\rangle_{h_1}| \le \sum_{i\in\mathbb{Z}^2\cap \operatorname{supp} g} O(1)e^{-m_1 \operatorname{dist}(\mathcal{O},\Delta_i)} \le O(1)e^{-m_1 d/2}.$$

As a special case, the cutoff Schwinger functions $S_h$ converge as $h \to 1$."

*(b) Proof of Theorem 1.1.7, pp. 596–597.* "Proof of Theorem 1.1.7, assuming Theorems 1.1.8 and 1.1.11. Take $n = 0$ in Theorem 1.1.11 Then $Q$ is a polynomial of degree zero, i. e., a constant. From the choice $B = I$, it follows that

$$Q = \int A\, dq_h + O(1)\|A\| e^{-m_0(1-\varepsilon)T}.$$

After rotating and translating the strip separating $Q_1$ and $Q_2$ in Theorem 1.1.7, and after choosing $A = Q_1$, $B_T = Q_2$ in Theorem 1.1.11, the proof follows from Theorem 1.1.8, (1.1.16) and the above determination of $Q$." The estimate (1.1.16), p. 596, reads "$|\int (A - Q) B_T\, dq_h| \leq O(1)\|A\|\, \|B\| e^{-\gamma T}$, uniformly in $h$."

*(c) p. 629.* "Theorem 1.1.11 follows from this inequality, while Theorem 1.1.8 is the special case $B = I$, $n = 0$." The uniformity sentence of the same page is Source Fact 2.4.

*(viii) Scaling.* The unitary space-time dilation of [7, p. 590] sends $(m_0, \lambda)$ to $(s^{-1}m_0, s^{-2}\lambda)$; the exact reduction of the dimensionless smallness hypothesis to the printed one is Proposition 3.2.

*Scope.* Every item of this assumption is a printed statement of [7], used at the scope at which it is printed; this is the standard under which Assumptions A1–A7 are imported, and no stricter one is available to this manuscript, since the printed proof of Theorem 1.1.7 itself obtains its $h$-independent constant from Theorem 1.1.8 at the same $h$ (p. 597) while Theorem 1.1.8 is printed only "uniformly as $h \to 1$" (p. 595). Four points of scope are recorded. *(a) Item (vi), retired.* From 16 August to 30 October 2026 the slab-uniform estimate ($U_{\mathrm{GJS}}$) was carried as a named hypothesis of Theorem 12.2, documented on 5 September 2026 as the quantifier the source states on p. 629 and consumes on pp. 594–597 (items (vii), (vii′)) but "assembled from four pages of the source rather than read off one theorem". The source fact of item (vi) shows that it is read off one printed theorem, Theorem 1.1.11, whose estimate (1.1.16) is printed "uniformly in $h$", in the specialization the source itself prints as its proof of Theorem 1.1.8; the reading of 5 September 2026 used the same pages as evidence for Theorem 1.1.8 and is the same mathematics with different bookkeeping. Accordingly, from 31 October 2026, item (vi) was imported as printed and Theorem 12.2 no longer carried it as a separate hypothesis; that was a re-sourcing, not a proof. Since 7 November 2026 item (vi) is no longer used: Appendix D derives the moment bounds of §§5 and 7 from item (iv) with Lemma C.2, Lemma C.4 and Theorem C.5, so that Theorem 1.1.7 is the one theorem of the source imported. The manuscript still does not re-derive the cluster expansion; Appendix C audits the source mechanism and proves the analytic prerequisites that Appendix D consumes. Items (v)–(vii′) are kept as provenance. *(b) Item (iv).* GJS Theorem 1.1.7 is printed for localized monomials separated by a strip of any orientation (p. 594). Its printed proof rotates the strip into the time direction (p. 597) and so reaches sharp-time monomials only across strips parallel to the time-zero line. Lemma 7.4 uses only that case; Appendix D also uses the transverse case, listed separately as A8(iv)-T. The price of retiring item (vi) is therefore that the printed statement of Theorem 1.1.7 is used in that case (Theorem D.4, hence Theorem 5.1 and all that consumes $M$); the uniform gap of Theorem 7.8 rests on the parallel case alone (Lemma 7.4, Theorem D.3). *(c)* The source's own proof of Theorem 1.1.7 obtains its constant from Theorem 1.1.8 at the same $h$ (p. 597); this manuscript does not re-verify that proof, exactly as it re-verifies no proof behind Assumptions A1–A7. *(d) Readings made explicit.* Theorem 1.1.7's "independent of $h$" is read over the class of p. 589 (compact support, $0 \leq h \leq 1$), the only class of space-time cutoffs $h$ of the measures $dq_h$ that the source defines; the quantifier "uniformly in $h$" of (1.1.16), in the retired item (vi), was read the same way, and p. 629 states the uniformity of Theorems 1.1.8 and 1.1.11 "in the space cutoff $h$" (Source Fact 2.4). The constants $O(1)$ of (1.1.10) and (1.1.16) are read as

depending on the standing thresholds (and, for (1.1.16), on $n$ and $\varepsilon$) only, not on $Q_1$, $Q_2$, $A$, $B$, $T$ or $h$, as the source's $O(1)$ is read throughout this manuscript (Source Facts 2.2–2.3); §7 needs this for observables of unbounded total degree. The stability input (2.4.34) of the source's proof (Appendix C, M6) enters Theorem 1.1.7; under the import standard it belongs to the proof this manuscript does not re-verify. The monomials (1.1.5)–(1.1.6), which GJS print as formal integrals with ": :" denoting "Wick order with respect to the measure $d\Phi$" (p. 589), are read as the objects constructed in Appendix D (Definition N.26), and the source's $V(h)$ of (1.8), p. 589, as the $\mathcal{U}_h$ of Assumption A3 (Remark N.2(iv)); items 1 and 10 of `part_i/C5_reading_record.md` ask a human reader to confirm the printed text these readings rest on (the definition (1.8) on p. 589; the forms (1.1.5)–(1.1.6) on pp. 593 and 595). The formerly illegible norm prefactor was settled by page-image inspection: $K_1$ enters to the first power and the local weight is $K_2^{2N(\Delta)}$ (gate R2, `m4/r2-check.tex`). All constants in this manuscript use the exact reading (1).

*Used in.* §3 (scaling), §4, §5 (items (ii)–(iii): Steps 1–4), §6 (norms), §7 (CE2, parallel case), Appendix C, Appendix D (CE2 in both orientations; items (i)–(iii)); items (v), (vi), (vii) and (vii′) are used in no proof.
*Dossier.* `gjs-source-05/06`, `gjs-p589-06`, `gjs-scaling-07`, `gjs-theorems-10/11/12`, `gjs-core-11/12`, `gjs-uniform-46`, `gjs-uniformity-46`, `gjs-proof-14` (E20–E26, E54); `gjs-thm1111-13` (p. 596), `gjs-proof118-17` (p. 600), `gjs-thm41-45` (p. 628), added 31 October 2026.

**Remark 2.5** (conventions lock)**.** The sharp-time restriction of the Euclidean covariance equals $C$: $\int \frac{dk_0}{2\pi}(k_0^2+\mu(k)^2)^{-1} = (2\mu(k))^{-1}$ exactly. Hence sharp-time $d\mu_0$-Wick ordering coincides, constant by constant, with the $C$-Wick ordering used in $V(h)$, and sharp-time functionals of $\Phi(0,\cdot)$ are multiplication operators on $\mathcal{H}$. No Wick-mass conversion arises anywhere below.

## 2.5 Immediate consequences of the imports

**Proposition 2.6** (domain smoothing at the ground state)**.** $\Omega_g \in D(V(h))$ *for every* $g \in \mathcal{C}_{\mathrm{H}}$ *and every real* $h \in L^1 \cap L^2$ *of compact support.*

*Proof.* By Assumption A5(i), for $t > 0$ the range of $e^{-tK(g)}$ is contained in $D(V(h))$. (Replacing $K(g)$ by $H(g) = K(g) - E(g)$ multiplies the semigroup by the nonzero scalar $e^{tE(g)}$ and does not change its range.) Since $K(g)\Omega_g = E(g)\Omega_g$,

$$\Omega_g = e^{tE(g)}e^{-tK(g)}\Omega_g \in \operatorname{Ran} e^{-tK(g)} \subseteq D(V(h)). \qquad \square$$

**Proposition 2.7** (cutoff propagation, uniformly in $g$)**.** $\beta_t^g(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|})$ *for every* $g \in \mathcal{C}_{\mathrm{H}}$ *and* $t \in \mathbb{R}$*, with propagation speed* $\leq 1$ *independent of* $g$.

*Proof.* The interaction automorphism $\operatorname{Ad} e^{itV(g)}$ has speed zero: let $A$ be generated by operators $W(f,0)$, $W(0,h)$ with $f$, $h$ supported in a compact set $B \subset O$; replacing $B$ by its convex hull, a compact interval contained in the interval $O$, we may take $B$ to be an interval. Pick $\chi \in C_c^\infty(O)$ with $\chi = 1$ on an open interval $O'' \supseteq B$; then $V(g) = V(\chi g) + V((1-\chi)g)$ with strongly commuting summands (multiplication operators), and $e^{itV(\chi g)} \in \mathfrak{A}(O)$ by *Locality* in §2.1, since $\operatorname{supp}(\chi g) \subseteq \operatorname{supp}\chi \subset O$. If $(1-\chi)g \neq 0$, its support is compact and does not meet $O''$, so $d := \operatorname{dist}(B, \operatorname{supp}((1-\chi)g)) > 0$; the open interval $O' := \{x : \operatorname{dist}(x,B) < d/2\}$ contains $B$, so $A \in \mathfrak{A}(O')$, and $\operatorname{dist}(\operatorname{supp}((1-\chi)g), O') \geq d/2 > 0$, so $e^{itV((1-\chi)g)} \in \mathfrak{A}(O')'$ by *Locality*. Hence $\operatorname{Ad} e^{itV(g)}(A) = e^{itV(\chi g)}Ae^{-itV(\chi g)} \in \mathfrak{A}(O)$; extend by strong density. The free dynamics has speed one (Assumption A6(i)). The $n$-th Trotter approximant of $\beta_t^g(A)$ then lies in $\mathfrak{A}(O_{|t|})$, which is strongly closed, and the Trotter formula converges strongly (Assumption A6(i)). $\square$

**Lemma 2.8** (covariance and parity of the cutoff states)**.** *For every $g \in \mathcal{C}_{\mathrm{H}}$ and $a \in \mathbb{R}$:*

$$\omega_{\tau_a g} = \omega_g \circ \tau_{-a}. \tag{T}$$

*For even $\mathscr{P}$: $\omega_g \circ \theta = \omega_g$.*

*Proof.* By Assumption A7, $U_a K(g) U_a^* = K(\tau_a g)$, so by simplicity (Assumption A1) $\Omega_{\tau_a g} = U_a \Omega_g$ up to a phase, which cancels in the state; this is (T).

For parity: $\Theta = \Gamma(-1)$ is unitary, self-adjoint, commutes with $H_0$, and for even $\mathscr{P}$ satisfies $\Theta V(g) \Theta = V(g)$ (Assumption A7); hence $\Theta\Omega_g = c\Omega_g$ with $c \in \{\pm 1\}$ by simplicity and $\Theta^2 = 1$. In $Q$-space $(\Theta\psi)(q) = \psi(-q)$, so $\Theta\Omega_g > 0$ a.e. together with $\Omega_g > 0$ a.e. (Assumption A1) forces $c = +1$. Thus $\omega_g(\theta(A)) = \langle \Theta\Omega_g, A\Theta\Omega_g \rangle = \omega_g(A)$. □

### 2.6 Standing weak-coupling assumption

Fix source-admissible constants $M_* > 0$ and $\lambda_* > 0$ such that the hypotheses printed for the one theorem imported from [7] hold at bare mass $M_*$ whenever $0 \le \lambda' < \lambda_*$: for Theorem 1.1.7, "Let $m_0$, $K_2$ and $K_1$ be sufficiently large and let $\lambda$ be sufficiently small" (p. 594). (Until 7 November 2026 the retired Theorem 1.1.11 was imported as well, whose statement prints no smallness hypothesis; the same hypotheses are printed with Theorem 4.1, from which it is proved, p. 628, and on p. 600.) Define

$$\varepsilon_{\mathscr{P}} := \frac{\lambda_*}{M_*^2}.$$

Throughout, $0 \le \lambda / m_0^2 < \varepsilon_{\mathscr{P}}$. By Proposition 3.2 and Remark 3.3 we may, and from now on do, assume that the pair $(m_0, \lambda)$ itself satisfies the printed source hypotheses ($m_0 = M_*$ and $\lambda < \lambda_*$), so that the *unit-lattice* statements of Assumption A8 apply literally in the working coordinates; $m_1 > 0$ and $C_7, K_1, K_2$ denote the printed constants. Remark 3.3 records how each conclusion transports back to the original coordinates.

## 3 Dimensionless weak-coupling reduction

**Lemma 3.1** (class compatibility)**.** (i) $\mathcal{G} \subset \mathcal{C}_{\mathrm{H}}$. (ii) *For every $g \in \mathcal{C}_{\mathrm{H}}$ and every compact interval $I \subset \mathbb{R}$, the slab cutoff $\mathbf{1}_I \otimes g$ belongs to $\mathcal{C}_{\mathrm{E}}$; in particular $h_{T,g} = \mathbf{1}_{[-T,T]} \otimes g \in \mathcal{C}_{\mathrm{E}}$ and $\mathbf{1}_{[0,T]} \otimes g \in \mathcal{C}_{\mathrm{E}}$.* (iii) *For $g, g' \in \mathcal{C}_{\mathrm{H}}$ and $s \in [0,1]$, the affine path $g_s := (1-s)g + sg'$ stays in $\mathcal{C}_{\mathrm{H}}$, and $\mathbf{1}_I \otimes g_s = (1-s)\,\mathbf{1}_I \otimes g + s\,\mathbf{1}_I \otimes g'$ for every compact interval $I$, so that $s \mapsto \mathcal{U}_{\mathbf{1}_I \otimes g_s} = (1-s)\mathcal{U}_{\mathbf{1}_I \otimes g} + s\mathcal{U}_{\mathbf{1}_I \otimes g'}$ is affine. If moreover $g, g' \in \mathcal{G}$ and $\{g = 1\}^\circ \cap \{g' = 1\}^\circ \neq \varnothing$, then $g_s \in \mathcal{G}$ with $\{g_s = 1\}^\circ \supseteq \{g = 1\}^\circ \cap \{g' = 1\}^\circ$; in particular $d(g_s, O) \ge \min\{d(g, O), d(g', O)\}$ for every bounded interval $O$.*

*Proof.* (i) A function in $C_c^\infty(\mathbb{R})$ with values in $[0,1]$ is measurable with compact support. (ii) $\mathbf{1}_I \otimes g$ is a product of measurable functions, takes values in $[0,1]$, and $\operatorname{supp}(\mathbf{1}_I \otimes g) \subseteq I \times \operatorname{supp} g$ is compact. (iii) A convex combination of $[0,1]$-valued measurable functions is $[0,1]$-valued and measurable, and $\operatorname{supp} g_s \subseteq \operatorname{supp} g \cup \operatorname{supp} g'$ is compact; the identity $\mathbf{1}_I \otimes g_s = (1-s)\,\mathbf{1}_I \otimes g + s\,\mathbf{1}_I \otimes g'$ holds pointwise, and $h \mapsto \mathcal{U}_h$ is linear in $h$ on the common $L^2(d\mu_0)$ domain guaranteed by Assumption A3. For the last statement: $g_s \in C_c^\infty$, $0 \le g_s \le 1$, and $\{g_s = 1\} \supseteq \{g = 1\} \cap \{g' = 1\}$ (for $s \in (0,1)$ a convex combination of two numbers in $[0,1]$ equals 1 if and only if both do, and for $s \in \{0,1\}$ the inclusion is trivial); passing to interiors gives the inclusion, which is a nonempty open set by hypothesis. The distance bound follows since $d(\cdot, O)$ is monotone in the plateau interior. □

**Proposition 3.2** (dimensionless weak-coupling reduction)**.** *The standing inequality* $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$ *implies the source smallness hypothesis of Assumption A8. More precisely, the GJS unitary dilation with* $s = m_0/M_*$ *sends*

$$(m_0, \lambda) \longmapsto (s^{-1}m_0, s^{-2}\lambda) = \left(M_*, \frac{M_*^2}{m_0^2}\lambda\right),$$

*and the transformed coupling is strictly smaller than* $\lambda_*$*.*

*Proof.* For $m > 0$, write the Euclidean covariance, as an identity of tempered distributions, as

$$C_m(z) = \frac{1}{(2\pi)^2}\int_{\mathbb{R}^2} \frac{e^{ip\cdot z}}{p^2 + m^2}\, dp.$$

Since $sM_* = m_0$, the substitution $q = sp$ gives, without an omitted power of $s$,

$$\begin{aligned} C_{M_*}(sz) &= \frac{1}{(2\pi)^2}\int_{\mathbb{R}^2} \frac{e^{ip\cdot sz}}{p^2 + M_*^2}\, dp \\ &= \frac{1}{(2\pi)^2}\int_{\mathbb{R}^2} \frac{s^{-2}e^{iq\cdot z}}{s^{-2}q^2 + M_*^2}\, dq = C_{m_0}(z). \end{aligned}$$

Thus the centered generalized Gaussian fields $\Phi_{m_0}(x)$ and $\Phi_{M_*}(sx)$ have identical joint covariances after smearing. This also fixes Wick ordering without a hidden constant. Indeed, for every regularized centered Gaussian variable $X$,

$$:e^{tX}: = \exp\bigl(tX - \tfrac{1}{2}t^2\mathbb{E}[X^2]\bigr),$$

and the coefficient of $t^n$ is $:X^n:/n!$. Apply the identity to a common mollifier, transported by $x \mapsto sx$, and then take the defining distributional limit. Equality of the regularized covariances shows that the unitary Gaussian identification sends $:\mathscr{P}(\Phi_{m_0}(x)):_{m_0}$ to $:\mathscr{P}(\Phi_{M_*}(sx)):_{M_*}$ with the coefficients of $\mathscr{P}$ unchanged. For a Euclidean cutoff $h$ the change of variables $y = sx$ therefore gives

$$\lambda\int_{\mathbb{R}^2} h(x):\mathscr{P}(\Phi_{m_0}(x)):_{m_0}\, dx = s^{-2}\lambda\int_{\mathbb{R}^2} h(y/s):\mathscr{P}(\Phi_{M_*}(y)):_{M_*}\, dy.$$

The transformed cutoff $h_s(y) := h(y/s)$ still satisfies $0 \le h_s \le 1$ and has compact support, so $h_s \in \mathcal{C}_{\mathrm{E}}$. Hence the transformed parameters are exactly those displayed in the proposition, as stated in [7, p. 590]. Finally,

$$s^{-2}\lambda = \frac{M_*^2}{m_0^2}\lambda < M_*^2\varepsilon_{\mathscr{P}} = \lambda_*.$$

This is precisely the source smallness condition. If the source decay rate in the scaled coordinate $y = sx$ is $m_{1,*} > 0$, then $e^{-m_{1,*}d_y} = e^{-sm_{1,*}d_x}$; thus the rate in the original coordinates is $m_1 := sm_{1,*} > 0$. The transport of the remaining structure is Remark 3.3. □

**Remark 3.3** (coordinate convention and transport of the conclusions)**.** The dilation behind Proposition 3.2 identifies the model at $(m_0, \lambda)$ with the model at $(M_*, \lambda')$, $\lambda' = (M_*/m_0)^2\lambda < \lambda_*$, as follows. Let $s = m_0/M_*$, $y = sx$, and let $\sigma_s$ be the scaling automorphism of the Weyl algebra, $\sigma_s(W(f,h)) = W(s^{-1}f(\cdot/s), h(\cdot/s))$ (a symplectic map). The sharp-time covariance is scale-covariant between the two masses, $C_{M_*}(s\,z) = C_{m_0}(z)$ (the one-dimensional analogue of the substitution in the proof of Proposition 3.2), so $\sigma_s$ carries the mass-$m_0$ Fock state onto the mass-$M_*$ Fock state and is unitarily implemented by some $U_s$ with $U_s\Omega_0^{(m_0)} = \Omega_0^{(M_*)}$;

moreover $\sigma_s(\mathfrak{A}(O)) = \mathfrak{A}(sO)$. On the generators, $\mu_{M_*}(k/s) = s^{-1}\mu_{m_0}(k)$ (since $sM_* = m_0$) gives $U_s H_0^{(m_0)} U_s^* = s\, H_0^{(M_*)}$, and the Euclidean Wick identification of Proposition 3.2, restricted to sharp time (Remark 2.5), gives $U_s V_{m_0,\lambda}(g) U_s^* = s\, V_{M_*,\lambda'}(g(\cdot/s))$; hence

$$U_s\, K_{m_0,\lambda}(g)\, U_s^* = s\, K_{M_*,\lambda'}(g(\cdot/s)), \qquad g \in \mathcal{C}_{\mathrm{H}}.$$

Consequently $g \mapsto g(\cdot/s)$ is an order isomorphism of $\mathcal{G}$ onto $\mathcal{G}$ (and a bijection of $\mathcal{C}_{\mathrm{H}}$), cutoff ground states correspond under $U_s$, and the two cutoff nets of states are carried onto each other. Every conclusion of Theorem 12.2, proved in the working (scaled) coordinates, therefore holds for the original model with the dictionary: localization $O \leftrightarrow sO$; plateau distance $d \leftrightarrow s\,d$, so the rate becomes $\Psi_{m_{1,*}}(s\,d)$, again decaying faster than every inverse power of $d$; dynamics $\alpha_t \leftrightarrow \alpha_{st}$, the light cone being form-invariant; gap $m_1 := s\, m_{1,*}$; and the local-moment clause of item (5) transports to supports of length $2/s$ — equivalently, it holds for supports of length 2 after splitting $h$ by a partition of unity into at most $\lceil s \rceil + 1$ pieces of support length $\le 2/s$ (finite additivity of $V$), with constant $(\lceil s \rceil + 1)$ times the transported one. All displayed constants of §§4–10 and of Appendices A, C and D are the constants of the working coordinates, in which the unit-lattice geometry of Assumption A8 is literal.

# 4 Slab transfer

Throughout this section and §§5, 7, a cutoff $g \in \mathcal{C}_{\mathrm{H}}$ is fixed where limits $T \to \infty$ are taken; *no uniformity in $g$ beyond the printed constants is used, and no gap is assumed.* Set

$$h_{T,g} := \mathbf{1}_{[-T,T]} \otimes g \in \mathcal{C}_{\mathrm{E}} \quad \text{(Lemma 3.1(ii))},$$
$$\psi_s := e^{-sH(g)}\Omega_0, \qquad \hat{\psi}_s := \psi_s/\|\psi_s\|, \qquad c_g := \langle \Omega_g, \Omega_0 \rangle > 0$$

($c_g > 0$ by Assumption A1: $\Omega_g > 0$ a.e., $\Omega_0 \equiv 1$).

**Lemma 4.1** (slab FKN; bounded nonnegative arguments)**.** *For $T > 0$, $t \in [0, 2T]$, $s \in [-T, T]$, and bounded nonnegative measurable functions $F_1, F_2$ of the time-zero field:*

$$\int F_1\big(\Phi(-\tfrac{t}{2}, \cdot)\big) F_2\big(\Phi(\tfrac{t}{2}, \cdot)\big)\, dq_{h_{T,g}} = \frac{\langle F_1 \psi_{T-t/2},\, e^{-tH(g)} F_2 \psi_{T-t/2} \rangle}{\|\psi_T\|^2}, \tag{2}$$

$$\int F_1\big(\Phi(s, \cdot)\big)\, dq_{h_{T,g}} = \frac{\langle \psi_{T-s},\, F_1\, \psi_{T+s} \rangle}{\|\psi_T\|^2}; \tag{3}$$

*all energy shifts cancel in the ratios.*

*Proof.* This is the Feynman–Kac–Nelson formula for $Z(h_{T,g})^{-1} e^{-\mathcal{U}_{h_{T,g}}}\, d\mu_0$ (Markov property of $d\mu_0$ along constant-time lines; Assumption A2; the measure is well-defined by Assumption A3 and Lemma 3.1(ii); the identification of sharp-time Wick functionals with $Q$-space multiplications is Remark 2.5). The unshifted formula contains $K(g)$ and the same factor $e^{-2TE(g)}$ in numerator and partition function; multiplying every semigroup by the corresponding $e^{aE(g)}$ replaces $K(g)$ by $H(g)$ and cancels that factor exactly. For bounded nonnegative $F_i$ both sides are finite. □

**Lemma 4.2** (signed bounded extension)**.** *Equations (2)–(3) hold for bounded signed measurable $F_1, F_2$.*

*Proof.* Write $F_i = F_i^+ - F_i^-$ and expand both sides into the same four (respectively two) finite nonnegative terms of Lemma 4.1; bilinearity (respectively linearity) proves the identities. □

**Lemma 4.3** (unbounded extension)**.** (i) *For nonnegative measurable $F_1$ and $F_2$, the identities (2)–(3) hold in $[0,\infty]$.* (ii) *For real measurable $F$, define the clipped truncation $F^{[\ell]} := (-\ell)\vee(F\wedge\ell)$, $\ell>0$. Whenever $F\psi_a\in\mathcal{H}$,*

$$\|(F-F^{[\ell]})\psi_a\|^2=\int|F-F^{[\ell]}|^2|\psi_a|^2\,d\phi_0\longrightarrow 0\qquad(\ell\uparrow\infty),$$

*by dominated convergence with dominator $|F|^2|\psi_a|^2$; and on path space*

$$|F_1(\Phi(s_1,\cdot))F_2(\Phi(s_2,\cdot))|\le\tfrac12\big(F_1(\Phi(s_1,\cdot))^2+F_2(\Phi(s_2,\cdot))^2\big)$$

*is a dominated-convergence majorant whenever the two square moments are finite. Consequently the signed identities used below are limits of the bounded signed identities of Lemma 4.2, with no formal manipulation of products of unbounded operators.*

*Proof.* (i) Apply Lemma 4.1 to $F_i\wedge\ell$ and let $\ell\uparrow\infty$; monotone convergence on both sides gives the identities in $[0,\infty]$. (ii) The Hilbert-space limit is dominated convergence as displayed. Membership $\psi_a\in D(F)$ is supplied at each application. For the interaction-type multiplications it follows from Assumption A5(i). For the cylinder monomials of §7, part (i) at $t=0$ over the $a$-slab gives $\|F\psi_a\|^2/\|\psi_a\|^2=\int F(\Phi(0,\cdot))^2\,dq_{h_{a,g}}$ in $[0,\infty]$; at every sharp time $s$ a cylinder monomial has under $d\mu_0$ the law of the same polynomial in the Gaussian variables $\varphi_0(f_i)$ under $d\phi_0$ (Lemma N.35), so $F(\Phi(s,\cdot))\in L^4(d\mu_0)$, and Hölder's inequality with exponents $(2,2)$ and Assumption A3 give, for every $h\in\mathcal{C}_{\mathrm{E}}$,

$$\int F(\Phi(s,\cdot))^2\,dq_h\le Z(h)^{-1}\,\|F(\Phi(s,\cdot))\|^2_{L^4(d\mu_0)}\,\|e^{-\mathcal{U}_h}\|_{L^2(d\mu_0)}<\infty,$$

since $Z(h)>0$ and $e^{-2\mathcal{U}_h}\in L^1(d\mu_0)$ by Assumption A3. With $h=h_{a,g}$ and $s=0$ this gives $F\psi_a\in\mathcal{H}$. On path space, $F_i^{[\ell]}(\Phi(s_i,\cdot))\to F_i(\Phi(s_i,\cdot))$ pointwise with the displayed majorant, whose integrability is exactly the finiteness of the two square moments; for the cylinder monomials of §7, at each fixed $T$ the same bound with $h=h_{T,g}$ supplies them, and no uniformity in $T$ is used here. Dominated convergence then passes Lemma 4.2 to the limit on both sides. □

**Lemma 4.4** (slab vacuum; simplicity only)**.** *$\hat\psi_s\to\Omega_g$ in norm, $\|\psi_s\|\downarrow c_g$, and $\|\psi_{s'}\|/\|\psi_s\|\in[1,1/c_g]$ for $0\le s'\le s$; the ratio tends to 1 whenever $s,s'\to\infty$.* No isolation of the ground-state eigenvalue is used.

*Proof.* Set $P_g=|\Omega_g\rangle\langle\Omega_g|$. Since $P_g\Omega_0=c_g\Omega_g$,

$$\psi_s=c_g\Omega_g+e^{-sH(g)}(1-P_g)\Omega_0,$$

and the two summands are orthogonal. If $\mu_g^{\perp}$ is the spectral measure of $(1-P_g)\Omega_0$, then

$$\|e^{-sH(g)}(1-P_g)\Omega_0\|^2=\int_{(0,\infty)}e^{-2sE}\,d\mu_g^{\perp}(E)\longrightarrow 0$$

by dominated convergence. The measure has no atom at 0 by simplicity alone (Assumption A1); no positive distance from zero is used. It follows that $\psi_s\to c_g\Omega_g$, $\|\psi_s\|\to c_g$, and $\hat\psi_s\to\Omega_g$. The spectral formula also shows that $s\mapsto\|\psi_s\|$ is decreasing. Since $c_g\le\|\psi_s\|\le\|\psi_0\|=1$, for $0\le s'\le s$,

$$1\le\frac{\|\psi_{s'}\|}{\|\psi_s\|}\le\frac{1}{c_g}.$$

If both indices tend to infinity, numerator and denominator tend to $c_g>0$, so their ratio tends to 1. □

# 5 The uniform local interaction moment

**Theorem 5.1** (uniform local interaction moment ($\mathrm{M}_2$)). *Let $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. For every $g \in \mathcal{C}_{\mathrm{H}}$ and every real measurable $h$ with $\|h\|_\infty \le 1$ and $\operatorname{supp} h$ contained in an interval of length 2,*

$$\|V(h)\Omega_g\| \le M, \tag{$\mathrm{M}_2$}$$

*where*

$$M^2 := M_B^2 = e^{15\lambda|a_0|}\, S_2(B_{3\times5})\, 9^{p'}\, V_* + 9\, C_7\, c_U\, \overline{\Sigma}_{3\times5}\, \mathfrak{K}(p')\, \lambda^2 \Big(\sum_{n=0}^{2p'} |a_n|\Big)^2,$$

$$\mathfrak{K}(p') := K_1 K_2^{8p'} (4p')!,$$

*the constants $S_2(B_{3\times5})$ (finite, not evaluated), $V_*$, $c_U$ and $\overline{\Sigma}_{3\times5}$ being those of Theorem D.4 (Appendix D); $M$ depends on $(\lambda, \mathscr{P}, m_0)$ through $C_7$, $m_1$, $K_1$, $K_2$, $S_2(B_{3\times5})$ and $V_*$, in working coordinates (Remark 3.3), and on nothing else.*

*Proof. Step 1: the squared interaction as a GJS observable.* Let $J_0 : L^2(d\phi_0) \to L^2(d\mu_0)$ be the sharp-time embedding at time 0, $F(\varphi_0) \mapsto F(\Phi(0,\cdot))$, a multiplicative isometry (Assumption A2; the proofs of Lemmas N.21(i) and N.22), and put

$$Q_V := \big(J_0 V(h)\big)^2 = J_0\big(V(h)^2\big) = \lambda^2 \sum_{n,n'=0}^{2p'} a_n a_{n'} \iint h(x)h(y)\, {:}\Phi(0,x)^n{:}\,{:}\Phi(0,y)^{n'}{:}\, dx\, dy, \tag{4}$$

where the last expression is the sum of the localized monomials of type (1.1.5) at time 0 with two factors, of degrees $n, n'$, and kernels $\lambda^2 a_n a_{n'}\, h\mathbf{1}_{[k-\frac12,k+\frac12)} \otimes h\mathbf{1}_{[k'-\frac12,k'+\frac12)}$ over the columns $k, k'$ meeting $\operatorname{supp} h$; that $Q_V$ equals this sum $d\mu_0$-almost everywhere is Lemma N.43(i) (through the identification of the record's $V(h)$ with the constructed object, Lemma N.23′, and the agreement of the two regularizations at sharp time, Lemma N.22). No Wick reordering of the product is performed or required; the observable class of Assumption A8(ii) permits products of individually Wick-ordered powers at equal times, and by Remark 2.5 the sharp-time Wick ordering coincides with the $C$-ordering of $V(h)$.

*Step 2: column count.* The support of $h$ meets at most three columns $[k-\frac12, k+\frac12)$ in positive measure. Indeed, if it met four distinct columns in positive measure, the extreme indices would satisfy $k_{\max} \ge k_{\min} + 3$, and the containing interval would have infimum $< k_{\min} + \frac12$ and supremum $> k_{\max} - \frac12 \ge k_{\min} + \frac52$, hence length $> 2$.

*Step 3: local weight of one ordered column pair.* For one ordered pair of columns and degrees $n, n'$, the GJS local weight of (1) is bounded by $K_1 K_2^{8p'} (4p')!$. Indeed the total local degree is $n+n' \le 4p'$. If both variables belong to one square, the weight is $K_1 K_2^{2(n+n')}(n+n')!$; if to two squares, it is $K_1 K_2^{2n} n!\, K_2^{2n'} n'!$. In both cases

$$K_2^{2(n+n')} \le K_2^{8p'} \quad (K_2 \ge 1), \qquad n!\, n'! \le (n+n')! \le (4p')!.$$

*Step 4: the GJS norm of $Q_V$.* Sum over the at most $3^2 = 9$ ordered column pairs and all coefficient pairs. The kernel of the $(n, n', \Delta, \Delta')$ piece is $w(x,y) = h(x)\mathbf{1}_\Delta(x)\, h(y)\mathbf{1}_{\Delta'}(y)$, and $\|w\|_2 = \|h\mathbf{1}_\Delta\|_2 \|h\mathbf{1}_{\Delta'}\|_2 \le 1$ because $\|h\mathbf{1}_\Delta\|_2^2 = \int_\Delta h^2 \le 1$ ($\|h\|_\infty \le 1$, unit column). Hence

$$\|Q_V\|_{\mathrm{GJS}} \le 9\, K_1 K_2^{8p'} (4p')!\, \lambda^2 \Big(\sum_{n=0}^{2p'} |a_n|\Big)^2. \tag{5}$$

(Steps 2–4 are Lemma N.43(ii)–(iii); Step 5 does not consume them, since Theorem D.4 bounds $\|Q_V\|_{\mathrm{GJS}}$ itself; they are kept because §6 and Lemma 7.4 use the same counting.)

*Step 5: transfer at time zero.* For every $T > 0$, apply (2) at $t = 0$ with the two *individually nonnegative* functions $F_1 = F_2 = |V(h)|$ (Lemma 4.3(i)); their product is $V(h)^2$, whose sharp-time version is $Q_V$. Choose $a \in \mathbb{R}$ with $\operatorname{supp} h \subseteq [a, a+2]$. Theorem D.4, applied with the Euclidean cutoff $h_{T,g} \in \mathcal{C}_{\mathrm{E}}$ (Lemma 3.1(ii)) and with $h' := h$, gives

$$\|V(h)\hat{\psi}_T\|^2 = \int V(h)^2 \, dq_{h_{T,g}} = \int Q_V \, dq_{h_{T,g}} \le M_B^2 = M^2. \tag{6}$$

In particular the left side is finite, so $\psi_T \in D(V(h))$. (Theorem D.4 uses GJS Theorem 1.1.7 also across strips transverse to the time-zero line, the case listed as A8(iv)-T in Assumption A8, which its printed proof does not reach.)

*Step 6: weak-graph passage to $\Omega_g$.* By (6), the family $(V(h)\hat{\psi}_T)_{T>0}$ is bounded by $M$; choose $T_k \to \infty$ with $V(h)\hat{\psi}_{T_k} \rightharpoonup y$ weakly. The graph of the closed operator $V(h)$ is a closed linear subspace of $\mathcal{H} \oplus \mathcal{H}$, hence weakly closed. Since $\hat{\psi}_{T_k} \to \Omega_g$ strongly (Lemma 4.4), the pair $(\Omega_g, y)$ lies in the graph; thus $\Omega_g \in D(V(h))$, $y = V(h)\Omega_g$, and weak lower semicontinuity of the norm gives

$$\|V(h)\Omega_g\| \le \liminf_k \|V(h)\hat{\psi}_{T_k}\| \le M. \qquad \square$$

**Corollary 5.2** ($\varphi^4$ instance)**.** *For $\mathscr{P}(\xi) = \xi^4$ ($p' = 2$, $a_4 = 1$, $a_0 = \cdots = a_3 = 0$, $\sum_n |a_n| = 1$; degrees exactly $4 + 4 = 8$),*

$$M^2 = 81\, S_2(B_{3\times 5})\, V_* + 8\,709\,120\, C_7 K_1^2 K_2^{24}\, \overline{\Sigma}_{3\times 5}\, \lambda^3, \qquad V_* \le 24\,\lambda^2 \int_{-1}^{1}\!\!\int_{-1}^{1} C(x-y)^4 \, dx\, dy,$$

*with $C$ the kernel of $(2\mu)^{-1}$ (Lemma N.19): here $e^{15\lambda|a_0|} = 1$, $9^{p'} = 81$, $c_U = 24\lambda K_1 K_2^8$ (Lemma N.18), $9\,\mathfrak{K}(2)\lambda^2 = 9 \cdot 8!\, K_1 K_2^{16} \lambda^2 = 362\,880\, K_1 K_2^{16} \lambda^2$, and $24 \cdot 362\,880 = 8\,709\,120$ (Remark N.51). The value is not a number, since $S_2$ is not evaluated. Through Theorem D.4 it rests on A8(iv)-T.*

## 6 GJS norms of sharp-time polynomial observables

Throughout this section, $Q = \prod_{i=1}^{r} \varphi_0(f_i)$ with real $f_i \in C_c^\infty(\mathbb{R})$, all supported in an interval $I$ of length $L$, viewed at a common sharp time as an observable of the class of Assumption A8(ii). We call such $Q$ *cylinder monomials* and their linear span, together with $\mathbf{1}$, *cylinder polynomials*. Set

$$N_I := \#\{k \in \mathbb{Z} : \ |I \cap [k - \tfrac{1}{2}, k + \tfrac{1}{2})| > 0\},$$

the number of columns of the squares of [7, p. 593] (Assumption A8(ii); Definition N.1) that meet $I$ in positive measure.

**Lemma 6.1** (cell count)**.** $N_I \le \lfloor L \rfloor + 2 \le L + 2$. *The bound $L + 1$ is false for nonintegral $L$: the interval $I = (0.4, 0.6)$ has length 0.2 and meets both $[-\frac{1}{2}, \frac{1}{2})$ and $[\frac{1}{2}, \frac{3}{2})$ in positive measure.*

*Proof.* Let $k_{\min}$ and $k_{\max}$ be the smallest and largest indices of columns meeting $I$ in positive measure, so $N_I \le k_{\max} - k_{\min} + 1$. Positive-measure intersection with $[k_{\min} - \frac{1}{2}, k_{\min} + \frac{1}{2})$ forces $\inf I < k_{\min} + \frac{1}{2}$, and with $[k_{\max} - \frac{1}{2}, k_{\max} + \frac{1}{2})$ forces $\sup I > k_{\max} - \frac{1}{2}$; hence

$$L = \sup I - \inf I > (k_{\max} - \tfrac{1}{2}) - (k_{\min} + \tfrac{1}{2}) = k_{\max} - k_{\min} - 1.$$

The integer $k_{\max} - k_{\min} - 1$ is strictly less than $L$, hence at most $\lceil L \rceil - 1 \le \lfloor L \rfloor$. Consequently $N_I \le (k_{\max} - k_{\min} - 1) + 2 \le \lfloor L \rfloor + 2$. $\square$

**Proposition 6.2** (GJS norms of polynomial observables)**.**

$$\|Q\|_{\mathrm{GJS}} \le K_1 K_2^{2r} r!\, N_I^{r/2} \prod_{i=1}^{r} \|f_i\|_2, \tag{7}$$

$$\|Q^2\|_{\mathrm{GJS}} \le K_1 K_2^{4r} (2r)!\, N_I^{r} \prod_{i=1}^{r} \|f_i\|_2^2 =: \mathfrak{n}(Q)^2. \tag{8}$$

*Proof.* Expand the kernel of $Q$ into column assignments: decompose each $f_i = \sum_k f_i \mathbf{1}_{[k-\frac{1}{2},k+\frac{1}{2})}$, where only the $N_I$ columns meeting $I$ in positive measure contribute (almost everywhere). For any fixed assignment of the $r$ factors to columns, the localized piece (localized in the squares of the common row and these columns, Remark N.2(ii)) has, by (1), the weight $K_1 \prod_\Delta K_2^{2N(\Delta)} N(\Delta)!$ with $\sum_\Delta N(\Delta) = r$; the product of local factorials is at most the factorial of the total degree, $\prod_\Delta N(\Delta)! \le r!$, and the $K_2$-exponents sum to $2r$. The localized kernel is the tensor product of the restrictions, with $L^2$ norm $\prod_i \|f_i \mathbf{1}_{[k_i-\frac{1}{2},k_i+\frac{1}{2})}\|_2$. Summing over assignments ((1.1.9) of Assumption A8(iii)) and using, for each $i$,

$$\sum_{k:\,|I\cap[k-\frac{1}{2},k+\frac{1}{2})|>0} \|f_i \mathbf{1}_{[k-\frac{1}{2},k+\frac{1}{2})}\|_2 \le N_I^{1/2} \|f_i\|_2$$

by Cauchy–Schwarz over the $N_I$ columns, proves (7). Applying the same argument to the $2r$ factors of $Q^2$ (degree $2r$, $K_2$-exponent $4r$, factorial $(2r)!$, and $N_I^{1/2}$ per factor) proves (8). (Remark N.3 shows that (7) fails for the count on the cells $[j, j+1)$ used until 7 November 2026: for $f$ supported in $(0.4, 0.6)$ and even about $\frac{1}{2}$, the two column pieces give $\|Q\|_{\mathrm{GJS}} = \sqrt{2} K_1 K_2^2 \|f\|_2$ against a cell count of 1.) □

**Remark 6.3** (row independence)**.** The norm of a sharp-time monomial is independent of the common sharp time $s$: at every $s$ all factors occupy one temporal row, and changing $s$ changes only that common row label, not the spatial cell assignments, local degrees, kernel norms, or the product of weights. This is the precise statement; arbitrary time translation does not literally preserve the fixed lattice of unit squares.

## 7 The uniform gap

Throughout this section $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$, $g \in \mathcal{C}_{\mathrm{H}}$ is arbitrary and fixed, and $Q = \prod_{i=1}^r \varphi_0(f_i)$ is as in §6. Write $Q(s)$ for the same monomial at sharp Euclidean time $s$.

**Lemma 7.1** (moment transfer)**.** *If the interval $I$ of §6 has length $L > 0$, let $A$ and $B'$ be the constants of Theorem D.3 for the cutoff $g$, the interval $I$ and the time $s = 0$; if $L = 0$, put $A := B' := 0$ (then every $f_i$ vanishes and $Q = 0$). Then* $\sup_{T>0} \|Q\hat{\psi}_T\|^2 \le A\,\|Q^2\|_{L^2(d\phi_0)} + B'\,\mathfrak{n}(Q)^2$*; moreover $\Omega_g \in D(Q)$ and $\|Q\Omega_g\|^2 \le A\,\|Q^2\|_{L^2(d\phi_0)} + B'\,\mathfrak{n}(Q)^2$.*

*Proof.* If $L = 0$, every $f_i \in C_c^\infty(\mathbb{R})$ vanishes outside a set of at most one point, hence everywhere, so $Q = 0$ and there is nothing to prove. Let $L > 0$. Apply (2) at $t = 0$ with $F_1 = F_2 = |Q|$ (Lemma 4.3(i)); the Euclidean integrand is $Q^2$ at time 0. $Q^2$ is a cylinder monomial supported in $I$ (its factors are $f_1, \dots, f_r, f_1, \dots, f_r$), so Theorem D.3 with $F = Q^2$ and $s = 0$, and (8), give for every $T > 0$

$$\begin{aligned} \|Q\hat{\psi}_T\|^2 = \int Q(0)^2\, dq_{h_{T,g}} &\le A\,\|Q^2\|_{L^2(d\phi_0)} + B'\,\|Q(0)^2\|_{\mathrm{GJS}} \\ &\le A\,\|Q^2\|_{L^2(d\phi_0)} + B'\,\mathfrak{n}(Q)^2, \end{aligned}$$

the right side being finite by the Gaussian moments of the free field. Choose from the bounded family $Q\hat{\psi}_T$ a weakly convergent subnet $Q\hat{\psi}_{T_\alpha} \rightharpoonup y$. Since $\hat{\psi}_{T_\alpha} \to \Omega_g$ strongly (Lemma 4.4) and the graph of the closed multiplication operator $Q$ is a closed linear subspace of $\mathcal{H} \oplus \mathcal{H}$, hence weakly closed, $(\Omega_g, y)$ lies in that graph. Thus $\Omega_g \in D(Q)$, $y = Q\Omega_g$, and weak lower semicontinuity gives $\|Q\Omega_g\| \le \liminf_\alpha \|Q\hat{\psi}_{T_\alpha}\| \le \big(A\|Q^2\|_{L^2(d\phi_0)} + B'\mathfrak{n}(Q)^2\big)^{1/2}$. □

**Lemma 7.2** (one-point limit)**.** *For every fixed $s \in \mathbb{R}$,*

$$\int Q(s)\, dq_{h_{T,g}} \longrightarrow \langle \Omega_g, Q\Omega_g \rangle \qquad (T \to \infty).$$

*Proof.* For $T > |s|$, (3) (extended to $Q$ by Lemma 4.3; the square moments $\int Q(s)^2\, dq_{h_{T,g}}$ at the needed times are finite at each $T$ by the Hölder bound in the proof of Lemma 4.3(ii), and Theorem D.3 with $F = Q^2$ at sharp time $s$ bounds them by $A\|Q^2\|_{L^2(d\phi_0)} + B'\mathfrak{n}(Q)^2$ uniformly in $T$, with $A$ and $B'$ as in Lemma 7.1, since the box $B_s(g, I)$ is $B_0(g, I)$ moved by whole rows and $S_2$ is translation invariant, Proposition N.15(iii)) gives

$$\int Q(s)\, dq_{h_{T,g}} = \Big\langle \hat{\psi}_{T-s},\, \frac{Q\psi_{T+s}}{\|\psi_T\|} \Big\rangle \frac{\|\psi_{T-s}\|}{\|\psi_T\|}. \tag{9}$$

The first vector tends strongly to $\Omega_g$ and the last factor tends to 1 (Lemma 4.4). Write the middle vector as

$$\frac{Q\psi_{T+s}}{\|\psi_T\|} = \frac{\|\psi_{T+s}\|}{\|\psi_T\|}\, Q\hat{\psi}_{T+s}. \tag{10}$$

If $s \ge 0$, the scalar ratio in (10) is at most 1; if $s < 0$, it is at most $c_g^{-1}$, both by Lemma 4.4. (The inequality $\|\psi_{T+s}\| \le \|\psi_T\|$ holds only for $s \ge 0$; the two signs must be treated separately.) Thus (10) is bounded in either case, by Lemma 7.1. Every weak subsequential limit of (10) equals $Q\Omega_g$, by weak closedness of the graph of $Q$, strong convergence $\hat{\psi}_{T+s} \to \Omega_g$, and because the scalar ratio tends to 1. Hence the whole middle vector converges weakly to $Q\Omega_g$, and (9) converges to $\langle \Omega_g, Q\Omega_g \rangle$. □

**Lemma 7.3** (two-point lower semicontinuity)**.** *For every $t \ge 0$, with the Euclidean integrals over slabs of half-width $T \ge t/2$ (so that (2) applies),*

$$\langle Q\Omega_g, e^{-tH(g)} Q\Omega_g \rangle \le \liminf_{T\to\infty} \int Q(-t/2)\, Q(t/2)\, dq_{h_{T,g}}.$$

*Proof.* With $u_T := Q\psi_{T-t/2}/\|\psi_T\|$, the extension of (2) to $Q$ (Lemma 4.3; square moments at the two times finite at each $T$ by the Hölder bound in the proof of Lemma 4.3(ii), and bounded uniformly in $T$ by Theorem D.3 at $s = \mp t/2$, as in Lemma 7.2) is the exact equality

$$\int Q(-t/2)\, Q(t/2)\, dq_{h_{T,g}} = \big\| e^{-tH(g)/2} u_T \big\|^2.$$

The ratio bound of Lemma 4.4 and Lemma 7.1 make $(u_T)_T$ bounded. Indeed,

$$u_T = \frac{\|\psi_{T-t/2}\|}{\|\psi_T\|}\, Q\widehat{\psi}_{T-t/2},$$

the scalar factor tends to 1 and is bounded by $c_g^{-1}$, and $\widehat{\psi}_{T-t/2} \to \Omega_g$ strongly. From any subnet of $(u_T)$ choose a weakly convergent further subnet, say $u_{T_\alpha} \rightharpoonup u$. Along a further subnet the bounded vectors $Q\widehat{\psi}_{T_\alpha - t/2}$ converge weakly to the same $u$. Since the graph of the closed linear operator $Q$ is a closed linear subspace of $\mathcal{H} \oplus \mathcal{H}$, it is weakly closed; hence

$(\Omega_g, u)$ belongs to that graph and $u = Q\Omega_g$. Every weak cluster point is therefore $Q\Omega_g$, so the full bounded net satisfies $u_T \rightharpoonup Q\Omega_g$. Boundedness of $e^{-tH(g)/2}$ preserves the weak convergence $e^{-tH(g)/2}u_T \rightharpoonup e^{-tH(g)/2}Q\Omega_g$, and weak lower semicontinuity of the norm gives the claim. □

**Lemma 7.4** (CE2 across a horizontal strip)**.** *For $t > 2$ and every $h \in \mathcal{C}_{\mathrm{E}}$,*

$$\Big|\int Q(-t/2)Q(t/2)\,dq_h - \int Q(-t/2)\,dq_h \int Q(t/2)\,dq_h\Big| \le C_7 e^{2m_1} e^{-m_1 t}\,\|Q\|_{\mathrm{GJS}}^2.$$

*Proof.* Use the finite localization decomposition prescribed by GJS (1.1.9):

$$Q = \sum_{a\in A} Q_a, \qquad \|Q\|_{\mathrm{GJS}} = \sum_{a\in A} \|Q_a\|_{\mathrm{GJS}},$$

where every $Q_a$ is a localized monomial and $A$ is finite because the kernels of the cylinder polynomial have compact spatial support and finite degree. By linearity of expectation,

$$\mathrm{Cov}_{dq_h}(Q(-t/2), Q(t/2)) = \sum_{a,b\in A} \mathrm{Cov}_{dq_h}(Q_a(-t/2), Q_b(t/2)).$$

The localization squares of $Q_a(-t/2)$ lie in the row of unit squares meeting $x^0 = -t/2$, and those of $Q_b(t/2)$ lie in the row meeting $x^0 = t/2$. The open strip $\{|x^0| < t/2 - 1\}$ has width $t - 2$ and separates every such pair. Source Fact 2.2, applied separately to each $(a, b)$, and the triangle inequality therefore give

$$\begin{aligned}
|\mathrm{Cov}_{dq_h}(Q(-t/2), Q(t/2))| &\le C_7 e^{-m_1(t-2)} \sum_{a,b\in A} \|Q_a(-t/2)\|_{\mathrm{GJS}} \|Q_b(t/2)\|_{\mathrm{GJS}} \\
&= C_7 e^{-m_1(t-2)} \left(\sum_a \|Q_a\|_{\mathrm{GJS}}\right)^2 \\
&= C_7 e^{2m_1} e^{-m_1 t} \|Q\|_{\mathrm{GJS}}^2,
\end{aligned}$$

where the second equality uses the row-independence proved in Remark 6.3. No application of the localized CE2 theorem to an unlocalized sum is made. □

**Proposition 7.5** (semigroup decay of centered observables)**.** *Let $\psi_Q := Q\Omega_g - \langle \Omega_g, Q\Omega_g\rangle \Omega_g$. Then*

$$\langle \psi_Q, e^{-tH(g)}\psi_Q\rangle \le \widehat{C}_Q\, e^{-m_1 t} \qquad (t \ge 0), \qquad \widehat{C}_Q := \max\{C_7 e^{2m_1}\|Q\|_{\mathrm{GJS}}^2,\ e^{3m_1}\|\psi_Q\|^2\}.$$

*Proof.* Let $t > 2$. Since $e^{-tH(g)}\Omega_g = \Omega_g$ and $\langle \Omega_g, Q\Omega_g\rangle$ is real,

$$\langle \psi_Q, e^{-tH(g)}\psi_Q\rangle = \langle Q\Omega_g, e^{-tH(g)}Q\Omega_g\rangle - \langle \Omega_g, Q\Omega_g\rangle^2.$$

Write $\int Q(-t/2)Q(t/2)\,dq_{h_{T,g}} = a_T + b_T$, where $b_T = \int Q(-t/2)\,dq_{h_{T,g}} \int Q(t/2)\,dq_{h_{T,g}}$ and $a_T$ is the connected difference. Then, by Lemma 7.3 and $\liminf_T (a_T + b_T) \le \sup_T |a_T| + \lim_T b_T$,

$$\langle Q\Omega_g, e^{-tH(g)}Q\Omega_g\rangle \le \sup_T |a_T| + \lim_T b_T \le C_7 e^{2m_1} e^{-m_1 t}\|Q\|_{\mathrm{GJS}}^2 + \langle \Omega_g, Q\Omega_g\rangle^2,$$

using Lemma 7.4 for $\sup_T |a_T|$ and Lemma 7.2 at $s = \mp t/2$ fixed for $\lim_T b_T$. Subtracting the square proves the bound for $t > 2$. For $0 \le t \le 3$, positivity of $H(g)$ gives $\langle \psi_Q, e^{-tH(g)}\psi_Q\rangle \le \|\psi_Q\|^2 \le e^{3m_1}\|\psi_Q\|^2 e^{-m_1 t}$. The two cases cover $[0, \infty)$ and both bounds are $\le \widehat{C}_Q e^{-m_1 t}$. □

**Lemma 7.6** (exponential moments)**.** *Let $d\mu_g := |\Omega_g|^2\, d\phi_0$ and $X_f := \varphi_0(f)$ for real $f \in C_c^\infty(\mathbb{R})$, $f \neq 0$, supported in an interval $I$ of length $L$, and put $K' := K_2^2 N_I^{1/2} \|f\|_2 > 0$, $\sigma_f^2 := \langle f, Cf\rangle$, $K_f := \max\{K', \sqrt{2}\,\sigma_f\}$ and $A_f := A + B' K_1$, with $A$ and $B'$ as in Lemma 7.1 for this $I$ (which has $L > 0$, since $f \neq 0$). Then*

$$\mathbb{E}_{\mu_g}\big[X_f^{2n}\big] \le A_f\, K_f^{2n}\,(2n)! \qquad (n \ge 1),$$

*and consequently*

$$\mathbb{E}_{\mu_g} e^{s|X_f|} \le 2\max\{1, A_f\} \sum_{n=0}^{\infty} (sK_f)^{2n} < \infty \qquad (0 \le s < 1/K_f).$$

*$A_f$ and $K_f$ do not depend on $n$; $A_f$ depends on $g$ (through $N_g$, $N_B$) and on $I$, $K_f$ on $f$, on $I$ (through $N_I$), on $K_2$ and on $m_0$, and not on $g$.*

*Proof.* $\mathbb{E}_{\mu_g}[X_f^{2n}] = \|X_f^n \Omega_g\|^2 \le A\,\|X_f^{2n}\|_{L^2(d\phi_0)} + B'\,\mathfrak{n}(X_f^n)^2$ by Lemma 7.1 applied to $Q = X_f^n$ ($r = n$, all $f_i = f$); and by (8) with $r = n$ and all $f_i = f$,

$$\mathfrak{n}(X_f^n)^2 = K_1 K_2^{4n} (2n)!\, N_I^n \|f\|_2^{2n} = K_1 (K')^{2n} (2n)!.$$

Under $d\phi_0$, $X_f$ is centred Gaussian with variance $\sigma_f^2$, so $\|X_f^{2n}\|_{L^2(d\phi_0)} = \sigma_f^{2n}((4n-1)!!)^{1/2} \le (\sqrt{2}\,\sigma_f)^{2n}(2n)!$, because $(4n-1)!! = (4n)!/(2^{2n}(2n)!)$ and $\binom{4n}{2n} \le 16^n$ (Lemma N.52). Hence $\mathbb{E}_{\mu_g}[X_f^{2n}] \le (A + B'K_1) K_f^{2n} (2n)!$. For the exponential bound: $e^{s|x|} \le e^{sx} + e^{-sx} = 2\cosh(sx)$, and by monotone convergence

$$\mathbb{E}_{\mu_g} \cosh(sX_f) = \sum_{n \ge 0} \frac{s^{2n}}{(2n)!} \mathbb{E}_{\mu_g}[X_f^{2n}] \le \max\{1, A_f\} \sum_{n \ge 0} (sK_f)^{2n},$$

where the maximum covers the $n = 0$ term $\mathbb{E}_{\mu_g}[1] = 1$. The series converges for $sK_f < 1$. (The radius $1/K_f$ comes from the route through Theorem D.3; whether $\mathbb{E}_{\mu_g} e^{s|X_f|} < \infty$ for larger $s$ is not decided here.) □

**Theorem 7.7** (density of polynomial vectors)**.** *Cylinder polynomials are dense in $L^2(d\mu_g)$. Equivalently, since $\Omega_g > 0$ a.e. makes multiplication by $\Omega_g$ a unitary from $L^2(d\mu_g)$ onto $\mathcal{H}$, the linear span of $\Omega_g$ and the vectors $Q\Omega_g$ ($Q$ a cylinder monomial) is dense in $\mathcal{H}$.*

*Proof. Step 1: vanishing conditional expectations.* Let $\zeta \in L^2(d\mu_g)$ be orthogonal to all cylinder polynomials. Fix real $e_1, \dots, e_k \in C_c^\infty(\mathbb{R})$ and define

$$F(z_1, \dots, z_k) := \mathbb{E}_{\mu_g}\Big[\overline{\zeta}\, \exp\Big(\sum_{j=1}^{k} z_j X_{e_j}\Big)\Big].$$

By Cauchy–Schwarz, Hölder, and Lemma 7.6, $F$ is defined and analytic in a nonempty polystrip $\{|\mathrm{Re}\, z_j| < s_0\}$ about $i\mathbb{R}^k$, with $s_0 := \min_{j:\, e_j \neq 0} (2kK_{e_j})^{-1}$, $K_{e_j}$ the constant $K_f$ of Lemma 7.6 at $f = e_j$ (any $s_0 > 0$ if all $e_j = 0$; Remark N.54 writes out the Cauchy–Schwarz and Hölder step with $k$ equal exponents); differentiation under the integral is justified by the exponential dominator of Lemma 7.6 (for $0 < s_1 < s_2 < s_0$ and integers $\ell \ge 0$, $|x|^\ell e^{s_1|x|} \le \ell!\,(s_2 - s_1)^{-\ell} e^{s_2|x|}$, so on $\{|\mathrm{Re}\, z_j| \le s_1\}$ every derivative's integrand is dominated by an integrable function). Every derivative of $F$ at zero is an inner product of $\zeta$ with a cylinder polynomial, hence zero. Repeated application of the one-variable identity theorem gives $F = 0$ throughout the connected polystrip, in particular on $i\mathbb{R}^k$. Thus the finite complex measure

$$B \longmapsto \mathbb{E}_{\mu_g}\big[\overline{\zeta}\, \mathbf{1}_{\{(X_{e_1}, \dots, X_{e_k}) \in B\}}\big]$$

on $\mathbb{R}^k$ has vanishing Fourier transform, so it is zero by uniqueness of Fourier transforms of finite measures; equivalently $\mathbb{E}_{\mu_g}[\zeta \mid \sigma(X_{e_1}, \dots, X_{e_k})] = 0$.

*Step 2: exhaustion.* Choose $(e_j)_{j\geq 1} \subseteq C_c^\infty(\mathbb{R})$, real and dense in real $L^2(\mathbb{R})$ ($C_c^\infty$ is dense there). The time-zero covariance $C = (2\mu)^{-1}$ is bounded, so

$$\mathbb{E}_{\phi_0}|X_f - X_e|^2 = \langle f - e, C(f-e)\rangle \leq \|C\|\,\|f-e\|_2^2.$$

For an approximating sequence $e_{j_l} \to f$ in $L^2$, a subsequence of $X_{e_{j_l}}$ converges $\phi_0$-almost surely, hence $\mu_g$-almost surely because $\mu_g \ll \phi_0$. Therefore every $X_f$ is measurable with respect to $\bigvee_k \sigma(X_{e_1}, \dots, X_{e_k})$ modulo null sets, and this sigma-algebra exhausts the cylinder sigma-algebra.

*Step 3: martingale convergence.* The conditional expectations of Step 1 vanish for every $k$ and converge in $L^2(d\mu_g)$ to the conditional expectation on the exhausted sigma-algebra, which is $\zeta$ itself. Hence $\zeta = 0$, proving density. □

**Theorem 7.8** (uniform gap (G))**.** *Let $0 \leq \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. For every $g \in \mathcal{C}_{\mathrm{H}}$,*

$$\operatorname{spec} H(g) \subseteq \{0\} \cup [m_1, \infty), \tag{G}$$

*and 0 is a simple eigenvalue.*

*Proof.* Fix $0 \leq \beta < m_1$ and let $P_\beta := \mathbf{1}_{[0,\beta]}(H(g))$.

*Step 1: $P_\beta \psi_Q = 0$ for every centered polynomial vector.* By the spectral theorem and Proposition 7.5, for every $t \geq 0$,

$$e^{-\beta t}\|P_\beta \psi_Q\|^2 \leq \langle \psi_Q, e^{-tH(g)}\psi_Q\rangle \leq \widehat{C}_Q\, e^{-m_1 t}.$$

Multiply by $e^{\beta t}$ and let $t \to \infty$; since $m_1 - \beta > 0$, $\|P_\beta \psi_Q\| = 0$. (The restriction $\beta \geq 0$ is necessary: for $\beta < 0$, $P_{[0,\beta]} = 0$, and Step 2 below fails.)

*Step 2: $P_\beta \Omega_g = \Omega_g$,* since $H(g)\Omega_g = 0$ and $0 \in [0, \beta]$.

*Step 3: $P_\beta = |\Omega_g\rangle\langle\Omega_g|$.* By Theorem 7.7, the complex span of $\Omega_g$ and the centered vectors $\psi_Q$ is dense in $\mathcal{H}$. Steps 1–2 show that $P_\beta$ maps this dense subspace into $\mathbb{C}\Omega_g$; continuity of the bounded projection extends the containment to $\mathcal{H}$, and $P_\beta \Omega_g = \Omega_g$ shows the range is exactly $\mathbb{C}\Omega_g$.

*Step 4: no spectrum in $(0, m_1)$.* If an open interval $J \Subset (0, m_1)$ carried a nonzero spectral projection, choose $\beta < m_1$ above the right endpoint of $J$; then $\mathbf{1}_J(H(g)) \leq P_\beta - |\Omega_g\rangle\langle\Omega_g| = 0$, a contradiction. Hence (G) holds; simplicity of 0 is Assumption A1. □

## 8 Affine-path regularity

Throughout §§8–10, $0 \leq \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$, and $\gamma > 0$ denotes a constant for which

$$\operatorname{spec} H(g) \subseteq \{0\} \cup [\gamma, \infty), \qquad \dim\ker H(g) = 1, \qquad \text{for every } g \in \mathcal{C}_{\mathrm{H}}; \tag{G$_\gamma$}$$

by Theorem 7.8 this holds with $\gamma = m_1$, and $\gamma$ is instantiated there in §11. Fix $g, g' \in \mathcal{G}$ equal to 1 on a common nonempty open interval, and put

$$g_s := (1-s)g + sg', \quad v := g' - g, \quad K_s := K(g_s), \quad E_s := E(g_s), \quad H_s := K_s - E_s,$$

$$\Omega_s := \Omega_{g_s}, \qquad P_s := |\Omega_s\rangle\langle\Omega_s|, \qquad Q_s := 1 - P_s.$$

By Lemma 3.1(iii), $g_s \in \mathcal{G} \subset \mathcal{C}_{\mathrm{H}}$ for every $s \in [0,1]$, so all results of §§4–7 apply to every $K_s$.

**Proposition 8.1** (path regularity of the semigroup)**.** *For each $T > 0$, the family $s \mapsto e^{-TK_s}$ is norm-continuous on $[0,1]$ and norm-$C^1$ on $(0,1)$, with derivative given by the FKN kernel of $-\mathcal{U}_v w_s$ in the notation of the proof.*

*Proof.* On free Euclidean path space over the slab $[0,T]$ define

$$\mathcal{U}_s := \lambda \int_0^T \int_{\mathbb{R}} g_s(x) :\mathscr{P}(\Phi(t,x)): dx\, dt, \qquad w_s := e^{-\mathcal{U}_s}, \qquad \mathcal{U}_v := \mathcal{U}_1 - \mathcal{U}_0,$$

so that $s \mapsto \mathcal{U}_s = (1-s)\mathcal{U}_0 + s\mathcal{U}_1$ is affine (Lemma 3.1(iii)). For bounded time-zero $f, h$, the interacting FKN formula of Assumption A2 reads

$$\langle f, e^{-TK_s} h\rangle = \int \overline{(J_T f)}\, (J_0 h)\, w_s \, d\mu_0, \tag{11}$$

with $J_t$ the sharp-time embeddings.

*Step 1: the pairing bound.* Set $a := e^{-m_0 T}$ and

$$q_T := \frac{2}{1-a}, \quad r_T := \frac{2}{1+a}, \quad p_T := 1+a, \quad p_T' := 1 + a^{-1}.$$

Then $1/r_T + 1/q_T = 1$, $r_T p_T = 2$, and $(p_T' - 1)/(p_T - 1) = a^{-2} = e^{2m_0 T}$, so Assumption A4 gives $\|\mathsf{P}_T F\|_{p_T'} \le \|F\|_{p_T}$ for the free Markov semigroup $\mathsf{P}_T$ on $L^2(d\phi_0)$. The Markov property of $d\mu_0$ along constant time lines and Hölder's inequality (note $1/p_T + 1/p_T' = 1$) yield

$$\|(J_T f)(J_0 h)\|_{r_T}^{r_T} = \int |h|^{r_T}\, \mathsf{P}_T\big(|f|^{r_T}\big)\, d\phi_0 \le \big\||h|^{r_T}\big\|_{p_T} \big\|\mathsf{P}_T(|f|^{r_T})\big\|_{p_T'} \le \|h\|_2^{r_T}\, \|f\|_2^{r_T},$$

using $\||h|^{r_T}\|_{p_T} = \|h\|_{r_T p_T}^{r_T} = \|h\|_2^{r_T}$ and hypercontractivity on the second factor. Hence

$$\|(J_T f)(J_0 h)\|_{r_T} \le \|f\|_2 \|h\|_2. \tag{12}$$

*Step 2: $L^a$ control and continuity of $s \mapsto w_s$.* Finite-volume stability (Assumption A3, applicable since the slab cutoffs lie in $\mathcal{C}_{\mathrm{E}}$ by Lemma 3.1(ii)) gives $\mathcal{U}_v \in L^p(d\mu_0)$ for every finite $p$ and $e^{-\varrho \mathcal{U}_i} \in L^1(d\mu_0)$ for every finite $\varrho > 0$, $i = 0, 1$. Since $\mathcal{U}_s$ is affine, Hölder's inequality with exponents $\frac{1}{1-s}, \frac{1}{s}$ gives

$$\|w_s\|_\varrho^\varrho \le \Big(\int e^{-\varrho \mathcal{U}_0} d\mu_0\Big)^{1-s} \Big(\int e^{-\varrho \mathcal{U}_1} d\mu_0\Big)^{s}, \tag{13}$$

a bound uniform in $s \in [0,1]$ for each $\varrho$. If $s_n \to s$, then $\mathcal{U}_{s_n} \to \mathcal{U}_s$ in probability (indeed in $L^2$), so $w_{s_n} \to w_s$ in probability; together with (13) at a larger exponent, Vitali's convergence theorem gives $w_{s_n} \to w_s$ in $L^\varrho$ for every finite $\varrho$.

*Step 3: the difference quotient.* The calculus identity

$$\frac{w_{s+\delta} - w_s}{\delta} = -\mathcal{U}_v \int_0^1 w_{s+\vartheta\delta}\, d\vartheta \tag{14}$$

holds pointwise on path space. Choose finite $p, \varrho$ with $q_T^{-1} = p^{-1} + \varrho^{-1}$. By Hölder, the $L^{q_T}$ distance of (14) from $-\mathcal{U}_v w_s$ is at most $\|\mathcal{U}_v\|_p \int_0^1 \|w_{s+\vartheta\delta} - w_s\|_\varrho \, d\vartheta$, which tends to 0 as $\delta \to 0$ for $0 < s < 1$ by Step 2 (dominated convergence in $\vartheta$, using (13)). Thus $s \mapsto w_s$ is $L^{q_T}$-differentiable on $(0,1)$ with derivative $-\mathcal{U}_v w_s$, and $L^{q_T}$-continuous on $[0,1]$.

*Step 4: duality.* By (11), (12), and Hölder with the conjugate pair $(r_T, q_T)$,

$$\big|\langle f, (e^{-TK_s} - e^{-TK_r}) h\rangle\big| \le \|(J_T f)(J_0 h)\|_{r_T}\, \|w_s - w_r\|_{q_T} \le \|f\|_2 \|h\|_2\, \|w_s - w_r\|_{q_T},$$

so $\|e^{-TK_s} - e^{-TK_r}\| \le \|w_s - w_r\|_{q_T}$, and the same estimate applied to the difference quotients transfers the $L^{q_T}$ regularity of Step 3 to the operator norm. This proves the asserted norm continuity and differentiability. □

**Proposition 8.2** (projection and eigenvector regularity; derivative formulas)**.** (i) $s \mapsto P_s$ *is norm-continuous on* $[0,1]$ *and norm-*$C^1$ *on* $(0,1)$*.* (ii) *Locally on* $(0,1)$ *one may choose normalized* $C^1$ *vectors* $\Omega_s$ *in the parallel gauge* $\langle \Omega_s, \dot\Omega_s\rangle = 0$*, and then*

$$\dot E_s = \langle \Omega_s, V(v)\Omega_s\rangle, \qquad \dot\Omega_s = -H_s^{-1} Q_s V(v)\Omega_s, \tag{15}$$

*with* $\|H_s^{-1}Q_s\| \le \gamma^{-1}$*.*

*Proof. (i)* Let $L_s := e^{-TK_s}$ for a fixed $T > 0$. By Assumption A1, $L_s$ has the simple largest eigenvalue $\lambda_s = e^{-TE_s} = \|L_s\|$, and by ($G_\gamma$),

$$\operatorname{spec} L_s \setminus \{\lambda_s\} \subseteq [0, e^{-T\gamma}\lambda_s].$$

The relative spectral separation $1 - e^{-T\gamma}$ is uniform in $s$. Since $s \mapsto L_s$ is norm-continuous, so is $s \mapsto \|L_s\| = \lambda_s$; a Riesz contour of relative radius independent of $s$ around $\lambda_s$ therefore represents $P_s$ as a norm-convergent Cauchy integral of the resolvent of $L_s$, locally in $s$, and the norm regularity of $L_s$ (Proposition 8.1) passes to $P_s$ through the contour integral. This gives exactly the stated regularity of $P_s$.

*(ii)* Let $\mathscr{C}$ be the common core of Assumption A5(ii). For $\xi \in \mathscr{C}$, $K_s\xi = K_{s_0}\xi + (s - s_0)V(v)\xi$. Differentiating the weak eigenvalue identity $\langle K_s\xi, \Omega_s\rangle = E_s\langle \xi, \Omega_s\rangle$ at $s = s_0$ (all terms differentiable: $\Omega_s$ is $C^1$ by (i) and the gauge; and $E_s = -T^{-1}\log\lambda_s$ is $C^1$ because $\lambda_s = \langle \Omega_s, L_s\Omega_s\rangle > 0$ is a composition of the $C^1$ maps of (i) and Proposition 8.1) gives

$$\langle H_s\xi, \dot\Omega_s\rangle = \langle \xi, (\dot E_s - V(v))\Omega_s\rangle, \tag{16}$$

where $\Omega_s \in D(V(v))$ by Proposition 2.6, so the right side is well-defined. Since $\mathscr{C}$ is a core for the self-adjoint $H_s$, (16) says $\dot\Omega_s \in D(H_s)$ and $H_s\dot\Omega_s = (\dot E_s - V(v))\Omega_s$. Pairing with $\Omega_s$ and using $H_s\Omega_s = 0$, $\langle \Omega_s, \dot\Omega_s\rangle = 0$ proves the first formula in (15). The parallel gauge puts $\dot\Omega_s \in Q_s\mathcal{H}$, and ($G_\gamma$) makes $H_s^{-1}Q_s$ bounded by $\gamma^{-1}$; applying $H_s^{-1}Q_s$ to $H_s\dot\Omega_s = (\dot E_s - V(v))\Omega_s$ and using $Q_s\Omega_s = 0$ proves the second formula. □

# 9 Filter, spectral separation, and clustering

**Lemma 9.1** (exact inverse-energy filter)**.** *Choose an even* $\chi \in C^\infty(\mathbb{R}; [0,1])$ *with* $\chi = 0$ *on* $[-\gamma/2, \gamma/2]$ *and* $\chi = 1$ *outside* $(-\gamma, \gamma)$*, and set*

$$\widehat{W}_\gamma(E) := \begin{cases} \chi(E)/E, & E \neq 0, \\ 0, & E = 0, \end{cases} \qquad W_\gamma(t) := \frac{1}{2\pi}\int_{\mathbb{R}} e^{-itE}\,\widehat{W}_\gamma(E)\,dE. \tag{17}$$

*Then:* (i) $W_\gamma \in L^1(\mathbb{R})$ *and, for every* $q \ge 0$, $\int_{\mathbb{R}}(1+|t|)^q|W_\gamma(t)|\,dt < \infty$*; in particular*

$$T_\gamma(r) := \int_{|t|>r} |W_\gamma(t)|\,dt \le C_{q,\gamma}(1+r)^{-q} \qquad (q \ge 0,\ r \ge 0).$$

(ii) *If* $H$ *is self-adjoint with* $\operatorname{spec} H \subseteq \{0\} \cup [\gamma, \infty)$*, then*

$$H^{-1}(1 - P_{\ker H}) = \int_{\mathbb{R}} W_\gamma(t)\, e^{itH}(1 - P_{\ker H})\,dt, \tag{18}$$

*a norm-convergent Bochner integral.*

*Proof.* Throughout, $\widehat{W}_\gamma \in L^2(\mathbb{R})$ (it vanishes on $[-\gamma/2, \gamma/2]$, is bounded by $2/\gamma$, and decays like $|E|^{-1}$), and (17) is understood as its $L^2$-inverse Fourier transform; the symmetric improper integrals below converge (Dirichlet's test) and represent it almost everywhere, which justifies the manipulations.

*(i)* Oddness of $\widehat{W}_\gamma$ gives

$$W_\gamma(t) = -\frac{i}{\pi}\int_0^\infty \frac{\sin(tE)\,\chi(E)}{E}\,dE.$$

For $0 < |t| \le 1$: split at $E = \gamma$. The transition part $\int_{\gamma/2}^{\gamma} |\sin(tE)|\chi(E)E^{-1}dE$ is absolutely bounded by a constant depending only on $\gamma$; for the tail $\int_\gamma^\infty \sin(tE)E^{-1}dE$, substitute $u = |t|E$ and use the uniform boundedness of $\int_c^d u^{-1}\sin u\,du$ over $0 < c < d$ (Dirichlet's test). Hence $W_\gamma$ is bounded near $t = 0$, so integrable there. For $|t| \ge 1$: every derivative $\widehat{W}_\gamma^{(k)}$ with $k \ge 1$ is integrable on $\mathbb{R}$ (it is smooth, supported off $(-\gamma/2, \gamma/2)$, and decays like $|E|^{-1-k}$), so $k$ integrations by parts in (17) give

$$|W_\gamma(t)| \le \frac{\|\widehat{W}_\gamma^{(k)}\|_{L^1}}{2\pi|t|^k}.$$

Choosing $k > q + 1$ proves the weighted integrability and, by Markov's inequality applied to the weight $(1 + |t|)^q$, the tail bound for $T_\gamma$.

*(ii)* By construction $\widehat{W}_\gamma(E) = E^{-1}$ on $[\gamma, \infty)$ and $\widehat{W}_\gamma(0) = 0$. Fourier inversion and the spectral theorem give (18) on the spectral subspace of $[\gamma, \infty)$ and on $\ker H$ separately; the integral converges in norm because $W_\gamma \in L^1$ and $\|e^{itH}(1 - P_{\ker H})\| \le 1$. □

**Lemma 9.2** (spectral separation)**.** *Let*

$$F(t) = \int_{[\gamma,\infty)} e^{itE}\,d\rho(E), \qquad G(t) = \int_{(-\infty,-\gamma]} e^{itE}\,d\sigma(E),$$

*with complex measures obeying* $|\rho|(\mathbb{R}), |\sigma|(\mathbb{R}) \le \mathfrak{m}$*, and suppose* $F = G$ *on* $(-R, R)$ *with* $\gamma R \ge 1$*. Then*

$$|F(0)| \le 2\,\mathfrak{m}\,e^{-\gamma R/2}. \tag{19}$$

*Proof. Step 1: the Gaussian separator.* For $a, \varepsilon > 0$ define

$$f_{a,\varepsilon}(t) := \frac{1}{2\pi i}\,\frac{e^{-t^2/(2a)}}{t - i\varepsilon}.$$

The transform of $(2\pi i)^{-1}(t - i\varepsilon)^{-1}$ is $\mathbf{1}_{(0,\infty)}(E)\,e^{-\varepsilon E}$ (one-pole contour integral); multiplication by the Gaussian in $t$ convolves this with the normalized Gaussian, so

$$\check{f}_{a,\varepsilon}(E) = \int_0^\infty \sqrt{\frac{a}{2\pi}}\,e^{-a(E-E')^2/2}\,e^{-\varepsilon E'}\,dE' \in [0, 1], \tag{20}$$

which tends to the standard normal distribution function $\Phi(\sqrt{a}E)$ as $\varepsilon \downarrow 0$.

*Step 2: the tail of the separator.* For $t \ge R$, $|t - i\varepsilon| \ge t$ and $t^{-1} \le t/R^2$, so

$$\int_{|t|\ge R} |f_{a,\varepsilon}(t)|\,dt \le \frac{1}{\pi R^2}\int_R^\infty te^{-t^2/(2a)}\,dt = \frac{a}{\pi R^2}\,e^{-R^2/(2a)}. \tag{21}$$

*Step 3: exchange of test function.* Pair $f_{a,\varepsilon}$ with $F - G$. By hypothesis the pairing is supported in $|t| \ge R$; Fubini (justified by the finite total variations and $f_{a,\varepsilon} \in L^1$) and then $\varepsilon \downarrow 0$ with dominated convergence in energy give

$$\left|\int \Phi(\sqrt{a}E)\,d\rho(E) - \int \Phi(\sqrt{a}E)\,d\sigma(E)\right| \le \frac{2\,\mathfrak{m}\,a}{\pi R^2}\,e^{-R^2/(2a)}. \tag{22}$$

*Step 4: optimization.* On the support of $\rho$, $E \geq \gamma$, so $1 - \Phi(\sqrt{a}E) \leq \frac{1}{2}e^{-a\gamma^2/2}$; on the support of $\sigma$, $E \leq -\gamma$, so $\Phi(\sqrt{a}E) \leq \frac{1}{2}e^{-a\gamma^2/2}$. Inserting and subtracting $\int d\rho = F(0)$ in (22),

$$|F(0)| \leq \mathfrak{m}\, e^{-a\gamma^2/2} + \frac{2\,\mathfrak{m}\, a}{\pi R^2}\, e^{-R^2/(2a)}.$$

Set $a = R/\gamma$: both exponentials become $e^{-\gamma R/2}$, and $2a/(\pi R^2) = 2/(\pi\gamma R) \leq 2/\pi < 1$ by $\gamma R \geq 1$; the two terms sum to at most $2\,\mathfrak{m}\, e^{-\gamma R/2}$, proving (19). □

**Lemma 9.3** (clustering with one unbounded local argument)**.** *Let $g \in \mathcal{C}_{\mathrm{H}}$, write $H = H(g)$, $\Omega = \Omega_g$, $\beta_t = \beta_t^g$, and assume ($\mathrm{G}_\gamma$). Let $B \subset \mathbb{R}$ be compact, let $X = X^*$ be affiliated with $\mathfrak{A}(O')'$ for every bounded open interval $O'$ with $\operatorname{dist}(B, O') > 0$, with $\Omega \in D(X)$, and let $A \in \mathfrak{A}(O)$ for a bounded interval $O$. If $R := \operatorname{dist}(B, O) > 0$ satisfies $\gamma R \geq 1$, then*

$$\big|\langle X\Omega, A\Omega\rangle - \langle X\Omega, \Omega\rangle\langle \Omega, A\Omega\rangle\big| \leq 2\,\|X\Omega\|\,\|A\|\,e^{-\gamma R/2}. \tag{23}$$

*Proof.* Put $P_0 = |\Omega\rangle\langle\Omega|$ and define

$$F(t) := \langle X\Omega, (e^{itH} - P_0)A\Omega\rangle, \qquad G(t) := \langle A^*\Omega, (e^{-itH} - P_0)X\Omega\rangle.$$

*Step 1: spectral supports and variations.* Inserting the spectral resolution of $H$ on $(1 - P_0)\mathcal{H}$ and using ($\mathrm{G}_\gamma$), $F$ and $G$ are Fourier transforms of complex measures supported in $[\gamma, \infty)$ and $(-\infty, -\gamma]$ respectively, with total variations at most $\mathfrak{m} := \|X\Omega\|\,\|A\|$ (Cauchy–Schwarz on the polar decomposition of the spectral measures).

*Step 2: locality identity.* For $|t| < R$: by Proposition 2.7, $\beta_t(A) \in \mathfrak{A}(O_{|t|})$, and $O_{|t|}$ is a bounded open interval with $\operatorname{dist}(B, O_{|t|}) \geq R - |t| > 0$. Affiliation of $X$ with $\mathfrak{A}(O_{|t|})'$ implies that $\beta_t(A) \in \mathfrak{A}(O_{|t|})$ preserves $D(X)$ and commutes with $X$ on it. Since $e^{-itH}\Omega = \Omega$,

$$\langle X\Omega, e^{itH}A\Omega\rangle = \langle \Omega, X\beta_t(A)\Omega\rangle = \langle \Omega, \beta_t(A)X\Omega\rangle = \langle A^*\Omega, e^{-itH}X\Omega\rangle.$$

The $P_0$ terms agree because $\langle \Omega, X\Omega\rangle$ is real ($X = X^*$, $\Omega \in D(X)$). Hence $F = G$ on $(-R, R)$.

*Step 3: conclusion.* Lemma 9.2 gives $|F(0)| \leq 2\mathfrak{m}\, e^{-\gamma R/2}$, and $F(0) = \langle X\Omega, A\Omega\rangle - \langle X\Omega, \Omega\rangle\langle \Omega, A\Omega\rangle$. □

## 10 The one-path LPPL estimate

The setting of §8 remains in force: $g, g' \in \mathcal{G}$ equal to 1 on a common nonempty open interval, $g_s = (1 - s)g + sg'$, $v = g' - g$; note $|v| \leq 1$.

**Lemma 10.1** (state derivative)**.** *For self-adjoint $A \in B(\mathcal{H})$ and $0 < s < 1$,*

$$\frac{d}{ds}\,\omega_{g_s}(A) = -2\,\mathrm{Re}\int_{\mathbb{R}} W_\gamma(t)\Big[\langle V(v)\Omega_s, \beta_t^{g_s}(A)\Omega_s\rangle - \langle V(v)\Omega_s, \Omega_s\rangle\,\omega_{g_s}(A)\Big]dt, \tag{24}$$

*and the integral converges absolutely.*

*Proof.* Since $A$ is bounded and self-adjoint, Proposition 8.2(ii) gives

$$\frac{d}{ds}\,\omega_{g_s}(A) = 2\,\mathrm{Re}\langle \dot\Omega_s, A\Omega_s\rangle = -2\,\mathrm{Re}\langle V(v)\Omega_s, H_s^{-1}Q_sA\Omega_s\rangle,$$

using (15) and self-adjointness of $H_s^{-1}Q_s$. Insert the exact filter (18) for $H = H_s$ (admissible by ($\mathrm{G}_\gamma$)), applied to the vector $Q_sA\Omega_s$: since $e^{itH_s}\Omega_s = \Omega_s$,

$$e^{itH_s}Q_sA\Omega_s = e^{itH_s}\big(A\Omega_s - \omega_{g_s}(A)\Omega_s\big) = \beta_t^{g_s}(A)\Omega_s - \omega_{g_s}(A)\Omega_s,$$

which yields (24). The integral is absolutely convergent because $W_\gamma \in L^1$ (Lemma 9.1) and the bracket is bounded by $2\|A\|\,\|V(v)\Omega_s\|$; here $V(v)\Omega_s$ exists by Proposition 2.6 ($v$ has compact support). □

**Lemma 10.2** (localization of the perturbation)**.** *Choose a smooth partition of unity* $\sum_{j\in\mathbb{Z}} \eta_j = 1$ *with* $0 \le \eta_j \le 1$ *and* $\operatorname{supp}\eta_j \subset (j-1, j+1)$*, and set* $h_j := \eta_j v$*,* $V_j := V(h_j)$*,* $B_j := \operatorname{supp} h_j$*. Then only finitely many* $h_j$ *are nonzero,*

$$V(v)\xi = \sum_j V_j\xi \quad \text{on common domains,} \qquad \|V_j\Omega_s\| \le M \quad (0 \le s \le 1). \tag{25}$$

*Proof.* Finitely many $h_j$ are nonzero because $v$ has compact support; the finite additivity is the convention of §2.1. Each $h_j$ is real and measurable with $\|h_j\|_\infty \le \|\eta_j\|_\infty\|v\|_\infty \le 1$ and $\operatorname{supp} h_j \subseteq [j-1, j+1]$, an interval of length 2. Since $g_s \in \mathcal{C}_{\mathrm{H}}$ (Lemma 3.1) and $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$, Theorem 5.1 applies to the cutoff $g_s$ and gives $\|V_j\Omega_s\| \le M$. □

**Lemma 10.3** (two-regime bound)**.** *Let* $A = A^* \in \mathfrak{A}(O)$ *with* $\|A\| \le 1$*, and for a nonzero* $h_j$ *put* $r_j := \operatorname{dist}(B_j, O)$ *and*

$$C_{j,s}(t) := \langle V_j\Omega_s, \beta_t^{g_s}(A)\Omega_s\rangle - \langle V_j\Omega_s, \Omega_s\rangle\,\omega_{g_s}(A).$$

*If* $\gamma r_j/2 \ge 1$*, then*

$$|C_{j,s}(t)| \le \begin{cases} 2Me^{-\gamma r_j/4}, & |t| \le r_j/2, \\ 2M, & t \in \mathbb{R}. \end{cases} \tag{26}$$

*Proof.* For arbitrary $t$, bound each of the two terms by $\|V_j\Omega_s\|\,\|A\| \le M$ (Lemma 10.2), giving $2M$. For $|t| \le r_j/2$: Proposition 2.7 places $\beta_t^{g_s}(A)$ in $\mathfrak{A}(O_{|t|})$, and $\operatorname{dist}(B_j, O_{|t|}) \ge r_j - |t| \ge r_j/2$, with $\gamma\cdot\operatorname{dist}(B_j, O_{|t|}) \ge \gamma r_j/2 \ge 1$. Apply Lemma 9.3 with $H = H(g_s)$, $X = V_j$ (affiliated with $\mathfrak{A}(O')'$ for every bounded open interval $O'$ with $\operatorname{dist}(B_j, O') > 0$, by *Locality* in §2.1, as $B_j = \operatorname{supp} h_j$; $\Omega_s \in D(V_j)$ by Proposition 2.6), and the observable $\beta_t^{g_s}(A) \in \mathfrak{A}(O_{|t|})$; note $\langle\Omega_s, \beta_t^{g_s}(A)\Omega_s\rangle = \omega_{g_s}(A)$ by invariance of $\Omega_s$. This gives

$$|C_{j,s}(t)| \le 2\|V_j\Omega_s\|\,\|\beta_t^{g_s}(A)\|\,e^{-\gamma\,\operatorname{dist}(B_j, O_{|t|})/2} \le 2Me^{-\gamma r_j/4}.$$ □

**Proposition 10.4** (one-path estimate)**.** *Let* $d \ge d_0 := 4 + 2/\gamma$ *and suppose* $g \equiv g' \equiv 1$ *on* $O_d$*. Then*

$$\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \le 32M \sum_{n \ge \lfloor d\rfloor} \Big(\|W_\gamma\|_1 e^{-\gamma n/4} + T_\gamma(n/2)\Big). \tag{27}$$

*Proof. Step 1: all perturbation cells are far from* $O$*.* On $O_d$, $v = g' - g = 0$, and $O_d$ is open, so $\operatorname{supp} v \subseteq \mathbb{R}\setminus O_d$; hence every nonzero $h_j$ has $r_j = \operatorname{dist}(B_j, O) \ge d \ge d_0$. In particular $\gamma r_j/2 \ge \gamma d_0/2 = 2\gamma + 1 \ge 1$, so Lemma 10.3 applies to every $j$.

*Step 2: derivative bound.* Fix $A = A^* \in \mathfrak{A}(O)$, $\|A\| \le 1$. By Lemmas 10.1 and 10.2,

$$\frac{d}{ds}\,\omega_{g_s}(A) = -2\,\mathrm{Re}\sum_j \int_{\mathbb{R}} W_\gamma(t)\,C_{j,s}(t)\,dt.$$

Split each $t$-integral at $|t| = r_j/2$ and use (26): the inner region contributes at most $\|W_\gamma\|_1 \cdot 2Me^{-\gamma r_j/4}$, the outer at most $T_\gamma(r_j/2)\cdot 2M$. With the overall factor 2,

$$\Big|\frac{d}{ds}\,\omega_{g_s}(A)\Big| \le 4M\sum_j \Big(\|W_\gamma\|_1 e^{-\gamma r_j/4} + T_\gamma(r_j/2)\Big). \tag{28}$$

*Step 3: shell count.* At most eight indices $j$ have $r_j \in [n, n+1)$, for any integer $n \geq \lfloor d \rfloor$. Indeed, write $O = (a, b)$. Since $B_j \subseteq [j-1, j+1]$ has length $\leq 2$, avoids $O_d$, and $O_d$ is an interval of length $> 2d \geq 8$, the set $B_j$ lies wholly on one side of $O_d$. On the right side, compactness of $B_j$ gives $x_j \in B_j$ with $x_j - b = r_j$; then $|x_j - j| \leq 1$ and $r_j \in [n, n+1)$ force $j \in [b+n-1, b+n+2]$, an interval containing at most four integers. The left side contributes at most four more.

*Step 4: integrate and sum shells.* Both functions of $r_j$ in (28) are decreasing in $r_j$, so grouping the indices into the shells of Step 3 gives

$$\left|\frac{d}{ds}\,\omega_{g_s}(A)\right| \leq 4M \cdot 8 \sum_{n \geq \lfloor d \rfloor} \Big( \|W_\gamma\|_1 e^{-\gamma n/4} + T_\gamma(n/2) \Big).$$

Integrate over $s \in [\varepsilon, 1-\varepsilon]$, where differentiability holds (Proposition 8.2), and let $\varepsilon \downarrow 0$ using the endpoint norm continuity of $s \mapsto P_s$; this bounds $|\omega_g(A) - \omega_{g'}(A)|$ by the right side of (27). Finally, $\omega_g - \omega_{g'}$ is a hermitian functional, so its norm on $\mathfrak{A}(O)$ is the supremum of $|\omega_g(A) - \omega_{g'}(A)|$ over self-adjoint contractions $A \in \mathfrak{A}(O)$. □

**Lemma 10.5** (the rate function is super-polynomial)**.** *Define*

$$\Psi_\gamma(d) := \begin{cases} 32M \displaystyle\sum_{n \geq \lfloor d \rfloor} \Big( \|W_\gamma\|_1 e^{-\gamma n/4} + T_\gamma(n/2) \Big), & d \geq d_0, \\ \max\Big\{ 2,\ \textit{the same expression} \Big\}, & 0 \leq d < d_0. \end{cases} \tag{29}$$

*Then for every $N \geq 1$ there is $C_{N,\gamma,M} < \infty$ with*

$$\Psi_\gamma(d) \leq C_{N,\gamma,M}\,(1+d)^{-N} \qquad (d \geq 0). \tag{30}$$

*In particular $\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \leq \Psi_\gamma(d)$ whenever $g \equiv g' \equiv 1$ on $O_d$, $d \geq 0$.*

*Proof.* For $d \geq d_0$: the exponential part sums to $\|W_\gamma\|_1 e^{-\gamma\lfloor d \rfloor/4}(1 - e^{-\gamma/4})^{-1} \leq C'_\gamma e^{-\gamma d/4}$, which is $O_N((1+d)^{-N})$ for every $N$. For the second part choose $q = N+2$ in the tail bound of Lemma 9.1(i): $T_\gamma(n/2) \leq C_{q,\gamma}(1+n/2)^{-q} \leq C_{q,\gamma} 2^q (1+n)^{-q}$, and

$$\sum_{n \geq \lfloor d \rfloor} (1+n)^{-N-2} \leq \int_{\lfloor d \rfloor - 1}^{\infty} (1+x)^{-N-2}\,dx = \frac{\lfloor d \rfloor^{-N-1}}{N+1} \leq \frac{C''_N}{(1+d)^{N+1}},$$

using $\lfloor d \rfloor \geq d - 1 \geq \frac{3}{5}(1+d)$ for $d \geq d_0 \geq 4$. For $0 \leq d < d_0$: $\Psi_\gamma$ is nonincreasing there, so the left side of (30) is bounded by $\Psi_\gamma(0) = \max\{2,\ 32M \sum_{n \geq 0}(\|W_\gamma\|_1 e^{-\gamma n/4} + T_\gamma(n/2))\} < \infty$, while the right side is bounded below by $C_{N,\gamma,M}(1+d_0)^{-N}$; enlarging $C_{N,\gamma,M}$ covers this range. The final assertion combines Proposition 10.4 with the trivial bound $\|\omega_g - \omega_{g'}\| \leq 2$ for $d < d_0$. □

## 11 Convergence of the full net; local normality

From here on the abstract gap parameter is instantiated: by Theorem 7.8, ($\mathrm{G}_\gamma$) holds with $\gamma = m_1$, and we write $\Psi_{m_1} := \Psi_\gamma|_{\gamma = m_1}$ with $M$ the constant of Theorem 5.1.

**Theorem 11.1** (full-net convergence with rate)**.** *Let $0 \leq \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. There is a state $\omega_\infty$ on $\mathfrak{A}$ such that, for every bounded open interval $O$,*

$$\lim_{g \in \mathcal{G}} \big\| (\omega_g - \omega_\infty)|_{\mathfrak{A}(O)} \big\| = 0,$$

*and*

$$\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| \leq \Psi_{m_1}(d) \qquad \textit{whenever } g \equiv 1 \textit{ on } O_d. \tag{31}$$

*Proof. Step 1: the net is locally Cauchy.* Fix $O$ and $\varepsilon > 0$. By Lemma 10.5, choose $d \geq d_0$ with $\Psi_{m_1}(d) < \varepsilon$, and choose $g_0 \in \mathcal{G}$ whose plateau interior contains $O_d$. If $g, g' \succcurlyeq g_0$, then $\{g = 1\}^\circ \supseteq \{g_0 = 1\}^\circ \supseteq O_d$ and likewise for $g'$; in particular $g \equiv g' \equiv 1$ on $O_d$ and their common plateau interior is nonempty, so Proposition 10.4 (with $\gamma = m_1$) applies and gives

$$\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \leq \Psi_{m_1}(d) < \varepsilon. \tag{32}$$

*Step 2: limit functionals and compatibility.* Each dual $\mathfrak{A}(O)^*$ is complete, so (32) produces a norm limit $\varphi_O$ of $(\omega_g|_{\mathfrak{A}(O)})_{g\in\mathcal{G}}$. If $O_1 \subseteq O_2$, then restriction to $\mathfrak{A}(O_1)$ of the $\mathfrak{A}(O_2)$-limit is again the norm limit of $\omega_g|_{\mathfrak{A}(O_1)}$, hence equals $\varphi_{O_1}$ by uniqueness of norm limits. The compatible family $(\varphi_O)_O$ defines a linear functional $\omega_\infty$ on the norm-dense union $\bigcup_O \mathfrak{A}(O)$, with $|\omega_\infty(A)| \leq \|A\|$, $\omega_\infty(1) = 1$, and $\omega_\infty(A^*A) = \lim_g \omega_g(A^*A) \geq 0$ for local $A$; it extends uniquely by norm continuity to $\mathfrak{A}$, and the extension is a state (a norm-one positive unital functional on a norm-dense subalgebra extends to a state).

*Step 3: the rate.* Fix $g \equiv 1$ on $O_d$. The subset $\{g' \in \mathcal{G} : g' \equiv 1 \text{ on } O_d\}$ is cofinal in $\mathcal{G}$ (every plateau can be enlarged to contain $O_d$). For every such $g'$, Proposition 10.4 gives $\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \leq \Psi_{m_1}(d)$ (via Lemma 10.5 for $d < d_0$); letting $g'$ run through this cofinal subset, $\omega_{g'} \to \omega_\infty$ in the norm of $\mathfrak{A}(O)^*$, and the bound passes to the limit, proving (31). □

**Proposition 11.2** (local normality)**.** *$\omega_\infty|_{\mathfrak{A}(O)}$ is normal for every bounded open interval $O$.*

*Proof.* Let $0 \leq A_i \uparrow A$ be a bounded increasing net in $\mathfrak{A}(O)$ with supremum $A$. Given $\delta > 0$, choose $g \in \mathcal{G}$ with $\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| < \delta$ (Theorem 11.1). Then

$$0 \leq \omega_\infty(A) - \omega_\infty(A_i) = \big[\omega_\infty - \omega_g\big](A - A_i) + \omega_g(A - A_i) \leq 2\delta\|A\| + \omega_g(A - A_i),$$

using $\|A - A_i\| \leq 2\|A\|$. The cutoff state $\omega_g$ is a vector state, hence normal on the von Neumann algebra $\mathfrak{A}(O)$, so $\omega_g(A - A_i) \to 0$ along the net. Therefore $\limsup_i[\omega_\infty(A) - \omega_\infty(A_i)] \leq 2\delta\|A\|$; let $\delta \downarrow 0$. Thus $\omega_\infty(A_i) \uparrow \omega_\infty(A)$, which is normality on $\mathfrak{A}(O)$. □

## 12 Ground state, invariances, and the Main Theorem

**Proposition 12.1** (exact spectral passage)**.** *If $(g_n) \subseteq \mathcal{G}$ satisfies $d(g_n, O) \to \infty$ for every bounded interval $O$, and $\omega_{g_n} \to \omega$ in norm on every $\mathfrak{A}(O)$, then $\omega$ is a ground state of $(\mathfrak{A}, \alpha)$ in the spectral sense: $\omega \circ \alpha_t = \omega$ for all $t$; the function $t \mapsto \omega(A\alpha_t(B))$ is continuous and bounded for all $A, B \in \mathfrak{A}$; and*

$$\int_{\mathbb{R}} f(t)\, \omega(A\alpha_t(B))\, dt = 0$$

*for every $f \in L^1(\mathbb{R})$ whose transform $\check{f}(E) = \int f(t)e^{itE}\, dt$ vanishes on $[0, \infty)$.*

*Proof. Exact identity.* Fix a bounded interval $O$, $A, B \in \mathfrak{A}(O)$, $\mathfrak{m} := \|A\|\|B\|$, and $g \in \mathcal{G}$; let $\rho_g(\cdot) := \langle A^*\Omega_g, P_{H(g)}(\cdot)B\Omega_g\rangle$, a complex measure with $\operatorname{supp}\rho_g \subseteq \operatorname{spec} H(g) \subseteq [0, \infty)$ and $|\rho_g|(\mathbb{R}) \leq \mathfrak{m}$, and $G_g(t) := \int e^{itE}\, d\rho_g(E)$. For $|t| < d(g, O)$, the cutoff equals one on $O_{|t|}$, so by (P) and $e^{-itH(g)}\Omega_g = \Omega_g$,

$$\omega_g\big(A\alpha_t(B)\big) = \langle \Omega_g, Ae^{itH(g)}Be^{-itH(g)}\Omega_g\rangle = \langle A^*\Omega_g, e^{itH(g)}B\Omega_g\rangle = G_g(t).$$

Moreover $\int_{\mathbb{R}} f\, G_g\, dt = \int_{[0,\infty)} \check{f}\, d\rho_g = 0$ for the stated class of $f$ (Fubini, justified by $|\rho_g|(\mathbb{R}) \leq \mathfrak{m}$ and $f \in L^1$; $\check{f} = 0$ on the support).

*Invariance.* For $A \in \mathfrak{A}(O)$ and fixed $t$, take $n$ so large that $d(g_n, O) > |t|$. Then (P) and $e^{-itH(g_n)}\Omega_{g_n} = \Omega_{g_n}$ give directly

$$\omega_{g_n}(\alpha_t(A)) = \langle \Omega_{g_n}, e^{itH(g_n)} A e^{-itH(g_n)} \Omega_{g_n} \rangle = \omega_{g_n}(A).$$

Both $\alpha_t(A)$ and $A$ belong to $\mathfrak{A}(O_{|t|})$, so norm convergence on $\mathfrak{A}(O_{|t|})$ passes the identity to the limit: $\omega(\alpha_t(A)) = \omega(A)$, and by density on $\mathfrak{A}$.

*Continuity and vanishing.* Fix $A, B \in \mathfrak{A}(O)$ and $T > 0$. For $n$ with $d(g_n, O) > T$ and $|t| \le T$: $A\alpha_t(B) \in \mathfrak{A}(O_T)$ with norm $\le \mathfrak{m}$, so

$$\sup_{|t|\le T} \left|\omega(A\alpha_t(B)) - \omega_{g_n}(A\alpha_t(B))\right| \le \mathfrak{m} \left\|(\omega - \omega_{g_n})|_{\mathfrak{A}(O_T)}\right\| \longrightarrow 0 :$$

$t \mapsto \omega(A\alpha_t(B))$ is a locally uniform limit of the continuous functions $G_{g_n}$, hence continuous, bounded by $\mathfrak{m}$. For $f$ as stated and $\varepsilon(T) := \int_{|t|>T} |f|$:

$$\int_{\mathbb{R}} f\, \omega(A\alpha_\cdot(B)) = \underbrace{\int_{|t|\le T} f\, [\omega - \omega_{g_n}](A\alpha_t(B))}_{\le \|f\|_1 \mathfrak{m} \|(\omega - \omega_{g_n})|_{\mathfrak{A}(O_T)}\|} + \underbrace{\int_{|t|\le T} f\, G_{g_n}}_{= -\int_{|t|>T} f\, G_{g_n}} + \int_{|t|>T} f\, \omega(A\alpha_t(B)),$$

where the middle identity uses $\int_{\mathbb{R}} f\, G_{g_n}\, dt = 0$. Hence the modulus is at most $\|f\|_1 \mathfrak{m} \|(\omega - \omega_{g_n})|_{\mathfrak{A}(O_T)}\| + 2\mathfrak{m}\, \varepsilon(T)$; let $n \to \infty$, then $T \to \infty$. Joint norm continuity in $(A, B)$ extends the conclusion from local $A, B$ to $\mathfrak{A}$. □

The Main Theorem now follows by assembly.

**Theorem 12.2** (Main Theorem)**.** *Let $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. Then there is a state $\omega_\infty$ on $\mathfrak{A}$ such that:*

(1) (full-net convergence with rate) *for every bounded open interval $O$, $\lim_{g\in\mathcal{G}} \|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| = 0$, and $\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| \le \Psi_{m_1}(d)$ whenever $g \equiv 1$ on $O_d$, with $\Psi_{m_1}$ decaying faster than every inverse power;*

(2) (local normality) *$\omega_\infty|_{\mathfrak{A}(O)}$ is normal for every $O$;*

(3) (ground state) *$\omega_\infty$ is a ground state of $(\mathfrak{A}, \alpha)$ in the spectral sense;*

(4) (invariances) *$\omega_\infty \circ \tau_a = \omega_\infty$ for all $a \in \mathbb{R}$; and for even $\mathscr{P}$, $\omega_\infty \circ \theta = \omega_\infty$;*

(5) (spectra and local moment)

$$\operatorname{spec} H(g) \subseteq \{0\} \cup [m_1, \infty) \quad (g \in \mathcal{C}_{\mathrm{H}}), \qquad \dim\ker H(g) = 1,$$

*and $\sup_{g\in\mathcal{C}_{\mathrm{H}}} \|V(h)\Omega_g\| \le M$ for real $h$ with $\|h\|_\infty \le 1$ supported in an interval of length 2.*

*The weak-coupling estimate imported is one printed theorem of [7]: Theorem 1.1.7 (Source Fact 2.2), used at its printed scope, including the transverse case A8(iv)-T of Assumption A8 for the sharp-time observable of Theorem 5.1, a case the printed statement covers and the printed proof does not reach. The uniform local moment and the moment bounds behind the uniform gap are theorems of this manuscript (Theorems D.3 and D.4, Appendix D), proved from Theorem 1.1.7, the normalization Lemma C.2, the ultraviolet Lemma C.4 and the stability Theorem C.5 of Appendix C, and the constructions of the Wick powers recorded in Assumptions A1 and A3; the former moment input, Theorem 1.1.11 (Assumption A8(vi)),*

*is retired. Appendix C does not re-derive the cluster expansion. In addition, the proof uses the separately identified standard cutoff-construction inputs of Assumptions A1–A7. Proposition 3.2 verifies that the dimensionless weak-coupling hypothesis implies the source smallness condition. The theorem is an implication from exactly this list of printed statements; no other external input is used. The constant $M$ of item (5), and through it the rate $\Psi_{m_1}$ of item (1), contain the finite, unevaluated constant $S_2(B_{3\times5})$ of Theorem C.5.*

*Proof.* (5) is Theorem 7.8 and Theorem 5.1. These supply ($G_\gamma$) with $\gamma = m_1$ and ($M_2$), so Theorem 11.1 and Proposition 11.2 give (1)–(2). For (3): choose $\widetilde{g}_n \in \mathcal{G}$ with $\widetilde{g}_n \equiv 1$ on $(-n, n)$; then $d(\widetilde{g}_n, O) \to \infty$ for every $O$ and, by (1), $\omega_{\widetilde{g}_n} \to \omega_\infty$ in norm on every $\mathfrak{A}(O)$; Proposition 12.1 applies and gives (3) together with the $\alpha$-invariance. For (4): translation is an order isomorphism of $\mathcal{G}$ with cofinal range, so by (T) and (1),

$$\omega_\infty(A) = \lim_g \omega_{\tau_a g}(A) = \lim_g \omega_g(\tau_{-a}A) = \omega_\infty(\tau_{-a}A)$$

for local $A$, and by density on $\mathfrak{A}$. The $\mathbb{Z}_2$ statement is the local norm limit of Lemma 2.8: $\omega_g(\theta(A)) = \omega_g(A)$ for $A \in \mathfrak{A}(O)$, and $\theta(A) \in \mathfrak{A}(O)$ ($\theta$ maps $W(f,h)$ to $W(-f,-h)$ and hence preserves each local algebra), so (1) passes the identity to $\omega_\infty$, and density extends it to $\mathfrak{A}$. □

**Corollary 12.3** (the Glimm–Jaffe vacuum is canonical at weak coupling)**.** *Every $w^*$ limit point of the (averaged or unaveraged) finite-volume ground states constructed in [3, §2] coincides on each $\mathfrak{A}(O)$ with $\omega_\infty$; in particular the construction of [3] is subsequence- and averaging-independent in the weak-coupling region.*

*Proof.* For the unaveraged net this is Theorem 12.2(1). For the averaged states of [3, (2.3)–(2.5)]: fix $g \in \mathcal{G}$ with $g \equiv 1$ on $[-3,3]$ and set $g_n := g(\cdot/n) \in \mathcal{G}$; let $\sigma_{a'}$ denote the space-translation automorphisms used there (our $\tau_{\epsilon a'}$, with $\epsilon \in \{\pm1\}$ the direction convention), and let $\bar{h} \geq 0$, $\operatorname{supp}\bar{h} \subseteq [-1,1]$, $\int \bar{h} = 1$, be the averaging weight. Then

$$\begin{aligned}\omega_n(A) &= \frac{1}{n}\int \langle \Omega_{g_n}, \sigma_{a'}(A)\Omega_{g_n}\rangle\, \bar{h}(a'/n)\, da' = \int_{-1}^{1} \bar{h}(u)\, \omega_{g_n}\big(\tau_{\epsilon n u}A\big)\, du \\ &= \int_{-1}^{1} \bar{h}(u)\, \omega_{\tau_{-\epsilon n u}g_n}(A)\, du,\end{aligned}$$

by (T) (the estimate below is invariant under $\epsilon \to -\epsilon$). Since $g_n \equiv 1$ on $[-3n, 3n]$, the shifted cutoff $\tau_{-\epsilon n u}g_n \in \mathcal{G}$ equals 1 on an interval containing $[-2n, 2n]$ for every $|u| \leq 1$; hence for $O \subseteq [-R, R]$, $d(\tau_{-\epsilon n u}g_n, O) \geq 2n - R$ *uniformly in* $u$, and Theorem 12.2(1) gives

$$\|(\omega_n - \omega_\infty)|_{\mathfrak{A}(O)}\| \leq \int_{-1}^{1} \bar{h}(u)\, \big\|(\omega_{\tau_{-\epsilon n u}g_n} - \omega_\infty)|_{\mathfrak{A}(O)}\big\|\, du \leq \Psi_{m_1}(2n - R) \longrightarrow 0.$$

No translation invariance of $\omega_\infty$ is needed for this bound. □

## 13 The two-phase dichotomy

The uniform gap of Theorem 7.8 is incompatible with a fully classified two-phase structure. The incompatibility is stated as a conditional theorem: *all four* hypotheses below are load-bearing, and (B4) in particular cannot be omitted — dropping it was the one proof-breaking defect (P1) found by the independent audit of the predecessor manuscript (Appendix B).

**Hypothesis 13.1** (structure of a broken phase)**.** *$\mathscr{P}$ is even, and:*

(B1) *there are translation-invariant ground states* $\omega_+$ *and* $\omega_-$ *of* $(\mathfrak{A},\alpha)$ *such that, for all bounded local* $X, Y$*,*

$$\omega_\pm(X\tau_a(Y)) - \omega_\pm(X)\,\omega_\pm(\tau_a(Y)) \longrightarrow 0 \qquad (|a|\to\infty);$$

(B2) $\omega_+ \circ \theta = \omega_-$ *and* $\omega_- \circ \theta = \omega_+$*;*

(B3) *there are a bounded local observable* $B = B^* \in \mathfrak{A}(O)$ *and* $m \neq 0$ *such that* $\omega_\pm(B) = \pm m$*;*

(B4) *every* $\theta$*-invariant ground state of* $(\mathfrak{A},\alpha)$ *equals* $\frac{1}{2}(\omega_+ + \omega_-)$*.*

**Theorem 13.2** (conditional exclusion of a two-phase structure)**.** *Assume that* $\mathscr{P}$ *is even and* $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$*. Then the assertions* (B1)–(B4) *of Hypothesis 13.1 cannot hold simultaneously.*

*Proof.* Assume that $\mathscr{P}$ is even and (B1)–(B4) hold. Choose $\widetilde{g}_n \in \mathcal{G}$ with $\widetilde{g}_n \equiv 1$ on $(-n,n)$ and write $\omega_n := \omega_{\widetilde{g}_n}$. Fix $a \in \mathbb{R}$ such that

$$R_a := \operatorname{dist}(O, O+a) > 0, \qquad m_1 R_a \ge 1,$$

where $O$ is the localization interval of the order parameter $B$ from (B3).

*Step 1: uniform cutoff clustering for the bounded order parameter.* For the cutoff Hamiltonian $H(\widetilde{g}_n)$ apply Lemma 9.3 with $\gamma = m_1$ (Theorem 7.8), unbounded-argument slot $X = B$, bounded-argument slot $A = \tau_a(B) \in \mathfrak{A}(O+a)$, and localization set $\overline{O}$. The bounded operator $B$ has domain $\mathcal{H}$; because $B \in \mathfrak{A}(O) \subseteq \mathfrak{A}(O')'$ for every bounded open interval $O'$ with $\operatorname{dist}(\overline{O}, O') > 0$ (such $O'$ is disjoint from $O$; Assumption A6(iii)), it is affiliated with every such $\mathfrak{A}(O')'$, and $\operatorname{dist}(\overline{O}, O+a) = R_a$. Thus all hypotheses of that lemma hold, and since $\|B\Omega_{\widetilde{g}_n}\| \le \|B\|$ and $\|\tau_a(B)\| = \|B\|$, it gives, uniformly in $n$,

$$\big|\omega_n\big(B\,\tau_a(B)\big) - \omega_n(B)\,\omega_n(\tau_a(B))\big| \le 2\|B\|^2 e^{-m_1 R_a/2}. \tag{33}$$

*Step 2: passage to the limit.* For fixed $a$, all three observables in (33) belong to the local algebra of one bounded interval containing $O \cup (O+a)$. The local-norm convergence of Theorem 12.2(1) therefore permits passage to $n \to \infty$ in (33); translation invariance of $\omega_\infty$ (Theorem 12.2(4)) then yields

$$\big|\omega_\infty\big(B\,\tau_a(B)\big) - \omega_\infty(B)^2\big| \le 2\|B\|^2 e^{-m_1 R_a/2} \xrightarrow[|a|\to\infty]{} 0. \tag{34}$$

*Step 3: the midpoint identification.* By Theorem 12.2(3)–(4), $\omega_\infty$ is a $\theta$-invariant ground state of $(\mathfrak{A},\alpha)$. Assumption (B4) consequently gives

$$\omega_\infty = \tfrac{1}{2}(\omega_+ + \omega_-), \qquad \omega_\infty(B) = \tfrac{1}{2}(m - m) = 0,$$

using (B3).

*Step 4: long-range order of the midpoint.* By (B1), translation invariance and clustering of each $\omega_\pm$ imply

$$\omega_\pm\big(B\,\tau_a(B)\big) - \omega_\pm(B)\,\omega_\pm(\tau_a(B)) \longrightarrow 0, \qquad \omega_\pm(\tau_a(B)) = \omega_\pm(B) = \pm m.$$

Hence

$$\omega_\infty\big(B\,\tau_a(B)\big) = \tfrac{1}{2}\big(\omega_+\big(B\,\tau_a(B)\big) + \omega_-\big(B\,\tau_a(B)\big)\big) \longrightarrow \tfrac{1}{2}(m^2 + m^2) = m^2 \neq 0,$$

contradicting (34), whose left side tends to zero because $\omega_\infty(B) = 0$. □

**Remark 13.3** (scope of the phase-transition literature)**.** The GJS phase-transition and mean-field papers [8, 9] prove the existence of phase transitions and of two pure phases in specified strong-coupling regimes. As cited here, they are context: they do not, by themselves, supply the full Hamiltonian phase-space classification (B1)–(B4) — in particular not the midpoint identification (B4) — for the state space of $(\mathfrak{A}, \alpha)$. This manuscript therefore makes no unconditional assertion that its state space carries the structure of Hypothesis 13.1; Theorem 13.2 says only that, at weak coupling, that structure cannot coexist with the cutoff-uniform gap. The two rigorous options recorded by the audit — state (B1)–(B4) explicitly, or delete the corollary — are resolved here by the first.

# 14 Discussion

*Relation to GRS Problems 1–2.* Problem 2 of [10, p. 126] asks for local convergence of Euclidean Gibbs states; Theorem 12.2(1) is a Hamiltonian counterpart, in norm on local von Neumann algebras, for the vacuum family. Problem 1 asks for uniform local $L^p$ bounds on the Euclidean vacuum density. The note added on [10, p. 128] records Newman's solution of the small-coupling $p = 1$ versions, so the printed problems must not be read as an unqualified current open-problem claim. The present proof neither establishes nor needs Problem 1: the identification step uses the exact spectral identity of Proposition 12.1, and the convergence step uses the Cauchy estimate.

*Sharpness.* In the exactly solvable quadratic model the true rate is $\mathrm{poly}(d)\, e^{-2m_1 d}$ (companion note `m2/`: a round trip through the region where the two Hamiltonians differ, at the local Agmon mass). $\Psi_{m_1}$ is super-polynomial but sub-exponential. The present filter argument does not obtain the sharp exponential rate; a proof of such a rate would require an additional resolvent or Combes–Thomas input not supplied here.

*Uniqueness.* $\omega_\infty$ is canonical for the cutoff family (Corollary 12.3), and translation invariant without averaging. The theorem does not identify every locally normal translation-invariant ground state of $(\mathfrak{A}, \alpha)$ with $\omega_\infty$. In particular, the Euclidean results [13, 14] do not by themselves provide the reverse reconstruction and state-space identification needed for that Hamiltonian conclusion; companion note `m3/` records that missing implication as its item (R3).

*Beyond weak coupling.* Theorem 13.2 is conditional on Hypothesis 13.1. Under exactly those hypotheses, a proof based on a cutoff-uniform gap cannot extend to that two-phase regime. The phase-transition results [8, 9], as cited here, are not used as a substitute for (B4) (Remark 13.3), and this manuscript makes no unconditional assertion that its Hamiltonian state space has the classification (B1)–(B4).

*Part II.* Uniformity in a lattice/wavelet resolution — the operator-algebraic renormalization programme over the maps of [18] — is a separate problem with no mathematical dependence in either direction, and its record lives in `part_ii/` and `strategy.md`. Its present state, in one paragraph: for the free nearest-neighbour model, with lattice vacua $\omega_N^{(0)}$ and continuum vacuum $\omega^{(0)}$, there are local Weyl unitaries with $\liminf_{N\to\infty} \|\omega_N^{(0)} - \omega^{(0)} \circ \beta_N\|_{\mathfrak{W}_N(O)} \geq e^{-\sqrt{2+\sqrt{2}}/4} - e^{-3\pi/16} > 0$ for the Daubechies observable embedding $\beta_N$, so no diagonal (finest-scale) local-state-norm error $C\varepsilon_N^{\vartheta}$, $\vartheta > 0$, is possible, and the meaningful comparison is at fixed coarse scale; there, the free full sequence converges on each fixed coarse Weyl generator with an explicit power rate, an interacting fixed-torus projective cluster state exists unconditionally along a cofinal subnet, and the full-sequence identified interacting theorem is reduced to a single scalar characteristic-functional limit, which remains open.

### Standing scope

Everything in this manuscript is restricted to the weak-coupling region $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$ of §2.6; no claim is made for the full noncritical region. The external residue — a second-reader verification of the GJS page images — is recorded in Appendix B.

# A Constants

All values below are fixed by the printed norms (1) (gate R2, page-image reading) and the displays of the body sections; the rows present before 7 November 2026 were re-verified against their defining displays at gates G3a/G3b (see `part_i/README.md`); the rows added or changed on 7 November 2026 are recomputed by hand in Appendix D, §D.g, and in Appendix C.

| constant | value / definition | defined in |
|---|---|---|
| $K_1, K_2, C_7, m_1$ | GJS constants; normalization $K_2 \ge 1$ (the constant of the retired A8(vi) is no longer listed) | Assumption A8 |
| $\varepsilon_{\mathscr{P}}$ | $\lambda_*/M_*^2$ | §2.6 |
| $N_I$ | columns of the GJS squares meeting $I$ in positive measure; $\le \lfloor L \rfloor + 2 \le L + 2$ ($\lvert I \rvert = L$) | Lemma 6.1 |
| $\mathfrak{K}(p')$ | $K_1 K_2^{8p'} (4p')!$ | Thm. 5.1 |
| $c_U$ | $\lambda \sum_{n=1}^{2p'} \lvert a_n \rvert K_1 K_2^{2n} n!$ (one square's interaction) | Lemma N.18 |
| $S_2(B)$ | $\sup\{\lVert e^{-\mathcal{U}_{h'}} \rVert_{L^2(d\mu_0)} : h' \in \mathcal{C}_{\mathrm{E}},\ \operatorname{supp} h' \subseteq \overline{B}\}$; finite by Thm. C.5, *not evaluated*; depends on the shape of $B$ only | Prop. N.15 |
| $\overline{\Sigma}_{3\times 5}$ | $\sum_{j\ge 1}(8j+12)e^{-m_1 j} = 4e^{-m_1}(5-3e^{-m_1})(1-e^{-m_1})^{-2}$ | Lemma N.11 |
| $V_*$ | $\sup_{h'} \lVert V(h') \rVert^2_{L^2(d\phi_0)} \le \lambda^2 \sum_{n=0}^{2p'} a_n^2 n!\, I_n$, $I_n = \iint_{[-1,1]^2} C(x-y)^n$ | Lemma N.20 |
| $M^2 = M_B^2$ | $e^{15\lambda\lvert a_0\rvert} S_2(B_{3\times 5})\, 9^{p'} V_* +$ $9\, C_7 c_U \overline{\Sigma}_{3\times 5}\, \mathfrak{K}(p') \lambda^2 \big(\sum_{n=0}^{2p'} \lvert a_n \rvert\big)^2$; contains the unevaluated $S_2$ | Thm. 5.1, Thm. D.4 |
| $M^2$ ($\varphi^4$) | $81\, S_2(B_{3\times 5}) V_* + 8\,709\,120\, C_7 K_1^2 K_2^{24} \overline{\Sigma}_{3\times 5} \lambda^3$, $V_* \le 24\lambda^2 I_4$ | Cor. 5.2 |
| $A$, $B'$ | $e^{3\lambda\lvert a_0\rvert N_B} S_2(B_s(g, I))$; $2C_7 c_U N_g e^{-m_1}(1-e^{-m_1})^{-1}$ (fixed $g$; parallel strips) | Thm. D.3 |
| $K'$ | $K_2^2 N_I^{1/2} \lVert f \rVert_2$ ($N_I$ on the columns) | Lemma 7.6 |
| $K_f$, $A_f$ | $\max\{K', \sqrt{2}\,\sigma_f\}$, $\sigma_f^2 = \langle f, Cf \rangle$; $A + B' K_1$ (exponential moments; radius $1/K_f$) | Lemma 7.6 |
| $s_0$ | $\min_{j:\, e_j \ne 0} (2k K_{e_j})^{-1}$ (polystrip of the density theorem) | Thm. 7.7, Rem. N.54 |
| $\widehat{C}_Q$ | $\max\{C_7 e^{2m_1} \lVert Q \rVert^2_{\mathrm{GJS}},\ e^{3m_1} \lVert \psi_Q \rVert^2\}$ | Prop. 7.5 |
| $\mathfrak{m}$ | total-variation bound, separation lemma | Lemma 9.2 |
| $d_0$ | $4 + 2/\gamma$ | Prop. 10.4 |
| shell count | $\le 8$ indices $j$ with $r_j \in [n, n+1)$ | Prop. 10.4 |
| $\Psi_\gamma(d)$ | $32 M \sum_{n \ge \lfloor d \rfloor} \big(\lVert W_\gamma \rVert_1 e^{-\gamma n/4} + T_\gamma(n/2)\big)$; $\max\{2, \cdot\}$ if $d < d_0$; $M = M_B$ | Lemma 10.5 |
| $\Psi_\gamma$ tail | $\le C_{N,\gamma,M}(1+d)^{-N}$ for every $N$; $M = M_B$ | Lemma 10.5 |
| $b_0$, $C_*$, $c_1$, $c_2$, $b_*$ | kernel and stability constants of Appendix C, all explicit ((C.a), (C.c), Lemmas C.3–C.4, Thm. C.5) | Appendix C |

The rows $S_2(B)$, $A$, $A_f$, $M$ and the $\Psi_\gamma$ rows contain the finite constant of Theorem C.5, which is not evaluated (its bound $(1+e^{2b_*})^{1/2}$ at $p=2$ is explicit but crude); every other constant is explicit. Until 7 November 2026 the $M$ rows contained the constant of the retired item A8(vi) (Appendix B, corrections history); the formulas of Lemma 10.5 are unchanged.

# B Provenance, corrections history, and dossier

## Frozen sources of this manuscript

| file | role |
|---|---|
| `m6/main.tex` (v2, audited repair) | predecessor manuscript; statements and compressed proofs |
| `audit_2026/critical_proof_audit.tex` | independent audit (frozen against the v1 SHA-256) |
| `audit_2026/lamport_euclidean_to_gap.tex` | fully explicit proof, GJS $\Rightarrow$ ($M_2$)+(G) |
| `audit_2026/lamport_gap_to_main.tex` | fully explicit proof, ($M_2$)+(G) $\Rightarrow$ main theorem |
| `audit_2026/citation_screenshot_dossier.tex` + `evidence/` | page images for every import |
| `m4/r1-check.tex` | gate R1: $h$-uniformity of the Euclidean inputs |
| `m4/r2-check.tex` | gate R2: printed norms and exact constants (page image, p. 594) |
| `m0/m0.tex`, `m2/m2.tex`, `m3/m3.tex` | spectral passage; quadratic-model rate; dichotomy |
| `m4/m2-audit.tex`, `m4/g-side.tex` | superseded derivations (with update notices) |
| `ugjs/comparison.tex` (with `ugjs/parts/`) | the comparison note, integrated as Appendix D on 7 November 2026; audit record `ugjs/council/` |
| `strategy.md` | programme record, incl. the unification plan and gates |

The two structured proof reconstructions follow the hierarchical proof format of [19].

## Corrections history (chronological)

- *v0 → v1 merge* (9 Aug 2026): filter-transform sign corrected to $+1/E$; spurious factor $\frac{1}{2}$ removed from the one-path bound; an unnecessary $g''$ detour eliminated; the averaging identity of the canonical-vacuum corollary repaired.

- *Gate R2* (9 Aug 2026, page-image inspection of [7, p. 594]): prefactor $K_1$ to the first power; local weight $K_2^{2N(\Delta)}$; the earlier "correction" $K_2^{4p} \to K_2^{2p}$ of `m4/m2-audit.tex`, Rem. 5.3(i), was an OCR artifact and is withdrawn; a half-degree slip in the v0 assembly ($(2p')!$, $K_2^{4p'}$ in place of $(4p')!$, $K_2^{8p'}$) found and fixed. The cell-pair count for ($M_2$) is 9, not 32.

- *Independent audit* (9 Aug 2026): severity register P1 (two-phase corollary missing premise (B4) — repaired by conditioning on Hypothesis 13.1); S1–S3 (citation pinpoints: FKN, hypercontractivity, domain smoothing — repaired in Assumptions A2, A4, A5); L1–L6 (cell count $\lfloor L \rfloor + 2$; negative-time norm ratio $c_g^{-1}$; nonnegative FKN inputs $F_1 = F_2 = |V(h)|$; signed/clipped truncations; exponential-moment prefactor $C_8 K_1$; spectral-parameter range $0 \le \beta < m_1$); C1 (dimensionless scaling reduction made explicit); C2 (all imports enumerated — Assumptions A1–A8); H1–H2 (literature claims narrowed; GRS $p = 1$ note recorded). No P0 defect was found.

- *Re-sourcing of A8(vi)* (31 October 2026, after the council record `ugjs/PLAN.md`): item (vi) is read off the printed Theorem 1.1.11 of [7] (p. 596, "uniformly in $h$") in the specialization $B = I$ ($T = 0$ on p. 600), $n = 0$ (named on p. 629), that the source prints as its proof of Theorem 1.1.8; the hypothesis clause naming $U_{\mathrm{GJS}}$ is removed from the

statements of Theorem 12.2 and Proposition C.7; no proof changed, no label moved. In the same pass: item (ii) names the time-integrated monomials (1.1.6) and the square convention of p. 593; the Scope paragraph records the one case of Theorem 1.1.7 that its printed proof does not reach (sharp-time monomials across strips transverse to the time-zero line, unused here); the mechanism audit of Appendix C gains item M6 (the stability input (2.4.34) of GJS p. 624, which its items M1 and M3–M5 had wrongly declared absent) and its covariance description is corrected from "Dirichlet" to "decoupled".

- *Integration of Appendix D* (7 November 2026, after the comparison note `ugjs/comparison.tex`, drafted 31 October–4 November 2026, and its four audit rounds of 2–4 November, `ugjs/council/`): the uniform local moment is derived from Theorem 1.1.7 (Lemmas D.1–D.2, Theorems D.3–D.4); the proofs of Lemma 4.3(ii), Theorem 5.1 (Steps 1 and 5), Corollary 5.2, Lemmas 7.1–7.3 and 7.6, and the radius in the proof of Theorem 7.7 are replaced, and the statements of Theorem 5.1 and Corollary 5.2 (their constant, $M := M_B$) and of Lemmas 7.1 and 7.6 (their form and constants) change; A8(vi) is retired and the transverse case is listed as A8(iv)-T; no label moved (the 125 labels of 2 November 2026 keep their numbers; Appendix D adds Lemmas D.1–D.2, Theorems D.3–D.4 and its auxiliary statements, numbered N.$k$). Three defects of the record found on the way were repaired in the same pass. *(i)* The V3 audit of Appendix C (2 November 2026; `ugjs/council/v3-appendix-c-report.md`) found Lemmas C.2–C.4 and Theorem C.5 true but resting on three unimported inputs (MAJOR): the kernel facts of Lemma C.4 and the undefined $K_0$, the undefined constant $C_*$, and the Wick-chaos structure of $\mathcal{U}_h$; repaired by the construction paragraph of Assumption A3, the kernel facts (C.a)–(C.b) (with Lemma N.19), the explicit constants $C_*$, $c_1$, $c_2$, $b_*$ and the thresholds of Theorem C.5. *(ii)* The count $N_I$ was taken on the cells $[j, j+1)$ while the norm of p. 594 sums over squares centred at lattice sites: the displayed bounds (7)–(8) were false as stated for that norm (counterexample in Remark N.3; CRITICAL for the display, MINOR for every theorem, which needs finiteness only); repaired by counting columns. *(iii)* The manuscript wrote $V(h)$ and $\mathcal{U}_h$ as formal integrals and stated no construction of their Wick powers, so that Step 1 of the proof of Theorem 5.1 identified $V(h)^2$ with a sum of Wick monomials without a definition behind it (MAJOR as a statement about the record; no step false); repaired by the Wick-power paragraphs of Assumptions A1 and A3 and Lemma N.23′; the affiliation of $V(h)$ with the local algebras, asserted in §2.1 without an argument, is now proved there (*Locality*), for every bounded open set containing $\operatorname{supp} h$; the hypothesis of Lemma 9.3 on its unbounded argument is restated to match, and the locality of the net, used before without being listed, is listed as Assumption A6(iii), with the generators of $\mathfrak{A}(O)$ written as in the sources, $W(f, 0)$ and $W(0, h)$ (§2.1). Also: A8(vii′)(a) transcribed (1.1.11) with "$O(1)$" where p. 594 prints "0(1)" with a digit zero; it now transcribes it as printed, marked [sic]; Lemma 7.1's misprint "le" for $\leq$ is gone with its replaced proof.

- *Withdrawn routes* (kept in the record, not used here): `m4/h6-attack.tex` Thm. 2.3 (withdrawn; UV-divergent double commutator); gap-based (H6) proofs; wavelet-MRA-as-Polchinski decomposition (see Part II documents).

### External residue

Audit closure item C5 — verification by a second human reader of the GJS page images — remains open at the level of this research record. The itemized reading checklist is

`part_i/C5_reading_record.md`: items 1–4 (the cutoff class, the localized monomials and squares, the norms, Theorem 1.1.7 and the dilation; pp. 589, 590, 593, 594) carry the imports of Assumption A8; items 10–11, item 12(a),(c) and item 9(c) (the two forms (1.1.5)–(1.1.6) with p. 595's multi-time notation; Theorem 1.1.7's strip without orientation; Theorem 1.1.11's "monomials of type (1.1.5)" and the rotation sentence of p. 597) carry the readings of the Scope paragraph, points (b)–(d), and the item A8(iv)-T; the rest of items 5–9 and 12, and item 13 (Theorem 1.1.8, Theorem 1.1.11 and the pages behind the retired item (vi)), are corroboration since 7 November 2026. Every status sentence of this manuscript is to be read with these human readings named as open beside it. The page images of Simon's book read for the Wick-power constructions on 7 November 2026 (pp. 10–12, 15–17, 25, 34, 38, 134–138, 152) are in `audit_2026/evidence/` and in the Dossier lines of Assumptions A1, A3 and A4; the dossier's index file has not yet been updated.

# C $h$-uniformity: source audit and analytic prerequisites

This appendix has two sharply separated purposes. First, it records the audit of the source mechanism that preceded the re-sourcing of Assumption A8(vi) on 31 October 2026 (items M1–M5, with the correction M6 added that day); that item was retired on 7 November 2026, and the audit is kept as a record. Second, it proves the $h$-sensitive analytic prerequisites over the full class $\mathcal{C}_{\mathrm{E}}$ that Appendix D consumes: normalization of $dq_h$ (Lemma C.2), the ultraviolet limit (Lemma C.4) and uniform $L^p$ stability under the free measure (Theorem C.5); since 7 November 2026 these are load-bearing for Theorems D.3–D.4 and hence for Theorem 12.2, and every input they use is derived here, in Appendix D (Lemmas N.19, N.23′(i) and N.25, which do not use this appendix), or imported in Assumptions A1, A2, A3 and A4 (the V3 audit of 2 November 2026 found them true but resting on three unimported inputs, repaired on 7 November 2026; Appendix B). The appendix does *not* reproduce the polymer expansion, its convergence summations, or the proof of the cluster-expansion constants: Theorem 1.1.7's own statement makes its constants independent of $h$; its printed proof consumes Theorem 1.1.8 at the same $h$ and reaches sharp-time monomials only across strips parallel to the time-zero line (Scope paragraph of Assumption A8, point (b); item A8(iv)-T).

## Mechanism audit

The expansion machinery of [7, §§2–4] manipulates integrals $\int R(\Phi)\, B\, e^{-\mathcal{U}_h}\, d\mu_{0,C}$, where $C$ is an interpolated covariance and $d\mu_{0,C}$ denotes the Gaussian measure with covariance $C$, through six devices; the following lists every point of contact with $h$ (item M6 was added on 31 October 2026, after the council of `ugjs/PLAN.md` found it missing).

**M1 (the norms are kernel norms).** The norm (2.4.3) of an expansion term $m_\alpha$ is $\int du\, ds\, \exp[\cdots]\, \|w(\alpha, u, s, y, \cdot)\|_{L^2}$ times combinatorial weights (2.4.4)–(2.4.5) built from $m_0$, $\varepsilon$, $K_2$, $N(\Delta)$ (the source's script $\mathcal{K}_2$ of (2.4.4) is $e^{-(1-\varepsilon)m_0 \sum \delta n}$, not the constant $K_2$), times $K_1^\nu e^{\gamma t}$ ((2.4.3), p. 613): a functional of the *kernel* $w$ alone. The norm itself contains no Gaussian expectation; the kernels $w$ it is applied to are, however, defined through integrals of $e^{-V(Y_j)}$ under the interpolated measures ((2.3.4), (2.3.8), pp. 611–612), see M6. The norm-decrease Theorems 2.4.1, 3.1, 4.1 are inequalities between such kernel functionals, and their proofs (Lemmas 2.4.2–2.4.4, (3.2)–(3.6), §4) estimate kernels: contraction kernels $\delta C$ against vertex kernels.

**M2 (covariances are $h$-free).** The interpolation $C(s) = sC' + (1-s)C''$ (2.0.1) is between covariances of the family generated from $C_0$ by "diagonalization" with respect to unions

$Y$ of unit squares, $C_Y = YCY + (1-Y)C(1-Y)$, and convex sums (p. 600); the kernels (2.2.2)–(2.2.3) are built from $(-d^2/dx^2 + m_0^2)$ and contour geometry. Verified against the legible scan (`m4/r1-check.tex`): no $h$ enters any covariance, any contraction kernel, or any of the Propositions 2.1.$x$.

**M3 ($h$ in vertices, majorised at first contact).** Vertices arise from functional derivatives of the spectator:

$$\frac{\delta}{\delta\Phi(x)} e^{-\mathcal{U}_h} = -\lambda\, h(x) :\mathscr{P}'(\Phi(x)): e^{-\mathcal{U}_h},$$

and iterates with $\mathscr{P}'', \dots$ A vertex therefore contributes to the *kernel* a factor $\lambda h(x) p_j$ supported in a lattice square. The very first estimate applied to it is the printed one (p. 626): "$v(x)$ is $O(\lambda)$ times the characteristic function of a lattice square" — that is, $|\lambda h(x) p_j| \le O(\lambda)\chi_\Delta(x)$, using $0 \le h \le 1$ and nothing else. Downstream of this line the vertex kernels are $h$-free. This is one of two points where $h$ touches a quantitative estimate; the other is M6.

**M4 (the identities are exact and $h$-blind).** The devices (2.0.4)–(2.0.5) (covariance interpolation), (2.2.1), (2.2.5), (2.2.7) (advancing annihilators), and (4.1)–(4.2), (4.5) (integration by parts) are identities applied to integrands $F = R(\Phi) B_T e^{-\mathcal{U}_h}$; $e^{-\mathcal{U}_h}$ rides along inside $F$. Their validity requires only that the integrals exist — integrability, not bounds. For the free-measure endpoint this is supplied, uniformly over $\mathcal{C}_{\mathrm{E}}$, by Theorem C.5; existence under the interpolated measures $d\mu_{0,C}$ is internal to the printed proof. A quantitative stability bound under those measures does enter the constants: see M6.

**M5 (the final division adds no stability constant).** The full expansion (4.3) reads

$$\int A B_T e^{-\mathcal{U}} d\mu_{0,C_0} = \sum_i \int \Big[ \int dy\, w_i(y) :\Phi(y_1)\cdots\Phi(y_i): \Big] B_T e^{-\mathcal{U}} d\mu_{0,C_0} + \sum_r w_r \int e^{-\mathcal{U}} d\mu_{0,C_0},$$

with $w_i, w_r$ *numerical kernels* controlled by the norm inequalities; the remainder terms are literally multiples of $\int e^{-\mathcal{U}} d\mu_0$ (their Table 1: final-$r$ terms have "no vertices, no annihilators"). The proof of Theorems 1.1.8/1.1.11 then *divides (4.3) by* $\int e^{-\mathcal{U}} d\mu_{0,C_0}$ (p. 628): the first sum reassembles into $dq_h$-integrals of the comparison polynomials, and the remainder becomes $\sum_r w_r$ exactly. Every factor $e^{-\mathcal{U}_h}$ cancels against the normalisation. Consequently the final division introduces no stability constant; the stability bound that does enter the constants is the one of M6, upstream of the division.

**M6 (stability under the interpolated measures; added 31 October 2026).** The kernels $w''$ of (2.3.8) are bounded on p. 624 through a stability estimate under the interpolated measures: "For $m_0 \ge 1$, $\lambda \le 1$, there is a constant $K_3$, independent of $C$, $m_0$ and $\lambda$ such that

$$(2.4.34) \qquad 1 \le \|e^{-V(Y)}\|_{L_2(d\Phi_C)} \le e^{K_3|Y|}$$

and (2.4.35) [a bound on $\|P_l''(u,s,y,x_B)\|_{L_2(d\Phi_C)}$ by $\prod_\Delta K_3^{N(\Delta)} N(\Delta)!$ times an $L_2$ kernel norm]. These are standard estimates for such integrals (see [1])." The bound "yields (2.4.28), for $K_2 \ge K_3^2$" (p. 624), and Lemma 2.4.6 (p. 622) takes "$K_2 = K_2(m_0, K_3)$"; so $K_3$ enters every constant chosen after $K_2$ in the printed order "$m_0$, $K_2$, $K_1$, and then $\lambda$" (p. 616), $C_7$ and $C_8$ included. The cutoff sits inside $V(Y)$: (2.0.2) on p. 601 factors $V(h)$ as $V(Yh) + V((1-Y)h)$. Reference [1] is Dimock–Glimm, Adv. Math. 12 (1974), not consulted here; $K_3$ is printed independent of $C$, $m_0$ and $\lambda$, and $h$ is not mentioned. Theorem C.5

does not supply (2.4.34): it is a bound under $d\mu_0$ for a fixed compact region, without linear growth in $|Y|$. The convergence argument on pp. 624–625 also uses $|m_\alpha| \le \|m_\alpha\|_{2K_2} e^{O(\mathrm{Vol})}$, "where the volume of interaction, denoted Vol, is the area of the set of points within unit distance of suppt $h$. Here we use (2.4.34), (2.4.35) and a Schwarz inequality to factor out vertices" (p. 625), a per-$h$ qualitative step. Items M1 and M3–M5, as transcribed from `m4/r1-check.tex` (Gate R1, 9 August 2026), declared this input absent; that statement was wrong about the source.

**Proposition C.1** (consequence of the audited source mechanisms)**.** *Within mechanisms M1–M5 as recorded above, and excluding the stability input of M6, which lies outside them, every displayed quantitative bound is unchanged when $h$ ranges over $0 \le h \le 1$: M3 replaces its only vertex occurrence by $|h| \le 1$, while M4–M5 use identities or exact cancellation. Consequently the constants visible in those mechanisms depend on $(m_0, K_2, K_1, \lambda, n, \varepsilon, \mathscr{P})$ and not on $h$; the $h$-uniformity of $K_3$ in M6 is not among the audited mechanisms.*

*Proof.* By direct substitution in M1–M5: the quantitative chain displayed there is entirely at the level of kernel norms (M1), whose ingredients are $h$-free covariance/contraction kernels (M2) and vertex kernels majorised by $O(\lambda)\chi_\Delta$ (M3); the measure-level steps are exact identities (M4) or exact cancellations (M5). This proposition is deliberately limited to the audited mechanisms; it makes no completeness claim about portions of the source expansion not reproduced here. □

### The two $h$-sensitive analytic facts, proved

Notation for this subsection: $G := (-\Delta_{\mathbb{R}^2} + m_0^2)^{-1}$ is the Euclidean covariance, with the kernel

$$G(\zeta) = \int_0^\infty \frac{e^{-m_0^2 t - |\zeta|^2/(4t)}}{4\pi t}\, dt \qquad (\zeta \neq 0)$$

of Lemma N.19(v) (Appendix D): positive, radial, nonincreasing in $|\zeta|$, continuous on $\mathbb{R}^2 \setminus \{0\}$, with $\int_{\mathbb{R}^2} G\, d\zeta = m_0^{-2}$ and

$$0 < G(\zeta) \le \frac{\ell(\zeta)}{4\pi}, \qquad \ell(\zeta) := b_0 + 2\big|\log|\zeta|\big|, \qquad b_0 := \log 4 + e^{-1} + m_0^{-2} \tag{C.a}$$

(the same lemma, part (v), through its part (iii); the substitution $t = (|\zeta|/2m_0)e^u$ writes $G(\zeta) = (2\pi)^{-1} K_0(m_0|\zeta|)$ with $K_0(x) := \int_0^\infty e^{-x\cosh u}\, du$, which is not used below). Further, for $\zeta \neq 0$,

$$|\nabla G(\zeta)| \le \frac{1}{2\pi|\zeta|}. \tag{C.b}$$

Indeed, on $\{|\zeta| \ge \varepsilon\}$, $\varepsilon > 0$, the $\zeta$-derivative of the integrand is dominated by $(4\pi t)^{-1}(|\zeta|/2t)e^{-\varepsilon^2/(4t)}$ for $|\zeta|$ in a bounded set, which is integrable over $t \in (0,\infty)$, so differentiation under the integral gives

$$\nabla G(\zeta) = -\int_0^\infty \frac{\zeta}{2t} \frac{e^{-m_0^2 t - |\zeta|^2/(4t)}}{4\pi t}\, dt, \qquad |\nabla G(\zeta)| \le \frac{|\zeta|}{8\pi} \int_0^\infty t^{-2} e^{-|\zeta|^2/(4t)}\, dt = \frac{|\zeta|}{8\pi} \cdot \frac{4}{|\zeta|^2}$$

(substitute $u = |\zeta|^2/(4t)$, so that $t^{-2}\, dt = -4|\zeta|^{-2}\, du$ and the integral is $4|\zeta|^{-2}\int_0^\infty e^{-u} du$). Let $h \in \mathcal{C}_{\mathrm{E}}$ with $\operatorname{supp} h \subseteq \Lambda$, $\Lambda$ compact, and $R := \operatorname{diam}\Lambda$ (so $|\Lambda| \le \pi R^2$); $\mathcal{U}_h = \lambda \sum_{n=0}^{2p'} a_n{:}\Phi^n(h)$: and $Z(h) = \int e^{-\mathcal{U}_h} d\mu_0$ as in Assumption A3, whose construction paragraph supplies the chaoses $\Gamma_n := \Gamma_n(\mathcal{H}_G)$, the pairing (W.2), the isometry (W.3), the orthogonality of the chaoses and the action of $\Gamma(e^{-t}\mathbf{1})$ on them. Choose once and for all

$$\delta \in C_c^\infty(\mathbb{R}^2), \qquad \delta \ge 0, \qquad \delta(z) = \delta(-z), \qquad \operatorname{supp}\delta \subseteq \{z : |z| \le 1\}, \qquad \int_{\mathbb{R}^2} \delta(z)\, dz = 1,$$

and, for $\kappa \geq 1$, define the two-dimensional rescaling $\delta_\kappa(z) := \kappa^2\delta(\kappa z)$, so that $\|\delta_\kappa\|_\infty = \kappa^2\|\delta\|_\infty$, $\|\delta_\kappa\|_1 = 1$ and $\operatorname{supp}\delta_\kappa \subseteq \{|z| \leq 1/\kappa\}$; $\|\delta\|_\infty \geq 1/\pi$, since $\delta$ is a probability density on the unit disc. Thus convolution by $\delta_\kappa$ is a positive average over a ball of radius $\kappa^{-1}$; a double convolution averages over displacements of radius at most $2\kappa^{-1}$. Set $\Phi_\kappa(z) := \Phi(\delta_\kappa(z-\cdot))$, a Gaussian random variable for each $z$, with $z \mapsto \Phi_\kappa(z)(q) = (q * \delta_\kappa)(z)$ smooth for every $q \in \mathcal{S}'(\mathbb{R}^2)$, and variance

$$c_\kappa := \mathbb{E}[\Phi_\kappa(z)^2] = \int (\delta_\kappa * \delta_\kappa)(w)\, G(w)\, dw,$$

which does not depend on $z$ for $\kappa \geq 2$, the only range used below (Lemma N.25(iii): $c_\kappa = G_{\kappa\kappa}(0)$). For every $\kappa \geq 2$,

$$c_\kappa \leq \frac{1}{2\pi}\log\kappa + C_*, \qquad C_* := \frac{b_0 + \pi\|\delta\|_\infty}{4\pi}. \tag{C.c}$$

Indeed $\rho := \delta_\kappa * \delta_\kappa$ is a probability density with $\operatorname{supp}\rho \subseteq \{|w| \leq 2/\kappa\}$ and $\|\rho\|_\infty \leq \|\delta_\kappa\|_\infty\|\delta_\kappa\|_1 = \kappa^2\|\delta\|_\infty$; on $\operatorname{supp}\rho$, $|w| \leq 2/\kappa \leq 1$, so (C.a) gives $4\pi c_\kappa \leq \int \rho(w)\big(b_0 + 2\log\frac{1}{|w|}\big)dw = b_0 + 2\log\kappa + 2\int\rho(w)\log\frac{1}{\kappa|w|}\,dw$, and, dropping the negative part of the last logarithm, $\int\rho(w)\log_+\frac{1}{\kappa|w|}\,dw \leq \kappa^2\|\delta\|_\infty\int_{|w|\leq 1/\kappa}\log\frac{1}{\kappa|w|}\,dw = \|\delta\|_\infty\int_{|v|\leq 1}\log\frac{1}{|v|}\,dv = 2\pi\|\delta\|_\infty\int_0^1 s\log\frac{1}{s}\,ds = \frac{\pi}{2}\|\delta\|_\infty$ (substitute $w = v/\kappa$; then polar coordinates; $\int_0^1 s\log\frac{1}{s}\,ds = \frac{1}{4}$). Let $\mathrm{He}_n$ be the Hermite polynomials with generating function $\exp(\alpha y - \frac{1}{2}\alpha^2) = \sum_n \alpha^n\,\mathrm{He}_n(y)/n!$, so that, for $c \geq 0$,

$$:y^n:_c := c^{n/2}\,\mathrm{He}_n(y/\sqrt{c}) = \sum_{0\leq j\leq n/2}\frac{n!}{j!\,(n-2j)!}\Big(-\frac{c}{2}\Big)^j y^{n-2j}$$

(the coefficient of $\alpha^n/n!$ in $\exp(\alpha y - \frac{1}{2}\alpha^2 c)$; the sum also defines it at $c = 0$, where $:y^n:_0 = y^n$), and $:\mathscr{P}:_c(y) := \sum_n a_n :y^n:_c$; for a centred Gaussian variable $X$ of variance $c$ this is the Wick power of Assumption A1(a), by (I.18a)–(I.18b). Put

$$\mathcal{U}_{h,\kappa} := \lambda\int h(z)\,:\mathscr{P}(\Phi_\kappa(z)):_{c_\kappa} dz,$$

the integral taken pointwise in $q$ (the integrand is continuous in $z$ and $h$ is integrable with bounded support). Its degree-$n$ part $X_{n,\kappa} := \int h(z)\,:\Phi_\kappa(z)^n:_{c_\kappa} dz$ lies in $\Gamma_n$ and equals the $L^2(d\mu_0)$-valued integral of the same integrand (Lemma N.23′(i), Appendix D: it is the $L^2$-limit of Riemann sums of the Wick monomials $:\Phi(\delta_\kappa(z-\cdot))^n:$), so inner products with it are computed under the integral sign; its degree-0 part is $\lambda a_0\int h$, the same as that of $\mathcal{U}_h$. By Lemma C.4 below and the comparison in the proof of Theorem C.5, $\mathcal{U}_{h,\kappa} \to \mathcal{U}_h$ in every $L^p(d\mu_0)$ as $\kappa \to \infty$; this is used only after Theorem C.5. (The constants $c_1$, $c_2$ of Lemmas C.3 and C.4 are not values of $\kappa \mapsto c_\kappa$.)

**Lemma C.2** (normalisation)**.** *For every $h \in \mathcal{C}_{\mathrm{E}}$ with $\operatorname{supp} h \subseteq \Lambda$: $Z(h) \geq e^{-\lambda|a_0||\Lambda|} > 0$, and $Z(h) < \infty$ by Theorem C.5. Hence $dq_h$ is a well-defined probability measure on the whole class $\mathcal{C}_{\mathrm{E}}$.*

*Proof.* $\mathcal{U}_h \in L^1(d\mu_0)$ by Assumption A3. For $n \geq 1$, $:\Phi^n(h):\, \in \Gamma_n$ is orthogonal to the constants $\Gamma_0$ (orthogonality of the chaoses, Assumption A1(c)), so $\mathbb{E}[:\Phi^n(h):] = \langle 1, :\Phi^n(h):\rangle = 0$; hence $\mathbb{E}[\mathcal{U}_h] = \lambda a_0\int h \in [-\lambda|a_0||\Lambda|, \lambda|a_0||\Lambda|]$, since $0 \leq h \leq \mathbf{1}_\Lambda$ (both $h \geq 0$ and $h \leq 1$ are used here), $\lambda \geq 0$ and $|\Lambda|$ is the Lebesgue measure. Jensen's inequality for the convex function exp and the probability measure $d\mu_0$ gives $Z(h) \geq e^{-\mathbb{E}[\mathcal{U}_h]} \geq e^{-\lambda|a_0||\Lambda|}$ (the inequality holds in $(0,\infty]$ even before finiteness is known). □

**Lemma C.3** (Wick lower bound; the $h \geq 0$ entry)**.** *With*

$$C(\mathscr{P}) := \sum_{n=0}^{2p'} |a_n| \max_{0 \leq j \leq n/2} \frac{n!}{j!\,(n-2j)!\,2^j},$$

$$C_{p'} := \max\Big\{1,\ \max_{1 \leq m < 2p'} \frac{2p'-m}{2p'}\,(2m)^{m/(2p'-m)}\Big\},$$

$$b(\mathscr{P}) := C_{p'} \sum_{m=0}^{2p'-1} C(\mathscr{P})^{2p'/(2p'-m)}\, a_{2p'}^{-m/(2p'-m)},$$

*one has* $:\mathscr{P}:_c(x) \geq -b(\mathscr{P})\,(1+c)^{p'}$ *for all* $c \geq 0$ *and* $x \in \mathbb{R}$*. Consequently, for every* $h \in \mathcal{C}_{\mathrm{E}}$ *with* $\operatorname{supp} h \subseteq \Lambda$ *and every* $\kappa \geq 2$*,*

$$\mathcal{U}_{h,\kappa} \ \geq\ -\lambda\, b(\mathscr{P})\,|\Lambda|\,\big(1 + C_* + \tfrac{1}{2\pi}\log\kappa\big)^{p'} \ \geq\ -c_1\,(\log\kappa)^{p'},$$

$$c_1 := \max\Big\{1,\ \lambda\, b(\mathscr{P})\,|\Lambda|\Big(\frac{1+C_*}{\log 2} + \frac{1}{2\pi}\Big)^{p'}\Big\} > 0,$$

*with* $C_*$ *of* (C.c)*;* $c_1$ *depends on* $\lambda$*,* $\mathscr{P}$*,* $|\Lambda|$*,* $m_0$ *and* $\|\delta\|_\infty$ *(the last two through* $C_*$*), and the maximum with* $1$ *makes* $c_1 \geq 1$*, which the proof of Theorem C.5 uses (for its threshold* $b_1$*).*

*Proof.* By the expansion of $:y^n:_c$ in the notation paragraph, $:\mathscr{P}:_c(x) = \sum_{m=0}^{2p'} \beta_m x^m$ with $\beta_{2p'} = a_{2p'} =: A > 0$ and, for $m < 2p'$, $\beta_m = \sum_{n \equiv m\ (2),\ n \geq m} a_n \frac{n!}{j!\,m!}(-\frac{c}{2})^j$ with $j = (n-m)/2 \leq (2p'-m)/2$; since $c^j \leq (1+c)^j \leq (1+c)^{(2p'-m)/2}$, $|\beta_m| \leq C(\mathscr{P})(1+c)^{(2p'-m)/2}$. For $1 \leq m < 2p'$, Young's inequality $uv \leq u^P/P + v^{P'}/P'$ with $P := 2p'/m$, $P' := 2p'/(2p'-m)$ (so $1/P + 1/P' = 1$), applied to $u := \eta|x|^m$, $v := |\beta_m|/\eta$ with $\eta^{2p'/m} := A/(2m)$, gives

$$\begin{aligned} |\beta_m||x|^m &\leq \frac{m}{2p'}\,\eta^{2p'/m}x^{2p'} + \frac{2p'-m}{2p'}\Big(\frac{|\beta_m|}{\eta}\Big)^{2p'/(2p'-m)} \\ &= \frac{A}{4p'}\,x^{2p'} + \frac{2p'-m}{2p'}(2m)^{m/(2p'-m)}\Big(\frac{|\beta_m|^{2p'}}{A^m}\Big)^{1/(2p'-m)}, \end{aligned}$$

and for $m = 0$ trivially $|\beta_0| \leq 1 \cdot (|\beta_0|^{2p'}/A^0)^{1/2p'}$. Summing over the $2p'$ values $m < 2p'$, the $x^{2p'}$ terms total at most $\frac{A}{2}x^{2p'}$, so $:\mathscr{P}:_c(x) \geq \frac{A}{2}x^{2p'} - C_{p'}\sum_{m<2p'}(|\beta_m|^{2p'}/A^m)^{1/(2p'-m)}$. With $|\beta_m| \leq C(\mathscr{P})(1+c)^{(2p'-m)/2}$ the exponent of $(1+c)$ is $\frac{2p'}{2p'-m}\cdot\frac{2p'-m}{2} = p'$, uniformly in $m$, which gives the first claim with the displayed $b(\mathscr{P})$. For the second: $h \geq 0$ (this is where the sign of $h$ is used), so the pointwise bound at $c = c_\kappa$ integrates to $\mathcal{U}_{h,\kappa} \geq -\lambda b(\mathscr{P})(1+c_\kappa)^{p'}\int h \geq -\lambda b(\mathscr{P})|\Lambda|(1+c_\kappa)^{p'}$, using $\int h \leq |\Lambda|$ ($h \leq 1$); then (C.c); and finally $1 + C_* + \frac{1}{2\pi}\log\kappa \leq \big(\frac{1+C_*}{\log 2} + \frac{1}{2\pi}\big)\log\kappa$ for $\kappa \geq 2$, since $\log\kappa \geq \log 2$. □

**Lemma C.4** (ultraviolet difference; the $h \leq 1$ entry)**.** *For* $n \geq 1$ *and* $R \geq 0$ *put*

$$A_n := 16\pi\Big[(4\pi)^{-n}2^{n-1}\big((b_0+2)^n + 2^{n+1}n!\big) + 3\big(\|\delta\|_\infty(b_0+2)\big)^n\Big],$$

$$L_m(R) := \int_0^R \big(b_0 + 2 + 2|\log r|\big)^m\,dr \quad (m \geq 0), \qquad B_n(R) := 12\,n\,(4\pi)^{-(n-1)}L_{n-1}(R),$$

$$c_2 := \lambda^2|\Lambda|\sum_{n=1}^{2p'} a_n^2\,n!\Big[\frac{A_n}{2}\Big(1 + \frac{1}{\log 2}\Big)^n(\log 2)^{n-2p'} + B_n(R)\,(\log 2)^{-2p'}\Big].$$

*Then for all* $h \in \mathcal{C}_{\mathrm{E}}$ *with* $\operatorname{supp} h \subseteq \Lambda$*,* $R = \operatorname{diam}\Lambda$*, and all* $\kappa \geq 2$*,*

$$\big\|\mathcal{U}_h - \mathcal{U}_{h,\kappa}\big\|^2_{L^2(d\mu_0)} \ \leq\ c_2\,\kappa^{-1}(\log\kappa)^{2p'}.$$

*The constant $c_2$ depends on $\lambda$, $\mathscr{P}$, $|\Lambda|$, $\operatorname{diam}\Lambda$, $m_0$ (through $b_0$) and $\|\delta\|_\infty$, and on nothing else; every $L_m(R)$ is finite, the logarithmic singularity being integrable.*

*Proof. Step 1: chaos decomposition and isometry.* Write $G_\kappa^{(1)} := \delta_\kappa * G$ and $G_\kappa := \delta_\kappa * \delta_\kappa * G$, both continuous on $\mathbb{R}^2$ as convolutions of the locally integrable $G$ with $C_c^\infty$ functions (cf. Lemma N.25(iii)). With $X_{n,\kappa}$ as in the notation paragraph, $\mathcal{U}_h - \mathcal{U}_{h,\kappa} = \lambda \sum_{n=1}^{2p'} a_n \big(:\Phi^n(h): - X_{n,\kappa}\big)$ (the degree-0 terms cancel), each summand lying in $\Gamma_n$; by the orthogonality of the chaoses, $\|\mathcal{U}_h - \mathcal{U}_{h,\kappa}\|_2^2 = \lambda^2 \sum_n a_n^2 \|:\Phi^n(h): - X_{n,\kappa}\|_2^2$. Expanding the square: $\|:\Phi^n(h):\|_2^2 = n! \iint h(z)h(z')G(z-z')^n$ is (W.3);

$$\langle :\Phi^n(h):, X_{n,\kappa}\rangle = \int h(z)\, \langle :\Phi^n(h):, :\Phi_\kappa(z)^n:\rangle \, dz$$

(the inner product passes under the $L^2$-valued integral), and for the Wick monomial $:\Phi_\kappa(z)^n: = :\Phi(f)\cdots\Phi(f):$ with $f = \delta_\kappa(z-\cdot)$ the pairing (W.2) gives $n!\int \big(\int \delta_\kappa(z-x)G(x-y)\,dx\big)^n h(y)\,dy = n!\int G_\kappa^{(1)}(z-y)^n h(y)\,dy$; and $\langle :\Phi_\kappa(z)^n:, :\Phi_\kappa(z')^n:\rangle = n!\,\mathbb{E}[\Phi_\kappa(z)\Phi_\kappa(z')]^n = n!\,G_\kappa(z-z')^n$ by (I.21) and Lemma N.25(iii), so $\|X_{n,\kappa}\|_2^2 = n!\iint h(z)h(z')G_\kappa(z-z')^n$. Hence

$$\|\mathcal{U}_h - \mathcal{U}_{h,\kappa}\|_{L^2}^2 = \lambda^2 \sum_{n=1}^{2p'} a_n^2\, n! \iint h(z)h(z') \big[G^n - 2(G_\kappa^{(1)})^n + G_\kappa^n\big](z-z')\,dz\,dz'.$$

Using $|h| \le \mathbf{1}_\Lambda$ (this is where the upper bound $h \le 1$ is used) and $|z-z'| \le R$ for $z, z' \in \Lambda$, the double integral is at most $|\Lambda| \int_{|\zeta|\le R} \big|G^n - 2J_1^n + J_2^n\big|(\zeta)\,d\zeta$ with $J_1 := G_\kappa^{(1)}$, $J_2 := G_\kappa$.

*Step 2: three facts about the averages.* Let $J = \nu * G$ with $\nu \in \{\delta_\kappa, \delta_\kappa * \delta_\kappa\}$: a probability density with $\operatorname{supp}\nu \subseteq \{|w| \le 2/\kappa\}$ and $\|\nu\|_\infty \le \kappa^2\|\delta\|_\infty$. *(J1)* $J(\zeta) \le \|\delta\|_\infty(b_0+1+2\log\kappa) \le \|\delta\|_\infty(b_0+2)(1+\log\kappa)$ for every $\zeta$. Indeed $J(\zeta) \le \|\nu\|_\infty \int_{B(\zeta,2/\kappa)} G \le \kappa^2\|\delta\|_\infty \int_{B(0,2/\kappa)} G$, because for the radial nonincreasing $G$ and $a > 0$ one has $\int_{B(\zeta,a)} G \le \int_{B(0,a)} G$ (the sets $B(\zeta,a)\setminus B(0,a)$ and $B(0,a)\setminus B(\zeta,a)$ have equal measure, $G \le G|_{|\cdot|=a}$ on the first and $G \ge G|_{|\cdot|=a}$ on the second); with $a := 2/\kappa \le 1$, (C.a) and polar coordinates, $\int_{B(0,a)} G \le \frac{1}{4\pi}\,2\pi\int_0^a r\big(b_0 + 2\log\frac{1}{r}\big)dr = \frac{a^2}{4}\big(b_0+1+2\log\frac{1}{a}\big) = \kappa^{-2}\big(b_0+1+2\log\frac{\kappa}{2}\big)$, and $(b_0+2)(1+\log\kappa) - (b_0+1+2\log\kappa) = 1 + b_0\log\kappa \ge 0$. *(J2)* For $|\zeta| \ge 4/\kappa$: $|J(\zeta) - G(\zeta)| \le \frac{2}{\pi\kappa|\zeta|}$. Indeed $J(\zeta) - G(\zeta) = \int \nu(w)\big(G(\zeta-w) - G(\zeta)\big)dw$, and for $|w| \le 2/\kappa \le |\zeta|/2$ the segment from $\zeta$ to $\zeta - w$ lies in $\{|\xi| \ge |\zeta|/2\}$, where $|\nabla G(\xi)| \le (\pi|\zeta|)^{-1}$ by (C.b); the mean value theorem gives $|G(\zeta-w) - G(\zeta)| \le |w|/(\pi|\zeta|) \le 2/(\pi\kappa|\zeta|)$. *(J3)* For $|\zeta| \ge 4/\kappa$: $\max\{G, J\}(\zeta) \le \frac{1}{4\pi}\big(\ell(\zeta)+2\big)$, since by (J2) and $\kappa|\zeta| \ge 4$, $J \le G + \frac{1}{2\pi}$, and $\frac{1}{2\pi} = \frac{2}{4\pi}$.

*Step 3: the region $|\zeta| < a$, $a := 4/\kappa \le 2$.* By (J1), $\int_{|\zeta|<a} J^n \le \pi a^2\big(\|\delta\|_\infty(b_0+2)\big)^n(1+\log\kappa)^n = 16\pi\kappa^{-2}\big(\|\delta\|_\infty(b_0+2)\big)^n(1+\log\kappa)^n$ for $J = J_1, J_2$. For $G$: on $|\zeta| \le 2$, $|\log|\zeta|| \le \log 2 + \log_+\frac{1}{|\zeta|}$, so $\ell(\zeta) \le (b_0+2) + 2\log_+\frac{1}{|\zeta|}$ and, by convexity, $\ell^n \le 2^{n-1}\big[(b_0+2)^n + 2^n(\log_+\frac{1}{|\zeta|})^n\big]$; and $\int_{|\zeta|<a}(\log_+\frac{1}{|\zeta|})^n\,d\zeta = 2\pi\int_0^{\min\{a,1\}} r(\log\frac{1}{r})^n\,dr \le 2\pi\cdot 16\kappa^{-2}(1+\log\kappa)^n\,n!$. (For $\kappa \ge 4$, i.e. $a \le 1$: the substitution $r = ae^{-s}$ gives $\int_0^a r(\log\frac{1}{r})^n dr = a^2\sum_{k=0}^n \binom{n}{k}(\log\frac{1}{a})^{n-k}\frac{k!}{2^{k+1}} \le a^2(1+\log\frac{1}{a})^n\sum_k \frac{n!}{(n-k)!\,2^{k+1}} \le a^2(1+\log\frac{1}{a})^n n!$, with $a^2 = 16/\kappa^2$ and $\log\frac{1}{a} = \log\frac{\kappa}{4} \le \log\kappa$; for $2 \le \kappa < 4$: $\int_0^1 r(\log\frac{1}{r})^n dr = n!/2^{n+1} \le n! \le 16\kappa^{-2}n!$, as $\kappa^2 < 16$.) Hence, with $\pi a^2 = 16\pi\kappa^{-2}$, $\int_{|\zeta|<a} G^n \le (4\pi)^{-n}\int_{|\zeta|<a}\ell^n \le (4\pi)^{-n}2^{n-1}\,16\pi\kappa^{-2}(1+\log\kappa)^n\big[(b_0+2)^n + 2^{n+1}n!\big]$, and altogether

$$\int_{|\zeta|<a} \big[G^n + 2J_1^n + J_2^n\big] \le \kappa^{-2}(1+\log\kappa)^n\, A_n.$$

*Step 4: the region $a \le |\zeta| \le R$* (empty if $R < a$). There $|G^n - 2J_1^n + J_2^n| \le 2|G^n - J_1^n| + |G^n - J_2^n|$, and for $J = J_1, J_2$, by (J2)–(J3), $|G^n - J^n| \le n \max\{G, J\}^{n-1}|G - J| \le n(4\pi)^{-(n-1)}(\ell+2)^{n-1} \cdot 2(\pi\kappa|\zeta|)^{-1}$. In polar coordinates, $|\zeta| = r$,

$$\int_{a\le|\zeta|\le R} \big|G^n - 2J_1^n + J_2^n\big| \le 2\pi \int_a^R 3n(4\pi)^{-(n-1)}\big(b_0 + 2 + 2|\log r|\big)^{n-1} \frac{2}{\pi\kappa r}\, r\, dr$$
$$\le \frac{12n}{\kappa}(4\pi)^{-(n-1)} L_{n-1}(R) = \kappa^{-1} B_n(R).$$

*Step 5: assembly.* By Steps 1, 3 and 4, $\|\mathcal{U}_h - \mathcal{U}_{h,\kappa}\|_2^2 \le \lambda^2|\Lambda| \sum_n a_n^2 n! \big[\kappa^{-2}(1+\log\kappa)^n A_n + \kappa^{-1}B_n(R)\big]$. For $\kappa \ge 2$ and $1 \le n \le 2p'$: $1 + \log\kappa \le (1 + \frac{1}{\log 2})\log\kappa$, $(\log\kappa)^n \le (\log 2)^{n-2p'}(\log\kappa)^{2p'}$ (as $\log\kappa \ge \log 2$ and $n - 2p' \le 0$), $\kappa^{-2} \le \frac{1}{2}\kappa^{-1}$, and $1 \le (\log\kappa/\log 2)^{2p'}$; inserting these gives $c_2\kappa^{-1}(\log\kappa)^{2p'}$. (Tests: at $n = 1$ the integrand of Step 4 is $|J_2 - 2J_1 + G| \le 3 \cdot 2(\pi\kappa r)^{-1}$ and $L_0(R) = R$; the $r$-integral in Step 3 at $n = 1$, $a \le 1$, is $\frac{a^2}{2}(\frac{1}{2} + \log\frac{1}{a}) \le a^2(1 + \log\frac{1}{a})$.) □

**Theorem C.5** ($h$-uniform stability)**.** *For every $p < \infty$ and compact $\Lambda$,*

$$\sup\big\{\|e^{-\mathcal{U}_h}\|_{L^p(d\mu_0)} \;:\; h \in \mathcal{C}_{\mathrm{E}},\ \operatorname{supp} h \subseteq \Lambda\big\} \;\le\; \big(1 + e^{pb_*}\big)^{1/p} < \infty,$$

*where $b_* = b_*(p, p', c_1, c_2)$ is the threshold defined in the proof, with $c_1$ of Lemma C.3 and $c_2$ of Lemma C.4. The bound depends on $\Lambda$ only through $|\Lambda|$ and $\operatorname{diam}\Lambda$ (monotonically), and otherwise on $p$, $\lambda$, $\mathscr{P}$, $m_0$ and $\|\delta\|_\infty$; in particular it is the same for all translates of $\Lambda$.*

*Proof.* (For $\lambda = 0$, $\mathcal{U}_h = 0$ and the supremum is 1; assume $\lambda > 0$. If $|\Lambda| = 0$, every $h$ of the class vanishes almost everywhere, $\mathcal{U}_h = 0$ and the supremum is 1 again; assume $|\Lambda| > 0$.) $Y := \mathcal{U}_h - \mathcal{U}_{h,\kappa}$ is a sum $Y = \sum_{r=0}^{2p'} Y_r$ with $Y_r := \lambda a_r(:\Phi^r(h):\, - X_{r,\kappa}) \in \Gamma_r$ (Step 1 of the proof of Lemma C.4), so Nelson's hypercontractive moment comparison applies in the form

$$\|Y\|_{L^q} \le (q-1)^{p'}\|Y\|_{L^2} \qquad (q \ge 2),$$

which is [12, Thm. I.22, (I.47), p. 38] with $m = 2p'$ (Assumption A4, Sources). (Proof from the abstract form of Assumption A4 [12, Thm. I.17]: the second-quantized contraction $\Gamma(e^{-t})$ on $L^2(d\mu_0)$ satisfies $\|\Gamma(e^{-t})X\|_q \le \|X\|_2$ for $q = 1 + e^{2t}$, and $\Gamma(e^{-t})Y_r = e^{-rt}Y_r$ by (I.32) on the Wick monomials spanning $\Gamma_r$ and continuity (Assumption A3, construction paragraph). Write $Y = \Gamma(e^{-t})Z$ with $Z = \sum_r e^{rt}Y_r$; by orthogonality of the chaoses, $\|Z\|_2 \le e^{2p't}\|Y\|_2$, whence $\|Y\|_q \le \|Z\|_2 \le e^{2p't}\|Y\|_2 = (q-1)^{p'}\|Y\|_2$.)

*The thresholds.* Put

$$b_0' := 2c_1\,(4p'\log 2)^{p'},$$
$$b_1 := \begin{cases} 2c_1\big(\max\{0,\, 4 + 2\log c_2\}\big)^{p'} & (c_2 > 0),\\ 0 & (c_2 = 0),\end{cases}$$

and, with $f(b) := \frac{1}{4p'}\big(\frac{b}{2c_1}\big)^{1/p'} - \log(2pb)$, which is nondecreasing on $[2c_1(4p'^2)^{p'}, \infty)$ (there $f'(b) = b^{-1}[(4p'^2)^{-1}(b/2c_1)^{1/p'} - 1] \ge 0$) and tends to $+\infty$,

$$b_2(p) := \min\big\{b \ge 2c_1(4p'^2)^{p'} \;:\; f(b) \ge 0\big\}, \qquad b_* := \max\{b_0', b_1, b_2(p)\}.$$

(The thresholds $b_0', b_1, b_2(p), b_*$ are local to this proof; the prime on $b_0'$ keeps it apart from the kernel constant $b_0$ of (C.a), and the constant $b_1$ of Lemma N.25 is unrelated.)

For $b \ge b_*$ choose $\kappa = \kappa(b)$ by $c_1(\log\kappa)^{p'} = b/2$, i.e. $L := \log\kappa = (b/2c_1)^{1/p'}$, and $2q := e^{L/(4p')}$. Since $b \ge b_0'$, $L \ge 4p'\log 2$, so $\kappa \ge 2^{4p'} \ge 2$ (Lemmas C.3 and C.4 apply)

and $2q \geq 2$ (the comparison applies at the exponent $2q$). By Lemma C.3, $\{\mathcal{U}_h \leq -b\} \subseteq \{\mathcal{U}_h - \mathcal{U}_{h,\kappa} \leq -b/2\}$, so by Chebyshev's inequality in $L^{2q}$, the comparison and Lemma C.4,

$$\begin{aligned}\Pr\big[\mathcal{U}_h \leq -b\big] &\leq \Big(\frac{2}{b}\Big)^{2q} (2q-1)^{2p'q} \left(c_2\, e^{-L} L^{2p'}\right)^q \\ &\leq \Big[\frac{2\sqrt{c_2}}{b}\,(2q)^{p'}\, L^{p'} e^{-L/2}\Big]^{2q} = \Big[\frac{2\sqrt{c_2}}{b}\, L^{p'} e^{-L/4}\Big]^{2q},\end{aligned}$$

using $(2q)^{p'} = e^{L/4}$. With $b = 2c_1 L^{p'}$ the bracket equals $(\sqrt{c_2}/c_1)\, e^{-L/4}$, which is $\leq e^{-1}$ because $b \geq b_1$ gives $L \geq 4 + 2\log c_2 \geq 4 + 2\log(c_2/c_1^2)$ (as $c_1 \geq 1$), i.e. $\log(\sqrt{c_2}/c_1) - L/4 \leq -1$ (for $c_2 = 0$ the bracket is 0). Hence

$$\Pr\big[\mathcal{U}_h \leq -b\big] \;\leq\; e^{-2q} = \exp\Big(-\exp\Big(\tfrac{1}{4p'}(b/2c_1)^{1/p'}\Big)\Big) \;\leq\; e^{-2pb} \qquad (b \geq b_*),$$

the last step because $b \geq b_2(p)$ gives $f(b) \geq 0$, i.e. $\exp\big(\frac{1}{4p'}(b/2c_1)^{1/p'}\big) \geq 2pb$: uniformly over the stated class. Then, by the layer-cake formula for the probability measure $d\mu_0$,

$$\|e^{-\mathcal{U}_h}\|_{L^p}^p = \int_{-\infty}^{\infty} p e^{pb} \Pr\big[-\mathcal{U}_h > b\big]\, db \leq e^{pb_*} + \int_{b_*}^{\infty} p e^{-pb}\, db \leq e^{pb_*} + 1,$$

bounding the probability by 1 for $b < b_*$ and by $e^{-2pb}$ for $b \geq b_*$. The threshold $b_*$ is a function of $p$, $p'$, $c_1$ and $c_2$ alone, nondecreasing in $c_1$ and $c_2$ ($b_0'$ and $b_1$ visibly; $b_2(p)$ because, when $c_1$ grows, $f$ decreases pointwise and the lower end of its range grows); $c_1$ depends on $\Lambda$ through $|\Lambda|$ and $c_2$ through $|\Lambda|$ and $\operatorname{diam}\Lambda$, both nondecreasingly, which gives the dependence stated. Every constant above came from Lemmas C.3–C.4, which used $h \geq 0$ and $h \leq 1$ (and compact support). □

Since Lemma C.4 gives $\mathcal{U}_{h,\kappa} \to \mathcal{U}_h$ in $L^2(d\mu_0)$ and the comparison above holds for $\mathcal{U}_h - \mathcal{U}_{h,\kappa}$ at every $\kappa$, the convergence holds in every $L^q(d\mu_0)$, $q < \infty$, as the notation paragraph announced.

**Remark C.6** (why the class is what it is)**.** The two halves of the constraint $0 \leq h \leq 1$ are used in complementary places: positivity in the pointwise lower bound (Lemma C.3 would fail for $h$ of variable sign — the interaction density is unbounded above *and* below), and boundedness in the $L^2$ kernel estimate (Lemma C.4) and in M3; the upper bound also enters, through $\int h \leq |\Lambda|$, in Lemmas C.2 and C.3. Nothing else about $h$ — smoothness, shape, time extent, monotonicity along a sequence — is touched in Lemmas C.3–C.4 or in M3; the stability input of the source's own proof (M6) is a separate matter. This is the structural content of Source Fact 2.4.

**Proposition C.7** (coverage of every CE application)**.** *CE1 is used nowhere in §§4–7: every use of Euclidean clustering (CE2) in this manuscript is Source Fact 2.2 as printed, whose constants are independent of $h$, at a cutoff of $\mathcal{C}_{\mathrm{E}}$, in the following orientations (§9 proves a Hamiltonian clustering lemma of its own, Lemma 9.3). In Lemma 7.4 and in Theorem D.3 (hence in §7) the strips are parallel to the time-zero line (for Lemma 7.4, with two sharp-time monomials, the case the printed proof reaches; for Theorem D.3, with a time-integrated partner, see Remark N.39); in Theorem D.4 (hence in Theorem 5.1 and in everything that consumes $M$) the strips are parallel for the squares of class (P) and transverse for the squares of class (T), the latter being A8(iv)-T. Lemma C.2, Lemma C.4 and Theorem C.5 supply the normalization, the ultraviolet limit and the integrability that Appendix D consumes. Hence no cutoff-uniformity assertion beyond the printed Theorem 1.1.7 is consumed by the proof.*

*Proof.* The moment bounds of Theorem 5.1 (Step 5) and Lemmas 7.1–7.3 are Theorems D.4 and D.3, whose proofs apply Source Fact 2.2 through Lemma D.2 at the cutoffs $h_\alpha = h^B + \alpha(h - h^B) \in \mathcal{C}_{\mathrm{E}}$ (Lemma N.5(ii)), with the strips named in their Steps 3 (Lemma N.38: parallel; the proof of Theorem D.4, Step 3: classes (P) and (T)); if $g$ is replaced by an affine path cutoff $g_s$, Lemma 3.1 gives $g_s \in \mathcal{C}_{\mathrm{H}}$, so $h_{T,g_s} \in \mathcal{C}_{\mathrm{E}}$. Lemma 7.4 is the one direct CE2 application, across the parallel strip $\{|x^0| < t/2 - 1\}$, and Source Fact 2.2 quantifies over its cutoff directly. The remaining assertions are exactly Lemma C.2, Lemma C.4 and Theorem C.5, consumed in Appendix D as listed in Remark N.34. □

# D The uniform local moment from the printed clustering theorem

*Purpose.* Theorem 5.1 and Lemmas 7.1–7.3 and 7.6 consume moment bounds for sharp-time observables under the interacting Euclidean measures $dq_h$ at slab cutoffs $h = h_{T,g}$, uniformly in $T > 0$. Until 7 November 2026 these were supplied by Assumption A8(vi), read off GJS Theorem 1.1.11 at $B = I$, $n = 0$. This appendix derives them from the printed clustering theorem, GJS Theorem 1.1.7, (1.1.10) [7, p. 594], imported as Assumption A8(iv), together with Lemma C.2, Lemma C.4 and Theorem C.5 of Appendix C and the imports A2, A3, A4 and A8(i)–(iii). The device is the source's own (p. 595): compare $dq_h$ with the measure of the cutoff restricted to a box of GJS squares around the observable; the squares outside the box are strip-separated from it, so that (1.1.10) bounds their terms, and the box term is bounded by Lemma C.2 and Theorem C.5. Theorem D.3 (Phase A of `ugjs/PLAN.md`) gives the bound at fixed spatial cutoff $g$, with constants depending on $g$, which is all §7 needs, across strips parallel to the time-zero line only; Theorem D.4 (Phase B) gives the bound of Theorem 5.1 uniformly in the cutoff, across transverse strips as well. Lemmas D.1 and D.2 serve both.

*Standard.* A printed statement is imported at the scope at which it is printed; this is the standard under which Assumptions A1–A8 are imported, and this appendix uses no other. None of GJS's proofs is verified here. A case of Theorem 1.1.7 is used that its printed proof does not reach: GJS Theorem 1.1.7 is printed for localized monomials separated by a strip of any orientation (p. 594). Its printed proof rotates the strip into the time direction (p. 597) and so reaches sharp-time monomials only across strips parallel to the time-zero line. Lemma 7.4 uses only that case; this appendix also uses the transverse case, listed separately as A8(iv)-T (Assumption A8, item (iv-T)). Theorem D.4 uses it, and every such use is flagged where it occurs; Theorem D.3 uses the parallel case only.

*What the appendix does not do.* It does not prove the transverse case of Theorem 1.1.7; it does not re-derive the cluster expansion; it does not evaluate the constant of Theorem C.5, which enters the constants below as $S_2(B)$; and it proves nothing about the net of ground states beyond what §§5–7 derive from its two theorems. The open human readings it rests on, items 1–4, 9(c) and 10–12 of `part_i/C5_reading_record.md`, are named beside every status word.

*Status and provenance.* Lemmas D.1–D.2 and Theorems D.3–D.4 are proved from Assumptions A2, A3, A4 and A8(i)–(iv) (for Theorem D.4 including A8(iv)-T), from Lemma C.2, Lemma C.4 and Theorem C.5 as repaired on 7 November 2026, and from the auxiliary statements of this appendix, under the reading of Remark N.2(iv) (that the objects constructed here are the monomials of p. 593 and the $\mathcal{U}_h$ of Assumption A3 is GJS's $V(h)$), which Assumption A8, Scope (d), records. The text was drafted as the comparison note `ugjs/comparison.tex` (31 October–4 November 2026) by three council members and audited in four rounds (`ugjs/council/`: V1 by an attacker and a quantifier checker, V2 by

a fresh auditor, re-checks V1b/V2b, closing pass V1c). At integration the conditions those audits attached to every statement (repairs R1–R10 of the V3 audit of Appendix C; the gap recorded after Lemma N.20) were discharged by the repairs of Appendix C, the Wick-power paragraphs of Assumptions A1 and A3, and the new Lemma N.23′; the integration gate is recorded in `part_i/README.md`. References to "the plan", and to its traps, findings, rulings and gates, are to `ugjs/PLAN.md`. The numbering N.$k$ of the note's auxiliary statements is kept, so that the audit reports apply to this text; the note's Lemma N.47 (a Gaussian moment inequality of its §8, which is Lemma N.52 here) is omitted, and Lemma N.23′ is new.

## D.a Notation and standing conventions

This section fixes the notation of `ugjs/PLAN.md`, Part 1, and proves the elementary facts that the later sections use: that every cutoff of the comparison lies in $\mathcal{C}_{\mathrm{E}}$ (Lemma N.5), the strip geometry (Lemma N.7), the ring count around the $3 \times 5$ box (Lemma N.11), the finiteness and the translation invariance of the stability constant $S_2$ (Proposition N.15), the norm of the interaction in one square (Lemma N.18), the Wick isometry with the constant $V_*$ and the comparison $\|Y\|_4 \le 3^{p'}\|Y\|_2$ (Lemmas N.20 and N.21), the identification of the constructed Wick powers with those of Assumptions A1 and A3 (Lemma N.23′), and the list of printed constants (§D.a.8).

*Appendix C.* Lemma C.2 (normalization), Lemma C.4 (ultraviolet difference) and Theorem C.5 (stability) are load-bearing in this appendix: Lemma C.4 in Lemma N.5(iv), Lemma N.13(iii)–(iv) and Lemma N.18(iii)–(iv), and through them in Proposition N.15(iii); Theorem C.5 in Proposition N.15(i); the moment comparison of its proof (Theorem I.22 of Simon's book, Assumption A4) in Lemma N.21(ii). Their proofs rest on the constructions of Assumptions A1 and A3, on Assumption A4 and on the kernel facts of this section, with every constant explicit (the V3 audit of 2 November 2026 found them true but resting on three unimported inputs; the repairs of 7 November 2026 are recorded in Appendix B).

*Inputs of this section.* §2.1, Definition 2.1, Assumptions A1 (the Wick powers of the time-zero field), A2, A3 (the construction of $\mathcal{U}_h$), A4 (its abstract form, in Lemma N.21) and A8(i)–(iv), Remark 2.5, §2.6, Remark 3.3, the proofs of Lemma 6.1 and Proposition 6.2 (on the columns of the GJS squares, Remark N.3), the regularization of Appendix C with Lemma C.4 and Theorem C.5; from GJS, pp. 589, 593 and 594 as quoted. No other source is used, and no import is made beyond Assumptions A1–A8.

### D.a.1 Coordinates, squares, rows and columns

*Working coordinates.* We work in the working coordinates, §2.6 and Remark 3.3: the bare mass is $m_0 = M_*$ and $\lambda < \lambda_*$, "so that the *unit-lattice* statements of Assumption A8 apply literally in the working coordinates" (§2.6). A point of the Euclidean plane is $z = (x^0, x^1) \in \mathbb{R}^2$, with $x^0$ the Euclidean time and $x^1$ the space coordinate; the slab cutoffs are $h_{T,g} = \mathbf{1}_{[-T,T]} \otimes g$, i.e. $h_{T,g}(z) = \mathbf{1}_{[-T,T]}(x^0)\, g(x^1)$ (Assumption A2). GJS also put time first: "$\Phi(\tau, f) \equiv \Phi(\delta_\tau \otimes f) = \int \Phi(\tau, \vec{x}) f(\vec{x}) d\vec{x}$" [7, p. 589].

*The squares of GJS.* The observables of GJS are the monomials (1.1.5)–(1.1.6), and p. 593 defines their localization as follows (read on the page image, quoted verbatim, misprints kept): "$Q$ is said to be localized if $w(x)$ is supported in a product $\Delta_{j_1} \times \cdots \times \Delta_{j_n} \subset R^{2r}$ where $\Delta_{j_i}$ is the unit lattice square in $R^2$ centered at the lattice side $j_i \in Z^2$. We assume a bound on the $n_i$ in (1.1.5), $n_i < \bar{n}$. Any choice $\bar{n} \geqq \deg P$ is sufficient. Since there is no restriction on $r$, $Q$ still may have an arbitrarily large degree." and "For a localized monomial $Q$ or $Q(x_0)$, let $N(\Delta)$ be the local degree, i.e. the sum of the $n_i$ for which $\Delta_{j_i} = \Delta$." [7,

p. 593]. ("side" stands for "site"; the last index of the product is printed $n$ although the product lies in $R^{2r}$.) Page 593 prints no convention for points on the lines of the lattice; the following definition supplies one, and it is used everywhere in this appendix.

**Definition N.1** (squares, rows, columns; the convention on lattice lines)**.** For $(l,k)\in\mathbb{Z}^2$ let

$$\Delta_{l,k} := [l-\tfrac12,\, l+\tfrac12)\times[k-\tfrac12,\, k+\tfrac12)\subset\mathbb{R}^2,$$

the first factor being time. *Row* $l$ is $\bigcup_{k\in\mathbb{Z}}\Delta_{l,k} = [l-\frac12, l+\frac12)\times\mathbb{R}$ and *column* $k$ is $\bigcup_{l\in\mathbb{Z}}\Delta_{l,k} = \mathbb{R}\times[k-\frac12, k+\frac12)$. For a time $s\in\mathbb{R}$,

$$l_s := \lfloor s+\tfrac12\rfloor$$

is the row containing $s$, i.e. the unique integer with $l_s-\frac12\le s< l_s+\frac12$; for $x^1\in\mathbb{R}$, $k(x^1) := \lfloor x^1+\frac12\rfloor$ is the column containing $x^1$. In particular a time on a lattice line, $s=l-\frac12$ with $l\in\mathbb{Z}$, lies in row $l_s=l$. The *sup-index distance* of $\Delta_{l,k}$ and $\Delta_{l',k'}$ is $\max\{|l-l'|,|k-k'|\}$.

**Remark N.2** (relation to the squares of p. 593 and to the former cells)**.** *(i)* The squares $\Delta_{l,k}$ partition $\mathbb{R}^2$: the point $z$ lies in exactly one of them, $\Delta_{l_{x^0},k(x^1)}$. The closure $\overline{\Delta}_{l,k} = [l-\frac12,l+\frac12]\times[k-\frac12,k+\frac12]$ is the unit lattice square centred at the lattice site $(l,k)\in\mathbb{Z}^2$ of p. 593. A kernel that vanishes almost everywhere outside a product of squares $\Delta_{j_1}\times\cdots\times\Delta_{j_r}$ has its (closed) support in the product of their closures, so the monomial is localized in the sense of p. 593; this is how "localized" is read in this appendix, for (1.1.5) and for (1.1.6) alike (Assumption A8(ii) defines "localized" for both forms).

*(ii)* For a sharp-time factor at time $x_{i,0}$, localization in a closed square requires $x_{i,0}\in[l-\frac12,l+\frac12]$; a time on a lattice line lies in two closed squares of the same column, and Definition N.1 assigns the factor to the square of row $l_{x_{i,0}}$. Thus a sharp-time monomial at time $s$ whose spatial kernel vanishes almost everywhere outside $\prod_i[k_i-\frac12,k_i+\frac12)$ is localized in the squares $\Delta_{l_s,k_i}$, and these are *its localization squares* in this appendix. The choice of row changes neither the local degrees nor $\|w(x_0,\cdot)\|_2$, hence not the norm (1.1.8) ((1); cf. Remark 6.3); the strips of §D.a.3 are computed for the row $l_s$, and every strip written there for that assignment separates the closed squares as well.

*(iii)* Until 7 November 2026, §§5–6 counted in the cells $[j,j+1)$, "this lattice shifted by one half unit in space" (the former wording of Assumption A8(ii)); since then they count, as this appendix does, on the columns of the squares of Definition N.1 (finding F3 of `ugjs/PLAN.md`). The next remark records the one place where the two lattices give different counts.

*(iv) A reading of the source.* GJS print (1.1.5)–(1.1.6) as formal integrals (p. 593), define the action "(1.8) $V(h) = \lambda\int_{R^2} : \mathscr{P}(\Phi(x)) : h(x)dx$ as a function on $\mathscr{S}'(R^2)$, where : : denotes Wick order with respect to the measure $d\Phi$, as in [9], [1]" (p. 589), and give no construction. That the random variables of this appendix — the one-square interactions $U_{\Delta,n}$ (Definition N.17), the sharp-time objects of Lemmas N.20–N.22, the regularized limits of §D.b and the pieces of $Q_V$ in §D.e — are GJS's monomials of p. 593 under GJS's $dq_h$, and that the $\mathcal{U}_h$ of Assumption A3 is GJS's $V(h)$, is a *reading* of the source, of the kind Assumption A8, Scope (d), names and records. It is named in the status lines of Lemma D.2 and Theorems D.3–D.4. Item 10 of `part_i/C5_reading_record.md`, still open, asks a human reader to confirm the printed text of (1.1.5)–(1.1.6) and of the multi-time notation (pp. 593, 595) on which the reading rests; it does not check the reading itself. At sharp time, Lemma N.22 shows that the two regularizations used in this appendix give the same objects.

**Remark N.3** (counts on the squares of p. 593)**.** For a bounded interval $I\subset\mathbb{R}$ of length $L$ put

$$N_I^{\square} := \#\{k\in\mathbb{Z}:\ |I\cap[k-\tfrac12,k+\tfrac12)|>0\},$$

the number of columns meeting $I$ in positive measure, and let $N_I^{\mathrm{cell}} := \#\{j \in \mathbb{Z} : |I \cap [j, j+1)| > 0\}$ be the count on the cells $[j, j+1)$ that §6 used until 7 November 2026 (since then $N_I = N_I^{\square}$ there).

*(i)* $N_I^{\square} \le \lfloor L \rfloor + 2$. The proof of Lemma 6.1 uses only that the cells are consecutive unit intervals partitioning $\mathbb{R}$; run with the columns: if $k_{\min} \le k_{\max}$ are the extreme columns meeting $I$ in positive measure, then $\inf I < k_{\min} + \frac{1}{2}$ and $\sup I > k_{\max} - \frac{1}{2}$, so $L > k_{\max} - k_{\min} - 1$, whence $k_{\max} - k_{\min} - 1 \le \lceil L \rceil - 1 \le \lfloor L \rfloor$ and $N_I^{\square} \le k_{\max} - k_{\min} + 1 \le \lfloor L \rfloor + 2$.

*(ii)* The two counts differ in general: $I = (-0.1, 0.1)$ gives $N_I^{\mathrm{cell}} = 2$ and $N_I^{\square} = 1$; $I = (0.4, 0.6)$ gives $N_I^{\mathrm{cell}} = 1$ and $N_I^{\square} = 2$.

*(iii)* Consequently the bounds of Proposition 6.2, (7)–(8), and the constant $K' = K_2^2 N_I^{1/2} \|f\|_2$ of Lemma 7.6 are stated, for norms computed on the squares of Definition N.1, with the column count $N_I = N_I^{\square}$ (as they are since 7 November 2026; until then they carried $N_I^{\mathrm{cell}}$). The proof of Proposition 6.2 decomposes $f_i = \sum_k f_i \mathbf{1}_{[k-\frac{1}{2}, k+\frac{1}{2})}$ over the columns and applies the Cauchy–Schwarz inequality over the $N_I = N_I^{\square}$ columns (with the former cells $[j, j+1)$ the same argument gives bounds with $N_I^{\mathrm{cell}}$, but for pieces localized in cells, which are not the localization squares of the norm of p. 594); it gives

$$\|Q\|_{\mathrm{GJS}} \le K_1 K_2^{2r} r! \, (N_I^{\square})^{r/2} \prod_{i=1}^{r} \|f_i\|_2, \qquad \|Q^2\|_{\mathrm{GJS}} \le K_1 K_2^{4r} (2r)! \, (N_I^{\square})^{r} \prod_{i=1}^{r} \|f_i\|_2^2 \quad \text{(N.a)}$$

for a cylinder monomial $Q = \prod_{i=1}^{r} \varphi_0(f_i)$ at one sharp time, $\operatorname{supp} f_i \subseteq I$. With the cell count $N_I^{\mathrm{cell}}$ the first bound can fail on these squares: for $r = 1$ and $f$ supported in $(0.4, 0.6)$ and even about $\frac{1}{2}$, the localized pieces of $Q = \varphi_0(f)$ on the squares of Definition N.1 are $\varphi_0(f\mathbf{1}_{[-1/2,1/2)})$ and $\varphi_0(f\mathbf{1}_{[1/2,3/2)})$, and (1.1.9) gives $\|Q\|_{\mathrm{GJS}} = K_1 K_2^2 (\|f\mathbf{1}_{[-1/2,1/2)}\|_2 + \|f\mathbf{1}_{[1/2,3/2)}\|_2) = \sqrt{2}\, K_1 K_2^2 \|f\|_2$, while (7) with $N_I^{\mathrm{cell}} = 1$ would assert $\|Q\|_{\mathrm{GJS}} \le K_1 K_2^2 \|f\|_2$ (this is the finding recorded in Appendix B, repaired by counting columns). Both counts obey the bound (i), so every constant written with $\lfloor L \rfloor + 2$ in place of the count is valid on both lattices.

### D.a.2 Boxes, the restricted cutoff $h^B$ and the path $h_\alpha$

**Definition N.4** (boxes; restriction; charged squares)**.** A *box* is a set

$$B = \bigcup_{l=l_1}^{l_2} \bigcup_{k=k_1}^{k_2} \Delta_{l,k} = [l_1 - \tfrac{1}{2}, l_2 + \tfrac{1}{2}) \times [k_1 - \tfrac{1}{2}, k_2 + \tfrac{1}{2}), \qquad l_1 \le l_2,\ k_1 \le k_2 \text{ integers};$$

its *shape* is the pair $(l_2 - l_1 + 1,\ k_2 - k_1 + 1)$ of the numbers of its rows and columns, and $|B| := (l_2 - l_1 + 1)(k_2 - k_1 + 1)$ is the number of its squares, which equals the area of $B$ and of its closure $\overline{B}$. Since the squares partition $\mathbb{R}^2$ and $B$ is a union of squares, a square $\Delta$ satisfies $\Delta \not\subseteq B$ if and only if $\Delta \cap B = \varnothing$. For a function $h$ on $\mathbb{R}^2$ and a box $B$,

$$h^B := h\, \mathbf{1}_B, \qquad h_\alpha := h^B + \alpha\,(h - h^B) \quad (\alpha \in [0, 1]),$$

and $\mathfrak{S}(h) := \{\Delta : \int_\Delta h \, dz > 0\}$ is the set of squares *charged* by $h$. For a measurable $h$, $\operatorname{supp} h$ denotes the closure of $\{z : h(z) \neq 0\}$.

The class $\mathcal{C}_{\mathrm{E}}$ is Definition 2.1:

$$\text{“}\mathcal{C}_{\mathrm{E}} := \{h : \mathbb{R}^2 \to [0, 1] \text{ measurable, } \operatorname{supp} h \text{ compact}\} \quad \text{(Euclidean class)”},$$

“the printed class[ ] of [7, p. 589]”; on p. 589 GJS write “Let $h$ be a function with compact support in $R^2$, and let $0 \leqq h \leqq 1$.” Assumption A8(i) reads: “For $h \in \mathcal{C}_{\mathrm{E}}$, $dq_h = Z(h)^{-1} e^{-\mathcal{U}_h} d\mu_0$ as in Assumption A3”; and Assumption A3 gives, for every $h \in \mathcal{C}_{\mathrm{E}}$, $\mathcal{U}_h \in L^p(d\mu_0)$ for every $p < \infty$ and $0 < Z(h) < \infty$.

**Lemma N.5** (the cutoffs of the comparison lie in $\mathcal{C}_{\mathrm{E}}$)**.** *Let $h \in \mathcal{C}_{\mathrm{E}}$ and let $B$ be a box.*

(i) *$h^B$, $h - h^B$ and $h\mathbf{1}_\Delta$ (for every square $\Delta$) belong to $\mathcal{C}_{\mathrm{E}}$, and $\operatorname{supp} h^B \subseteq \overline{B} \cap \operatorname{supp} h$.*

(ii) *For every $\alpha \in [0,1]$, $h_\alpha = (1-\alpha)h^B + \alpha h = h\,(\mathbf{1}_B + \alpha \mathbf{1}_{\mathbb{R}^2\setminus B})$ belongs to $\mathcal{C}_{\mathrm{E}}$; moreover $h^B \le h_\alpha \le h$ pointwise, $h_0 = h^B$, $h_1 = h$, and $\operatorname{supp} h_\alpha \subseteq \operatorname{supp} h$.*

(iii) *$\mathfrak{S}(h)$ is finite, every $\Delta \in \mathfrak{S}(h)$ meets $\operatorname{supp} h$, and $h - h^B = \sum_{\Delta\in\mathfrak{S}(h),\ \Delta\cap B=\varnothing} h\mathbf{1}_\Delta$ almost everywhere.*

(iv) *If $h_1, \dots, h_m \in \mathcal{C}_{\mathrm{E}}$ and $c_1, \dots, c_m \in \mathbb{R}$ satisfy $\sum_i c_i h_i = 0$ almost everywhere, then $\sum_i c_i \mathcal{U}_{h_i} = 0$ in $L^2(d\mu_0)$. In particular*

$$\mathcal{U}_{h_\alpha} = \mathcal{U}_{h^B} + \alpha\,\mathcal{U}_{h-h^B} \quad (\alpha \in [0,1]), \qquad \mathcal{U}_{h-h^B} = \sum_{\Delta\in\mathfrak{S}(h),\ \Delta\cap B=\varnothing} \mathcal{U}_{h\mathbf{1}_\Delta}.$$

*Proof.* (i) $B$ and $\Delta$ are Borel sets, so $h\mathbf{1}_B$, $h\mathbf{1}_{\mathbb{R}^2\setminus B} = h - h^B$ and $h\mathbf{1}_\Delta$ are products of measurable functions, hence measurable; they take values in $[0,1]$ because $h$ does and the indicators take values in $\{0,1\}$. Each vanishes wherever $h$ does, so its set of non-zeros lies in $\{h \neq 0\}$ and its support is a closed subset of the compact set $\operatorname{supp} h$, hence compact. For $h^B$ the non-zeros lie in $B \cap \{h \neq 0\}$, whose closure lies in $\overline{B} \cap \operatorname{supp} h$.

(ii) Pointwise, $h^B + \alpha(h - h^B) = h\mathbf{1}_B + \alpha h \mathbf{1}_{\mathbb{R}^2\setminus B}$, which equals $(1-\alpha)h^B + \alpha h$. For $0 \le \alpha \le 1$ and $0 \le h \le 1$ this gives $0 \le h\mathbf{1}_B \le h_\alpha \le h \le 1$; $h_\alpha$ is measurable, and $\{h_\alpha \neq 0\} \subseteq \{h \neq 0\}$ gives $\operatorname{supp} h_\alpha \subseteq \operatorname{supp} h$, compact. At $\alpha = 0, 1$ the formula gives $h^B$ and $h$.

(iii) $\operatorname{supp} h$ is compact, hence contained in $[-\rho,\rho]^2$ for some $\rho$, which meets only finitely many squares. If $\int_\Delta h > 0$, the set $\Delta \cap \{h > 0\}$ has positive measure, so $\Delta$ meets $\{h \neq 0\} \subseteq \operatorname{supp} h$; hence $\mathfrak{S}(h)$ is finite. Since the squares partition $\mathbb{R}^2$ and $B$ is a union of squares, $\mathbf{1}_{\mathbb{R}^2\setminus B} = \sum_{\Delta\cap B=\varnothing} \mathbf{1}_\Delta$ pointwise, so $h - h^B = \sum_{\Delta\cap B=\varnothing} h\mathbf{1}_\Delta$ pointwise, with only the finitely many squares meeting $\{h \neq 0\}$ contributing. If $\Delta \notin \mathfrak{S}(h)$, then $h \ge 0$ and $\int_\Delta h = 0$ give $h\mathbf{1}_\Delta = 0$ almost everywhere, and these terms may be dropped.

(iv) We use the regularization of Appendix C (the notation paragraph before Lemma C.2): $\Phi_\kappa = \delta_\kappa * \Phi$ with an even mollifier $\delta_\kappa$, the variance $c_\kappa$ of $\Phi_\kappa(z)$, and $\mathcal{U}_{h,\kappa} = \lambda \int h :\!\mathscr{P}(\Phi_\kappa)\!:_{c_\kappa} dz$. For every $q \in \mathcal{S}'(\mathbb{R}^2)$ the function $z \mapsto \Phi_\kappa(z)(q) = (q * \delta_\kappa)(z)$ is smooth, and $:\!\mathscr{P}(y)\!:_c = \sum_n a_n c^{n/2} \operatorname{He}_n(y/\sqrt{c})$ is a polynomial in $y$; so $\mathcal{U}_{h,\kappa}(q)$ is the integral of $h$ against a function bounded on the compact $\operatorname{supp} h$, linear in $h$ and unchanged when $h$ is changed on a null set. Hence $\sum_i c_i \mathcal{U}_{h_i,\kappa} = \lambda \int (\sum_i c_i h_i) :\!\mathscr{P}(\Phi_\kappa)\!:_{c_\kappa} dz = 0$ pointwise. Lemma C.4 states: for all $h \in \mathcal{C}_{\mathrm{E}}$ with $\operatorname{supp} h \subseteq \Lambda$ and all $\kappa \ge 2$, $\|\mathcal{U}_h - \mathcal{U}_{h,\kappa}\|^2_{L^2(d\mu_0)} \le c_2\, \kappa^{-1} (\log\kappa)^{2p'}$, with the explicit $c_2 = c_2(\lambda, \mathscr{P}, \Lambda)$ given there. Its hypotheses hold for each $h_i$, with $\Lambda := \bigcup_i \operatorname{supp} h_i$ (compact) and every $\kappa \ge 2$; $\mathcal{U}_h$ is the element of $L^2(d\mu_0)$ constructed in Assumption A3. Therefore

$$\Big\| \sum_i c_i \mathcal{U}_{h_i} \Big\|_{L^2(d\mu_0)} \le \sum_i |c_i|\, \|\mathcal{U}_{h_i} - \mathcal{U}_{h_i,\kappa}\|_{L^2(d\mu_0)} \le \Big(\sum_i |c_i|\Big) \big(c_2 \kappa^{-1} (\log\kappa)^{2p'}\big)^{1/2} \xrightarrow[\kappa\to\infty]{} 0.$$

The two displayed identities are the cases $h_\alpha - h^B - \alpha(h - h^B) = 0$ (all three in $\mathcal{C}_{\mathrm{E}}$ by (i)–(ii)) and (iii). The same identities also follow directly from the linearity of $h \mapsto :\!\Phi^n(h)\!:$ in the construction of Assumption A3, and without Lemma C.4 from Step 1 of the proof of Lemma D.1 (§D.b), which proves from Lemma N.28(a) that the regularizations converge in $L^2(d\mu_0)$ for every bounded kernel of compact support and that the limit is linear in the kernel; that this limit is the $\mathcal{U}_h$ of Assumption A3 is Lemma N.23′(ii) (until 7 November 2026 the manuscript stated no construction of $\mathcal{U}_h$, and the identification was a condition of this appendix). □

In Lemma D.1 the path is used with $h_1 = h^B$ and $h_2 = h$; there $h_2 - h_1 = h - h^B \geq 0$, and it enters only as the observable $\mathcal{U}_{h-h^B}$, never as a cutoff.

### D.a.3 Strips and the strip width

The exponent of (1.1.10) is a strip width, printed on p. 594 as follows (read on the page image, quoted verbatim; the sentence ends without a period in the source): "For two localized monomials $Q_1$ and $Q_2$, with disjoint localization squares $\{\Delta_{j_i}^{(1)}\}$ and $\{\Delta_{j_l}^{(2)}\}$ respectively, let $d$ be the width of the widest strip bounded by two parallel lines, and separating the $\{\Delta_{j_i}^{(1)}\}$ from the $\{\Delta_{j_l}^{(2)}\}$" [7, p. 594]. No orientation is prescribed.

**Definition N.6** (strips; separation; orientation; strip width)**.** Let $L_1, L_2$ be parallel lines in $\mathbb{R}^2$. If $L_1 \neq L_2$, the open region between them is a *strip* of width $\operatorname{dist}(L_1, L_2)$, and its complement is the union of two disjoint closed half-planes $H_1, H_2$ with boundaries $L_1, L_2$. If $L_1 = L_2 =: L$, the strip is the line $L$ itself, of width 0, and $H_1, H_2$ are the two closed half-planes bounded by $L$. The strip *separates* $X \subseteq \mathbb{R}^2$ from $Y \subseteq \mathbb{R}^2$ if $X \subseteq H_1$ and $Y \subseteq H_2$, or the other way round; since $H_1, H_2$ are closed, it then also separates $\overline{X}$ from $\overline{Y}$. The strip is *parallel* (to the time-zero line $\{x^0 = 0\}$) if $L_1, L_2$ are lines $\{x^0 = \text{const}\}$, and *transverse* otherwise. For a square $\Delta$ and a non-empty finite set $R$ of squares with $\Delta \notin R$,

$$d(\Delta, R) := \sup\Big\{\text{width of } S:\ S \text{ a strip separating } \Delta \text{ from } \textstyle\bigcup_{\Delta' \in R} \Delta'\Big\},$$

defined whenever some strip separates them.

If $Q_2$ is localized in $\Delta$ and $Q_1$ in squares of $R$, then every strip separating $\Delta$ from $\bigcup R$ separates the localization squares of $Q_2$ from those of $Q_1$, so its width $w$ is at most the $d$ of p. 594, and $e^{-m_1 d} \leq e^{-m_1 w}$ because $m_1 > 0$. Every application of (1.1.10) in this appendix is made in this weakened form, with the width of one explicit separating strip.

**Lemma N.7** (strip width)**.** *Let $S := \{\Delta_{l',k'} : l_1 \leq l' \leq l_2,\ k_1 \leq k' \leq k_2\}$ be a rectangle of squares, let $R \subseteq S$ be non-empty, and let $\Delta = \Delta_{l,k} \notin S$. Put*

$$\rho_t := \max\{l_1 - l,\, 0,\, l - l_2\}, \qquad \rho_x := \max\{k_1 - k,\, 0,\, k - k_2\}, \qquad m := \max\{\rho_t, \rho_x\},$$

*so that $m \geq 1$ is the sup-index distance from $\Delta$ to the nearest square of $S$.*

(i) *If $\rho_t \geq 1$, the strip bounded by the lines $x^0 = l_2 + \frac{1}{2}$ and $x^0 = l - \frac{1}{2}$ (if $l > l_2$), resp. $x^0 = l + \frac{1}{2}$ and $x^0 = l_1 - \frac{1}{2}$ (if $l < l_1$), is* parallel*, has width $\rho_t - 1$, and separates $\Delta$ from $\bigcup S$.*

(ii) *If $\rho_x \geq 1$, the strip bounded by the lines $x^1 = k_2 + \frac{1}{2}$ and $x^1 = k - \frac{1}{2}$ (if $k > k_2$), resp. $x^1 = k + \frac{1}{2}$ and $x^1 = k_1 - \frac{1}{2}$ (if $k < k_1$), is* transverse *(it is perpendicular to the time-zero line), has width $\rho_x - 1$, and separates $\Delta$ from $\bigcup S$.*

(iii) *$d(\Delta, R)$ is defined and $d(\Delta, R) \geq m - 1$. If $m \geq 2$, the strip of* (i) *or* (ii) *realising $m - 1$ has width $\geq 1$.*

(iv) (Rows next to $R$.) *If $R$ contains a square in a row $l'$ with $|l - l'| \leq 1$, then no parallel strip of positive width separates $\Delta$ from $\bigcup R$: every strip of positive width that separates them is transverse.*

(v) (Monotonicity.) *If $\varnothing \neq R_1 \subseteq R$, then $d(\Delta, R_1) \geq d(\Delta, R)$.*

*Proof.* Every square $\Delta_{l',k'}$ is contained in $\{l'-\frac{1}{2} \le x^0 < l'+\frac{1}{2}\}$ and in $\{k'-\frac{1}{2} \le x^1 < k'+\frac{1}{2}\}$.

(i) Let $l > l_2$, so $\rho_t = l - l_2 \ge 1$. Then $\bigcup S \subseteq \{x^0 \le l_2 + \frac{1}{2}\}$ and $\Delta \subseteq \{x^0 \ge l - \frac{1}{2}\}$. If $\rho_t \ge 2$, the lines $x^0 = l_2 + \frac{1}{2}$ and $x^0 = l - \frac{1}{2}$ are distinct, the open strip $\{l_2 + \frac{1}{2} < x^0 < l - \frac{1}{2}\}$ has width $(l - \frac{1}{2}) - (l_2 + \frac{1}{2}) = \rho_t - 1$, and the two closed half-planes just named are the complement of that strip; if $\rho_t = 1$ the two lines coincide and the strip of width 0 separates. The case $l < l_1$ is the same with $\bigcup S \subseteq \{x^0 \ge l_1 - \frac{1}{2}\}$, $\Delta \subseteq \{x^0 \le l + \frac{1}{2}\}$ and width $(l_1 - \frac{1}{2}) - (l + \frac{1}{2}) = l_1 - l - 1 = \rho_t - 1$. The bounding lines are lines of constant $x^0$, so the strip is parallel.

(ii) The same computation in the coordinate $x^1$, with columns in place of rows; the bounding lines $\{x^1 = \text{const}\}$ are perpendicular to $\{x^0 = 0\}$, so the strip is transverse.

(iii) $\Delta \notin S$ means $l \notin [l_1, l_2]$ or $k \notin [k_1, k_2]$, i.e. $m \ge 1$. If $m = \rho_t$, (i) gives a strip of width $m - 1$ separating $\Delta$ from $\bigcup S$; otherwise $m = \rho_x$ and (ii) does. As $\bigcup R \subseteq \bigcup S$, the same strip separates $\Delta$ from $\bigcup R$. So $d(\Delta, R)$ is defined and at least $m - 1$, which is $\ge 1$ when $m \ge 2$.

(iv) Let $\Delta' \in R$ lie in row $l'$ with $|l - l'| \le 1$, and suppose that the parallel strip $\{\alpha < x^0 < \beta\}$, $\beta > \alpha$, separates $\Delta$ from $\bigcup R$. If $\Delta \subseteq \{x^0 \le \alpha\}$ and $\Delta' \subseteq \{x^0 \ge \beta\}$, then, because $\Delta$ contains points with $x^0$ arbitrarily close to $l + \frac{1}{2}$ and $\Delta'$ contains points with $x^0 = l' - \frac{1}{2}$, we get $l + \frac{1}{2} \le \alpha$ and $l' - \frac{1}{2} \ge \beta$, whence $l' - l \ge 1 + (\beta - \alpha) > 1$, a contradiction. The other case, $\Delta \subseteq \{x^0 \ge \beta\}$ and $\Delta' \subseteq \{x^0 \le \alpha\}$, gives $l - l' > 1$ in the same way.

(v) A strip separating $\Delta$ from $\bigcup R$ separates $\Delta$ from the subset $\bigcup R_1$, so the supremum defining $d(\Delta, R_1)$ is over a larger set. □

**Remark N.8** (the bound fails for a general $R$)**.** The rectangle $S \supseteq R$ in Lemma N.7 cannot be dropped. Let $R = \{\Delta_{0,0}, \Delta_{0,4}\}$ and $\Delta = \Delta_{0,2}$ (verifier, item m2 of the R3 report). Each square of $R$ is at sup-index distance 2 from $\Delta$, but no strip, of positive or of zero width, separates $\Delta$ from $\Delta_{0,0} \cup \Delta_{0,4}$: a closed half-plane containing $\Delta_{0,0} \cup \Delta_{0,4}$ contains the closures and their convex hull $[-\frac{1}{2}, \frac{1}{2}] \times [-\frac{1}{2}, \frac{9}{2}]$, which contains $\overline{\Delta}_{0,2} = [-\frac{1}{2}, \frac{1}{2}] \times [\frac{3}{2}, \frac{5}{2}]$; so a separating strip would place $\Delta$ in $H_1 \cap H_2$, which is empty for positive width and a line for width 0, and a square lies in neither. Thus $d(\Delta, R)$ is undefined, (1.1.10) does not apply to a pair localized in $R$ and in $\Delta$, and the inequality "$d(\Delta, R) \ge$ (sup-index distance to the nearest square of $R$) $- 1$" is false. Lemma N.7 measures the distance to a rectangle containing $R$; here every such rectangle contains $\Delta$.

**Remark N.9** (tilted strips)**.** Lemma N.7 uses axis-parallel strips and gives lower bounds only; the widest separating strip may be tilted and wider. For $R = \{\Delta_{0,0}\}$ and $\Delta = \Delta_{3,3}$ the axis-parallel strips have width 2, while the strip between the lines $x^0 + x^1 = 1$ and $x^0 + x^1 = 5$ separates $\overline{\Delta}_{0,0}$ (on which $x^0 + x^1 \le 1$) from $\overline{\Delta}_{3,3}$ (on which $x^0 + x^1 \ge 5$) and has width $4/\sqrt{2} = 2\sqrt{2}$. A tilted strip is transverse in the sense of Definition N.6; this appendix never needs one.

### D.a.4 The $3 \times 5$ box and its rings

**Definition N.10** (the box $B_{3\times5}$ and its rings)**.** For $a \in \mathbb{R}$ let $k_a := \lfloor a + \frac{1}{2} \rfloor$ and

$$S_a := \{\Delta_{0,k} : \ k_a \le k \le k_a + 2\},$$

$$B^{(j)} := \bigcup \{\Delta_{l,k} : \ |l| \le 1 + j, \ k_a - 1 - j \le k \le k_a + 3 + j\} \qquad (j \ge 0),$$

and $B_{3\times5} = B_{3\times5}(a) := B^{(0)}$, the rows $-1, 0, 1$ times the columns $k_a - 1, \dots, k_a + 3$. For $j \ge 1$ the *ring* $\mathcal{L}_j$ is the set of squares contained in $B^{(j)}$ and not in $B^{(j-1)}$. For a square $\Delta_{l,k}$ put $\delta(\Delta_{l,k}) := \max\{|l| - 1, \ k_a - 1 - k, \ k - k_a - 3\}$.

**Lemma N.11** (ring count and strip widths around $B_{3\times5}$)**.** *Let $a \in \mathbb{R}$ and let $R \subseteq S_a$ be non-empty.*

(i) *The columns $k$ with $[k-\frac{1}{2}, k+\frac{1}{2}) \cap [a, a+2] \neq \varnothing$ are exactly $k_a, k_a+1, k_a+2$; the columns meeting $[a, a+2]$ in positive measure are among them. Hence every square of row 0 whose column meets $[a, a+2]$ lies in $S_a$, and $|B_{3\times5}| = 15$.*

(ii) *$\Delta \subseteq B^{(j)}$ if and only if $\delta(\Delta) \le j$. The rings $\mathcal{L}_j = \{\Delta : \delta(\Delta) = j\}$, $j \ge 1$, are pairwise disjoint, their union is the set of all squares $\Delta \not\subseteq B_{3\times5}$, and $|\mathcal{L}_j| = 8j+12$.*

(iii) *The 20 squares of $\mathcal{L}_1$ are $\Delta_{2,k}$ and $\Delta_{-2,k}$ for $k_a - 2 \le k \le k_a + 4$ (fourteen squares) and $\Delta_{l,k_a-2}$, $\Delta_{l,k_a+4}$ for $|l| \le 1$ (six squares).*

(iv) *Every $\Delta = \Delta_{l,k} \in \mathcal{L}_j$ satisfies $d(\Delta, R) \ge j$. If $|l| = 1+j$, a parallel strip of width $j$ separates $\Delta$ from $\bigcup R$. If $|l| \le j$, then $\Delta$ lies in column $k_a - 1 - j$ or $k_a + 3 + j$, a transverse strip of width $j$ separates it, and every parallel strip separating it has width at most $|l| - 1 \le j - 1$ (none of positive width if $|l| \le 1$). The four corner squares ($|l| = 1+j$ in the two side columns) admit both strips of width $j$.*

(v) *For the squares of rows $-1, 0, 1$ outside $B_{3\times5}$, i.e. $\Delta_{l,k}$ with $|l| \le 1$ and $k \le k_a - 2$ or $k \ge k_a + 4$, no parallel strip of positive width separates $\Delta_{l,k}$ from $\bigcup R$: every strip of positive width separating them is transverse.*

(vi) *For every $m_1 > 0$, with $x := e^{-m_1} \in (0,1)$, put*

$$\overline{\Sigma}_{3\times5} := \sum_{j\ge1} (8j+12)\, e^{-m_1 j} = \frac{4x\,(5-3x)}{(1-x)^2} = \frac{4e^{-m_1}\,(5-3e^{-m_1})}{(1-e^{-m_1})^2}.$$

*With the strips of* (iv)*, of width $w_\Delta := j$ for $\Delta \in \mathcal{L}_j$, $\sum_{\Delta\not\subseteq B_{3\times5}} e^{-m_1 w_\Delta} = \overline{\Sigma}_{3\times5}$; and the sum over the widest strips, which depends on $R$, satisfies*

$$\Sigma_{3\times5} := \sum_{\Delta\not\subseteq B_{3\times5}} e^{-m_1 d(\Delta,R)} \le \overline{\Sigma}_{3\times5}.$$

*Proof.* (i) $[k-\frac{1}{2}, k+\frac{1}{2}) \cap [a, a+2] \neq \varnothing$ if and only if $k - \frac{1}{2} \le a+2$ and $k+\frac{1}{2} > a$, i.e. $a - \frac{1}{2} < k \le a + \frac{5}{2}$. The interval $(a-\frac{1}{2}, a+\frac{5}{2}]$ is half-open of length 3, so it contains exactly three integers; the smallest integer $> a - \frac{1}{2}$ is $\lfloor a - \frac{1}{2} \rfloor + 1 = \lfloor a + \frac{1}{2} \rfloor = k_a$. A column meeting $[a, a+2]$ in positive measure meets it. The rows of $S_a$ and the columns of $B^{(0)}$ give $|B_{3\times5}| = 3 \cdot 5 = 15$.

(ii) $\Delta_{l,k} \subseteq B^{(j)}$ iff $|l| \le 1+j$, $k \ge k_a - 1 - j$ and $k \le k_a + 3 + j$, i.e. iff $|l| - 1 \le j$, $k_a - 1 - k \le j$ and $k - k_a - 3 \le j$, i.e. iff $\delta(\Delta_{l,k}) \le j$. Since $\delta$ takes integer values, $\mathcal{L}_j = \{\delta = j\}$; these sets are disjoint for different $j$, and $\Delta \not\subseteq B_{3\times5} = B^{(0)}$ iff $\delta(\Delta) \ge 1$. $B^{(j)}$ has $2(1+j)+1 = 3+2j$ rows and $(k_a+3+j) - (k_a-1-j) + 1 = 5+2j$ columns, and $B^{(j-1)} \subseteq B^{(j)}$, so

$$|\mathcal{L}_j| = (3+2j)(5+2j) - (1+2j)(3+2j) = (3+2j)\big[(5+2j) - (1+2j)\big] = 4(3+2j) = 8j+12.$$

(iii) $\delta(\Delta_{l,k}) = 1$ means: either $|l| = 2$ and $k_a - 2 \le k \le k_a + 4$ ($2 \times 7$ squares), or $|l| \le 1$ and $k \in \{k_a - 2, k_a + 4\}$ ($3 \times 2$ squares); $14 + 6 = 20 = 8 \cdot 1 + 12$.

(iv) $S_a$ is the rectangle of Lemma N.7 with $l_1 = l_2 = 0$, $k_1 = k_a$, $k_2 = k_a + 2$, and $\Delta \notin S_a$ since $\delta(\Delta) \ge 1$. Let $\Delta = \Delta_{l,k} \in \mathcal{L}_j$. If $|l| - 1 = j$, then $\rho_t = |l| = j+1 \ge 2$ and Lemma N.7(i) gives a parallel strip of width $j$. If $k_a - 1 - k = j$, then $\rho_x = k_a - k = j+1$ and Lemma N.7(ii) gives a transverse strip of width $j$; if $k - k_a - 3 = j$, then $\rho_x = k - (k_a + 2) = j+1$, and

again a transverse strip of width $j$. At least one of the three cases holds because $\delta(\Delta) = j$; if $|l| \le j$ the first fails, so $\Delta$ lies in a side column. Each strip separates $\Delta$ from $\bigcup S_a \supseteq \bigcup R$, so $d(\Delta, R) \ge j$. For a parallel strip $\{\alpha < x^0 < \beta\}$ separating $\Delta$ from a square $\Delta' \in R$ of row 0, the computation in the proof of Lemma N.7(iv) gives $\beta - \alpha \le |l| - 1$; for $|l| \le 1$ this excludes positive width. The corners have $|l| = 1 + j$ and $k \in \{k_a - 1 - j, k_a + 3 + j\}$.

(v) $R$ is a non-empty subset of row 0, and $|l - 0| \le 1$; apply Lemma N.7(iv).

(vi) By (ii) the sum over $\Delta \not\subseteq B_{3\times5}$ is the sum over $j \ge 1$ of the sums over $\mathcal{L}_j$, which have $8j + 12$ terms each, all equal to $e^{-m_1 w_\Delta} = x^j$; this gives $\sum_\Delta e^{-m_1 w_\Delta} = \sum_{j\ge1}(8j + 12)x^j$. By (iv), $d(\Delta, R) \ge w_\Delta = j$ for $\Delta \in \mathcal{L}_j$, and $m_1 > 0$, so $e^{-m_1 d(\Delta,R)} \le x^j$ and $\Sigma_{3\times5} \le \overline{\Sigma}_{3\times5}$. The closed form: for $0 \le x < 1$ and $N \ge 1$,

$$(1 - x)\sum_{j=1}^{N} x^j = x - x^{N+1}, \qquad (1 - x)\sum_{j=1}^{N} jx^j = \sum_{j=1}^{N} x^j - Nx^{N+1},$$

and $x^{N+1} \to 0$, $Nx^{N+1} \to 0$ as $N \to \infty$; hence $\sum_{j\ge1} x^j = x/(1 - x)$ and $\sum_{j\ge1} jx^j = x/(1 - x)^2$, and

$$\sum_{j\ge1}(8j + 12)x^j = \frac{8x}{(1 - x)^2} + \frac{12x}{1 - x} = \frac{8x + 12x - 12x^2}{(1 - x)^2} = \frac{4x(5 - 3x)}{(1 - x)^2}.$$

Test by hand: $4x(5 - 3x)(1 - x)^{-2} = 4x(5 - 3x)(1 + 2x + 3x^2 + \cdots) = 20x + 28x^2 + \cdots$, and $8 \cdot 1 + 12 = 20$, $8 \cdot 2 + 12 = 28$. □

**Remark N.12** (comparison and check)**.** (a) The bound $(2j + 3)(2j + 5)$ of the plan (`aqft`) is $|B^{(j)}|$, the number of squares of the box that contains $\mathcal{L}_j$; it is valid and cruder. (b) *Where the transverse case enters.* The squares of rows $-1, 0, 1$ outside $B_{3\times5}$ admit no parallel strip of positive width (Lemma N.11(v)). The other side squares of $\mathcal{L}_j$, in rows $2 \le |l| \le j$, admit parallel strips, but only of width $\le |l| - 1 < j$; the bound $d(\Delta, R) \ge j$ uses the transverse strip for them too. It cannot be replaced by the parallel width without losing the uniformity: for fixed $|l| \ge 2$ the parallel width is the same in every column, and the number of squares of row $l$ charged by $h$ is not bounded uniformly in $h \in \mathcal{C}_\mathrm{E}$. So the uniform bound $\overline{\Sigma}_{3\times5}$ of (vi), with the strips of (iv), uses the transverse case of Theorem 1.1.7 for every square of $\mathcal{L}_j$ with $|l| \le j$, not only for rows $-1, 0, 1$; each such use is the case A8(iv)-T of Assumption A8, item (iv-T). (c) *Numerical check* (`ugjs/parts/cqft2_ringcheck.py`; nothing is deduced from it): for $j = 1, \ldots, 12$ and several $a$, the script lists the squares with $\delta = j$, confirms $8j + 12$ of them, computes for each the axis-parallel width of Lemma N.7 and confirms it is $\ge j$, and compares partial sums of $\sum(8j + 12)x^j$ with the closed form in exact rational arithmetic. Run record: 3 November 2026, `/usr/local/bin/python3` (Python 3.14.7, numpy 2.5.2, scipy 1.18.1), output `ugjs/parts/cqft2_ringcheck.output.txt`, "ALL CHECKS PASSED"; the quadrature there uses the relative tolerance $10^{-11}$, set in the script and not printed in the output; it also checks the kernel bound of Lemma N.19 against quadrature (relative tolerance $10^{-11}$) and the inequality $(4n - 1)!! \le 4^n(2n)!$ of §D.g for $n \le 40$.

### D.a.5 Translation covariance and the stability constant $S_2$

The free Euclidean measure $d\mu_0$ is the centred Gaussian measure on $\mathcal{S}'(\mathbb{R}^2)$, with the $\sigma$-algebra generated by the coordinate functions $\Phi(f)(q) = \langle q, f\rangle$, and covariance $G := (-\Delta_{\mathbb{R}^2} + m_0^2)^{-1}$ (Assumption A2; [7, p. 589]: "$d\Phi$ is the Gaussian measure with mean zero and covariance $(-\Delta + m_0^2)^{-1}$").

**Lemma N.13** (translation covariance)**.** *For $a \in \mathbb{R}^2$ let $(\tau_a f)(z) := f(z - a)$ for functions on $\mathbb{R}^2$, and $(T_a q)(f) := q(\tau_a f)$ for $q \in \mathcal{S}'(\mathbb{R}^2)$, $f \in \mathcal{S}(\mathbb{R}^2)$.*

(i) *$T_a$ is a measurable bijection of $\mathcal{S}'(\mathbb{R}^2)$ with inverse $T_{-a}$, and $(T_a)_*\mu_0 = \mu_0$. Hence $\int F \circ T_a \, d\mu_0 = \int F \, d\mu_0$ for every measurable $F \geq 0$, and $F \mapsto F \circ T_a$ is an isometry of every $L^p(d\mu_0)$.*

(ii) *$\tau_a$ maps $\mathcal{C}_{\mathrm{E}}$ bijectively onto $\mathcal{C}_{\mathrm{E}}$, with $\operatorname{supp} \tau_a h = \operatorname{supp} h + a$.*

(iii) *For every $h \in \mathcal{C}_{\mathrm{E}}$, $\mathcal{U}_{\tau_a h} = \mathcal{U}_h \circ T_a$ almost everywhere.*

(iv) *For every $h \in \mathcal{C}_{\mathrm{E}}$, $Z(\tau_a h) = Z(h)$, and $\int F \circ T_a \, dq_{\tau_a h} = \int F \, dq_h$ for every measurable $F \geq 0$ and every $F \in L^1(dq_h)$.*

*Proof.* (i) $\Phi(f) \circ T_a = \Phi(\tau_a f)$, since $\Phi(f)(T_a q) = (T_a q)(f) = q(\tau_a f)$; so the preimage of a cylinder set is a cylinder set, and $T_a$ is measurable, with inverse $T_{-a}$. For $f_1, \dots, f_m \in \mathcal{S}(\mathbb{R}^2)$, the vector $(\Phi(f_i) \circ T_a)_i = (\Phi(\tau_a f_i))_i$ is centred Gaussian under $\mu_0$ with covariance matrix $\langle \tau_a f_i, G\, \tau_a f_j \rangle$. In Fourier space $\widehat{\tau_a f}(p) = e^{-ip\cdot a} \hat{f}(p)$ and $G$ is multiplication by $(p^2 + m_0^2)^{-1}$, so the phases cancel and $\langle \tau_a f_i, G\, \tau_a f_j \rangle = \langle f_i, G f_j \rangle$. Thus $(T_a)_*\mu_0$ and $\mu_0$ have the same finite-dimensional distributions; they agree on the cylinder sets, a $\pi$-system generating the $\sigma$-algebra, hence everywhere. The integral identity follows for indicators, then for simple functions, then for $F \geq 0$ by monotone convergence; applied to $|F|^p$ it gives the isometry.

(ii) $\tau_a h$ is measurable with values in $[0,1]$, and $\{\tau_a h \neq 0\} = \{h \neq 0\} + a$, so $\operatorname{supp} \tau_a h = \operatorname{supp} h + a$ is compact; $\tau_{-a}$ is the inverse.

(iii) In the regularization of Appendix C, $\Phi_\kappa(z) = \Phi(\delta_\kappa(z - \cdot))$ with $\delta_\kappa$ even, so $\delta_\kappa(z - \cdot) = \tau_z \delta_\kappa$ and, by (i), $\Phi_\kappa(z) \circ T_a = \Phi(\tau_a \tau_z \delta_\kappa) = \Phi_\kappa(z + a)$. The Wick constant $c_\kappa = \langle \tau_z \delta_\kappa, G\, \tau_z \delta_\kappa \rangle = \langle \delta_\kappa, G \delta_\kappa \rangle$ does not depend on $z$ (by the computation in (i); this is the fact used in (C.c)). Hence, for every $q$,

$$\begin{aligned} \mathcal{U}_{h,\kappa}(T_a q) &= \lambda \int h(z) :\!\mathscr{P}(\Phi_\kappa(z + a)(q))\!:_{c_\kappa} dz \\ &= \lambda \int h(w - a) :\!\mathscr{P}(\Phi_\kappa(w)(q))\!:_{c_\kappa} dw = \mathcal{U}_{\tau_a h,\kappa}(q). \end{aligned}$$

Lemma C.4 (quoted in the proof of Lemma N.5), applied to $h$ with $\Lambda = \operatorname{supp} h$ and to $\tau_a h \in \mathcal{C}_{\mathrm{E}}$ with $\Lambda = \operatorname{supp} h + a$, gives $\mathcal{U}_{h,\kappa} \to \mathcal{U}_h$ and $\mathcal{U}_{\tau_a h,\kappa} \to \mathcal{U}_{\tau_a h}$ in $L^2(d\mu_0)$ as $\kappa \to \infty$; by (i), $\mathcal{U}_{h,\kappa} \circ T_a \to \mathcal{U}_h \circ T_a$ in $L^2(d\mu_0)$. The sequence $\mathcal{U}_{h,\kappa} \circ T_a = \mathcal{U}_{\tau_a h,\kappa}$ has both limits, so they agree almost everywhere.

(iv) By (iii), $e^{-\mathcal{U}_{\tau_a h}} = e^{-\mathcal{U}_h} \circ T_a$ almost everywhere, and (i) gives $Z(\tau_a h) = Z(h)$ and, for $F \geq 0$, $\int (F \circ T_a) e^{-\mathcal{U}_{\tau_a h}} d\mu_0 = \int (F e^{-\mathcal{U}_h}) \circ T_a \, d\mu_0 = \int F e^{-\mathcal{U}_h} d\mu_0$. Divide by $Z(h)$; for $F \in L^1(dq_h)$ apply this to $F^{\pm}$. □

**Definition N.14** (the stability constant)**.** For a compact set $\Lambda \subset \mathbb{R}^2$,

$$S_2(\Lambda) := \sup\big\{ \|e^{-\mathcal{U}_{h'}}\|_{L^2(d\mu_0)} : \ h' \in \mathcal{C}_{\mathrm{E}}, \ \operatorname{supp} h' \subseteq \Lambda \big\},$$

and for a box $B$, $S_2(B) := S_2(\overline{B})$.

**Proposition N.15** ($S_2$ is finite and depends on the shape only)**.** (i) *$1 \leq S_2(\Lambda) < \infty$ for every compact $\Lambda$.*

(ii) *$\Lambda \subseteq \Lambda'$ implies $S_2(\Lambda) \leq S_2(\Lambda')$.*

(iii) *$S_2(\Lambda + a) = S_2(\Lambda)$ for every $a \in \mathbb{R}^2$. In particular, two boxes of the same shape have the same $S_2$: $S_2(B)$ depends on the box $B$ only through its shape.*

(iv) *For every box $B$ and every $h \in \mathcal{C}_{\mathrm{E}}$, $\|e^{-\mathcal{U}_{h_B}}\|_{L^2(d\mu_0)} \leq S_2(B)$.*

*Proof.* (i) Theorem C.5 reads: "For every $p < \infty$ and compact $\Lambda$, $\sup\{\|e^{-\mathcal{U}_h}\|_{L^p(d\mu_0)} : h \in \mathcal{C}_{\mathrm{E}},\ \operatorname{supp} h \subseteq \Lambda\} \le (1 + e^{pb_*})^{1/p} < \infty$". Its hypotheses, checked: $p = 2 < \infty$; $\Lambda$ compact. Its standing data are those of §2.1 and Appendix C, which hold here: $m_0 > 0$; $\mathscr{P}(\xi) = \sum_{n=0}^{2p'} a_n \xi^n$ with $a_{2p'} > 0$ and $p' \ge 1$ (used in Lemma C.3); $\lambda \ge 0$ (its proof treats $\lambda = 0$ separately). The supremum in Definition N.14 is the supremum of Theorem C.5 at $p = 2$, so $S_2(\Lambda) \le (1 + e^{2b_*})^{1/2} < \infty$. (The statement of Theorem C.5 records that its bound depends on $\Lambda$ only through $|\Lambda|$ and $\operatorname{diam}\Lambda$; (iii) below proves the stronger fact that the supremum itself is translation invariant.) The function $h' \equiv 0$ is measurable, $[0,1]$-valued and has empty, hence compact, support contained in $\Lambda$; $\mathcal{U}_0 = 0$ by Lemma N.5(iv) ($m = 1$, $c_1 = 1$, $h_1 = 0$), so $\|e^{-\mathcal{U}_0}\|_2 = 1$ and $S_2(\Lambda) \ge 1$.

(ii) The supremum is over a larger set.

(iii) By Lemma N.13(ii), $h' \mapsto \tau_a h'$ is a bijection from $\{h' \in \mathcal{C}_{\mathrm{E}} : \operatorname{supp} h' \subseteq \Lambda\}$ onto $\{h'' \in \mathcal{C}_{\mathrm{E}} : \operatorname{supp} h'' \subseteq \Lambda + a\}$, and by (iii) and (i) of that lemma $\|e^{-\mathcal{U}_{\tau_a h'}}\|_{L^2(d\mu_0)} = \|e^{-\mathcal{U}_{h'}} \circ T_a\|_{L^2(d\mu_0)} = \|e^{-\mathcal{U}_{h'}}\|_{L^2(d\mu_0)}$. Two boxes of the same shape differ by a translation $a \in \mathbb{Z}^2$, and so do their closures.

(iv) $h^B \in \mathcal{C}_{\mathrm{E}}$ and $\operatorname{supp} h^B \subseteq \overline{B}$ by Lemma N.5(i). □

**Remark N.16** (what $S_2$ is not)**.** $S_2(B)$ is not evaluated. Theorem C.5 bounds it by $(1 + e^{2b_*})^{1/2}$, with $b_*$ explicit but crude; every constant below that contains $S_2$ is finite and not evaluated, and the constant $M_B$ of Theorem D.4 is explicit only up to $S_2$ (plan, Part 2). Theorem C.5 is a bound under $d\mu_0$ for a fixed compact region; it is not the stability estimate (2.4.34) of GJS under interpolated measures and has no growth linear in the volume (plan, trap 14). By Proposition N.15(iii), $S_2(B_s(g, I))$ in Theorem D.3 depends only on $N_B$, and $S_2(B_{3\times 5}(a))$ in Theorem D.4 is one number for all $a$.

### D.a.6 The interaction in one square and the constant $c_U$

The printed norms are used exactly as (1), Assumption A8(iii), transcribes them:

$$\|Q\|_{\mathrm{GJS}} = K_1 \Big[\prod_{\Delta} K_2^{2N(\Delta)} N(\Delta)!\Big] \|w\|_2, \tag{1.1.7}$$

$$\|Q(x_0)\|_{\mathrm{GJS}} = K_1 \Big[\prod_{\Delta} K_2^{2N(\Delta)} N(\Delta)!\Big] \|w(x_0, \cdot)\|_2, \tag{1.1.8}$$

"where $N(\Delta)$ is the total degree localized in the unit square $\Delta$, and (1.1.9): a nonlocalized polynomial is normed by summing its localized pieces", with $K_2 \ge 1$ fixed there; $K_1$ enters to the first power and the local weight is $K_2^{2N(\Delta)} N(\Delta)!$. On p. 594 GJS print (1.1.9) as: "If $Q$ is not localized, or is a general polynomial, it is written as a sum of localized monomials $Q_i$, and then we set (1.1.9) $\|Q\| = \sum_i \|Q_i\|$." [7, p. 594]. The interaction is $\mathcal{U}_h = \lambda \int h(z) {:}\mathscr{P}(\Phi(z)){:}\, dz$ with $\mathscr{P}(\xi) = \sum_{n=0}^{2p'} a_n \xi^n$ (Assumption A2, §2.1).

**Definition N.17** (the interaction in one square)**.** For $h \in \mathcal{C}_{\mathrm{E}}$, a square $\Delta$ and $0 \le n \le 2p'$, let

$$U_{\Delta,n} := \lambda a_n \int_{\Delta} h(z)\, {:}\Phi(z)^n{:}\, dz, \qquad U_{\Delta} := \sum_{n=1}^{2p'} U_{\Delta,n},$$

so that $U_{\Delta,n}$ ($n \ge 1$) is the monomial of type (1.1.6) with $r = 1$, one Wick factor of degree $n$, and kernel $w_{\Delta,n} := \lambda a_n\, h\mathbf{1}_{\Delta} \in L^2(\mathbb{R}^2)$, while $U_{\Delta,0} = \lambda a_0 \int_{\Delta} h\, dz$ is a real number. Expanding ${:}\mathscr{P}(\Phi){:} = \sum_n a_n {:}\Phi^n{:}$ in the definition of $\mathcal{U}$, $\mathcal{U}_{h\mathbf{1}_{\Delta}} = U_{\Delta,0} + U_{\Delta}$.

**Lemma N.18** (the constant $c_U$)**.** *Let $h \in \mathcal{C}_{\mathrm{E}}$ and let $\Delta$ be a square.*

(i) *For $1 \le n \le 2p'$, $U_{\Delta,n}$ is a localized monomial of type (1.1.6) with localization square $\Delta$, local degree $N(\Delta) = n < \bar{n} = 2p'+1$, and*

$$\|U_{\Delta,n}\|_{\mathrm{GJS}} = K_1 K_2^{2n} n!\, \lambda|a_n|\, \|h\mathbf{1}_\Delta\|_{L^2(\mathbb{R}^2)} \le \lambda|a_n| K_1 K_2^{2n} n!.$$

(ii) *With (1.1.9) applied to $U_\Delta = \sum_{n=1}^{2p'} U_{\Delta,n}$,*

$$\|U_\Delta\|_{\mathrm{GJS}} = \sum_{n=1}^{2p'} \|U_{\Delta,n}\|_{\mathrm{GJS}} \le c_U := \lambda \sum_{n=1}^{2p'} |a_n|\, K_1 K_2^{2n}\, n!.$$

*The constant $c_U$ depends on $\lambda$, $a_1, \dots, a_{2p'}$, $K_1$, $K_2$ only; it is uniform in $\Delta$ and in $h \in \mathcal{C}_{\mathrm{E}}$.*

(iii) (The term $n = 0$.) *For every $h' \in \mathcal{C}_{\mathrm{E}}$ and every $F \in L^1(dq_{h'})$, $\mathrm{Cov}_{h'}(F, U_{\Delta,0}) = 0$. The measures $dq_h$, $h \in \mathcal{C}_{\mathrm{E}}$, do not depend on $a_0$. As a coefficient of the interaction, $a_0$ enters the estimates of this appendix only through the lower bound of Lemma C.2, $Z(h') \ge e^{-\lambda|a_0||\Lambda|}$, and through $\mathcal{U}_{h'}$ inside $S_2$, i.e. through the base terms of Theorems D.3 and D.4. As a coefficient of the observable $V(h')$ of Theorem D.4 it also enters $V_*$ (the term $n = 0$ of Lemma N.20(iv)) and the bound $\mathfrak{Q}$ on $\|Q_V\|_{\mathrm{GJS}}$ (the sum over $n \ge 0$, §D.g). It does not enter $c_U$.*

(iv) *For every box $B$, $\mathcal{U}_{h-h^B} = \sum_{\Delta\in\mathfrak{S}(h),\, \Delta\cap B=\varnothing} (U_{\Delta,0} + U_\Delta)$ in $L^2(d\mu_0)$; and for every $h' \in \mathcal{C}_{\mathrm{E}}$ and every $F$ such that $F$ and the finitely many $U_{\Delta,n}$ involved lie in $L^2(dq_{h'})$,*

$$\mathrm{Cov}_{h'}\big(F, \mathcal{U}_{h-h^B}\big) = \sum_{\Delta\in\mathfrak{S}(h),\ \Delta\cap B=\varnothing} \sum_{n=1}^{2p'} \mathrm{Cov}_{h'}\big(F, U_{\Delta,n}\big).$$

*Proof.* (i) In the notation of p. 593, $U_{\Delta,n} = \int dx_0\, Q(x_0)$ with $Q(x_0) = \int d\vec{x}\, w_{\Delta,n}(x)\, {:}\Phi(x)^n{:}$, $r = 1$, $n_1 = n$, and the kernel $w_{\Delta,n} = \lambda a_n h \mathbf{1}_\Delta$ vanishes outside $\Delta$; so the monomial is localized in $\Delta$ (Remark N.2(i)), with $N(\Delta) = n$ and $N(\Delta') = 0$ for every other square $\Delta'$, whose factors in (1.1.7) are $K_2^0\, 0! = 1$. The degree bound: $n \le 2p' < 2p'+1 = \bar{n}$ (Assumption A8(ii)). By (1.1.7), $\|U_{\Delta,n}\|_{\mathrm{GJS}} = K_1 K_2^{2n} n!\, \|w_{\Delta,n}\|_2$, and $\|w_{\Delta,n}\|_2 = \lambda|a_n|\, (\int_\Delta h^2\, dz)^{1/2} \le \lambda|a_n|\, |\Delta|^{1/2} = \lambda|a_n|$ because $0 \le h \le 1$ and $|\Delta| = 1$. (Test: for $h = \mathbf{1}_\Delta$ equality holds, $\|U_{\Delta,n}\|_{\mathrm{GJS}} = \lambda|a_n| K_1 K_2^{2n} n!$.)

(ii) The decomposition $U_\Delta = \sum_n U_{\Delta,n}$ writes the polynomial $U_\Delta$ as a sum of the localized monomials of (i), and (1.1.9) sets its norm to the sum of theirs; sum the bounds of (i). The bound contains no $\Delta$ and no $h$.

(iii) $dq_{h'}$ is a probability measure (Assumption A3), so for a constant $c$, $\int F c\, dq_{h'} - \int F\, dq_{h'} \int c\, dq_{h'} = c \int F\, dq_{h'} - c \int F\, dq_{h'} = 0$. At the regularized level ${:}\mathscr{P}(y){:}_c = a_0 + {:}(\mathscr{P} - a_0){:}(y){:}_c$, since the Wick polynomial of degree 0 is the constant; hence $\mathcal{U}_{h,\kappa} = \lambda a_0 \int h\, dz + \mathcal{U}^{(0)}_{h,\kappa}$, where $\mathcal{U}^{(0)}$ is the interaction of $\mathscr{P} - a_0$. Lemma C.4, which applies to $\mathscr{P} - a_0$ as well (same degree $2p'$, same leading coefficient), passes this to the limit: $\mathcal{U}_h = \lambda a_0 \int h\, dz + \mathcal{U}^{(0)}_h$ in $L^2(d\mu_0)$. The factor $e^{-\lambda a_0 \int h}$ of $e^{-\mathcal{U}_h}$ then cancels against the same factor of $Z(h)$. The two places where $a_0$ survives as a coefficient of the interaction are the separate bounds of the numerator $\|e^{-\mathcal{U}_{h^B}}\|_2 \le S_2(B)$ and of the denominator $Z(h^B) \ge e^{-\lambda|a_0||B|}$ in the base term; the ratio itself does not depend on $a_0$. In $V_*$ and $\mathfrak{Q}$ it is a coefficient of the observable $V(h') = \lambda \sum_{n\ge 0} a_n W_n$, not of the measure; $c_U$ sums over $n \ge 1$ only.

(iv) Lemma N.5(iv) and Definition N.17 give the first identity. The covariance is bilinear on $L^2(dq_{h'})$, and the constants $U_{\Delta,0}$ drop out by (iii). The integrability of $F$ and of the $U_{\Delta,n}$ under $dq_{h'}$ is part of the chain of Lemma D.1 (§D.b). □

### D.a.7 The kernel $C$, the Wick isometry and $V_*$

The time-zero measure $d\phi_0$ is "Gaussian with covariance $C = (2\mu)^{-1}$, $\mu = (-\partial_x^2 + m_0^2)^{1/2}$" (§2.1). Fourier convention: $\hat f(k) = \int e^{-ikx} f(x)\,dx$, so that $\langle f, g\rangle = (2\pi)^{-1}\int \overline{\hat f}\,\hat g\,dk$ and $\mu$ acts as multiplication by $(k^2+m_0^2)^{1/2}$.

**Lemma N.19** (the kernel of $(2\mu)^{-1}$)**.** *For $u \in \mathbb{R}\setminus\{0\}$ put*

$$C(u) := \frac{1}{4\pi}\int_0^\infty t^{-1}\, e^{-tm_0^2 - u^2/(4t)}\,dt, \qquad b_0 := \log 4 + e^{-1} + m_0^{-2}.$$

(i) *$0 < C(u) < \infty$; $C$ is even, nonincreasing in $|u|$, and continuous on $\mathbb{R}\setminus\{0\}$.*

(ii) *For real $f, g \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, $\langle f, (2\mu)^{-1} g\rangle = \iint f(x)\, C(x-y)\, g(y)\,dx\,dy$, the double integral converging absolutely. Thus $C$ is the kernel of $(2\mu)^{-1}$, and $\mathbb{E}_{\phi_0}[\varphi_0(f)\varphi_0(g)] = \iint f(x)C(x-y)g(y)\,dx\,dy$.*

(iii) *$C(u) \le (4\pi)^{-1}\big(b_0 + 2|\log|u||\big)$ for every $u \neq 0$.*

(iv) *For $n \ge 0$ let $I_n := \iint_{[-1,1]^2} C(x-y)^n\,dx\,dy = \int_{-2}^{2}(2-|u|)\,C(u)^n\,du$. Then $I_0 = 4$, and for $n \ge 1$*

$$I_n \le (4\pi)^{-n}\, 2^{n+1}\big(2b_0^n + 2^n(n! + (\log 2)^n)\big) < \infty.$$

(v) (The Euclidean kernel.) *For real $f, g \in L^1(\mathbb{R}^2) \cap L^2(\mathbb{R}^2)$, $\langle f, (-\Delta_{\mathbb{R}^2} + m_0^2)^{-1} g\rangle = \iint f(z)\, C(|z-z'|)\, g(z')\,dz\,dz'$, the double integral converging absolutely. So the covariance $G$ of $d\mu_0$ has the kernel $G(\zeta) = C(|\zeta|)$, $\zeta \neq 0$, which is positive, radial, nonincreasing in $|\zeta|$, continuous on $\mathbb{R}^2\setminus\{0\}$, bounded by $(4\pi)^{-1}(b_0 + 2|\log|\zeta||)$, and satisfies $\int_{\mathbb{R}^2} G = m_0^{-2}$.*

*Proof.* (ii) For $A > 0$, the substitution $s = tA$ gives $\int_0^\infty t^{-1/2}e^{-tA}dt = \Gamma(\frac12)A^{-1/2} = \sqrt{\pi}A^{-1/2}$; with $A = k^2 + m_0^2$,

$$\big(2(k^2+m_0^2)^{1/2}\big)^{-1} = (2\sqrt{\pi})^{-1}\int_0^\infty t^{-1/2}e^{-tm_0^2}e^{-tk^2}dt.$$

The heat kernel $p_t(u) := (4\pi t)^{-1/2}e^{-u^2/(4t)}$ has $\hat p_t(k) = e^{-tk^2}$ (complete the square), so for $f \in L^1 \cap L^2$, by Plancherel and the convolution theorem,

$$(2\pi)^{-1}\int |\hat f(k)|^2 e^{-tk^2}dk = \iint f(x)\, p_t(x-y)\, f(y)\,dx\,dy.$$

Insert the $t$-integral into $\langle f, (2\mu)^{-1}f\rangle = (2\pi)^{-1}\int |\hat f(k)|^2\big(2(k^2+m_0^2)^{1/2}\big)^{-1}dk$ and exchange $k$ and $t$ (Tonelli; the integrand is $\ge 0$):

$$\begin{aligned}\langle f, (2\mu)^{-1}f\rangle &= (2\sqrt{\pi})^{-1}\int_0^\infty t^{-1/2}e^{-tm_0^2}\,(2\pi)^{-1}\int |\hat f(k)|^2 e^{-tk^2}\,dk\,dt\\ &= (2\sqrt{\pi})^{-1}\int_0^\infty t^{-1/2}e^{-tm_0^2}\iint f(x)\, p_t(x-y)\, f(y)\,dx\,dy\,dt.\end{aligned}$$

For $f \ge 0$ the integrand in $(t,x,y)$ is nonnegative and Tonelli exchanges again; as

$$(2\sqrt{\pi})^{-1}\, t^{-1/2}\,(4\pi t)^{-1/2} = (4\pi t)^{-1},$$

the result is

$$\langle f, (2\mu)^{-1}f\rangle = \iint f(x)\, C(x-y)\, f(y)\,dx\,dy,$$

which is therefore at most $\|f\|_2^2/(2m_0) < \infty$, since $\|(2\mu)^{-1}\| \le (2m_0)^{-1}$. For real $f, g$ apply this to $F := |f| + |g| \ge 0$:

$$\iint F(x)\, C(x-y)\, F(y)\, dx\, dy < \infty$$

dominates every double integral of $|f|, |g|$ against $C$, and hence (by the same Tonelli step) against $t^{-1/2} e^{-tm_0^2} p_t$ in $(t, x, y)$; so Fubini applies to the display for signed $f$, giving $\langle f, (2\mu)^{-1} f\rangle = \iint fCf$; polarization (both sides are symmetric bilinear forms, $C$ being even) gives the statement for $f, g$.

(i) The integrand is positive, so $C(u) > 0$; finiteness is (iii). $C$ depends on $u^2$ only, and $e^{-u^2/(4t)}$ is nonincreasing in $|u|$. For $|u| \ge u_0 > 0$ the integrand is dominated by $t^{-1} e^{-tm_0^2 - u_0^2/(4t)}$, whose integral is $4\pi C(u_0) < \infty$; dominated convergence gives continuity on $\{|u| > u_0\}$ for every $u_0 > 0$.

(iii) Drop $e^{-tm_0^2} \le 1$ where convenient. The substitution $s = u^2/(4t)$ gives, for $\tau > 0$,

$$\int_0^\tau t^{-1} e^{-u^2/(4t)} dt = \int_{u^2/(4\tau)}^\infty s^{-1} e^{-s} ds,$$

$$\int_{1/4}^\infty s^{-1} e^{-s} ds \le \int_{1/4}^1 \frac{ds}{s} + \int_1^\infty e^{-s} ds = \log 4 + e^{-1};$$

also $\int_1^\infty t^{-1} e^{-tm_0^2} dt \le \int_1^\infty e^{-tm_0^2} dt \le m_0^{-2}$. If $0 < |u| \le 1$, split $\int_0^\infty = \int_0^{u^2} + \int_{u^2}^1 + \int_1^\infty$: the three pieces are at most $\log 4 + e^{-1}$ ($\tau = u^2$), $\int_{u^2}^1 t^{-1} dt = 2|\log|u||$, and $m_0^{-2}$. If $|u| > 1$, split $\int_0^1 + \int_1^\infty$: at most $\int_{u^2/4}^\infty s^{-1} e^{-s} ds \le \log 4 + e^{-1}$ (as $u^2/4 > \frac{1}{4}$) and $m_0^{-2}$. In both cases $4\pi C(u) \le b_0 + 2|\log|u||$.

(iv) For $\varphi \ge 0$ measurable, $\iint_{[-1,1]^2} \varphi(x-y)\, dx\, dy = \int_{-2}^2 (2 - |u|)\varphi(u)\, du$, since $\{x \in [-1,1] : x - u \in [-1,1]\}$ has length $2 - |u|$ for $|u| \le 2$. For $n = 0$ this is 4. For $n \ge 1$, by (iii) and $(\alpha+\beta)^n \le 2^{n-1}(\alpha^n + \beta^n)$ ($\alpha, \beta \ge 0$, convexity), and $2 - |u| \le 2$,

$$I_n \le 4 \int_0^2 C(u)^n\, du \le 4(4\pi)^{-n} 2^{n-1} \int_0^2 \big(b_0^n + 2^n |\log u|^n\big) du,$$

and $\int_0^2 |\log u|^n du \le \int_0^1 |\log u|^n du + (\log 2)^n = n! + (\log 2)^n$ (substitute $u = e^{-s}$ in the first integral). Collect: $4 \cdot 2^{n-1}(2b_0^n + 2^n(n! + (\log 2)^n))(4\pi)^{-n}$.

(v) The computation of (ii) in two dimensions. With $\hat f(p) := \int e^{-ip\cdot z} f(z)\, dz$, $\langle f, (-\Delta + m_0^2)^{-1} f\rangle = (2\pi)^{-2} \int |\hat f(p)|^2 (|p|^2 + m_0^2)^{-1} dp$, and $(|p|^2 + m_0^2)^{-1} = \int_0^\infty e^{-tm_0^2} e^{-t|p|^2} dt$. The heat kernel $P_t(\zeta) := (4\pi t)^{-1} e^{-|\zeta|^2/(4t)}$ has $\hat P_t(p) = e^{-t|p|^2}$, so $(2\pi)^{-2} \int |\hat f|^2 e^{-t|p|^2} dp = \iint f(z) P_t(z - z') f(z')\, dz\, dz'$. Tonelli for $f \ge 0$, then $|f| + |g|$ for absolute convergence (with $\|(-\Delta + m_0^2)^{-1}\| \le m_0^{-2}$), Fubini and polarization, exactly as in (ii), give the kernel $\int_0^\infty e^{-tm_0^2} (4\pi t)^{-1} e^{-|\zeta|^2/(4t)} dt = C(|\zeta|)$. The properties follow from (i) and (iii), and by Tonelli $\int_{\mathbb{R}^2} G = (4\pi)^{-1} \int_0^\infty t^{-1} e^{-tm_0^2} (4\pi t)\, dt = m_0^{-2}$. □

*Chaoses.* Let $\mathcal{H}_{\le n}$ be the closure in $L^2(d\phi_0)$ of the polynomials of degree $\le n$ in the variables $\varphi_0(f)$, and $\mathcal{H}_n := \mathcal{H}_{\le n} \ominus \mathcal{H}_{\le n-1}$ (the $n$-th Wick chaos); the same definitions are made in $L^2(d\mu_0)$ with the variables $\Phi(f)$. These are definitions and need no import. The fact that the second quantization $\Gamma(e^{-t}\mathbf{1})$ acts on the $n$-th chaos as multiplication by $e^{-nt}$, which the proof of Theorem C.5 uses ("$\Gamma(e^{-t})Y_r = e^{-rt}Y_r$"), is imported in Assumption A3 ((I.32) of Simon's book) on Simon's $\Gamma_n$, the closed span of the Wick monomials $:\Phi(f_1)\cdots\Phi(f_n):$, and is used here only in Lemma N.21; every object of degree $n$ to which this appendix applies it is an $L^2$-limit of integrals of Wick monomials $:\Phi(g)^n:$ with all $n$ test functions

equal (Lemma N.21(i)), so it lies in $\Gamma_n$ directly (Lemma N.23′(i) makes this precise). For a centred Gaussian variable $X$, $:X^n: := c^{n/2}\,\mathrm{He}_n(X/\sqrt{c})$, $c = \mathbb{E}[X^2]$, is the coefficient of $s^n/n!$ in $:e^{sX}: := e^{sX - s^2c/2}$ (as in the proof of Proposition 3.2). Fix $\rho \in C_c^\infty(\mathbb{R})$, $\rho \ge 0$, even, $\operatorname{supp}\rho \subseteq [-1,1]$, $\int \rho = 1$, put $\rho_\kappa(x) := \kappa\rho(\kappa x)$ and $\varphi_\kappa(x) := \varphi_0(\rho_\kappa(x - \cdot))$ for $\kappa \ge 1$.

**Lemma N.20** (Wick isometry; the constant $V_*$)**.** *Let $h'$ be real and measurable with $\|h'\|_\infty \le 1$ and $\operatorname{supp} h' \subseteq [a, a+2]$, $a \in \mathbb{R}$, and let $0 \le n, m \le 2p'$.*

(i) *For $\kappa, \kappa' \ge 1$: $\mathbb{E}_{\phi_0}\big[:\varphi_\kappa(x)^n: :\varphi_{\kappa'}(y)^m:\big] = \delta_{nm}\, n!\, C_{\kappa,\kappa'}(x-y)^n$, where $C_{\kappa,\kappa'}(u) := \int \nu(v) C(u-v)\,dv > 0$ and $\nu := \rho_\kappa * \rho_{\kappa'}$.*

(ii) *If $\min\{\kappa,\kappa'\} \ge 6$, then $C_{\kappa,\kappa'}(u) \le c_\rho\,(b_0 + 2 + 2|\log|u||)$ for all $u \ne 0$, with $c_\rho := 3\|\rho\|_\infty/\pi$; and $C_{\kappa,\kappa'}(u) \to C(u)$ for every $u \ne 0$ as $\min\{\kappa,\kappa'\} \to \infty$.*

(iii) *$W_{n,\kappa} := \int h'(x) :\varphi_\kappa(x)^n: dx$ lies in $\mathcal{H}_n$, and converges in $L^2(d\phi_0)$, as $\kappa \to \infty$, to some $W_n \in \mathcal{H}_n$ with $\langle W_n, W_m\rangle = \delta_{nm}\, n! \iint h'(x)h'(y)C(x-y)^n\,dx\,dy$.*

(iv) *$V(h') := \lambda \sum_{n=0}^{2p'} a_n W_n$ satisfies*

$$\|V(h')\|^2_{L^2(d\phi_0)} = \lambda^2 \sum_{n=0}^{2p'} a_n^2\, n! \iint h'(x)h'(y)\,C(x-y)^n\,dx\,dy \le \overline{V}_* := \lambda^2 \sum_{n=0}^{2p'} a_n^2\, n!\, I_n.$$

*Hence $V_* := \sup_{h'} \|V(h')\|^2_{L^2(d\phi_0)} \le \overline{V}_* < \infty$, the supremum over all $h'$ as above and all $a \in \mathbb{R}$.*

*Proof.* (i) $X := \varphi_\kappa(x)$ and $Y := \varphi_{\kappa'}(y)$ are jointly centred Gaussian, with $\mathbb{E}[XY] = \langle \rho_\kappa(x - \cdot), (2\mu)^{-1}\rho_{\kappa'}(y - \cdot)\rangle$; by Lemma N.19(ii) and the substitutions $x' = x - \alpha$, $y' = y - \beta$ this is $\iint \rho_\kappa(\alpha)\rho_{\kappa'}(\beta)C(x - y - \alpha + \beta)\,d\alpha\,d\beta = C_{\kappa,\kappa'}(x-y)$, with $\nu$ the density of $\alpha - \beta$, which is $\rho_\kappa * \rho_{\kappa'}$ because $\rho_{\kappa'}$ is even; $C > 0$ gives $C_{\kappa,\kappa'} > 0$. The Gaussian moment generating function gives $\mathbb{E}[:e^{sX}::e^{tY}:] = e^{-(s^2\mathbb{E}X^2 + t^2\mathbb{E}Y^2)/2}\mathbb{E}[e^{sX+tY}] = e^{st\,\mathbb{E}[XY]}$. Both sides are entire in $(s,t)$; the left side may be differentiated under $\mathbb{E}$ (for $|s|, |t| \le 1$ the derivatives are dominated by polynomials in $|X|, |Y|$ times $e^{|X|+|Y|}$, which are integrable), and its coefficient of $s^n t^m/(n!\,m!)$ is $\mathbb{E}[:X^n::Y^m:]$; on the right it is $\delta_{nm} n!\,\mathbb{E}[XY]^n$.

(ii) $\nu \ge 0$, $\int \nu = 1$, $\operatorname{supp}\nu \subseteq [-\varepsilon, \varepsilon]$ with $\varepsilon := \kappa^{-1} + \kappa'^{-1} \le 2/K$, $K := \min\{\kappa,\kappa'\}$, and $\|\nu\|_\infty \le \min\{\|\rho_\kappa\|_\infty, \|\rho_{\kappa'}\|_\infty\} = K\|\rho\|_\infty$. *Case $|u| \ge 2\varepsilon$.* For $|v| \le \varepsilon$, $|u - v| \ge |u|/2$, so by Lemma N.19(i),(iii) $C(u-v) \le C(u/2) \le (4\pi)^{-1}(b_0 + 2\log 2 + 2|\log|u||)$; average over $\nu$. *Case $0 < |u| < 2\varepsilon$.* Then $C_{\kappa,\kappa'}(u) \le K\|\rho\|_\infty \int_{|w|\le 3\varepsilon} C(w)\,dw$, and $3\varepsilon \le 6/K \le 1$. For $0 < r \le 1$, Lemma N.19(iii) and $\int_0^r |\log w|\,dw = r(1 + |\log r|)$ give $\int_{|w|\le r} C \le (4\pi)^{-1} 2r(b_0 + 2 + 2|\log r|)$, which is increasing in $r \in (0,1]$ (derivative $(4\pi)^{-1}2(b_0 - 2\log r) > 0$); at $r = 6/K$ it is $(4\pi)^{-1}(12/K)(b_0 + 2 + 2\log(K/6))$. Since $|u| < 4/K$, $\log(K/6) < \log(2/3) - \log|u| \le |\log|u||$. So $C_{\kappa,\kappa'}(u) \le (3\|\rho\|_\infty/\pi)(b_0 + 2 + 2|\log|u||)$. As $\int \rho = 1$ on $[-1,1]$, $\|\rho\|_\infty \ge \frac{1}{2}$, so $c_\rho \ge (4\pi)^{-1}$ and $2\log 2 < 2$ make $c_\rho(b_0 + 2 + 2|\log|u||)$ a bound in both cases. Convergence: $C_{\kappa,\kappa'}(u) - C(u) = \int \nu(v)(C(u-v) - C(u))dv \to 0$ as $\varepsilon \to 0$, by the continuity of $C$ at $u \ne 0$.

(iii) $(x,q) \mapsto :\varphi_\kappa(x)^n:(q)$ is continuous in $x$ and measurable in $q$, and its moments are bounded uniformly in $x$; Fubini therefore applies below. For a polynomial $P$ of degree $< n$ in variables $\varphi_0(g_i)$, the coefficient of $s^n$ in $\mathbb{E}[:e^{sX}: e^{\sum_i t_i\varphi_0(g_i)}] = e^{s\sum_i t_i \mathbb{E}[X\varphi_0(g_i)]}\,\mathbb{E}[e^{\sum_i t_i\varphi_0(g_i)}]$ is a power series in the $t_i$ all of whose terms have degree $\ge n$, so $\mathbb{E}[:X^n:P] = 0$; with Fubini, $W_{n,\kappa} \perp \mathcal{H}_{\le n-1}$. For $Q \perp \mathcal{H}_{\le n}$, $\mathbb{E}[W_{n,\kappa}Q] = \int h'(x)\mathbb{E}[:\varphi_\kappa(x)^n:Q]dx = 0$, so $W_{n,\kappa} \in \mathcal{H}_{\le n}$; hence $W_{n,\kappa} \in \mathcal{H}_n$. By (i) and Fubini,

$$\|W_{n,\kappa} - W_{n,\kappa'}\|_2^2 = n! \iint h'(x)h'(y)\big[C_{\kappa,\kappa}^n - 2C_{\kappa,\kappa'}^n + C_{\kappa',\kappa'}^n\big](x-y)\,dx\,dy.$$

By (ii), for $\min\{\kappa,\kappa'\}\ge 6$ each of the three integrands is dominated by

$$\mathbf{1}_{[a,a+2]}(x)\,\mathbf{1}_{[a,a+2]}(y)\;c_\rho^n\big(b_0+2+2|\log|x-y||\big)^n,$$

which is integrable (as in Lemma N.19(iv)), and each converges pointwise off the diagonal to $h'(x)h'(y)C(x-y)^n$. By dominated convergence the three integrals have the same limit, so $(W_{n,\kappa})$ is Cauchy; its limit $W_n$ lies in the closed subspace $\mathcal{H}_n$, and $\langle W_n,W_m\rangle=\lim_\kappa\langle W_{n,\kappa},W_{m,\kappa}\rangle$ is given by (i) and the same dominated convergence.

(iv) Expand the square with (iii). Since $|h'|\le\mathbf{1}_{[a,a+2]}$ and $C^n\ge 0$ (Lemma N.19(i): here the positivity of $C$ is used, for odd $n$), $|\iint h'(x)h'(y)C(x-y)^n|\le\iint_{[a,a+2]^2}C(x-y)^n=I_n$, the last step by the translation $x\mapsto x-a-1$, $y\mapsto y-a-1$. The bound does not depend on $h'$ or $a$. □

*The gap recorded here until 7 November 2026, and its closure.* The comparison note recorded at this place a GAP: the manuscript then wrote $V(h)$ as the formal integral $\lambda\int h(x){:}\mathscr{P}(\varphi_0(x)){:}_C\,dx$, "a self-adjoint multiplication operator in $Q$-space", and stated no construction of its Wick powers, so that the identification of that $V(h)$ with the limit $\lambda\sum_n a_nW_n$ of Lemma N.20 was assumed; it entered at Step 1 of the proof of Theorem 5.1 and at the transfer of Theorem D.4 to it. Since 7 November 2026, $V(h)$ is defined in §2.1 through the Wick powers ${:}\varphi_0^n(h){:}_C$ of Assumption A1 (the elements of the chaoses fixed by the pairings (W.1)), and Lemma N.23′ below proves that $W_n(h')={:}\varphi_0^n(h'){:}_C$, so that $V(h')=\lambda\sum_n a_nW_n(h')$ is a theorem; the corresponding identification for $\mathcal{U}_h$ (Assumption A3) is part (iii) of the same lemma. Lemmas N.20–N.22, the constants $V_*$, $9^{p'}$ and $M_B$ of §D.g, and the Euclidean bound of Theorem D.4 concern the constructed objects (Remark N.23(b)) and never needed the identification.

**Lemma N.21** (the comparison $\|Y\|_4\le 3^{p'}\|Y\|_2$)**.** *Let $h'$ be as in Lemma N.20, and let*

$$Y:=V(h')(0)=\lambda\sum_{n=0}^{2p'}a_n\int h'(x)\,{:}\Phi(0,x)^n{:}\,dx$$

*be the sharp-time observable at time* 0*, as an element of $L^2(d\mu_0)$.*

(i) *$Y=\sum_{r=0}^{2p'}Y_r$, where each $Y_r$ lies in the $r$-th Wick chaos of $L^2(d\mu_0)$; moreover $\|Y\|_{L^2(d\mu_0)}=\|V(h')\|_{L^2(d\phi_0)}$.*

(ii) *$\|Y\|_{L^4(d\mu_0)}\le 3^{p'}\|Y\|_{L^2(d\mu_0)}$. Consequently*

$$\|Y^2\|_{L^2(d\mu_0)}=\|Y\|^2_{L^4(d\mu_0)}\le 9^{p'}\|V(h')\|^2_{L^2(d\phi_0)}\le 9^{p'}V_*,$$

*and the same bound holds for $\|V(h')^2\|_{L^2(d\phi_0)}$, which equals $\|Y^2\|_{L^2(d\mu_0)}$.*

*Proof.* (i) Remark 2.5: "The sharp-time restriction of the Euclidean covariance equals $C$" and "sharp-time $d\mu_0$-Wick ordering coincides, constant by constant, with the $C$-Wick ordering used in $V(h)$". GJS write the sharp-time field as "$\Phi(\tau,f)\equiv\Phi(\delta_\tau\otimes f)$" [7, p. 589] and print no construction; we read $\Phi(0,f)$ as the extension of the linear map $f\mapsto\Phi(f)$ by continuity in the covariance norm, which is finite at $\delta_0\otimes f$ because $\langle\delta_0\otimes f,G\,\delta_0\otimes f\rangle=\langle f,Cf\rangle$ (Remark 2.5). So $\Phi(0,f)$ is the $L^2(d\mu_0)$-limit of $\Phi(f_\varepsilon)$ along test functions $f_\varepsilon\to\delta_0\otimes f$ in that norm, a centred Gaussian variable in the closed first chaos of $L^2(d\mu_0)$, and the family $(\Phi(0,f))_f$ has the covariance $C$ of $(\varphi_0(f))_f$ under $d\phi_0$. So $J_0:F(\varphi_0)\mapsto F(\Phi(0,\cdot))$ preserves the joint laws of finitely many field variables and is an isometry $L^2(d\phi_0)\to L^2(d\mu_0)$; it maps ${:}\varphi_\kappa(x)^n{:}$ to the Wick power, at its own variance (the same $c$), of the Gaussian variable

$\Phi(0,\rho_\kappa(x-\cdot))$. The generating-function argument of Lemma N.20(iii) holds in any Gaussian space: this Wick power is orthogonal to every polynomial of degree $<n$ in the variables $\Phi(f)$, and it is a polynomial of degree $n$ in a first-chaos variable, hence (Gaussian $L^2$-limits converge in every $L^q$) in the closure of the polynomials of degree $\le n$. So $J_0W_{n,\kappa}$ lies in the $n$-th Wick chaos of $L^2(d\mu_0)$, and so does its limit $J_0W_n$. Moreover $J_0W_{n,\kappa}$ is the integral over $x$ of the Wick monomials $:\Phi(0,\rho_\kappa(x-\cdot))^n:$, each the $L^2$-limit of $:\Phi(f_\varepsilon)^n:$ for test functions $f_\varepsilon\to\delta_0\otimes\rho_\kappa(x-\cdot)$ in the covariance norm; so $J_0W_n$ also lies in Simon's $\Gamma_n$, the closed span of the Wick monomials, on which Assumption A3 imports the action of $\Gamma$ ((I.32) of Simon's book). Put $Y_r:=\lambda a_r J_0W_r$; then $Y=J_0V(h')=\sum_r Y_r$, and $\|Y\|_{L^2(d\mu_0)}=\|V(h')\|_{L^2(d\phi_0)}$.

(ii) The proof of Theorem C.5 proves the comparison under $d\mu_0$ (its first paragraph): for $Y$ a sum $\sum_{r=0}^{2p'}Y_r$ of Wick chaoses, $\|Y\|_{L^q}\le(q-1)^{p'}\|Y\|_{L^2}$ $(q\ge2)$, from the abstract form of Assumption A4 (the second-quantized contraction $\Gamma(e^{-t})$ on $L^2(d\mu_0)$ satisfies $\|\Gamma(e^{-t})X\|_q\le\|X\|_2$ for $q=1+e^{2t}$) and $\Gamma(e^{-t})Y_r=e^{-rt}Y_r$: writing $Y=\Gamma(e^{-t})Z$ with $Z=\sum_r e^{rt}Y_r$, orthogonality of the chaoses gives $\|Z\|_2\le e^{2p't}\|Y\|_2$, whence $\|Y\|_q\le\|Z\|_2\le e^{2p't}\|Y\|_2=(q-1)^{p'}\|Y\|_2$. The argument uses of its $Y$ only that it is a sum of Wick chaoses of degree $r\le2p'$ in $L^2(d\mu_0)$, which our $Y$ is by (i); so it applies verbatim, with the action $\Gamma(e^{-t})Y_r=e^{-rt}Y_r$ imported in Assumption A3 (part (i) above does not need it); it is also Theorem I.22 of Simon's book at $p=4$, $m=2p'$ (Assumption A4). With $q=4$: $t=\frac12\log3$, the contraction $e^{-t}\mathbf{1}$ has $\|e^{-t}\mathbf{1}\|^2=\frac13=(2-1)/(4-1)$ as the abstract form of Assumption A4 requires, and $e^{2p't}=3^{p'}$:

$$\|Y\|_4=\|\Gamma(e^{-t})Z\|_4\le\|Z\|_2=\Big(\sum_r e^{2rt}\|Y_r\|_2^2\Big)^{1/2}\le3^{p'}\|Y\|_2.$$

(Test: $p'=1$, $Y$ centred Gaussian with variance $\sigma^2$: $\|Y\|_4=3^{1/4}\sigma\le3\sigma$.) Finally $\|Y^2\|_2=(\mathbb{E}Y^4)^{1/2}=\|Y\|_4^2$, Lemma N.20(iv) bounds $\|V(h')\|^2_{L^2(d\phi_0)}$ by $V_*$, and $J_0$ maps $V(h')^2$ to $Y^2$ isometrically. □

By Lemma N.22 below and Remark N.23(b), $Y$ is also the sharp-time observable built with the regularization of Appendix C, so the bound applies to $Q_V$ in the form in which Lemma D.2 uses it.

**Lemma N.22** (the two regularizations of a sharp-time Wick power agree)**.** *Let $t\in\mathbb{R}$, let $f$ be real, bounded and measurable with compact support in $\mathbb{R}$, and let $0\le n\le2p'$. Let $\delta_\kappa$ be the mollifier of Appendix C (notation paragraph: $\delta\in C_c^\infty(\mathbb{R}^2)$, $\delta\ge0$, $\delta(z)=\delta(-z)$, $\operatorname{supp}\delta\subseteq\{|z|\le1\}$, $\int\delta=1$, $\delta_\kappa(z)=\kappa^2\delta(\kappa z)$), let $\Phi_\kappa(\zeta)=\Phi(\delta_\kappa(\zeta-\cdot))$, with variance $c_\kappa$, and*

$$X_\kappa:=\int f(x)\,:\Phi_\kappa(t,x)^n:_{c_\kappa}dx\qquad(\kappa\ge1),$$

*the integral taken pointwise in $q\in\mathcal{S}'(\mathbb{R}^2)$. Let $W_n(f)$ be the limit of Lemma N.20(iii) with $f$ in place of $h'$ (its proof uses only that the kernel is bounded with compact support), and let $J_t$ be the sharp-time map at time $t$, $F(\varphi_0)\mapsto F(\Phi(t,\cdot))$; $J_0$ is the map of Lemma N.21. Then $X_\kappa\to J_tW_n(f)$ in $L^2(d\mu_0)$ as $\kappa\to\infty$.*

*Proof. The map $J_t$.* $J_t$ is defined on cylinder functions,

$$J_t:\ F(\varphi_0(g_1),\dots,\varphi_0(g_k))\longmapsto F(\Phi(t,g_1),\dots,\Phi(t,g_k)),$$

and preserves their laws: by Lemma N.13(i) with $a=(t,0)$, $\Phi(g)\circ T_a=\Phi(\tau_ag)$ and $(T_a)_*\mu_0=\mu_0$; the extension by continuity in the covariance norm, which $\tau_a$ preserves, gives $\Phi(0,g)\circ T_a=\Phi(t,g)$, so $(\Phi(t,g))_g$ has the law of $(\Phi(0,g))_g$, which is that of $(\varphi_0(g))_g$ (proof

of Lemma N.21(i)). Hence $J_t$ extends to an isometry of every $L^p(d\phi_0)$ into $L^p(d\mu_0)$, is multiplicative, and maps a Wick power at its own variance to the Wick power at its own variance of the image variable.

*Covariances.* By Lemma N.19(v) and the evenness of $\delta$, $\mathbb{E}[\Phi_\kappa(t,x)\Phi_\kappa(t,y)] = (\delta_\kappa * \delta_\kappa * G)((0,x-y))$. The variable $\Phi(t,g)$, $g := \rho_{\kappa'}(y-\cdot)$, is the $L^2(d\mu_0)$-limit of $\Phi(\eta_\varepsilon \otimes g)$ for $\eta_\varepsilon \in C_c^\infty(\mathbb{R})$, $\eta_\varepsilon \geq 0$, $\int \eta_\varepsilon = 1$, $\operatorname{supp}\eta_\varepsilon \subseteq [t-\varepsilon, t+\varepsilon]$ (the reading of Lemma N.21(i)); since $\delta_\kappa * G$ is continuous, $\mathbb{E}[\Phi_\kappa(t,x)\Phi(\eta_\varepsilon \otimes g)] = \int (\delta_\kappa * G)((t,x)-z')\,\eta_\varepsilon(z'^0)\,g(z'^1)\,dz' \to \int (\delta_\kappa * G)((0,x-y'))\,g(y')\,dy'$, so

$$\mathbb{E}\big[\Phi_\kappa(t,x)\,\Phi(t,\rho_{\kappa'}(y-\cdot))\big] = \int (\delta_\kappa * G)((0,x-y'))\,\rho_{\kappa'}(y-y')\,dy' =: \Gamma_{\kappa,\kappa'}(x,y).$$

*Inner products.* Put $Y_{\kappa'} := J_t W_{n,\kappa'}(f) = \int f(y)\,{:}\Phi(t,\rho_{\kappa'}(y-\cdot))^n{:}\,dy$ and $Y := J_t W_n(f)$, so that $Y_{\kappa'} \to Y$ in $L^2(d\mu_0)$ (Lemma N.20(iii) and the isometry). The generating-function identity in the proof of Lemma N.20(i) holds for any two jointly centred Gaussian variables, and Fubini applies as there: the moments are bounded uniformly in $x, y$, and $|f| \leq \|f\|_\infty \mathbf{1}_K$ with $K$ bounded. Hence

$$\mathbb{E}[X_\kappa^2] = n! \iint f(x)f(y)\,(\delta_\kappa * \delta_\kappa * G)((0,x-y))^n\,dx\,dy,$$

$$\mathbb{E}[X_\kappa Y_{\kappa'}] = n! \iint f(x)f(y)\,\Gamma_{\kappa,\kappa'}(x,y)^n\,dx\,dy,$$

and $\mathbb{E}[Y^2] = I := n! \iint f(x)f(y)\,C(x-y)^n\,dx\,dy$ by Lemma N.20(iii).

*A majorant.* Let $\kappa \geq 6$, and let $\nu$ be a probability density on $\mathbb{R}^2$ with $\operatorname{supp}\nu \subseteq \{|v| \leq \varepsilon\}$, $\varepsilon \leq 2/\kappa$, and $\|\nu\|_\infty \leq \kappa^2\|\delta\|_\infty$. Then, for $u \neq 0$,

$$(\nu * G)((0,u)) \leq 9\|\delta\|_\infty\big(b_0 + 2 + 2|\log|u||\big),$$

and $(\nu * G)((0,u)) \to G((0,u)) = C(u)$ as $\varepsilon \to 0$. Indeed, if $|u| \geq 2\varepsilon$, then $|(0,u)-v| \geq |u|/2$ on $\operatorname{supp}\nu$, and by Lemma N.19(v), (i) and (iii), $G((0,u)-v) \leq C(|u|/2) \leq (4\pi)^{-1}(b_0 + 2\log 2 + 2|\log|u||)$. If $0 < |u| < 2\varepsilon$, then $|(0,u)-v| < 3\varepsilon \leq 6/\kappa \leq 1$ on $\operatorname{supp}\nu$, so, in polar coordinates and by (iii),

$$\begin{aligned}(\nu * G)((0,u)) &\leq \kappa^2\|\delta\|_\infty\, 2\pi \int_0^{6/\kappa} rC(r)\,dr \leq \kappa^2\|\delta\|_\infty \frac{2\pi}{4\pi}\frac{(6/\kappa)^2}{2}\Big(b_0 + 1 + 2\log\frac{\kappa}{6}\Big)\\ &= 9\|\delta\|_\infty\Big(b_0 + 1 + 2\log\frac{\kappa}{6}\Big),\end{aligned}$$

using $\int_0^{r_0} r(b_0 + 2|\log r|)\,dr = \frac{r_0^2}{2}(b_0 + 1 + 2|\log r_0|)$, which increases in $r_0 \in (0,1]$ (derivative $r_0(b_0 - 2\log r_0) > 0$); and $\log(\kappa/6) < |\log|u||$ because $|u| < 4/\kappa$. Both cases are at most $9\|\delta\|_\infty(b_0 + 2 + 2|\log|u||)$, since $\|\delta\|_\infty \geq 1/\pi$ ($\delta$ is a probability density on the unit disc) and $2\log 2 < 2$. The convergence holds because $G$ is continuous at $(0,u) \neq 0$. The majorant applies to $\nu = \delta_\kappa$ ($\varepsilon = 1/\kappa$) and to $\nu = \delta_\kappa * \delta_\kappa$ ($\varepsilon = 2/\kappa$, $\|\delta_\kappa * \delta_\kappa\|_\infty \leq \|\delta_\kappa\|_\infty = \kappa^2\|\delta\|_\infty$).

*Limits.* (1) For fixed $\kappa$, $\delta_\kappa * G$ is continuous and bounded by $\|\delta_\kappa\|_\infty \int G = \kappa^2\|\delta\|_\infty m_0^{-2}$ (Lemma N.19(v)), so $\Gamma_{\kappa,\kappa'}(x,y) \to (\delta_\kappa * G)((0,x-y))$ as $\kappa' \to \infty$, boundedly; by dominated convergence and $Y_{\kappa'} \to Y$,

$$\mathbb{E}[X_\kappa Y] = n! \iint f(x)f(y)\,(\delta_\kappa * G)((0,x-y))^n\,dx\,dy.$$

(2) As $\kappa \to \infty$, the integrands of $\mathbb{E}[X_\kappa^2]$ and of $\mathbb{E}[X_\kappa Y]$ converge to $f(x)f(y)C(x-y)^n$ off the diagonal and are dominated by $\|f\|_\infty^2 \mathbf{1}_K(x)\mathbf{1}_K(y)(9\|\delta\|_\infty)^n(b_0 + 2 + 2|\log|x-y||)^n$, which is integrable (as in Lemma N.19(iv)). So $\mathbb{E}[X_\kappa^2] \to I$ and $\mathbb{E}[X_\kappa Y] \to I$, and

$$\|X_\kappa - Y\|_{L^2(d\mu_0)}^2 = \mathbb{E}[X_\kappa^2] - 2\,\mathbb{E}[X_\kappa Y] + \mathbb{E}[Y^2] \to I - 2I + I = 0. \qquad \square$$

**Remark N.23** (what Lemma N.22 identifies)**.** (a) The localized monomial of type (1.1.5) with one factor of degree $n \le 2p'$ and a bounded kernel $f$ at time $t$, which Definition N.26(b) of §D.b constructs with the mollifier of Appendix C, is the limit of $X_\kappa$ (Lemma N.28(a)), hence equals $J_t W_n(f)$. For a product kernel $f_1 \otimes \cdots \otimes f_r$ at one time $t$, with every $f_i$ bounded and vanishing outside one column, Lemma N.28(c) makes the monomial the product of its one-factor monomials, hence equal to $\prod_i J_t W_{n_i}(f_i) = J_t(\prod_i W_{n_i}(f_i))$ almost everywhere. (b) In particular the pieces $Q_{n,n',k,k'}$ of $Q_V$ (§D.e), to which Theorem D.4 applies Lemma D.2, are $\lambda^2 a_n a_{n'} J_0 W_n(h'_k) J_0 W_{n'}(h'_{k'})$, and their sum is $Q_V = (J_0 V(h'))^2 = Y^2$ with $Y$ of Lemma N.21, because $W_n(h') = \sum_k W_n(h'_k)$ (the regularized integrals, hence their limits, are linear in the kernel). So the base term of Theorem D.4, bounded through Lemma N.21, and its comparison term, bounded through Lemma D.2, concern one and the same element of $L^2(d\mu_0)$. (c) At degree 1 and $f \in C_c^\infty(\mathbb{R})$, $X_\kappa = \Phi(\delta_\kappa * (\delta_t \otimes f)) \to \Phi(t, f) = J_t W_1(f)$; this is also immediate from $\delta_\kappa * (\delta_t \otimes f) \to \delta_t \otimes f$ in the covariance norm. It covers the unsplit factors $\Phi(s, f_i)$, $f_i \in C_c^\infty(\mathbb{R})$, of the cylinder monomials of Theorem D.3, but not the pieces to which Lemma D.2 is applied there (Lemma N.36(vi)): their kernels $f_i \mathbf{1}_{[k-\frac{1}{2}, k+\frac{1}{2})}$ are bounded, measurable and of compact support, but need not be smooth. Those pieces are covered by Lemma N.22 with $n = 1$, through (a), which identifies a one-factor piece of degree 1 with kernel $g$ at time $t$ with $J_t W_1(g)$, and by Lemma N.28(f)(1), which identifies it with $\Phi(t, g)$; Lemma N.36(vi) uses the latter, with Lemma N.28(c) for the products. (d) Lemma N.22 uses of Appendix C only its regularization, none of Lemmas C.2–C.4 and Theorem C.5.

**Lemma N.23′** (identification of the constructed Wick powers; added 7 November 2026)**.** *Let $(\mathcal{H}, \Phi, S, d)$ be either $(\mathcal{H}_C, \varphi_0, C, 1)$, the time-zero field of Assumption A1, or $(\mathcal{H}_G, \Phi, G, 2)$, the Euclidean field of Assumption A3, so that $(\Phi(f))_f$ is the Gaussian process indexed by $\mathcal{H}$ with $\mathbb{E}[\Phi(f)\Phi(g)] = \iint f(x) S(x-y) g(y)\, dx\, dy$ (Lemma N.19(ii),(v)); $S > 0$ is locally integrable, with $\int_{\mathbb{R}^d} S = (2m_0)^{-1}$ for $d = 1$ and $m_0^{-2}$ for $d = 2$. Let $\nu \in C_c^\infty(\mathbb{R}^d)$ be a probability density, even, with $\operatorname{supp} \nu \subseteq \{|x| \le 1\}$; put $\nu_\kappa(x) := \kappa^d \nu(\kappa x)$ and $\Phi_\kappa(x) := \Phi(\nu_\kappa(x - \cdot))$ for $\kappa \ge 1$, with variance $c_\kappa = \mathbb{E}[\Phi_\kappa(x)^2] = \iint \nu_\kappa(u) S(u - u') \nu_\kappa(u')\, du\, du'$, independent of $x$ (substitute $y \mapsto x - y$). Let $w$ be a real bounded measurable function on $\mathbb{R}^d$ with compact support, let $n \ge 0$, and let*

$$X_\kappa := \int w(x) :\Phi_\kappa(x)^n:_{c_\kappa} dx,$$

*the integral taken pointwise in $q$.*

(i) *$X_\kappa \in \Gamma_n(\mathcal{H})$, the closed span of the Wick monomials of degree $n$ (Assumption A1(c)); $X_\kappa$ equals almost everywhere the $L^2$-valued (Bochner) integral of the same integrand, so that $\langle Y, X_\kappa \rangle = \int w(x) \langle Y, :\Phi_\kappa(x)^n:_{c_\kappa} \rangle\, dx$ for every $Y \in L^2$.*

(ii) *If $X_\kappa \to X$ in $L^2$ as $\kappa \to \infty$ (which holds for $0 \le n \le 2p'$: for $d = 1$ and $\nu = \rho$ by Lemma N.20(iii), after splitting $w$ into pieces supported in intervals of length 2 and scaling them to $\|\cdot\|_\infty \le 1$, by linearity in $w$; for $d = 2$ and $\nu = \delta$ by Lemma N.28(a) with $r = 1$, after splitting $w$ over the GJS squares), then $X = :\Phi^n(w):$, the Wick power of Assumption A1(d) ($d = 1$, $:\varphi_0^n(w):_C$), resp. of Assumption A3 ($d = 2$): the unique element of $\Gamma_n(\mathcal{H})$ with the pairings (W.1), resp. (W.2).*

(iii) *Consequently $W_n(h') = :\varphi_0^n(h'):_C$ for every $h'$ as in Lemma N.20, so that $V(h') = \lambda \sum_n a_n W_n(h')$ is the $V(h')$ of §2.1; and for $h \in \mathcal{C}_{\mathrm{E}}$ the degree-$n$ part $X_{n,\kappa}$ of the mollified interaction $\mathcal{U}_{h,\kappa}$ of Appendix C converges in $L^2(d\mu_0)$ to $:\Phi^n(h):$, so that $\mathcal{U}_{h,\kappa} \to \mathcal{U}_h$ of Assumption A3, as Lemma C.4 quantifies.*

*Proof.* (i) For fixed $\kappa$ and $x$, $\Phi_\kappa(x) = \Phi(g_x)$ with $g_x := \nu_\kappa(x - \cdot) \in C_c^\infty(\mathbb{R}^d)$, so $:\Phi_\kappa(x)^n:_{c_\kappa}$ is the Wick monomial $:\Phi(g_x)\cdots\Phi(g_x):$ (Assumption A1(a) with $k = 1$: the recursion gives the one-variable Wick power (I.18b), which is $:X^n:_c$ at $c = \mathbb{E}[\Phi(g_x)^2] = c_\kappa$) and lies in $\Gamma_n(\mathcal{H})$. The map $x \mapsto \Phi_\kappa(x)$ is continuous into $L^2$: $\mathbb{E}[(\Phi_\kappa(x) - \Phi_\kappa(x'))^2] = 2c_\kappa - 2(\nu_\kappa * \nu_\kappa * S)(x - x') \to 0$ as $x' \to x$, since $\nu_\kappa * \nu_\kappa * S$ is continuous (a convolution of the locally integrable $S$ with a $C_c^\infty$ function); Gaussian variables converging in $L^2$ converge in every $L^p$, and $:X^n:_{c_\kappa}$ is a polynomial in $X$ with fixed coefficients, so $F : x \mapsto :\Phi_\kappa(x)^n:_{c_\kappa}$ is continuous into $L^2$ and bounded on compact sets. For a step function $v = \sum_i c_i \mathbf{1}_{I_i}$ ($I_i$ bounded cubes), the Bochner integral $\int vF\,dx = \sum_i c_i \int_{I_i} F\,dx$ is the $L^2$-limit of Riemann sums of $F$ over the cubes ($F$ is uniformly continuous on each $\overline{I_i}$), each in $\Gamma_n(\mathcal{H})$, hence it lies in the closed $\Gamma_n(\mathcal{H})$; for each $q$ the same Riemann sums converge to the pointwise integral $\int v(x)F(x)(q)\,dx$ ($x \mapsto F(x)(q)$ is continuous), and a subsequence converges almost everywhere to the Bochner integral, so the two integrals agree almost everywhere. Choose step functions $v_j \to w$ in $L^1$ with $|v_j| \le \|w\|_\infty \mathbf{1}_K$, $K$ a cube containing $\operatorname{supp} w$: the Bochner integrals $\int v_j F$ converge in $L^2$ to $\int wF$ (dominated by $\|w\|_\infty \mathbf{1}_K \sup_K \|F\|_2$), which therefore lies in $\Gamma_n(\mathcal{H})$, and the pointwise integrals converge to $X_\kappa(q)$ for every $q$ (dominated convergence, $F(\cdot)(q)$ being bounded on $K$); so $X_\kappa$ equals the Bochner integral almost everywhere. A bounded linear functional commutes with a Bochner integral, which gives the formula for $\langle Y, X_\kappa\rangle$.

(ii) *The pairing of a Wick monomial with a Wick power.* Let $A_1, \dots, A_n, X$ be jointly centred Gaussian variables. For a centred Gaussian $A$ put $:e^A: := e^{A - \frac{1}{2}\mathbb{E}[A^2]}$ (this is (I.18a)); for $A = \sum_j s_j A_j$ it is the generating function of the Wick products in the sense of Assumption A1(a): its coefficient of $\prod_j s_j^{m_j}/m_j!$ has expectation 0 unless all $m_j = 0$ (since $\mathbb{E}[:e^A:] = 1$), equals 1 if all $m_j = 0$, and, because $:e^{\sum s_j A_j}: = e^{\sum_j s_j A_j} \cdot e^{-\frac{1}{2}\sum_{j,k} s_j s_k \mathbb{E}[A_j A_k]}$ as a function of the variables $A_j$ satisfies $\frac{\partial}{\partial A_i} :e^{\sum s_j A_j}: = s_i :e^{\sum s_j A_j}:$, its coefficients obey the recursion of Assumption A1(a); that recursion determines the Wick products uniquely by induction on the total degree, so $:A_1 \cdots A_n:$ is the coefficient of $s_1 \cdots s_n$ in $:e^{\sum_j s_j A_j}:$, and $:X^n:$ that of $t^n/n!$ in $:e^{tX}:$. Now $\mathbb{E}[:e^A: :e^B:] = \mathbb{E}[e^{A+B}] e^{-\frac{1}{2}\mathbb{E}A^2 - \frac{1}{2}\mathbb{E}B^2} = e^{\frac{1}{2}\mathbb{E}(A+B)^2 - \frac{1}{2}\mathbb{E}A^2 - \frac{1}{2}\mathbb{E}B^2} = e^{\mathbb{E}[AB]}$ for jointly centred Gaussian $A, B$, so with $A = \sum_j s_j A_j$ and $B = tX$,

$$\mathbb{E}\big[:e^{\sum_j s_j A_j}: :e^{tX}:\big] = \exp\Big(t \sum_j s_j\, \mathbb{E}[A_j X]\Big),$$

both sides entire in $(s, t)$ (the left may be expanded under $\mathbb{E}$: for $|s_j|, |t| \le 1$ the terms are dominated by polynomials in $|A_j|, |X|$ times $e^{\sum |A_j| + |X|}$, which are integrable). Comparing the coefficients of $s_1 \cdots s_n t^n$: on the left it is $\mathbb{E}[:A_1 \cdots A_n: :X^n:]/n!$, on the right $\prod_j \mathbb{E}[A_j X]$ (the term of degree $n$ in $t$ is $t^n(\sum_j s_j c_j)^n/n!$, whose $s_1 \cdots s_n$ coefficient is $t^n \prod_j c_j$). Hence

$$\mathbb{E}\big[:A_1 \cdots A_n: :X^n:\big] = n! \prod_{j=1}^n \mathbb{E}[A_j X]. \tag{W.4}$$

(Test, $n = 1$: $\mathbb{E}[A_1 X]$ on both sides.) *The pairings of $X_\kappa$.* Take $A_j = \Phi(f_j)$, $f_j \in \mathcal{S}(\mathbb{R}^d)$ real, and $X = \Phi_\kappa(x)$. Then $\mathbb{E}[\Phi(f_j)\Phi_\kappa(x)] = \iint f_j(y) S(y - y') \nu_\kappa(x - y')\,dy\,dy' = (\nu_\kappa * Sf_j)(x)$, where $(Sf)(y') := \int f(y) S(y - y')\,dy = \int f(y' + u) S(u)\,du$ is continuous and bounded by $\|f\|_\infty \|S\|_{L^1}$ ($S \in L^1(\mathbb{R}^d)$: for $d = 2$, $\int G = m_0^{-2}$, Lemma N.19(v); for $d = 1$, by Tonelli and $\int_{\mathbb{R}} (4\pi t)^{-1} e^{-u^2/(4t)} du = (4\pi)^{-1/2}$, $\int_{\mathbb{R}} C = \int_0^\infty (4\pi t)^{-1/2} e^{-t m_0^2} dt = (2m_0)^{-1}$), and $\nu_\kappa$ is even. By (i) and (W.4),

$$\langle :\Phi(f_1)\cdots\Phi(f_n):, X_\kappa\rangle = n! \int w(x) \prod_{j=1}^n (\nu_\kappa * Sf_j)(x)\,dx \xrightarrow[\kappa\to\infty]{} n! \int w(x) \prod_{j=1}^n (Sf_j)(x)\,dx,$$

by dominated convergence: $\nu_\kappa * Sf_j \to Sf_j$ pointwise ($Sf_j$ is continuous and $\nu_\kappa$ is a probability density supported in $\{|u| \le 1/\kappa\}$), $|\nu_\kappa * Sf_j| \le \|f_j\|_\infty \|S\|_1$, and $w \in L^1$. Since $X_\kappa \to X$ in $L^2$, the left side tends to $\langle :\Phi(f_1)\cdots\Phi(f_n):, X\rangle$; and $X \in \Gamma_n(\mathcal{H})$ by (i), $\Gamma_n(\mathcal{H})$ being closed. The limit is the right side of (W.1), resp. (W.2), with $g = w$ (there $\int f_j(x_j)S(x_j - y)\,dx_j = (Sf_j)(y)$), and $w \in L^1 \cap L^{1+\varepsilon}$ since it is bounded with compact support, so $:\Phi^n(w):$ exists (for $n = 0$ it is the constant $\int w$ of the convention in Assumption A1(d), resp. Assumption A3); an element of $\Gamma_n(\mathcal{H})$ is determined by its pairings with the Wick monomials with $f_j \in \mathcal{S}(\mathbb{R}^d)$, as the source's "unique" in (V.3) asserts (Assumption A1(d)). Hence $X = :\Phi^n(w):$.

(iii) $W_n(h')$ is the limit of $X_\kappa$ with $d = 1$, $\nu = \rho$ and $w = h'$ (Lemma N.20(iii)), so (ii) gives $W_n(h') = :\varphi_0^n(h'):_C$, and $\lambda \sum_n a_n W_n(h') = \lambda \sum_n a_n :\varphi_0^n(h'):_C = V(h')$ by §2.1 ($n = 0$: $W_0 = \int h' = :\varphi_0^0(h'):$). For $h \in \mathcal{C}_{\mathrm{E}}$ the degree-$n$ part of $\mathcal{U}_{h,\kappa}$ is $X_\kappa$ with $d = 2$, $\nu = \delta$ and $w = h$ (Appendix C, notation paragraph), which converges by Lemma N.28(a) (applied square by square, as in Step 1 of the proof of Lemma D.1); by (ii) its limit is $:\Phi^n(h):$, and summing over $n$ with the coefficients $\lambda a_n$ gives $\mathcal{U}_{h,\kappa} \to \mathcal{U}_h$. □

### D.a.8 The printed constants

The constants $C_7$, $m_1$, $K_1$, $K_2$ and $\bar{n}$ are fixed once, with the working constants of §2.6: "$m_1 > 0$ and $C_7, K_1, K_2$ denote the printed constants". The source prints $C_7$ and $m_1$ as "independent of $h$, $Q_1$ and $Q_2$"; how they depend on $(m_0, \lambda, \mathscr{P}, K_1, K_2)$ is not printed, and this appendix uses no such dependence. The constant $C_8$ of the retired items Assumption A8(v)–(vi) is not used in this appendix, nor anywhere else since 7 November 2026.

| constant | printed in [7] (read on the page images) | in this manuscript |
|---|---|---|
| $\bar{n}$ | p. 593: "We assume a bound on the $n_i$ in (1.1.5), $n_i < \bar{n}$. Any choice $\bar{n} \geqq \deg P$ is sufficient." | A8(ii), p. 10: the fixed choice $\bar{n} = 2p' + 1$; every Wick factor used here has degree $\le 2p' < \bar{n}$ |
| $K_1$, $K_2$ | (1.1.7)–(1.1.8), p. 594; Theorem 1.1.7, p. 594: "Let $m_0$, $K_2$ and $K_1$ be sufficiently large and let $\lambda$ be sufficiently small." | A8(iii), (1), p. 10; "we fix $K_2 \geq 1$" |
| $C_7$, $m_1$ | Theorem 1.1.7, p. 594: "There are constants $0 < m_1$ and $O(1)$ independent of $h$, $Q_1$ and $Q_2$ such that" (1.1.10); $C_7$ is this manuscript's name for this $O(1)$ | Source Fact 2.2 (A8(iv)), p. 10: "with $m_1 > 0$ and $C_7$ independent of $h$, $Q_1$, $Q_2$ (so stated in the source)" |
| $m_0$, $\lambda$ | hypotheses of Theorem 1.1.7, p. 594 (quoted above) | §2.6, p. 14: $m_0 = M_*$, $0 \le \lambda < \lambda_*$ in working coordinates |
| $\mathscr{P}$, $p'$, $a_n$ | (1.8), p. 589: "$V(h) = \lambda \int_{R^2} : \mathscr{P}(\Phi(x)) : h(x)dx$" | §2.1, p. 6: $\mathscr{P}(\xi) = \sum_{n=0}^{2p'} a_n \xi^n$, $a_{2p'} > 0$, $p' \geq 1$ |

The derived constants of this section are $c_U$ (Lemma N.18), $S_2$ (Definition N.14; finite and unevaluated), $b_0$ and $I_n$ (Lemma N.19), $\overline{V}_* \geq V_*$ (Lemma N.20), and the ring constant $\overline{\Sigma}_{3\times 5} \geq \Sigma_{3\times 5}$ (Lemma N.11(vi)); §D.g tabulates them with the constants of Theorems D.3–D.4 and of §§4–7.

## D.b The affine-path identity

This section proves Lemma D.1. It is the identity that GJS print as the first two equalities of (1.1.12) [7, p. 595],

$$\text{“}\langle Q\rangle_{h_2} - \langle Q\rangle_{h_1} = \int_0^1 d\alpha\, \frac{d}{d\alpha}\langle Q\rangle_{h_1+\alpha g} = -\int_0^1 d\alpha \big[\langle QV(g)\rangle_{h_1+\alpha g} - \langle Q\rangle_{h_1+\alpha g}\langle V(g)\rangle_{h_1+\alpha g}\big]\text{”},$$

where "To prove (1.1.11) let $g = h_2 - h_1$" (p. 594) and $\langle F\rangle_h = \int F\, dq_h$ (ibid.); the pages give no integrability argument, and this section supplies one. Its inputs are: Assumption A3, as stated there, including the linearity of $h \mapsto \mathcal{U}_h$ of its construction (Step 1 of the proof below also obtains the linearity from the identification of $\mathcal{U}_h$ with the limit of its regularizations, Lemma N.23′(iii), quantified by Lemma C.4); the Gaussian moments of the observables, computed explicitly in Lemmas N.24, N.25 and N.28 from Assumption A2 alone (neither the decomposition of $L^2(d\mu_0)$ into Wick chaoses nor hypercontractivity is used); Hölder's inequality, with the exponents written at each use; dominated convergence; Fatou's lemma; and the fundamental theorem of calculus. No smallness hypothesis is used: everything in this section holds for every $\lambda \geq 0$ and every $m_0 > 0$.

**The observables and their Gaussian moments**

*Gaussian variables.* As in §D.a, $d\mu_0$ is the centred Gaussian measure on $\mathcal{S}'(\mathbb{R}^2)$ with covariance $(-\Delta_{\mathbb{R}^2} + m_0^2)^{-1}$ (Assumption A2), and $\Phi(f)(q) = \langle q, f\rangle$. Let $\mathscr{H}_1 \subset L^2(d\mu_0)$ be the closed linear span of the $\Phi(f)$, $f \in C_c^\infty(\mathbb{R}^2)$ real. Its elements are jointly Gaussian and centred, and $\mathbb{E}[\Phi(f)\Phi(g)] = \langle f, (-\Delta_{\mathbb{R}^2} + m_0^2)^{-1} g\rangle_{L^2(\mathbb{R}^2)}$.

*Regularization and Wick powers.* Appendix C fixes $\delta \in C_c^\infty(\mathbb{R}^2)$ with $\delta \geq 0$, $\delta(z) = \delta(-z)$, $\operatorname{supp}\delta \subseteq \{z : |z| \leq 1\}$ and $\int \delta = 1$, and puts $\delta_\kappa(z) := \kappa^2\delta(\kappa z)$. For $\kappa \geq 2$ and $\zeta \in \mathbb{R}^2$ let $\Phi_\kappa(\zeta) := \Phi(\delta_\kappa(\zeta - \cdot)) \in \mathscr{H}_1$, the $\delta_\kappa * \Phi$ of Appendix C at $\zeta$; for every $q \in \mathcal{S}'(\mathbb{R}^2)$ the function $\zeta \mapsto \Phi_\kappa(\zeta)(q) = \langle q, \delta_\kappa(\zeta - \cdot)\rangle$ is smooth, being the convolution of a tempered distribution with a test function. Let $c_\kappa := \mathbb{E}[\Phi_\kappa(\zeta)^2]$, which does not depend on $\zeta$ (Lemma N.25(iii)). For a centred Gaussian variable $X$ and $c \geq 0$ the Wick powers $:X^n:_c$ are the coefficients in $\sum_{n\geq 0} \frac{s^n}{n!}:X^n:_c = \exp(sX - \frac{1}{2}s^2 c)$, the convention (proof of Proposition 3.2); expanding the two exponentials,

$$:X^n:_c = \sum_{0\leq j\leq n/2} \frac{n!}{j!\,(n-2j)!}\Big(-\frac{c}{2}\Big)^j X^{n-2j}. \tag{35}$$

**Lemma N.24** (moments of products of Wick powers)**.** *Let $X_1, \dots, X_\ell \in \mathscr{H}_1$, $c_a := \mathbb{E}[X_a^2]$, $\gamma_{ab} := \mathbb{E}[X_a X_b]$, and let $m_1, \dots, m_\ell \geq 0$ be integers. Then*

$$\mathbb{E}\Big[\prod_{a=1}^{\ell} :X_a^{m_a}:_{c_a}\Big] = \mathcal{P}_m(\gamma) := \sum_{(k_{ab})} \prod_{a=1}^{\ell} m_a! \prod_{1\leq a<b\leq \ell} \frac{\gamma_{ab}^{k_{ab}}}{k_{ab}!}, \tag{36}$$

*the sum running over all families $(k_{ab})_{1\leq a<b\leq \ell}$ of nonnegative integers with $\sum_{b\neq a} k_{ab} = m_a$ for every $a$, where $k_{ba} := k_{ab}$ (an empty sum is 0). Thus $\mathcal{P}_m$ is a polynomial with nonnegative coefficients in the off-diagonal covariances $\gamma_{ab}$, $a < b$, alone, and each of its monomials has degree $\frac{1}{2}\sum_a m_a$. For $\ell = 2$: $\mathbb{E}[:X_1^n:_{c_1}:X_2^{n'}:_{c_2}] = \delta_{nn'}\, n!\, \gamma_{12}^n$ (Wick orthogonality and the Wick isometry).*

*Proof.* For real $s_1, \dots, s_\ell$ the variable $\sum_a s_a X_a$ is centred Gaussian with variance $\sum_a s_a^2 c_a + 2\sum_{a<b} s_a s_b \gamma_{ab}$, so

$$\mathbb{E}\Big[\prod_a \exp\big(s_a X_a - \tfrac{1}{2}s_a^2 c_a\big)\Big] = \exp\Big(\sum_{a<b} s_a s_b \gamma_{ab}\Big) = \prod_{a<b}\sum_{k\geq 0} \frac{(s_a s_b \gamma_{ab})^k}{k!}.$$

On the left expand each factor by the generating function. The expansion may be integrated term by term: by (35), $\sum_n \frac{|s|^n}{n!}|:X^n:_c| \leq \sum_n \sum_{j\leq n/2} \frac{|s|^n (c/2)^j |X|^{n-2j}}{j!\,(n-2j)!} = e^{|s||X| + s^2 c/2}$, and $\mathbb{E}\prod_a e^{|s_a||X_a|} \leq \prod_a (\mathbb{E}e^{\ell|s_a||X_a|})^{1/\ell} < \infty$ by Hölder's inequality with $\ell$ factors, each with exponent $\ell$, and the Gaussian exponential moments. Fubini's theorem then turns the left

side into $\sum_m \prod_a \frac{s_a^{m_a}}{m_a!}\,\mathbb{E}[\prod_a :X_a^{m_a}:_{c_a}]$, a power series converging for all $s$, and comparing the coefficients of $\prod_a s_a^{m_a}$ gives (36). No $\gamma_{aa}$ occurs, and every term has degree $\sum_{a<b} k_{ab} = \frac{1}{2}\sum_a m_a$. For $\ell = 2$ the only family is $k_{12} = n = n'$. □

**Lemma N.25** (the free covariance)**.** *Let $G$ be the kernel of $(-\Delta_{\mathbb{R}^2} + m_0^2)^{-1}$. Put $g_0 := (4 + m_0^{-2})/(4\pi)$, $\log_+ t := \max\{\log t, 0\}$ and $M_0(\varrho) := g_0 + \frac{1}{2\pi}\log_+ \frac{1}{\varrho}$ for $\varrho > 0$.*

*(i) For $\zeta \neq 0$, $G(\zeta) = \int_0^\infty (4\pi s)^{-1} \exp\bigl(-\frac{|\zeta|^2}{4s} - m_0^2 s\bigr)\, ds$, and $\mathbb{E}[\Phi(f)\Phi(g)] = \iint f(\zeta)\, G(\zeta - \zeta')\, g(\zeta')\, d\zeta\, d\zeta'$ for real $f, g \in C_c^\infty(\mathbb{R}^2)$. Likewise the time-zero covariance $C = (2\mu)^{-1}$ has the kernel $C(x) = G(0, x)$, $x \in \mathbb{R}\backslash\{0\}$: $\langle f, Cg\rangle_{L^2(\mathbb{R})} = \iint f(x)\, G(0, x-y)\, g(y)\, dx\, dy$ for real $f, g \in C_c^\infty(\mathbb{R})$.*

*(ii) $G$ is positive, continuous on $\mathbb{R}^2 \setminus \{0\}$, a nonincreasing function of $|\zeta|$, and $G(\zeta) \le M_0(|\zeta|)$. In particular the kernel $C(x) = G(0, x)$ of $(2\mu)^{-1}$ is positive and $C(x) \le M_0(|x|)$; this is the kernel form's conventions lock, Remark 2.5.*

*(iii) For $\kappa, \kappa' \ge 2$ and $\zeta, \zeta' \in \mathbb{R}^2$: $\mathbb{E}[\Phi_\kappa(\zeta)\Phi_{\kappa'}(\zeta')] = G_{\kappa\kappa'}(\zeta - \zeta')$ with $G_{\kappa\kappa'} := \delta_\kappa * \delta_{\kappa'} * G$, a continuous function on $\mathbb{R}^2$; $0 < G_{\kappa\kappa'}(\zeta) \le M_1(|\zeta|) := g_1 + b_1 \log_+ \frac{1}{|\zeta|}$ for $\zeta \neq 0$, where $g_1 := \max\bigl\{g_0 + \frac{\log 2}{2\pi},\ \|\delta\|_\infty(4\pi g_0 + 1 + 2\log 4)\bigr\}$ and $b_1 := \max\{\frac{1}{2\pi}, 2\|\delta\|_\infty\}$; and $G_{\kappa\kappa'}(\zeta) \to G(\zeta)$ as $\min\{\kappa, \kappa'\} \to \infty$, for every $\zeta \neq 0$. In particular $c_\kappa = G_{\kappa\kappa}(0)$.*

*(iv) For $R > 0$, $q \in [1, \infty)$ and $d \in \{1, 2\}$: $\int_{\{t \in \mathbb{R}^d :\, |t| \le R\}} M_1(|t|)^q\, dt < \infty$.*

*Proof.* (ii) The integrand of the formula in (i), taken as the definition of $G$ on $\mathbb{R}^2 \setminus \{0\}$, is positive and nonincreasing in $|\zeta|$; on $\{|\zeta| \ge \varepsilon\}$ it is bounded by $(4\pi s)^{-1} e^{-\varepsilon^2/(4s) - m_0^2 s}$, which is integrable over $(0, \infty)$, so $G$ is continuous there by dominated convergence. Let $0 < |\zeta| \le 1$ and split the $s$-integral at $s = |\zeta|^2$ and $s = 1$. With $s = |\zeta|^2 v$ and $e^{-y} \le y^{-1}$ $(y > 0)$, $\int_0^{|\zeta|^2} \frac{e^{-|\zeta|^2/(4s)}}{4\pi s} ds = \frac{1}{4\pi}\int_0^1 \frac{e^{-1/(4v)}}{v} dv \le \frac{1}{4\pi}\int_0^1 4\, dv = \frac{1}{\pi}$; next $\int_{|\zeta|^2}^1 \frac{ds}{4\pi s} = \frac{1}{2\pi}\log\frac{1}{|\zeta|}$; and $\int_1^\infty \frac{e^{-m_0^2 s}}{4\pi s} ds \le \frac{1}{4\pi m_0^2}$. Dropping the factors $e^{-m_0^2 s} \le 1$ and $e^{-|\zeta|^2/(4s)} \le 1$ where they were not used, $G(\zeta) \le \frac{1}{\pi} + \frac{1}{4\pi m_0^2} + \frac{1}{2\pi}\log\frac{1}{|\zeta|} = M_0(|\zeta|)$. For $|\zeta| > 1$, monotonicity gives $G(\zeta) \le \sup_{|\eta|=1} G(\eta) \le g_0 = M_0(|\zeta|)$. The statement about $C$ follows, since $C(x) = G(0, x)$ by (i).

(iv) Since $\log_+(1/|t|) = 0$ for $|t| \ge 1$ and $(a+b)^q \le 2^{q-1}(a^q + b^q)$ for $a, b \ge 0$, it suffices that $\int_0^1 (\log\frac{1}{t})^q dt = \Gamma(q+1)$ and $\int_0^1 t(\log\frac{1}{t})^q dt = 2^{-q-1}\Gamma(q+1)$ are finite (substitute $t = e^{-u}$).

(i) By Assumption A2 and Parseval's identity, with $\hat f(p) := \int e^{-ip\cdot\zeta} f(\zeta)\, d\zeta$, $\mathbb{E}[\Phi(f)\Phi(g)] = (2\pi)^{-2}\int \overline{\hat f(p)}\, \hat g(p)\, (|p|^2 + m_0^2)^{-1} dp$. Insert $(|p|^2 + m_0^2)^{-1} = \int_0^\infty e^{-s(|p|^2 + m_0^2)} ds$; Fubini's theorem applies, because the double integral of $|\hat f \hat g| e^{-s(|p|^2+m_0^2)}$ is $\int |\hat f \hat g| (|p|^2 + m_0^2)^{-1} dp < \infty$. For fixed $s > 0$, expanding $\hat f, \hat g$ (all integrals absolutely convergent) and using $(2\pi)^{-2}\int e^{ip\cdot\zeta - s|p|^2} dp = (4\pi s)^{-1} e^{-|\zeta|^2/(4s)}$ (two one-dimensional Gaussian integrals) gives $(2\pi)^{-2}\int \overline{\hat f}\hat g\, e^{-s|p|^2} dp = \iint f(\zeta)(4\pi s)^{-1} e^{-|\zeta-\zeta'|^2/(4s)} g(\zeta')\, d\zeta\, d\zeta'$. A second use of Fubini's theorem, legitimate since by (ii) and (iv) the absolute integrand integrates to $\iint |f(\zeta)|\, G(\zeta - \zeta')\, |g(\zeta')| < \infty$ ($f, g$ bounded with bounded support), gives the kernel formula with $G$ as displayed in (i). The time-zero statement is the same computation in one dimension: $(2\mu(k))^{-1} = \frac{1}{2}\pi^{-1/2}\int_0^\infty s^{-1/2} e^{-s(k^2+m_0^2)} ds$ (since $\int_0^\infty s^{-1/2} e^{-sa} ds = \sqrt{\pi}\, a^{-1/2}$), $(2\pi)^{-1}\int e^{ikx - sk^2} dk = (4\pi s)^{-1/2} e^{-x^2/(4s)}$, and $\frac{1}{2}\pi^{-1/2} s^{-1/2} (4\pi s)^{-1/2} = (4\pi s)^{-1}$; the two uses of Fubini's theorem are justified as before, by $\int |\hat f \hat g| (2\mu)^{-1} dk < \infty$ and by (ii) and (iv) with $d = 1$.

(iii) By (i) with $f = \delta_\kappa(\zeta - \cdot)$, $g = \delta_{\kappa'}(\zeta' - \cdot)$, the substitutions $\eta = \zeta - a$, $\eta' = \zeta' + b$ and $\delta_{\kappa'}(-b) = \delta_{\kappa'}(b)$, $\mathbb{E}[\Phi_\kappa(\zeta)\Phi_{\kappa'}(\zeta')] = \iint \delta_\kappa(a)\delta_{\kappa'}(b)\, G(\zeta - \zeta' - a - b)\, da\, db = G_{\kappa\kappa'}(\zeta - \zeta')$; it is continuous as the convolution of the locally integrable $G$ (by (ii) and (iv)) with $\tilde\delta := \delta_\kappa * \delta_{\kappa'} \in C_c^\infty(\mathbb{R}^2)$. Now $\tilde\delta \ge 0$, $\int \tilde\delta = 1$, $\operatorname{supp}\tilde\delta \subseteq \{|c| \le \varrho\}$ with $\varrho := \kappa^{-1} + \kappa'^{-1} \le 1$, and $\|\tilde\delta\|_\infty \le \min\{\|\delta_\kappa\|_\infty\|\delta_{\kappa'}\|_1, \|\delta_{\kappa'}\|_\infty\|\delta_\kappa\|_1\} = \kappa_m^2\|\delta\|_\infty$ with $\kappa_m := \min\{\kappa, \kappa'\}$, and $\kappa_m^{-1} \le \varrho \le 2\kappa_m^{-1}$. Positivity follows from $G > 0$. If $|\zeta| \ge 2\varrho$, every $c$ with $|c| \le \varrho$ has $|\zeta - c| \ge |\zeta|/2$, so by (ii) $G_{\kappa\kappa'}(\zeta) \le M_0(|\zeta|/2) \le g_0 + \frac{\log 2}{2\pi} + \frac{1}{2\pi}\log_+\frac{1}{|\zeta|} \le M_1(|\zeta|)$. If $0 < |\zeta| < 2\varrho$, then $G_{\kappa\kappa'}(\zeta) \le \kappa_m^2\|\delta\|_\infty \int_{|\theta-\zeta|\le\varrho} M_0(|\theta|)\, d\theta \le \kappa_m^2\|\delta\|_\infty \int_{|\theta|\le\varrho} M_0(|\theta|)\, d\theta$. The second inequality holds for every nonincreasing $M_0$: the sets $A := \{|\theta - \zeta| \le \varrho < |\theta|\}$ and $A' := \{|\theta| \le \varrho < |\theta - \zeta|\}$ have equal measure, $M_0(|\theta|) \le M_0(\varrho)$ on $A$ and $M_0(|\theta|) \ge M_0(\varrho)$ on $A'$. Since $\varrho \le 1$, $\int_{|\theta|\le\varrho} M_0(|\theta|)\, d\theta = \pi\varrho^2 g_0 + \int_0^\varrho t\log\frac{1}{t}\, dt = \pi\varrho^2 g_0 + \frac{\varrho^2}{2}\log\frac{1}{\varrho} + \frac{\varrho^2}{4}$, and with $\kappa_m^2\varrho^2 \le 4$ and $\log\frac{1}{\varrho} \le \log\kappa_m$ this gives $G_{\kappa\kappa'}(\zeta) \le \|\delta\|_\infty(4\pi g_0 + 1 + 2\log\kappa_m)$. Finally $|\zeta| < 2\varrho \le 4/\kappa_m$ gives $\log\kappa_m < \log 4 + \log_+\frac{1}{|\zeta|}$, so $G_{\kappa\kappa'}(\zeta) \le M_1(|\zeta|)$ in this case too. For $\zeta \ne 0$ and $\varrho \le |\zeta|/2$, $|G_{\kappa\kappa'}(\zeta) - G(\zeta)| \le \sup_{|c|\le\varrho}|G(\zeta - c) - G(\zeta)|$, which tends to 0 as $\varrho \to 0$, i.e. as $\kappa_m \to \infty$, by the continuity of $G$ at $\zeta$. Taking $\zeta = \zeta'$ gives $c_\kappa = G_{\kappa\kappa}(0)$. □

**Definition N.26** (localized monomials, made precise)**.** Let $r \ge 0$ and let $n_1, \dots, n_r$ be integers with $0 \le n_i < \bar n$; $N := n_1 + \cdots + n_r$ is the total degree, and $\Delta_{j_1}, \dots, \Delta_{j_r}$ are GJS squares. For $\kappa \ge 2$:

(a) *type (1.1.6):* for real $w \in L^2(\mathbb{R}^{2r})$ vanishing outside $\Delta_{j_1} \times \cdots \times \Delta_{j_r}$, and $x = (x_1, \dots, x_r)$ with $x_i \in \mathbb{R}^2$,

$$Q^{(\kappa)} := \int_{\mathbb{R}^{2r}} w(x)\, A_\kappa(x)\, dx, \qquad A_\kappa(x) := \prod_{i=1}^r :\Phi_\kappa(x_i)^{n_i}:_{c_\kappa};$$

(b) *type (1.1.5):* for times $x_0 = (x_{1,0}, \dots, x_{r,0}) \in \mathbb{R}^r$, equal or not, and real $w(x_0, \cdot) \in L^2(\mathbb{R}^r)$ vanishing at $\vec x = (x_1, \dots, x_r)$ unless $(x_{i,0}, x_i) \in \Delta_{j_i}$ for every $i$,

$$Q^{(\kappa)}(x_0) := \int_{\mathbb{R}^r} w(x_0, \vec x)\, A_\kappa(\vec x)\, d\vec x, \qquad A_\kappa(\vec x) := \prod_{i=1}^r :\Phi_\kappa(x_{i,0}, x_i)^{n_i}:_{c_\kappa}.$$

The integrals are taken pointwise in $q \in \mathcal{S}'(\mathbb{R}^2)$: for each $q$ the integrand $A_\kappa(\cdot)(q)$ is continuous, by the smoothness of $\zeta \mapsto \Phi_\kappa(\zeta)(q)$ and (35), and $w$ is integrable, its support being bounded; the result is measurable in $q$ because $(q, x) \mapsto A_\kappa(x)(q)$ is measurable in $q$ and continuous in $x$. The localized monomial $Q$, resp. $Q(x_0)$, is the limit of $Q^{(\kappa)}$, resp. $Q^{(\kappa)}(x_0)$, in $L^2(d\mu_0)$ as $\kappa \to \infty$, which exists by Lemma N.28(a). The class $\mathfrak{F}$ consists of the finite real linear combinations of localized monomials of either type.

**Remark N.27** (provenance of the definition)**.** GJS print (1.1.5)–(1.1.6) as formal integrals, "$Q(x_0) = \int d\vec x\, w(x) \prod_{i=1}^r :\Phi(x_i)^{n_i}:$" and "$Q = \int dx_0 Q(x_0)$", with "localized" meaning that "$w(x)$ is supported in a product $\Delta_{j_1} \times \cdots \times \Delta_{j_n} \subset R^{2r}$ where $\Delta_{j_i}$ is the unit lattice square in $R^2$ centered at the lattice side [sic] $j_i \in Z^{2}$" (p. 593; the index $n$ of the product is printed against $r$, also [sic]); the class is imported as printed (Assumption A8(ii)), with no construction. Definition N.26 makes the integrals precise in the way Appendix C treats the interaction: for $h \in \mathcal{C}_{\mathrm{E}}$, the degree-$n$ part of $\mathcal{U}_{h,\kappa}$ is the sum, over the finitely many squares $\Delta$ meeting $\operatorname{supp} h$, of the $Q^{(\kappa)}$ with $r = 1$ and kernel $\lambda a_n h \mathbf{1}_\Delta$, and $\mathcal{U}_h$ of Assumption A3 is the limit of $\mathcal{U}_{h,\kappa}$ (Lemma N.23′(iii); quantified by Lemma C.4). For a product kernel, $Q$ is the product of the individually Wick-ordered smeared powers (Lemma N.28(c)), which is how A8(ii) reads p. 593 ("products of individually Wick-ordered powers are admissible without

reordering"). Case (b) admits distinct times $x_{i,0}$, the reading of p. 595 ("In the monomial $Q(x_0)$ of (1.1.5), suppose $x_0 = (x_{1,0}, \cdots, x_{r,0})$"); nothing below depends on whether the times are equal. The bound $n_i < \bar{n}$ is carried because p. 593 prints it ("We assume a bound on the $n_i$ in (1.1.5), $n_i < \bar{n}$"); nothing in this section uses it. That these limits are the monomials of p. 593, which GJS print as formal integrals with ": :" denoting "Wick order with respect to the measure $d\Phi$, as in [9], [1]" (p. 589), is a reading of the source, recorded in Assumption A8, Scope (d); so is the identification of the $\mathcal{U}_h$ of Assumption A3 with GJS's $V(h)$ of (1.8), p. 589, which the V3 audit (its §12) left unchecked. The limits do not depend on the mollifier $\delta$ (Lemma N.28(e)), and at sharp times they agree with the constructions of §D.a (Lemma N.28(f) and Lemma N.22).

**Lemma N.28** (Gaussian moments of the observables)**.** *In the setting of Definition N.26 write $x$ for the integration variable ($x \in \mathbb{R}^{2r}$ in case (a), $x = \vec{x} \in \mathbb{R}^r$ in case (b)), $w(x)$ for the kernel and $K$ for a bounded set outside which $w$ vanishes.*

*(a) $Q^{(\kappa)} \in L^2(d\mu_0)$ for $\kappa \geq 2$, and $Q^{(\kappa)}$ converges in $L^2(d\mu_0)$ as $\kappa \to \infty$; the limit is $Q$.*

*(b) For every integer $j \geq 1$, $\mathbb{E}[Q^{2j}] \leq \|\Pi_{Q,j}\|_{L^2(K^{2j})} \|w\|_{L^2}^{2j} < \infty$, with $\Pi_{Q,j}$ as in (38); $\|\Pi_{Q,j}\|_{L^2(K^{2j})}$ depends only on $j$, $n_1, \ldots, n_r$, $K$, $m_0$ and $\delta$. Hence $Q \in L^q(d\mu_0)$ for every $q < \infty$, and so is every $F \in \mathfrak{F}$, by Minkowski's inequality.*

*(c) If $w(x) = \prod_{i=1}^r f_i(x_i)$, with each $f_i$ vanishing outside (the spatial projection of) $\Delta_{j_i}$, then $Q = \prod_{i=1}^r X_i$ almost everywhere, where $X_i$ is the monomial of Definition N.26 with one factor, degree $n_i$ and kernel $f_i$ (in case (b) at the time $x_{i,0}$), and $\|X_i\|^2_{L^2(d\mu_0)} = n_i! \iint f_i(u) f_i(v)\, \Gamma(u-v)^{n_i}\, du\, dv$ with $\Gamma = G$ in case (a) and $\Gamma = C$ in case (b).*

*(d)* (Wick orthogonality and isometry for the limits.) *If $X$ and $X'$ are monomials with one factor, of degrees $n, n'$ and kernels $f, f'$, both of type (a), or both of type (b) at one time $t$, then $\mathbb{E}[XX'] = \delta_{nn'}\, n! \iint f(u) f'(v)\, \Gamma(u-v)^n\, du\, dv$, with $\Gamma$ as in (c).*

*(e)* (Independence of the mollifier.) *If $\delta'$ has the properties of $\delta$ ($\delta' \in C_c^\infty(\mathbb{R}^2)$, $\delta' \geq 0$, $\delta'(z) = \delta'(-z)$, $\operatorname{supp} \delta' \subseteq \{|z| \leq 1\}$, $\int \delta' = 1$), the limits of Definition N.26 formed with $\delta$ and with $\delta'$ coincide almost everywhere.*

*(f)* (Agreement with the sharp-time constructions of §D.a.) *Let $\Phi(t, f)$ be the sharp-time field defined in the proof of Lemma N.21(i) (the extension of $f \mapsto \Phi(f)$ by continuity in the covariance norm, there at $t = 0$).*

*(1) For real $f \in L^2(\mathbb{R})$ with bounded support and every $t \in \mathbb{R}$, the monomial of Definition N.26(b) with one factor of degree 1 and kernel $f$ at the time $t$ is $\Phi(t, f)$.*

*(2) For real bounded measurable $f$ vanishing outside one column, $0 \leq n \leq 2p'$ and every $t \in \mathbb{R}$, the monomial with one factor of degree $n$ and kernel $f$ at the time $t$ equals $J_t W_n(f)$ almost everywhere, and a product-kernel monomial at one time $t$ with factors of this kind equals $J_t(\prod_i W_{n_i}(f_i))$ almost everywhere: this is Lemma N.22 with Remark N.23(a) of §D.a ($W_n(f)$ the limit of Lemma N.20(iii) for $h' = f$, $J_t$ the sharp-time map there), the monomial being, by (a), the $L^2(d\mu_0)$-limit of that lemma's $X_\kappa$.*

*Only (1) is proved here, since Lemma N.22 does not cover it: that lemma needs a bounded kernel and names the degree-1 limit $J_t W_1(f)$, while (1) holds for every kernel in $L^2$ with bounded support and names it $\Phi(t, f)$ (for $n \geq 2$ and an unbounded kernel, §D.a has no $W_n(f)$ to compare with).*

*Proof. The moment formula.* For $x \in K$ let $\zeta_i(x) := x_i$ in case (a) and $\zeta_i(x) := (x_{i,0}, x_i)$ in case (b). Let $j \geq 1$, let $\kappa_1, \dots, \kappa_{2j} \geq 2$ and let $x^1, \dots, x^{2j}$ be configurations. Lemma N.24 with the $\ell = 2jr$ variables $\Phi_{\kappa_c}(\zeta_i(x^c))$, $c \leq 2j$, $i \leq r$, the degrees $m = (n, \dots, n)$ and the variances $c_{\kappa_c}$, together with Lemma N.25(iii), gives

$$\mathbb{E}\Big[\prod_{c=1}^{2j} A_{\kappa_c}(x^c)\Big] = \mathcal{P}_{(n,\dots,n)}\big(\gamma(x^1, \dots, x^{2j})\big), \tag{37}$$

where the entries of $\gamma$ are $G_{\kappa_c \kappa_{c'}}(\zeta_i(x^c) - \zeta_{i'}(x^{c'}))$, for $c < c'$ and all $i, i'$, and for $c = c'$ and $i < i'$. Each entry is a continuous function of the configurations (Lemma N.25(iii)), so $x \mapsto \mathbb{E}[A_\kappa(x)^{2j}]$ is continuous and bounded on the compact set $\bar{K}$.

*The majorant.* By Lemma N.25(iii) every entry of $\gamma$ is positive and at most $M_1$ of the distance of its two points; in case (b) that distance is at least the distance $|u - v|$ of the two spatial integration variables $u, v$ attached to the entry, and $M_1$ is nonincreasing. As $\mathcal{P}$ has nonnegative coefficients, for all $\kappa_1, \dots, \kappa_{2j} \geq 2$

$$0 \leq \mathcal{P}_{(n,\dots,n)}\big(\gamma(x^1, \dots, x^{2j})\big) \leq \Pi_{Q,j}(x^1, \dots, x^{2j}) := \mathcal{P}_{(n,\dots,n)}\big(\mu(x^1, \dots, x^{2j})\big), \tag{38}$$

where $\mu$ has the entry $M_1(|u - v|)$ in place of each entry of $\gamma$, $(u, v)$ the pair of integration variables (in $\mathbb{R}^2$ in case (a), in $\mathbb{R}$ in case (b)) attached to it. Each monomial of $\Pi_{Q,j}$ is a product of $jN$ factors $M_1(|u_e - v_e|)$, $e = 1, \dots, jN$, in which $u_e$ and $v_e$ are two distinct integration variables, because $\mathcal{P}$ contains no diagonal covariance. By Hölder's inequality with the $jN$ exponents $2jN$, $\| \prod_e M_1(|u_e - v_e|)\|_{L^2(K^{2j})} \leq \prod_e \|M_1(|u_e - v_e|)\|_{L^{2jN}(K^{2j})}$, and each factor is finite: integrate first over $u_e$, which ranges over a bounded set, so that $|u_e - v_e| \leq R$ for some $R < \infty$, and use Lemma N.25(iv); the other variables range over bounded sets. Hence $\Pi_{Q,j} \in L^2(K^{2j})$ (if $N = 0$, $\Pi_{Q,j} = 1$).

*Moments of the regularized observables.* For each $q$, $Q^{(\kappa)}(q)^{2j}$ is the product of $2j$ absolutely convergent integrals, so

$$Q^{(\kappa)}(q)^{2j} = \int_{K^{2j}} \prod_{c=1}^{2j} w(x^c) \prod_{c=1}^{2j} A_\kappa(x^c)(q)\, dx^1 \cdots dx^{2j}.$$

Fubini's theorem on $\mathcal{S}'(\mathbb{R}^2) \times K^{2j}$ applies: by Hölder's inequality with the $2j$ exponents $2j$, $\mathbb{E}\prod_c |A_\kappa(x^c)| \leq \sup_{x \in \bar{K}} \mathbb{E}[A_\kappa(x)^{2j}] < \infty$, and $\int \prod_c |w(x^c)| \leq (|K|^{1/2}\|w\|_2)^{2j}$. With (37), (38) and the Cauchy–Schwarz inequality on $K^{2j}$,

$$\begin{aligned} \mathbb{E}\big[(Q^{(\kappa)})^{2j}\big] &= \int_{K^{2j}} \prod_c w(x^c)\, \mathcal{P}\big(\gamma(x^1, \dots, x^{2j})\big)\, dx^1 \cdots dx^{2j} \\ &\leq \|w^{\otimes 2j}\|_{L^2}\, \|\Pi_{Q,j}\|_{L^2(K^{2j})} = \|\Pi_{Q,j}\|_{L^2(K^{2j})}\, \|w\|_2^{2j}. \end{aligned} \tag{39}$$

(a) The case $j = 1$ of (39) gives $Q^{(\kappa)} \in L^2(d\mu_0)$, and the same computation with two regularizations $\kappa, \kappa'$ gives $\mathbb{E}[Q^{(\kappa)} Q^{(\kappa')}] = \iint w(x) w(y)\, \mathcal{P}_{(n,n)}(\gamma^{\kappa\kappa'}(x, y))\, dx\, dy$, the entries of $\gamma^{\kappa\kappa'}$ being as in (37) with $(\kappa_1, \kappa_2) = (\kappa, \kappa')$, dominated by $|w(x)||w(y)|\Pi_{Q,1}(x, y)$, an integrable function. As $\kappa, \kappa' \to \infty$, every entry converges to the corresponding value of $G$, for every $(x, y)$ at which the relevant differences are nonzero, that is off a null set (Lemma N.25(iii)); by dominated convergence $\mathbb{E}[Q^{(\kappa)} Q^{(\kappa')}]$ converges to a limit $I_Q$ independent of how $\kappa, \kappa' \to \infty$. Therefore $\|Q^{(\kappa)} - Q^{(\kappa')}\|_2^2 = \mathbb{E}[Q^{(\kappa)} Q^{(\kappa)}] - 2\mathbb{E}[Q^{(\kappa)} Q^{(\kappa')}] + \mathbb{E}[Q^{(\kappa')} Q^{(\kappa')}] \to I_Q - 2I_Q + I_Q = 0$: the family $Q^{(\kappa)}$ is Cauchy in $L^2(d\mu_0)$ and its limit $Q$ exists, with $\|Q\|_2^2 = I_Q$.

(b) Choose $\kappa_k \to \infty$ with $Q^{(\kappa_k)} \to Q$ almost everywhere (possible by (a)). Fatou's lemma and (39) give $\mathbb{E}[Q^{2j}] \le \liminf_k \mathbb{E}[(Q^{(\kappa_k)})^{2j}] \le \|\Pi_{Q,j}\|_{L^2(K^{2j})}\|w\|_2^{2j}$. For $q < \infty$ take $2j \ge q$ and use $\|Q\|_q \le \|Q\|_{2j}$, $d\mu_0$ being a probability measure.

(c) For each $q$ the integral defining $Q^{(\kappa)}(q)$ factorizes by Fubini's theorem, $Q^{(\kappa)}(q) = \prod_i X_i^{(\kappa)}(q)$ with $X_i^{(\kappa)}(q) := \int f_i(x_i) :\Phi_\kappa(\zeta_i)^{n_i}:_{c_\kappa}(q)\, dx_i$. Take any $\kappa_k \to \infty$. By (a), $X_i^{(\kappa_k)} \to X_i$ and $Q^{(\kappa_k)} \to Q$ in $L^2(d\mu_0)$, hence in probability; a finite product of sequences converging in probability converges in probability to the product of the limits, and limits in probability are unique almost everywhere; so $Q = \prod_i X_i$ almost everywhere. For one factor, $\mathcal{P}_{(n_i,n_i)}(\gamma) = n_i!\,\gamma^{n_i}$ (Lemma N.24, $\ell = 2$), and the limiting covariance is $G(u-v)$ in case (a) and $G((x_{i,0},u)-(x_{i,0},v)) = G(0,u-v) = C(u-v)$ in case (b) (Lemma N.25(i)); so $\|X_i\|_2^2 = I_{X_i}$ is the displayed integral.

(d) As in (a), Lemma N.24 with $\ell = 2$ gives

$$\mathbb{E}\big[X^{(\kappa)}X'^{(\kappa')}\big] = \delta_{nn'}\, n! \iint f(u)f'(v)\, G_{\kappa\kappa'}(\zeta(u)-\zeta(v))^n\, du\, dv,$$

with $\zeta(u) = u$ in case (a) and $\zeta(u) = (t,u)$ in case (b); the integrand is dominated by $n!\,|f(u)||f'(v)|M_1(|u-v|)^n$, which is integrable by the Cauchy–Schwarz inequality and Lemma N.25(iv). Let $\kappa,\kappa' \to \infty$: the left side tends to $\mathbb{E}[XX']$ because $X^{(\kappa)} \to X$ and $X'^{(\kappa')} \to X'$ in $L^2(d\mu_0)$, and the right side to the stated integral by dominated convergence and Lemma N.25(i),(iii).

(e) Write $\Phi'_{\kappa'}(\zeta) := \Phi(\delta'_{\kappa'}(\zeta - \cdot))$. As in the proof of Lemma N.25(iii), $\mathbb{E}[\Phi_\kappa(\zeta)\Phi'_{\kappa'}(\zeta')] = (\tilde\delta * G)(\zeta - \zeta')$ with $\tilde\delta := \delta_\kappa * \delta'_{\kappa'}$, which is nonnegative, has integral 1 and support in $\{|c| \le \kappa^{-1} + \kappa'^{-1}\}$, and satisfies $\|\tilde\delta\|_\infty \le \min\{\kappa^2\|\delta\|_\infty, \kappa'^2\|\delta'\|_\infty\} \le \min\{\kappa,\kappa'\}^2 \max\{\|\delta\|_\infty, \|\delta'\|_\infty\}$. Those are the only properties of $\tilde\delta$ that proof uses, so $\tilde\delta * G$ is positive, continuous, bounded by $M_1$ with $\|\delta\|_\infty$ replaced by $\max\{\|\delta\|_\infty, \|\delta'\|_\infty\}$, and tends to $G$ off 0. The computation of (a), with one regularization formed with $\delta$ and the other with $\delta'$, each Wick ordered at its own variance, then gives $\mathbb{E}[Q^{(\kappa)}Q'^{(\kappa')}] \to I_Q$, the common limit of $\mathbb{E}[(Q^{(\kappa)})^2]$ and $\mathbb{E}[(Q'^{(\kappa')})^2]$, since $I_Q$ involves $G$ only. Hence $\|Q - Q'\|_2^2 = I_Q - 2I_Q + I_Q = 0$.

(f)(1) For each $q$, $\int f(x)\,\Phi_\kappa(t,x)(q)\,dx = \langle q, \delta_\kappa * (\delta_t \otimes f)\rangle$, the integral of the test functions $f(x)\delta_\kappa((t,x)-\cdot)$ being one in $\mathcal{S}(\mathbb{R}^2)$; so the regularized monomial is $\Phi(\delta_\kappa * (\delta_t \otimes f))$. In the covariance norm $\delta_\kappa * (\delta_t \otimes f) \to \delta_t \otimes f$: the squared distance is $(2\pi)^{-2}\int |\hat\delta(p/\kappa) - 1|^2 |\hat f(p_1)|^2 (|p|^2 + m_0^2)^{-1} dp$, with $|\hat\delta| \le 1$ and the integrable dominator $4|\hat f(p_1)|^2(p_0^2 + p_1^2 + m_0^2)^{-1}$ (its $p_0$-integral is $4\pi|\hat f(p_1)|^2(p_1^2 + m_0^2)^{-1/2}$), so it tends to 0. Hence the limit is $\Phi(t,f)$.

(f)(2) is Lemma N.22 with Remark N.23(a), as stated; it is not proved again here. □

### The identity

Throughout this subsection $h_1, h_2 \in \mathcal{C}_{\mathrm{E}}$ are fixed, and for $\alpha \in [0,1]$

$$h_\alpha := h_1 + \alpha(h_2 - h_1) = (1-\alpha)h_1 + \alpha h_2, \qquad \mathcal{U}_{h_2-h_1} := \mathcal{U}_{h_2} - \mathcal{U}_{h_1}.$$

The symbol $h_\alpha$ is used only with the letter $\alpha$: the numerals in $h_1, h_2$ always denote the two given cutoffs, and the path equals $h_1$ at $\alpha = 0$ and $h_2$ at $\alpha = 1$. At $h_1 = h^B$, $h_2 = h$ it is the path $h_\alpha = h^B + \alpha(h - h^B)$ of §D.a. $\mathcal{U}_{h_2-h_1}$ lies in $L^q(d\mu_0)$ for every $q < \infty$ (Step 2 below); it is an *observable*, the covariance partner of $F$, and the function $h_2 - h_1$, which may take both signs, is nowhere used as a cutoff: every measure in this section is $dq_{h_\alpha}$ with $h_\alpha \in \mathcal{C}_{\mathrm{E}}$.

**Lemma D.1** (affine-path identity)**.** *Let $h_1, h_2 \in \mathcal{C}_{\mathrm{E}}$, and let $F$ be a finite sum of localized monomials of type* (1.1.5) *or* (1.1.6)*, with kernels in $L^2$ and $n_i < \bar n$, that is, $F \in \mathfrak{F}$ (Definition N.26). Then $h_\alpha \in \mathcal{C}_{\mathrm{E}}$ for every $\alpha \in [0,1]$; the functions $F$, $\mathcal{U}_{h_2-h_1}$ and $F\,\mathcal{U}_{h_2-h_1}$*

*are $dq_{h_\alpha}$-integrable for every $\alpha \in [0,1]$; the map $\alpha \mapsto \langle F\rangle_{h_1+\alpha(h_2-h_1)}$ is $C^1$ on $[0,1]$ (one-sided derivatives at the endpoints), with*

$$\frac{d}{d\alpha}\langle F\rangle_{h_\alpha} = -\operatorname{Cov}_{h_\alpha}\big(F, \mathcal{U}_{h_2-h_1}\big); \tag{40}$$

*and*

$$\langle F\rangle_{h_2} - \langle F\rangle_{h_1} = -\int_0^1 \operatorname{Cov}_{h_\alpha}\big(F, \mathcal{U}_{h_2-h_1}\big)\, d\alpha. \tag{41}$$

Here $\langle F\rangle_h = \int F\, dq_h$ and $\operatorname{Cov}_h(F,G) = \langle FG\rangle_h - \langle F\rangle_h\langle G\rangle_h$; the measures $dq_h = Z(h)^{-1}e^{-\mathcal{U}_h}d\mu_0$ are those of Assumption A3.

*Status of Lemma D.1. Proved*, from Assumptions A2 and A3 and the Gaussian computations below. Step 1 of the proof needs that $h \mapsto \mathcal{U}_h$ is linear for the $\mathcal{U}_h$ of Assumption A3; this holds by its construction ($:\Phi^n(g):$ is fixed by pairings linear in $g$) and also follows from the identification of $\mathcal{U}_h$ with the $L^2(d\mu_0)$-limit of its regularizations (Lemma N.23′(iii); Lemma C.4 quantifies the convergence), which is the route written in Step 1. (Until 7 November 2026 the manuscript stated no construction of $\mathcal{U}_h$, and the comparison note recorded this lemma as proved conditional on an import of that construction; the import is now Assumption A3.)

*Proof. Step 0: the path stays in $\mathcal{C}_{\mathrm{E}}$.* For $\alpha \in [0,1]$, $h_\alpha$ is measurable; $0 \le h_\alpha \le 1$ pointwise, as a convex combination of two numbers in $[0,1]$; and, since $h_1, h_2 \ge 0$, $\{h_\alpha \ne 0\} \subseteq \{h_1 \ne 0\} \cup \{h_2 \ne 0\}$, so $\operatorname{supp} h_\alpha \subseteq \Lambda := \operatorname{supp} h_1 \cup \operatorname{supp} h_2$, which is compact (supp the closure of the set of non-zeros, as in §D.a; the inclusion holds equally for essential supports). By Definition 2.1 ($\mathcal{C}_{\mathrm{E}}$: measurable $h : \mathbb{R}^2 \to [0,1]$ with compact support), $h_\alpha \in \mathcal{C}_{\mathrm{E}}$. For $h_1 = h^B$, $h_2 = h$ this is Lemma N.5(ii) of §D.a.

*Step 1: linearity, and representatives.* $\mathcal{U}_h$, $h \in \mathcal{C}_{\mathrm{E}}$, of Assumption A3 is the limit in $L^2(d\mu_0)$ of $\mathcal{U}_{h,\kappa} = \lambda\int h :\mathscr{P}(\Phi_\kappa):_{c_\kappa} dz$ (Lemma N.23′(iii); Appendix C, notation paragraph and Lemma C.4, which quantifies the convergence in $L^2$). For every bounded measurable $w$ of compact support, of either sign, $\lambda\int w :\mathscr{P}(\Phi_\kappa):_{c_\kappa} dz$ converges in $L^2(d\mu_0)$ as $\kappa \to \infty$: write $w = \sum_\Delta w\mathbf{1}_\Delta$ over the finitely many squares meeting its support, and apply Lemma N.28(a) with $r = 1$ to each kernel $\lambda a_n w\mathbf{1}_\Delta$, $0 \le n \le 2p'$. The regularized integral is linear in $w$ (pointwise in $q$), hence so is the limit. With $w = h_\alpha, h_1, h_2$ and $h_\alpha - (1-\alpha)h_1 - \alpha h_2 = 0$,

$$\mathcal{U}_{h_\alpha} = (1-\alpha)\mathcal{U}_{h_1} + \alpha\mathcal{U}_{h_2} = \mathcal{U}_{h_1} + \alpha\,\mathcal{U}_{h_2-h_1} \qquad d\mu_0\text{-a.e., for each } \alpha \in [0,1], \tag{42}$$

where $\mathcal{U}_{h_2-h_1}$, the limit for $w = h_2 - h_1$, equals $\mathcal{U}_{h_2} - \mathcal{U}_{h_1}$. This is the linearity of $h \mapsto \mathcal{U}_h$ recorded in the proof of Lemma 3.1; for $h_1 = h^B$, $h_2 = h$ it is Lemma N.5(iv) of §D.a, proved there from Lemma C.4; by the construction of Assumption A3, which defines $\mathcal{U}_h$ linearly in $h$, (42) also holds directly. The null set in (42) may depend on $\alpha$. Fix therefore finite real-valued representatives $v$ of $\mathcal{U}_{h_1}$ and $d$ of $\mathcal{U}_{h_2-h_1}$, and put $u_\alpha := v + \alpha d$ pointwise, for all $\alpha \in [0,1]$ at once. By (42), $u_\alpha$ represents $\mathcal{U}_{h_\alpha}$ for every $\alpha$; $u_0 = v$ represents $\mathcal{U}_{h_1}$ and $u_1 = v + d$ represents $\mathcal{U}_{h_2}$. Every pointwise statement below concerns these representatives.

*Step 2: integrability, with the exponents.* Assumption A3 reads: "For every $h \in \mathcal{C}_{\mathrm{E}}$: $\mathcal{U}_h \in L^p(d\mu_0)$ for every $p < \infty$, and $e^{-\varrho\mathcal{U}_h} \in L^1(d\mu_0)$ for every finite $\varrho > 0$. In particular $0 < Z(h) := \int e^{-\mathcal{U}_h}d\mu_0 < \infty$". We use it at $h_1$, $h_2$ and $h_\alpha$, all in $\mathcal{C}_{\mathrm{E}}$ by Step 0.

(I1) $F \in L^2(d\mu_0)$, by Lemma N.28(b).

(I2) $d \in L^q(d\mu_0)$ for every $q < \infty$: A3 gives $\mathcal{U}_{h_1}, \mathcal{U}_{h_2} \in L^q$, and $\|d\|_q \le \|\mathcal{U}_{h_1}\|_q + \|\mathcal{U}_{h_2}\|_q$.

(I3) $W := e^{-u_0} + e^{-u_1} \in L^q(d\mu_0)$ for every $q < \infty$: A3 with $\varrho := q$ gives $\|e^{-\mathcal{U}_{h_i}}\|_q^q = \int e^{-q\mathcal{U}_{h_i}}d\mu_0 < \infty$ for $i = 1, 2$, and Minkowski's inequality.

(I4) For every point and every $\beta \in [0,1]$, by the convexity of the exponential function, $e^{-u_\beta} = e^{-(1-\beta)u_0 - \beta u_1} \le (1-\beta)e^{-u_0} + \beta e^{-u_1} \le W$.

(I5) Hölder's inequality with the exponents $(2,4,4)$, where $\frac{1}{2}+\frac{1}{4}+\frac{1}{4}=1$, and with the exponents $(2,2)$:

$$\|F\,d\,W\|_1 \le \|F\|_2\|d\|_4\|W\|_4, \quad \|F\,W\|_1 \le \|F\|_2\|W\|_2, \quad \|d\,W\|_1 \le \|d\|_2\|W\|_2,$$

and $\|W\|_1 \le \|W\|_2$.

Put $Z(\alpha) := \int e^{-u_\alpha} d\mu_0 = Z(h_\alpha)$ and $N(\alpha) := \int F\, e^{-u_\alpha} d\mu_0$. By (I4), $|F|e^{-u_\alpha} \le |F|W$, $|d|e^{-u_\alpha} \le |d|W$ and $|F\,d|e^{-u_\alpha} \le |F||d|W$, all in $L^1(d\mu_0)$ by (I5); dividing by $Z(h_\alpha) > 0$ (A3) shows that $F$, $d = \mathcal{U}_{h_2-h_1}$ and $F\,d$ are $dq_{h_\alpha}$-integrable for every $\alpha$, and $\langle F\rangle_{h_\alpha} = N(\alpha)/Z(\alpha)$.

*Step 3: differentiation under the integral.* For $\alpha, \alpha+\varepsilon \in [0,1]$, $\varepsilon \neq 0$, pointwise

$$\frac{e^{-u_{\alpha+\varepsilon}} - e^{-u_\alpha}}{\varepsilon} = -\,d \int_0^1 e^{-u_{\alpha+\theta\varepsilon}}\, d\theta, \tag{43}$$

since $u_{\alpha+\theta\varepsilon} = u_\alpha + \theta\varepsilon d$ and $\frac{d}{d\theta} e^{-u_\alpha - \theta\varepsilon d} = -\varepsilon\, d\, e^{-u_{\alpha+\theta\varepsilon}}$. By (I4) the modulus of (43) is at most $|d|\,W$, and (43) tends to $-d\,e^{-u_\alpha}$ pointwise as $\varepsilon \to 0$. Multiply by $F$, resp. by 1, and apply dominated convergence with the dominator $|F||d|W$, resp. $|d|W$, integrable by (I5): $N$ and $Z$ are differentiable on $[0,1]$ (one-sided at the endpoints), with

$$N'(\alpha) = -\int F\, d\, e^{-u_\alpha}\, d\mu_0, \qquad Z'(\alpha) = -\int d\, e^{-u_\alpha}\, d\mu_0.$$

If $\alpha_k \to \alpha$ in $[0,1]$, then $F\,d\,e^{-u_{\alpha_k}} \to F\,d\,e^{-u_\alpha}$ and $d\,e^{-u_{\alpha_k}} \to d\,e^{-u_\alpha}$ pointwise, with the same dominators; so $N'$ and $Z'$ are continuous, and $N$, $Z$ are $C^1$ on $[0,1]$.

*Step 4: normalization.* $Z(\alpha) = Z(h_\alpha) > 0$ for every $\alpha \in [0,1]$, by A3 at $h_\alpha \in \mathcal{C}_{\mathrm{E}}$; $Z$ is continuous on the compact interval $[0,1]$ (Step 3), so $\min_{[0,1]} Z > 0$. (Lemma C.2 gives the explicit bound $Z(\alpha) \ge e^{-\lambda|a_0||\Lambda|}$, since $\operatorname{supp} h_\alpha \subseteq \Lambda$ by Step 0; this proof does not use it.)

*Step 5: the quotient rule.* By Steps 3 and 4, $\alpha \mapsto \langle F\rangle_{h_\alpha} = N/Z$ is $C^1$ on $[0,1]$, and, since $Z(\alpha)^{-1}e^{-u_\alpha}d\mu_0 = dq_{h_\alpha}$ ($u_\alpha$ represents $\mathcal{U}_{h_\alpha}$),

$$\frac{d}{d\alpha}\langle F\rangle_{h_\alpha} = \frac{N'}{Z} - \frac{N\,Z'}{Z^2} = -\int F\,d\,dq_{h_\alpha} + \int F\,dq_{h_\alpha} \int d\,dq_{h_\alpha} = -\operatorname{Cov}_{h_\alpha}\big(F, \mathcal{U}_{h_2-h_1}\big),$$

which is (40).

*Step 6: integration.* The fundamental theorem of calculus for the $C^1$ function $\alpha \mapsto \langle F\rangle_{h_\alpha}$ on $[0,1]$, whose values at $\alpha = 0$ and $\alpha = 1$ are $\langle F\rangle_{h_1}$ and $\langle F\rangle_{h_2}$ ($u_0$ and $u_1$ represent $\mathcal{U}_{h_1}$ and $\mathcal{U}_{h_2}$), gives (41). □

**Remark N.29** (what the proof uses, and what it does not)**.** (a) *The class of $F$.* $F$ enters only through $F \in L^2(d\mu_0)$. For any $F \in L^{p_0}(d\mu_0)$ with $1 < p_0 < \infty$ the proof runs verbatim with the exponents $(p_0, 2p_0', 2p_0')$, $p_0' := p_0/(p_0-1)$, in place of $(2,4,4)$ in (I5) ($\frac{1}{p_0}+\frac{1}{2p_0'}+\frac{1}{2p_0'}=1$), and $(p_0, p_0')$ in place of $(2,2)$; so Lemma D.1 holds for every $F \in \bigcup_{p_0>1} L^{p_0}(d\mu_0)$. Neither $n_i < \bar{n}$, nor the localization squares, nor the GJS norm enters.

(b) *No constant, no uniformity in $h$.* Neither Theorem C.5 nor Lemma C.2 nor Assumption A4 is used: the dominator involves only the endpoints $h_1, h_2$, at which A3 applies, and positivity of $Z$ on the path comes from A3 and continuity (Step 4). Lemma D.1 is qualitative and carries no constant. The Gaussian input is computed explicitly (Lemmas N.24, N.25 and N.28); it does not use the chaos structure of $\mathcal{U}_h$ or the Wick isometry for $\mathcal{U}_h$. The proof is not independent of Appendix C, though: the linearity used in Step 1 rests on the

construction of Assumption A3 or, on the route written in Step 1, on Lemma C.4 with Lemma N.23′(iii).

(c) *The partner is an observable.* $\mathcal{U}_{h_2-h_1}$ is an element of $\bigcap_{q<\infty} L^q(d\mu_0)$, built from $\mathcal{U}_{h_1}$ and $\mathcal{U}_{h_2}$. A3 is applied only at $h_1$, $h_2$ and $h_\alpha$, all in $\mathcal{C}_{\mathrm{E}}$; Step 1 takes the regularized limit for the kernel $h_2 - h_1$ as for any observable; and no statement about measures is applied at $h_2 - h_1$, for which, as Remark C.6 states, the Wick lower bound of Lemma C.3 would fail when $h_2 - h_1$ takes both signs. In the application in Lemma D.2, $h_1 = h^B$ and $h_2 = h$, so that $h_2 - h_1 = h\,\mathbf{1}_{\mathbb{R}^2\setminus B} \geq 0$.

(d) *Relation to §8 and to GJS.* Proposition 8.1, Steps 2–3, runs the same argument for slab cutoffs, with the log-convexity bound (13) in place of the convexity bound (I4) and with the difference quotient (14), which is (43). GJS's (1.1.12) states the identity without the integrability argument; its third equality, the decomposition of $V(g)$ into the pieces $V(g_i)$, is the business of Lemma D.2.

**Remark N.30** (a check of the bookkeeping)**.** In the one-variable model in which $d\mu_0$ is replaced by the standard Gaussian law of a real variable $X$, $\mathcal{U}_h$ by $hX^2/2$ for $h \in [0,1]$, $F = X^2$, $h_1 = 0$ and $h_2 = 1$, one has $dq_h = N(0,(1+h)^{-1})$, $\langle F\rangle_h = (1+h)^{-1}$ and $\mathrm{Cov}_h(X^2, X^2/2) = \frac{1}{2}\mathrm{Var}_h(X^2) = (1+h)^{-2}$, so that $-\int_0^1 (1+\alpha)^{-2} d\alpha = -\frac{1}{2} = \langle F\rangle_{h_2} - \langle F\rangle_{h_1}$, as (41) asserts. This tests the signs and the normalization only; it is not part of the proof.

*Numerical record* (*numerical*; nothing in this section is deduced from it). The script `parts/check_d1.py`, run on 3 November 2026 with the system `python3` 3.14.7, numpy 2.5.2 and scipy 1.18.1 (output and run header in `parts/check_d1.output.txt`), checks: the moment formula of Lemma N.24 against Isserlis's theorem in exact rational arithmetic, for twelve degree patterns with random rational covariance matrices (seed 20261102), all equal; the bound $G \leq M_0$ of Lemma N.25(ii) on 4001 log-spaced points $|\zeta| \in [10^{-10}, 10^3]$ for six values of $m_0 \in [0.05, 10]$, with margin at least 0.319; the formula of Lemma N.25(i) against the value $K_0(m_0|\zeta|)/(2\pi)$ at three points, to twelve digits (default tolerances of `scipy.integrate.quad`); and the model of this remark by quadrature.

## D.c Localized comparison

This section proves Lemma D.2: the expectation of a localized observable under $dq_h$ differs from its expectation under the cutoff restricted to a box of squares containing the observable's squares by a sum of clustering terms, one for each square outside the box, and each term is bounded by GJS Theorem 1.1.7. The device is the source's own: in the proof of its (1.1.11), GJS write the difference of two expectations as an integral along an affine path of cutoffs of a sum over unit squares and state "We apply Theorem 1.1.7 to each term in the sum to obtain" [7, p. 595].

### D.c.1 Standing assumption and notation

*Standing assumption.* In §§D.c–D.e and §§4–7, $0 \leq \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$, and we work in the working coordinates of §2.6 (Remark 3.3 transports the conclusions): the pair $(m_0, \lambda)$ satisfies the hypotheses printed for GJS Theorem 1.1.7, "Let $m_0$, $K_2$ and $K_1$ be sufficiently large and let $\lambda$ be sufficiently small" [7, p. 594], and $m_1 > 0$, $C_7$, $K_1$ and $K_2 \geq 1$ are the printed constants of Assumption A8(iii)–(iv) (§D.a.8). The interaction polynomial is $\mathscr{P}(\xi) = \sum_{n=0}^{2p'} a_n \xi^n$, $a_{2p'} > 0$, $p' \geq 1$, and $\bar{n} := 2p' + 1$ (Assumption A8(ii)). For $h \in \mathcal{C}_{\mathrm{E}}$, $\mathcal{U}_h = \lambda \int h : \mathscr{P}(\Phi) : dz$, $Z(h) = \int e^{-\mathcal{U}_h} d\mu_0$, $dq_h = Z(h)^{-1} e^{-\mathcal{U}_h} d\mu_0$, $\langle F\rangle_h = \int F\, dq_h$ (Assumption A3) and $\mathrm{Cov}_h(F,G) = \langle FG\rangle_h - \langle F\rangle_h \langle G\rangle_h$.

*Notation.* We use the notation of §D.a: the squares $\Delta_{l,k}$, rows and columns (Definition N.1); boxes $B$, $|B|$, the restriction $h^B$, the path $h_\alpha$ and the set $\mathfrak{S}(h)$ of squares charged

by $h$ (Definition N.4); strips, separation, the parallel and the transverse orientation, and $d(\Delta, R)$ (Definition N.6); $U_{\Delta,n}$, $U_\Delta$ and $c_U$ (Definition N.17, Lemma N.18). For $h \in \mathcal{C}_{\mathrm{E}}$ and a box $B$ we write

$$\mathcal{D}(h, B) := \{\Delta \in \mathfrak{S}(h) : \ \Delta \cap B = \varnothing\}$$

for the squares outside $B$ charged by $h$; it is finite, and each of its squares meets $\operatorname{supp} h$ (Lemma N.5(iii)).

**Lemma N.31** (the widest strip exists)**.** *Let $X, Y \subseteq \mathbb{R}^2$ be bounded and non-empty, and suppose that some strip separates $X$ from $Y$. Then the supremum of the widths of the strips separating $X$ from $Y$ is attained. In particular, for two localized monomials whose localization squares are separated by some strip, "the widest strip bounded by two parallel lines, and separating" them [7, p. 594] exists, its width is the supremum $d$ of Definition N.6, and every separating strip has width at most $d$.*

Definition N.6 defines the width $d$ as a supremum; the lemma shows that it is attained, as the wording of p. 594 presupposes.

*Proof.* For a unit vector $\nu$, $c \in \mathbb{R}$ and $w \geq 0$, the strip bounded by $\{\nu \cdot z = c\}$ and $\{\nu \cdot z = c+w\}$ has the closed half-planes $H_1 = \{\nu \cdot z \leq c\}$ and $H_2 = \{\nu \cdot z \geq c + w\}$ (Definition N.6). Put $a(\nu) := \sup_{x \in X} \nu \cdot x$ and $b(\nu) := \inf_{y \in Y} \nu \cdot y$. Then $X \subseteq H_1$ and $Y \subseteq H_2$ if and only if $a(\nu) \leq c$ and $c + w \leq b(\nu)$; the other assignment of the half-planes corresponds to $-\nu$. So a strip with normal $\nu$, oriented from $X$ to $Y$, separates them exactly when $b(\nu) - a(\nu) \geq 0$, its width is then at most $b(\nu) - a(\nu)$, and the choice $c = a(\nu)$, $w = b(\nu) - a(\nu)$ attains this. Hence the supremum of the separating widths is $\sup_{|\nu|=1}(b(\nu) - a(\nu))$, which is $\geq 0$ by hypothesis. With $\varrho := \sup\{|z| : z \in X \cup Y\} < \infty$, $|a(\nu) - a(\nu')| \leq \varrho|\nu - \nu'|$ and $|b(\nu) - b(\nu')| \leq \varrho|\nu - \nu'|$, so $\nu \mapsto b(\nu) - a(\nu)$ is continuous on the unit circle and attains its supremum at some $\nu_*$; the strip with normal $\nu_*$, $c = a(\nu_*)$ and $w = b(\nu_*) - a(\nu_*)$ separates and has the maximal width. □

**Lemma N.32** (square integrability of the pieces)**.** *Let $h \in \mathcal{C}_{\mathrm{E}}$. Every $X \in L^4(d\mu_0)$ lies in $L^2(dq_h)$. On $L^2(dq_h)$ the covariance $\mathrm{Cov}_h$ is defined, bilinear and symmetric, and $\mathrm{Cov}_h(X, c) = 0$ for constants $c$. Every finite sum of localized monomials of type (1.1.5) or (1.1.6) with kernels in $L^2$ and individual degrees $n_i < \bar{n}$ lies in $L^4(d\mu_0)$, hence in $L^2(dq_h)$; so do, in particular, every piece $F_a$ of Lemma D.2 below and every $U_{\Delta,n}$, $1 \leq n \leq 2p'$.*

This also supplies the hypothesis "$F$ and the finitely many $U_{\Delta,n}$ involved lie in $L^2(dq_{h'})$" of Lemma N.18(iv), which §D.a refers to the chain of Lemma D.1; that chain gives the $dq_h$-integrability of $F$ and of $\mathcal{U}_{h_2 - h_1}$, not of a single $U_{\Delta,n}$.

*Proof.* By Assumption A3, $Z(h) > 0$ and $e^{-2\mathcal{U}_h} \in L^1(d\mu_0)$, i.e. $e^{-\mathcal{U}_h} \in L^2(d\mu_0)$. Hölder's inequality with exponents $(2, 2)$ gives

$$\int X^2 e^{-\mathcal{U}_h} d\mu_0 \leq \|X^2\|_{L^2(d\mu_0)} \|e^{-\mathcal{U}_h}\|_{L^2(d\mu_0)} = \|X\|^2_{L^4(d\mu_0)} \|e^{-\mathcal{U}_h}\|_{L^2(d\mu_0)} < \infty;$$

divide by $Z(h)$. For $X, Y \in L^2(dq_h)$, $XY \in L^1(dq_h)$ by the Cauchy–Schwarz inequality, and $L^2(dq_h) \subseteq L^1(dq_h)$ because $dq_h$ is a probability measure; so $\mathrm{Cov}_h(X, Y)$ is defined, and it is bilinear and symmetric by the linearity of the integral; for a constant $c$, $\langle Xc \rangle_h - \langle X \rangle_h c = 0$. The membership in $L^4(d\mu_0)$ is Lemma N.28(b) of §D.b ("$Q \in L^q(d\mu_0)$ for every $q < \infty$, and so is every $F \in \mathfrak{F}$"); $U_{\Delta,n}$ is a localized monomial of type (1.1.6) with kernel $\lambda a_n h \mathbf{1}_\Delta \in L^2(\mathbb{R}^2)$ and degree $n < \bar{n}$ (Lemma N.18(i)). □

### D.c.2 The lemma

**Lemma D.2** (localized comparison)**.** *Assume the standing assumption of §D.c.1. Let $F=\sum_{a\in\mathfrak{A}}F_a$ be a finite sum of localized monomials $F_a$ of type (1.1.5) or (1.1.6) [7, p. 593], with kernels in $L^2$ and every individual degree $n_i<\bar{n}$, and put $\|F\|_{\mathrm{GJS}}:=\sum_{a\in\mathfrak{A}}\|F_a\|_{\mathrm{GJS}}$, the norm (1.1.9) of this decomposition. Let $R$ be a non-empty finite set of squares containing every localization square of every $F_a$, let $B\supseteq R$ be a box, let $h\in\mathcal{C}_{\mathrm{E}}$, and let $\mathcal{D}:=\mathcal{D}(h,B)$.*

(a) *Suppose that to every $\Delta\in\mathcal{D}$ a strip $S_\Delta$ of positive width $w_\Delta$ separating $\Delta$ from $\bigcup R$ is assigned. Then*

$$\left|\langle F\rangle_h-\langle F\rangle_{h^B}\right|\le C_7\,c_U\,\|F\|_{\mathrm{GJS}}\sum_{\Delta\in\mathcal{D}}e^{-m_1w_\Delta}.$$

*Every estimate behind this bound is Source Fact 2.2 (GJS (1.1.10)) for a pair $(F_a,U_{\Delta,n})$, across the strip $S_\Delta$, at a cutoff $h_\alpha\in\mathcal{C}_{\mathrm{E}}$; the orientation of $S_\Delta$ is the orientation of that application. If $S_\Delta$ is transverse to the time-zero line and $F_a$ is of type (1.1.5), that application is the transverse case of GJS Theorem 1.1.7, used at the printed statement's scope and listed as A8(iv)-T (Assumption A8, item (iv-T)).*

(b) *(A form of the statement of `ugjs/PLAN.md`, Part 1, with positive widths; the plan's form does not require positive width.) If every square $\Delta\not\subseteq B$ meeting $\operatorname{supp}h$ is separated from $\bigcup R$ by a strip of positive width, then*

$$\left|\langle F\rangle_h-\langle F\rangle_{h^B}\right|\le C_7\,c_U\,\|F\|_{\mathrm{GJS}}\sum_{\Delta\not\subseteq B,\ \Delta\cap\operatorname{supp}h\neq\varnothing}e^{-m_1d(\Delta,R)}.$$

*The widest strips used here are not controlled in orientation (Remark N.33(2)); where one is transverse to the time-zero line and some $F_a$ is of type (1.1.5), form (b) uses the transverse case of GJS Theorem 1.1.7 at the printed statement's scope (A8(iv)-T, Assumption A8, item (iv-T)).*

*The constants $C_7$ and $m_1$ are those of Theorem 1.1.7 in working coordinates (Remark 3.3), and $c_U$ is that of Lemma N.18; none depends on $h$, $B$, $R$ or $F$.*

*Status of Lemma D.2. Proved*, from Lemma D.1, Lemma C.4 with Lemma N.23′(iii) (which supply, in Step 3, the identification of $\mathcal{U}_{h-h^B}=\mathcal{U}_h-\mathcal{U}_{h^B}$ with the $L^2(d\mu_0)$-limit of its regularizations) and Assumption A8(iv) at the printed statement's scope. Lemma D.2 uses neither Lemma C.2 nor Theorem C.5. It applies GJS Theorem 1.1.7 to the localized monomials of Definition N.26; that these are the monomials of p. 593, and that the $\mathcal{U}_h$ of Assumption A3 is GJS's $V(h)$, is the reading of the source of Remark N.2(iv) (Remark N.27; Step 4), recorded in Assumption A8, Scope (d), with the open human reading C5 item 10 named there. The orientation of each application is that of the assigned strip, and a transverse strip with a sharp-time $F_a$ is the case A8(iv)-T.

*Proof. Step 1: the path.* By Lemma N.5(i)–(ii), $h^B\in\mathcal{C}_{\mathrm{E}}$ and $h_\alpha=h^B+\alpha(h-h^B)\in\mathcal{C}_{\mathrm{E}}$ for every $\alpha\in[0,1]$, with $h_0=h^B$ and $h_1=h$.

*Step 2: the identity.* $F$ is a finite sum of localized monomials of type (1.1.5) or (1.1.6) with kernels in $L^2$ and $n_i<\bar{n}$, which is the class of Lemma D.1. Lemma D.1 with $h_1=h^B$, $h_2=h$ gives that $\alpha\mapsto\langle F\rangle_{h_\alpha}$ is $C^1$ on $[0,1]$ and

$$\langle F\rangle_h-\langle F\rangle_{h^B}=-\int_0^1\operatorname{Cov}_{h_\alpha}\!\left(F,\mathcal{U}_{h-h^B}\right)d\alpha;$$

the covariance is taken under $dq_{h_\alpha}$, $h_\alpha = h^B + \alpha(h - h^B)$, and $\mathcal{U}_{h-h^B} = \mathcal{U}_h - \mathcal{U}_{h^B}$ is the observable of Lemma D.1. Here $h - h^B = h\mathbf{1}_{\mathbb{R}^2\setminus B} \geq 0$ lies in $\mathcal{C}_{\mathrm{E}}$ (Lemma N.5(i)); it enters only as the observable $\mathcal{U}_{h-h^B}$, not as a cutoff.

*Step 3: decomposition into squares.* Put $w := h - h^B$, a bounded measurable function with compact support, and, for $\kappa \geq 2$, $\mathcal{U}_w^{(\kappa)} := \lambda \int w :\mathscr{P}(\Phi_\kappa):_{c_\kappa} dz$, pointwise in $q$, with the regularization of §D.b. For each $q$, $\mathcal{U}_w^{(\kappa)}(q)$ is the integral of $w$ against a function continuous in $z$ (Definition N.26: $z \mapsto \Phi_\kappa(z)(q)$ is smooth), so it is linear in $w$ and unchanged when $w$ is changed on a null set. In particular $\mathcal{U}_w^{(\kappa)} = \mathcal{U}_{h,\kappa} - \mathcal{U}_{h^B,\kappa}$, the difference of the regularized interactions (Appendix C, notation paragraph). Lemma C.4 reads: for all $h \in \mathcal{C}_{\mathrm{E}}$ with $\operatorname{supp} h \subseteq \Lambda$ and all $\kappa \geq 2$, $\|\mathcal{U}_h - \mathcal{U}_{h,\kappa}\|^2_{L^2(d\mu_0)} \leq c_2\, \kappa^{-1} (\log \kappa)^{2p'}$, with the explicit $c_2 = c_2(\lambda, \mathscr{P}, \Lambda)$ given there. Its hypotheses hold for $h$ and for $h^B$ with the compact set $\Lambda := \operatorname{supp} h$ (both lie in $\mathcal{C}_{\mathrm{E}}$ and have support in $\Lambda$, Lemma N.5(i)) and every $\kappa \geq 2$. Hence $\mathcal{U}_{h,\kappa} \to \mathcal{U}_h$ and $\mathcal{U}_{h^B,\kappa} \to \mathcal{U}_{h^B}$ in $L^2(d\mu_0)$, and $\mathcal{U}_w^{(\kappa)} \to \mathcal{U}_h - \mathcal{U}_{h^B} = \mathcal{U}_{h-h^B}$. (Step 1 of the proof of Lemma D.1 shows from Lemma N.28(a) alone that $\mathcal{U}_w^{(\kappa)}$ converges; Lemma C.4, or Lemma N.23′(iii), identifies the limit with the $\mathcal{U}_h - \mathcal{U}_{h^B}$ of Assumption A3.) By Lemma N.5(iii), $w = \sum_{\Delta\in\mathcal{D}} h\mathbf{1}_\Delta$ almost everywhere; with $:\mathscr{P}(y):_c = \sum_n a_n :y^n:_c$ and $:y^0:_c = 1$ the linearity in $w$ gives, pointwise in $q$,

$$\mathcal{U}_w^{(\kappa)} = \sum_{\Delta\in\mathcal{D}} \Big( \lambda a_0 \int_\Delta h\, dz + \sum_{n=1}^{2p'} U_{\Delta,n}^{(\kappa)} \Big), \qquad U_{\Delta,n}^{(\kappa)} := \lambda a_n \int h\mathbf{1}_\Delta :\Phi_\kappa^n:_{c_\kappa} dz.$$

$U_{\Delta,n}^{(\kappa)}$ is the regularized monomial of Definition N.26(a) with $r = 1$, degree $n$ and kernel $\lambda a_n h\mathbf{1}_\Delta$, so by Lemma N.28(a) it converges in $L^2(d\mu_0)$ to the localized monomial $U_{\Delta,n}$ of Definition N.17. A finite sum of convergent sequences converges to the sum of the limits; hence

$$\mathcal{U}_{h-h^B} = \sum_{\Delta\in\mathcal{D}} \Big( U_{\Delta,0} + \sum_{n=1}^{2p'} U_{\Delta,n} \Big) \quad \text{in } L^2(d\mu_0), \qquad U_{\Delta,0} = \lambda a_0 \int_\Delta h\, dz.$$

This is the first identity of Lemma N.18(iv), which §D.a proves through Lemma N.5(iv), also from Lemma C.4. By Lemma N.32, $F$, every $F_a$ and every $U_{\Delta,n}$ lie in $L^2(dq_{h_\alpha})$, where $\mathrm{Cov}_{h_\alpha}$ is bilinear and vanishes against the constants $U_{\Delta,0}$; therefore

$$\mathrm{Cov}_{h_\alpha}\big(F, \mathcal{U}_{h-h^B}\big) = \sum_{\Delta\in\mathcal{D}} \sum_{n=1}^{2p'} \sum_{a\in\mathfrak{A}} \mathrm{Cov}_{h_\alpha}\big(F_a, U_{\Delta,n}\big),$$

a finite sum.

*Step 4: one term, by Source Fact 2.2.* Fix $\alpha \in [0,1]$, $\Delta \in \mathcal{D}$, $1 \leq n \leq 2p'$ and $a \in \mathfrak{A}$. Source Fact 2.2 reads: "For localized monomials $Q_1, Q_2$ with disjoint localization squares separated by a strip of width $d$ bounded by two parallel lines (any orientation), $\big|\int Q_1 Q_2\, dq_h - \int Q_1 dq_h \int Q_2 dq_h\big| \leq C_7 e^{-m_1 d} \|Q_1\|_{\mathrm{GJS}} \|Q_2\|_{\mathrm{GJS}}$, with $m_1 > 0$ and $C_7$ independent of $h$, $Q_1$, $Q_2$ (so stated in the source)." The source states: "THEOREM 1.1.7. *Let $m_0$, $K_2$ and $K_1$ be sufficiently large and let $\lambda$ be sufficiently small. There are constants $0 < m_1$ and $O(1)$ independent of $h$, $Q_1$ and $Q_2$ such that* (1.1.10) $\big|\int Q_1 Q_2 dq_h - \int Q_1 dq_h \int Q_2 dq_h\big| \leqq O(1) e^{-m_1 d} \|Q_1\|\, \|Q_2\|$" [7, p. 594], with $d$ the width of the widest separating strip, which exists by Lemma N.31; every separating strip has width at most that $d$, and $m_1 > 0$, so the printed bound implies the bound with the width of any separating strip, which is the form of Source Fact 2.2 (and the form in which §D.a applies it, after Definition N.6). We apply it with:

- $Q_1 := F_a$: a localized monomial of type (1.1.5) or (1.1.6), as given; kernel in $L^2$; every $n_i < \bar{n}$; localization squares in $R$.
- $Q_2 := U_{\Delta,n}$: a localized monomial of type (1.1.6) with $r = 1$, localized in the single square $\Delta$, degree $n \le 2p' < \bar{n}$, kernel $\lambda a_n h \mathbf{1}_\Delta \in L^2(\mathbb{R}^2)$ (Lemma N.18(i)). Page 593 defines "localized" for both forms ("For a localized monomial $Q$ or $Q(x_0)$"), and Theorem 1.1.7 is stated for "two localized monomials $Q_1$ and $Q_2$" [7, p. 594]; the mixed pair is covered by the statement.
- The objects: $Q_1$ and $Q_2$ are the limits of Definition N.26, and the measure is $dq_{h_\alpha}$ of Assumption A3. That these limits are the monomials of p. 593, which GJS print as formal integrals, and that the $\mathcal{U}_h$ of Assumption A3 is GJS's $V(h)$, is a reading of the source, recorded in Assumption A8, Scope (d) (Remark N.2(iv), Remark N.27); it is not proved here.
- The strip: $S_\Delta$ separates $\Delta$ from $\bigcup R$, hence from the localization squares of $F_a$, which lie in $\bigcup R$; its width is $w_\Delta > 0$ and its orientation is the one assigned. Its two closed half-planes are disjoint because $w_\Delta > 0$ (Definition N.6), so the localization squares of $F_a$ and $\Delta$ are disjoint, even as closed squares. If $S_\Delta$ is transverse and $F_a$ is of type (1.1.5), this is the transverse case A8(iv)-T (Assumption A8, item (iv-T)).
- The cutoff: $h_\alpha \in \mathcal{C}_{\mathrm{E}}$ (Step 1), the class of [7, p. 589], over which "independent of $h$" is read (Assumption A8, Scope (d)).
- The printed hypotheses on $m_0$, $K_2$, $K_1$, $\lambda$: the standing assumption.

Source Fact 2.2 gives

$$\left|\mathrm{Cov}_{h_\alpha}(F_a, U_{\Delta,n})\right| \le C_7\, e^{-m_1 w_\Delta}\, \|F_a\|_{\mathrm{GJS}}\, \|U_{\Delta,n}\|_{\mathrm{GJS}}.$$

*Step 5: summation.* We sum over $a$ and $n$. By Lemma N.18(ii), $\sum_n \|U_{\Delta,n}\|_{\mathrm{GJS}} \le c_U$, and $\sum_a \|F_a\|_{\mathrm{GJS}} = \|F\|_{\mathrm{GJS}}$; hence

$$\begin{aligned}\left|\mathrm{Cov}_{h_\alpha}\left(F, \mathcal{U}_{h-h^B}\right)\right| &\le \sum_{\Delta\in\mathcal{D}} C_7 e^{-m_1 w_\Delta} \Big(\sum_a \|F_a\|_{\mathrm{GJS}}\Big)\Big(\sum_n \|U_{\Delta,n}\|_{\mathrm{GJS}}\Big) \\ &\le C_7\, c_U\, \|F\|_{\mathrm{GJS}} \sum_{\Delta\in\mathcal{D}} e^{-m_1 w_\Delta}.\end{aligned}$$

The right side does not depend on $\alpha$; integrating over $\alpha \in [0,1]$ in Step 2 proves (a).

*Step 6: the form (b).* Every $\Delta \in \mathcal{D}$ is a square outside $B$ meeting $\operatorname{supp} h$ (Lemma N.5(iii)), so by hypothesis a strip of positive width separates it from $\bigcup R$, and by Lemma N.31 a widest one exists; its width is $d(\Delta, R) > 0$. Assign it as $S_\Delta$ and apply (a); since every term $e^{-m_1 d(\Delta,R)}$ is positive and $\mathcal{D} \subseteq \{\Delta \not\subseteq B : \Delta \cap \operatorname{supp} h \ne \varnothing\}$, the sum over $\mathcal{D}$ is at most the sum in (b). □

**Remark N.33** (what the lemma uses, and the orientation)**.** *(1)* The lemma uses of the cutoff $h$ exactly: $h \ge 0$ (so that $h - h^B \ge 0$ and the charged squares are defined by $\int_\Delta h > 0$), $h \le 1$ (in $c_U$ and in $h_\alpha \in \mathcal{C}_{\mathrm{E}}$), and compact support (finiteness of $\mathcal{D}$). It is a statement for $h \in \mathcal{C}_{\mathrm{E}}$ only (trap 20 of the plan); Theorem 1.1.7 and Lemma D.1 are used at cutoffs of $\mathcal{C}_{\mathrm{E}}$ only. *(2)* In (b) the widest strips are not controlled in orientation: for a square $\Delta$ far to the side of $R$ in a row near that of $R$, the widest separating strip is transverse to the time-zero line even when a narrower parallel one exists (Remark N.9: it may also be tilted). Where such a strip is used with a sharp-time $F_a$, form (b) rests on the transverse case of

Theorem 1.1.7, used at the printed statement's scope and listed as A8(iv)-T (Assumption A8, item (iv-T)). Theorems D.3 and D.4 therefore use (a), with the strips assigned explicitly and their orientation stated, so that every application of Theorem 1.1.7 has a named orientation (gate GA1 of the plan). Form (b) is recorded because it is a form of the statement of the plan; it is not used below. *(3)* The sum in (a) is over the charged squares $\mathcal{D}(h, B)$, the squares outside $B$ on which $h$ is not almost everywhere zero; squares that meet $\operatorname{supp} h$ only in a null set carry $U_\Delta = 0$ and need no strip. This matters in §D.d. An example: for $g = \mathbf{1}_{[0,1/10]} + \mathbf{1}_{\{3\}} \in \mathcal{C}_{\mathrm{H}}$, $\operatorname{supp} g = [0, \frac{1}{10}] \cup \{3\}$ meets column 3 in one point only, so $N_g = 1$; for $I \subseteq (0, \frac{1}{10})$ the box $B_s(g, I)$ of §D.d has the single column 0, and for $T \geq |l_s| + 2$ the squares $\Delta_{l,3}$ with $|l - l_s| \leq 1$ meet $\operatorname{supp} h_{T,g} \supseteq [-T, T] \times \{3\}$, lie outside $B$, and are separated from row $l_s$ only by transverse strips (Lemma N.7(iv)). With the plan's index set $\{\Delta \not\subseteq B : \Delta \cap \operatorname{supp} h \neq \varnothing\}$ they would enter the sum, and the parallel-only claim and the count $2N_g$ of Theorem D.3 would fail; they are not charged ($h_{T,g}\mathbf{1}_{\Delta_{l,3}} = 0$ almost everywhere), so (a) omits them.

## D.d Phase A: fixed cutoff, parallel strips

Theorem D.3 bounds the expectation of a sharp-time cylinder observable under the slab measures $dq_{h_{T,g}}$ uniformly in $T > 0$, for one fixed spatial cutoff $g$, with constants that depend on $g$; that is all that §7 consumes (§§4–7). Every use of GJS Theorem 1.1.7 in this section is across a strip parallel to the time-zero line.

### D.d.1 Setting

The standing assumption of §D.c.1 holds. Let $g \in \mathcal{C}_{\mathrm{H}}$ (Definition 2.1: $g : \mathbb{R} \to [0, 1]$ measurable with compact support) and, for $T > 0$, $h_{T,g} := \mathbf{1}_{[-T,T]} \otimes g$, which lies in $\mathcal{C}_{\mathrm{E}}$ by Lemma 3.1(ii). Let $I \subset \mathbb{R}$ be a bounded interval of length $L > 0$, and $s \in \mathbb{R}$, with $l_s$ the row containing time $s$ (Definition N.1). Put

$$K_g := \big\{k \in \mathbb{Z} : \ |\operatorname{supp} g \cap [k - \tfrac{1}{2}, k + \tfrac{1}{2})| > 0\big\}, \qquad N_g := \#K_g,$$
$$K_I := \big\{k \in \mathbb{Z} : \ |I \cap [k - \tfrac{1}{2}, k + \tfrac{1}{2})| > 0\big\}, \qquad N_I^{\square} = \#K_I,$$

($|\cdot|$ the Lebesgue measure on $\mathbb{R}$; $N_I^{\square}$ is the column count of Remark N.3), so that $N_g$ is the number of columns meeting $\operatorname{supp} g$ in positive measure; both sets are finite, and $K_I \neq \varnothing$ because $L > 0$. Let $k_- := \min(K_g \cup K_I)$, $k_+ := \max(K_g \cup K_I)$, $N_B := k_+ - k_- + 1$, and

$$B = B_s(g, I) := \bigcup\{\Delta_{l,k} : \ |l - l_s| \leq 1, \ k_- \leq k \leq k_+\}, \qquad R_s(I) := \{\Delta_{l_s,k} : \ k \in K_I\}.$$

So $B$ is rows $l_s - 1$ to $l_s + 1$ times the smallest interval of $N_B$ columns containing every column that meets $\operatorname{supp} g \cup I$ in positive measure, $|B| = 3N_B$, and the squares of $R_s(I)$ lie in $B$.

A *cylinder monomial* (§6) is $Q = \prod_{i=1}^r \varphi_0(f_i)$ with $r \geq 1$ and real $f_i \in C_c^\infty(\mathbb{R})$; it is *supported in* $I$ if every $f_i$ is. At sharp Euclidean time $s$ it is $Q(s) := \prod_{i=1}^r \Phi(s, f_i)$, with "$\Phi(\tau, f) \equiv \Phi(\delta_\tau \otimes f) = \int \Phi(\tau, \vec{x}) f(\vec{x}) d\vec{x}$" [7, p. 589]; GJS print no construction, and $\Phi(s, f)$ is read, as in the proof of Lemma N.21(i), as the extension of $f \mapsto \Phi(f)$ to $\delta_s \otimes f$ by continuity in the covariance norm, which is finite there (Lemma N.35). For a finite real linear combination $F = \sum_{\beta \in \mathfrak{B}} c_\beta Q_\beta$ of cylinder monomials, $F(s) := \sum_\beta c_\beta Q_\beta(s)$.

### D.d.2 Auxiliary facts

**Remark N.34** (Lemma C.2, Lemma C.4 and Theorem C.5 are load-bearing)**.** This section is the first of this appendix to consume Lemma C.2 and Theorem C.5; Theorem D.4 and

the proofs of §§5–7 that cite Theorems D.3–D.4 consume them too, and from here on they are load-bearing (trap 12 of the plan). So is Lemma C.4, which Lemma D.2 uses in Step 3 of its proof and §D.a uses for the translation covariance of $\mathcal{U}_h$ (Lemma N.13(iii)) and hence for the shape-only property of $S_2$ (Proposition N.15(iii)); and so are the structural facts of Gaussian Wick powers (the chaos decomposition, the Wick isometry, $\Gamma(e^{-t})$ on chaoses), imported in Assumptions A1 and A3. The independent audit V3 of these results (`ugjs/council/v3-appendix-c-report.md`, 2 November 2026) found them true, with every displayed computation re-derived, but resting on three inputs that were neither proved nor imported (the kernel facts of Lemma C.4, the constant $C_*$, the Wick-chaos structure of $\mathcal{U}_h$); the repairs of 7 November 2026 supply all three (Appendix C; corrections history in Appendix B). Every later use says "as in Remark N.34".

**Lemma N.35** (the sharp-time field under $d\mu_0$)**.** *For every $s \in \mathbb{R}$ and every real $f \in C_c^\infty(\mathbb{R})$, $\delta_s \otimes f$ has finite covariance norm, so $\Phi(s,f)$ of §D.d.1 is defined. The family $(\Phi(s,f))_f$ is under $d\mu_0$ a centred Gaussian family with covariance $\langle f, Cf'\rangle$, $C = (2\mu)^{-1}$, $\mu = (-\partial_x^2 + m_0^2)^{1/2}$ (§2.1); it therefore has the law of $(\varphi_0(f))_f$ under $d\phi_0$. In particular $\|F(s)\|_{L^p(d\mu_0)} = \|F\|_{L^p(d\phi_0)}$ for every $p \in [1,\infty)$ and every $F$ as in §D.d.1.*

At $s = 0$ this is the identification of the sharp-time field used in Lemma N.21(i); the lemma records it at every time $s$, which Theorem D.3 needs.

*Proof.* $d\mu_0$ is the centred Gaussian measure with covariance $(-\Delta_{\mathbb{R}^2} + m_0^2)^{-1}$ (Assumption A2). The Fourier transform of $\delta_s \otimes f$ is $e^{-ik_0 s}\hat f(k)$, and $|e^{-ik_0 s}| = 1$, so for real $f, f' \in C_c^\infty(\mathbb{R})$ the covariance form at $\delta_s \otimes f$ and $\delta_s \otimes f'$ is

$$\begin{aligned}\langle \delta_s \otimes f, (-\Delta_{\mathbb{R}^2} + m_0^2)^{-1}\, \delta_s \otimes f'\rangle &= \int \frac{dk}{2\pi}\, \overline{\hat f(k)}\hat f'(k) \int \frac{dk_0}{2\pi}\, \frac{1}{k_0^2 + \mu(k)^2} \\ &= \int \frac{dk}{2\pi}\, \overline{\hat f(k)}\hat f'(k)\, \frac{1}{2\mu(k)} = \langle f, Cf'\rangle,\end{aligned}$$

by the identity $\int \frac{dk_0}{2\pi}(k_0^2 + \mu(k)^2)^{-1} = (2\mu(k))^{-1}$ of Remark 2.5, whose computation does not involve the value of $s$; the integrals converge absolutely because $\hat f, \hat f'$ are Schwartz functions and $(2\mu)^{-1} \le (2m_0)^{-1}$. With $f = f'$ this shows that the covariance norm of $\delta_s \otimes f$ is finite. The extension of the linear isometry $f \mapsto \Phi(f)$ (from test functions with the covariance norm into $L^2(d\mu_0)$) by continuity preserves inner products, so $\mathbb{E}_{\mu_0}[\Phi(s,f)\Phi(s,f')] = \langle f, Cf'\rangle$; and $L^2$-limits of centred jointly Gaussian variables are centred and jointly Gaussian. A centred Gaussian family is determined in law by its covariance, and $d\phi_0$ is the centred Gaussian measure with covariance $C$ (§2.1). $F(s)$ is the same polynomial in the family at time $s$ as $F$ is in $(\varphi_0(f))_f$. □

**Lemma N.36** (cylinder observables as GJS monomials)**.** *Let $Q = \prod_{i=1}^r \varphi_0(f_i)$ be a cylinder monomial supported in $I$, and $s \in \mathbb{R}$.* (i) *$Q(s) = \int d\vec{x}\, w(x) \prod_{i=1}^r {:}\Phi(x_i)^{n_i}{:}$ with $n_i = 1$, all times $x_{i,0} = s$, and kernel $w = f_1 \otimes \cdots \otimes f_r \in L^2(\mathbb{R}^r)$: a monomial of type (1.1.5) [7, p. 593], since ${:}\Phi^1{:} = \Phi$; no Wick reordering is made (Assumption A8(ii): "products of individually Wick-ordered powers are admissible without reordering"). Its individual degrees are $n_i = 1 < \bar n$.* (ii) *For $\kappa = (\kappa_1, \dots, \kappa_r) \in K_I^r$ let $Q_\kappa(s)$ be the monomial of type (1.1.5) at time $s$ with kernel $w_\kappa := \bigotimes_i f_i \mathbf{1}_{[\kappa_i - \frac12, \kappa_i + \frac12)}$. It is localized in the squares $\Delta_{l_s,\kappa_1}, \dots, \Delta_{l_s,\kappa_r}$, which belong to $R_s(I)$, and $Q(s) = \sum_{\kappa \in K_I^r} Q_\kappa(s)$.* (iii) *For the norm $\|Q(s)\|_{\mathrm{GJS}} := \sum_\kappa \|Q_\kappa(s)\|_{\mathrm{GJS}}$ of this decomposition, and for $Q(s)^2$ (the cylinder monomial with the $2r$ factors $f_1, \dots, f_r, f_1, \dots, f_r$, decomposed in the same way), the two bounds (N.a) of Remark N.3(iii) hold, at every time $s$.* (iv) *$N_I^\square \le \lfloor L \rfloor + 2$ (Remark N.3(i)).* (v) *For*

*$F = \sum_\beta c_\beta Q_\beta$ as in §D.d.1, $F(s) = \sum_\beta \sum_\kappa c_\beta Q_{\beta,\kappa}(s)$ is a finite sum of localized monomials of type (1.1.5) with kernels in $L^2$, individual degrees $1 < \bar{n}$ and localization squares in $R_s(I)$, and the norm of this decomposition is*

$$\|F(s)\|_{\mathrm{GJS}} = \sum_\beta |c_\beta| \, \|Q_\beta(s)\|_{\mathrm{GJS}} \le \sum_\beta |c_\beta| \, K_1 K_2^{2r_\beta} \, r_\beta! \, (N_I^{\square})^{r_\beta/2} \prod_{i=1}^{r_\beta} \|f_{\beta,i}\|_2 .$$

(vi) *As an element of $L^2(d\mu_0)$, the localized monomial $Q_\kappa(s)$ of Definition N.26(b) (the object to which Lemma D.2 is applied) equals $\prod_i \Phi(s, f_i \mathbf{1}_{[\kappa_i - \frac12, \kappa_i + \frac12)})$ almost everywhere, and $\sum_{\kappa \in K_I^\tau} Q_\kappa(s) = Q(s)$ almost everywhere, with $Q(s)$ as in §D.d.1.*

Items (iii)–(iv) are those of Remark N.3, applied to the explicit decomposition (ii); items (i), (ii), (v) and (vi), which name the localized pieces and their squares for gate GA1 and identify them with the observable, are not in §D.a.

*Proof.* (i) is the definition (1.1.5) with $n_i = 1$. (ii) A point $(s, \vec{x}_i)$ with $\vec{x}_i \in [k - \frac12, k + \frac12)$ lies in $\Delta_{l_s,k}$, so $w_\kappa$ vanishes outside the product of the squares $\Delta_{l_s,\kappa_i}$, and the monomial is localized in them (Remark N.2(i)–(ii)). Since $f_i$ vanishes outside $I$ and $I$ meets the columns $k \notin K_I$ in null sets, $f_i = \sum_{k \in K_I} f_i \mathbf{1}_{[k-\frac12, k+\frac12)}$ almost everywhere; expanding the tensor product gives $w = \sum_\kappa w_\kappa$ a.e., hence the decomposition (made precise in (vi)). (iii) The pieces of (ii) are the decomposition $f_i = \sum_k f_i \mathbf{1}_{[k-\frac12, k+\frac12)}$ with which Remark N.3(iii) runs the proof of Proposition 6.2 on the columns; its bounds use the local degrees and kernel norms of the pieces only, which do not depend on the row $l_s$. (iv) is Remark N.3(i). (v) follows from (i)–(iii) applied to each $Q_\beta$, with $\|cQ_\kappa(s)\|_{\mathrm{GJS}} = |c| \, \|Q_\kappa(s)\|_{\mathrm{GJS}}$ by (1.1.8) (the kernel is multiplied by $c$). (vi) The kernel $w_\kappa$ is a product of the functions $f_i \mathbf{1}_{[\kappa_i - \frac12, \kappa_i + \frac12)} \in L^2(\mathbb{R})$, each with bounded support and vanishing outside the spatial projection of $\Delta_{l_s,\kappa_i}$. By Lemma N.28(c), $Q_\kappa(s)$ is almost everywhere the product of the one-factor monomials of degree 1 at time $s$ with these kernels, and by Lemma N.28(f)(1) each of them is $\Phi(s, f_i \mathbf{1}_{[\kappa_i - \frac12, \kappa_i + \frac12)})$. The map $f \mapsto \Phi(s, f)$ is linear and, being isometric for the covariance norm, takes almost everywhere equal functions to the same element; so, with $f_i = \sum_{k \in K_I} f_i \mathbf{1}_{[k-\frac12, k+\frac12)}$ a.e., $\sum_\kappa \prod_i \Phi(s, f_i \mathbf{1}_{[\kappa_i - \frac12, \kappa_i + \frac12)}) = \prod_i \Phi(s, f_i) = Q(s)$. (Remark N.23(c) records the degree-1 identification for $f \in C_c^\infty(\mathbb{R})$; the pieces have the kernels $f_i \mathbf{1}_{[\kappa_i - \frac12, \kappa_i + \frac12)}$, which are not smooth, and are covered by Lemma N.28(f)(1), and also by Lemma N.22 with $n = 1$.) □

**Remark N.37** (the former cell count fails in (7) and (8); a finding, repaired)**.** Remark N.3(ii)–(iii) shows that the cell count $N_I^{\mathrm{cell}}$, which §6 used until 7 November 2026, and the column count $N_I^{\square}$ differ, and that (7) fails with $N_I^{\mathrm{cell}}$ on GJS's squares ($I = (0.4, 0.6)$, $Q = \varphi_0(f)$: $\sqrt{2} K_1 K_2^2 \|f\|_2$ against $K_1 K_2^2 \|f\|_2$). On GJS's squares the cells $[j, j+1)$ are not localization regions, so the decomposition of the former proof of Proposition 6.2 is not available there; the bound fails for the decomposition by columns, which is what that proof becomes on GJS's squares, and, for $r = 1$, for every decomposition of $\Phi(s, f)$ into one-factor pieces of degree 1 at time $s$: such pieces are $\Phi(s, f_a)$ with $f_a$ vanishing outside one column, $\sum_a f_a = f$ almost everywhere because $f \mapsto \Phi(s, f)$ is injective ($C$ is positive definite), so the kernels of the pieces in column $c$ sum to $f \mathbf{1}_c$, and the triangle inequality gives a norm sum $\ge K_1 K_2^2 (\|f \mathbf{1}_{[-\frac12, \frac12)}\|_2 + \|f \mathbf{1}_{[\frac12, \frac32)}\|_2) = \sqrt{2} K_1 K_2^2 \|f\|_2$ for $f$ even about $\frac12$. (Decompositions with pieces of other degrees are not examined here.) The same example fails (8), which is the bound Lemma 7.1 consumes: for $f \in C_c^\infty(I)$ real, even about $\frac12$, with $\|f\|_2 = 1$, the column decomposition (ii) of $Q(s)^2 = \Phi(s, f)^2$ has the four pieces $\kappa \in \{0, 1\}^2$, with norms $K_1 K_2^4 \cdot 2! \cdot \frac12$ for $\kappa = (0, 0)$ and $(1, 1)$ and $K_1 K_2^4 \cdot \frac12$ for $\kappa = (0, 1)$ and $(1, 0)$, in total $3 K_1 K_2^4$, while the

right side of (8) with $N_I^{\mathrm{cell}} = 1$ is $K_1 K_2^4 \cdot 2! \cdot 1 \cdot 1 = 2K_1 K_2^4$. *Severity, as a finding about the former text:* CRITICAL for the display of Proposition 6.2 as then stated, on the printed norm of Assumption A8(iii) (GJS's squares); the displayed bounds of Lemmas 7.1 and 7.6, whose proofs consume (8), failed as stated; MINOR downstream, for Theorems 7.7, 7.8 and 12.2, which use finiteness only. Repaired on 7 November 2026 by counting columns (§6, the notation table and Assumption A8(ii)), which changes the statements of Lemma 6.1 and Proposition 6.2 and the displayed constants $\mathfrak{n}(Q)$ and $K'$. (The counts and the value $3K_1K_2^4$ for a bump function are reproduced by `ugjs/parts/aqft_checks.py`, check 6, *numerical*.)

**Lemma N.38** (geometry of Phase A: parallel strips)**.** *Let $T > 0$, $h := h_{T,g}$, $B := B_s(g, I)$ and $R := R_s(I)$.* (i) *If $\Delta_{l,k} \in \mathcal{D}(h, B)$, then $k \in K_g$, $|l - l_s| \geq 2$, and row $l$ meets $[-T, T]$ in positive measure.* (ii) *For every $k \in \mathbb{Z}$ and every $l$ with $|l - l_s| \geq 2$, the strip*

$$P_l := \{z : \ \min(l, l_s) + \tfrac{1}{2} < x^0 < \max(l, l_s) - \tfrac{1}{2}\},$$

*bounded by two lines of constant time, hence* parallel to the time-zero line*, of width $|l - l_s| - 1 \geq 1$, separates $\Delta_{l,k}$ from $\bigcup R$ (indeed from the whole row $l_s$).* (iii) *With $w_{\Delta_{l,k}} := |l - l_s| - 1$,*

$$\sum_{\Delta \in \mathcal{D}(h,B)} e^{-m_1 w_\Delta} \leq 2N_g \sum_{j \geq 1} e^{-m_1 j} = \frac{2N_g\, e^{-m_1}}{1 - e^{-m_1}}.$$

*Proof.* (i) By Tonelli's theorem, $\int_{\Delta_{l,k}} h\, dz = \big|[-T, T] \cap [l - \tfrac{1}{2}, l + \tfrac{1}{2})\big| \cdot \int_{k-1/2}^{k+1/2} g(x^1)\, dx^1$. If this is positive, both factors are; the second gives $|\{g > 0\} \cap [k - \tfrac{1}{2}, k + \tfrac{1}{2})| > 0$, and $\{g > 0\} \subseteq \operatorname{supp} g$, so $k \in K_g$, hence $k_- \leq k \leq k_+$. A square of a column in $[k_-, k_+]$ that is not in $B$ has $l \notin \{l_s - 1, l_s, l_s + 1\}$, i.e. $|l - l_s| \geq 2$. (ii) Lemma N.7(i), applied to the rectangle $S := \{\Delta_{l_s,k'} : \min K_I \leq k' \leq \max K_I\} \supseteq R$ (one row, $l_1 = l_2 = l_s$) and to $\Delta = \Delta_{l,k} \notin S$, whose row distance from $S$ is $\rho_t = |l - l_s| \geq 2$: the strip bounded by $x^0 = l_s + \frac{1}{2}$ and $x^0 = l - \frac{1}{2}$ (if $l > l_s$), resp. $x^0 = l + \frac{1}{2}$ and $x^0 = l_s - \frac{1}{2}$ (if $l < l_s$), which is $P_l$, is parallel, has width $\rho_t - 1$, and separates $\Delta$ from $\bigcup S \supseteq \bigcup R$. The same two half-planes contain the whole row $l_s$. (iii) By (i) every $\Delta_{l,k} \in \mathcal{D}(h, B)$ has $k \in K_g$ and $|l - l_s| \geq 2$; the terms are positive, so the sum is at most $\sum_{k \in K_g} \sum_{|l - l_s| \geq 2} e^{-m_1(|l - l_s| - 1)}$, in which $j := |l - l_s| - 1$ takes every value $j \geq 1$ for exactly two $l$. The geometric series with ratio $e^{-m_1} \in (0, 1)$ gives the closed form. (Test: $N_g = 1$, $e^{-m_1} = \frac{1}{2}$: both sides equal 2.) □

### D.d.3 The theorem

**Theorem D.3** (Phase A: fixed cutoff, parallel strips)**.** *Assume the standing assumption of §D.c.1. Let $g \in \mathcal{C}_{\mathrm{H}}$, let $I \subset \mathbb{R}$ be a bounded interval of positive length, let $s \in \mathbb{R}$, and let $l_s$, $N_g$, $N_B$, $B = B_s(g, I)$ be as in §D.d.1. Let $F = \sum_\beta c_\beta Q_\beta$ be a finite real linear combination of cylinder monomials supported in $I$, and $F(s)$ its value at sharp time $s$. Then for every $T > 0$,*

$$\Big| \int F(s)\, dq_{h_{T,g}} \Big| \leq A\, \|F\|_{L^2(d\phi_0)} + B'\, \|F(s)\|_{\mathrm{GJS}},$$

$$A := e^{3\lambda |a_0| N_B}\, S_2(B), \qquad B' := 2\, C_7\, c_U\, N_g\, \frac{e^{-m_1}}{1 - e^{-m_1}},$$

*where $S_2(B)$ is the constant of Definition N.14. The norm $\|F(s)\|_{\mathrm{GJS}}$ is that of the decomposition in Lemma N.36(v), which bounds it in terms of the functions $f_{\beta,i}$. In working coordinates (Remark 3.3), the constant $A$ depends only on $(\lambda, \mathscr{P}, m_0)$ and on $N_B$, through the unevaluated constant $S_2$; the constant $B'$ depends only on $C_7$, $m_1$, $K_1$, $K_2$, $\lambda$, $\mathscr{P}$ and $N_g$. Both are independent of $T$ and $s$, and $B'$ is independent of $I$.*

*Status of Theorem D.3. Proved*, from Lemma D.1 and Lemma D.2, Lemma C.2, Theorem C.5 with Proposition N.15 (whose part (iii), resting on Lemma C.4, gives the independence of $A$ from $s$), all as in Remark N.34, under the reading of the source of Remark N.2(iv), which it inherits through Lemma D.2 (the pieces of Lemma N.36(vi) and the $U_{\Delta,n}$ are taken as GJS's monomials, and the $\mathcal{U}_h$ of Assumption A3 as GJS's $V(h)$); C5 items 1–4 and 10 are open. It uses GJS Theorem 1.1.7 in the parallel case only, with a time-integrated partner (Remark N.39); it does not use A8(iv)-T or GJS Theorem 1.1.11.

*Proof. Step 1: the observable.* By Lemma N.36(v), $F(s) = \sum_{\beta,\kappa} c_\beta Q_{\beta,\kappa}(s)$ is a finite sum of localized monomials of type (1.1.5), with kernels in $L^2$, individual degrees $n_i = 1 < \bar{n}$, and localization squares in $R := R_s(I)$, and by Lemma N.36(vi) the sum of these pieces, as objects of Definition N.26, is the observable $F(s)$ almost everywhere; $B$ is a box containing the squares of $R$; $\|F(s)\|_{\mathrm{GJS}} = \sum_{\beta,\kappa} |c_\beta| \, \|Q_{\beta,\kappa}(s)\|_{\mathrm{GJS}}$. Put $h := h_{T,g} \in \mathcal{C}_{\mathrm{E}}$.

*Step 2: the base term.* By Lemma N.5(i), $h^B \in \mathcal{C}_{\mathrm{E}}$ and $\operatorname{supp} h^B \subseteq \overline{B}$, a compact set of Lebesgue measure $|B| = 3N_B$. Lemma C.2 (as in Remark N.34) states: for every $h \in \mathcal{C}_{\mathrm{E}}$ with $\operatorname{supp} h \subseteq \Lambda$, $Z(h) \geq e^{-\lambda |a_0| |\Lambda|} > 0$; its hypotheses ($h \in \mathcal{C}_{\mathrm{E}}$, $\operatorname{supp} h \subseteq \Lambda$, $\Lambda$ compact as in its notation) hold for $h^B$ and $\Lambda = \overline{B}$, so $Z(h^B) \geq e^{-3\lambda |a_0| N_B}$. Proposition N.15(iv) (as in Remark N.34) gives $\|e^{-\mathcal{U}_{h^B}}\|_{L^2(d\mu_0)} \leq S_2(B)$. By Cauchy–Schwarz in $L^2(d\mu_0)$ and Lemma N.35,

$$\left|\langle F(s)\rangle_{h^B}\right| = \frac{\left|\int F(s)\, e^{-\mathcal{U}_{h^B}}\, d\mu_0\right|}{Z(h^B)} \leq e^{3\lambda |a_0| N_B} \, \|F(s)\|_{L^2(d\mu_0)} \, \|e^{-\mathcal{U}_{h^B}}\|_{L^2(d\mu_0)} \leq A \, \|F\|_{L^2(d\phi_0)}.$$

*Step 3: the comparison term.* Apply Lemma D.2(a) with $F(s)$ and its pieces (Step 1), $R = R_s(I)$, the box $B = B_s(g, I)$, the cutoff $h = h_{T,g} \in \mathcal{C}_{\mathrm{E}}$, and, for $\Delta = \Delta_{l,k} \in \mathcal{D}(h, B)$, the strip $S_\Delta := P_l$ of Lemma N.38(ii), of width $w_\Delta = |l - l_s| - 1$ (by Lemma N.38(i), $|l - l_s| \geq 2$, so $P_l$ is defined and separates $\Delta$ from $\bigcup R$). Each estimate inside that lemma is Source Fact 2.2 with (gate GA1):

- $Q_1 = c_\beta Q_{\beta,\kappa}(s)$: type (1.1.5), $r_\beta$ factors of degree $1 < \bar{n}$ at time $s$, localized in $\Delta_{l_s,\kappa_1}, \ldots, \Delta_{l_s,\kappa_{r_\beta}} \in R_s(I)$, kernel $c_\beta \bigotimes_i f_{\beta,i} \mathbf{1}_{[\kappa_i - \frac{1}{2}, \kappa_i + \frac{1}{2})} \in L^2$;
- $Q_2 = U_{\Delta,n}$, $1 \leq n \leq 2p'$: type (1.1.6), localized in the one square $\Delta = \Delta_{l,k}$, $k \in K_g$, $|l - l_s| \geq 2$, degree $n < \bar{n}$, kernel $\lambda a_n h \mathbf{1}_\Delta \in L^2(\mathbb{R}^2)$, norm $\leq \lambda |a_n| K_1 K_2^{2n} n!$ (Lemma N.18(i));
- the cutoff $h_\alpha = h^B + \alpha(h - h^B) \in \mathcal{C}_{\mathrm{E}}$, $0 \leq \alpha \leq 1$ (Lemma N.5(ii));
- the strip $P_l$: *parallel to the time-zero line*, width $|l - l_s| - 1 \geq 1$;
- the resulting term, summed over $n$: $C_7 \, c_U \, |c_\beta| \, \|Q_{\beta,\kappa}(s)\|_{\mathrm{GJS}} \, e^{-m_1(|l - l_s| - 1)}$.

Lemma D.2(a) and Lemma N.38(iii) give

$$\begin{aligned} \left|\langle F(s)\rangle_h - \langle F(s)\rangle_{h^B}\right| &\leq C_7 \, c_U \, \|F(s)\|_{\mathrm{GJS}} \sum_{\Delta \in \mathcal{D}(h,B)} e^{-m_1 w_\Delta} \\ &\leq C_7 \, c_U \, \|F(s)\|_{\mathrm{GJS}} \, \frac{2 N_g e^{-m_1}}{1 - e^{-m_1}} = B' \, \|F(s)\|_{\mathrm{GJS}}. \end{aligned}$$

*Step 4.* The triangle inequality and Steps 2–3 give the bound. The constants: $S_2(B)$ depends on $(\lambda, \mathscr{P}, m_0)$ through Theorem C.5 and on $B$ only through its shape $3 \times N_B$ (Proposition N.15(iii)), so $A$ does not depend on $s$ (which moves $B$ by whole rows) or on $T$; $B'$ contains $N_g$ and the printed constants and nothing that depends on $T$, $s$ or $I$. □

### D.d.4 Remarks on Theorem D.3

**Remark N.39** (orientation, and what the printed proof reaches)**.** Every application of Source Fact 2.2 in the proof of Theorem D.3 pairs a sharp-time piece at time $s$ with a time-integrated $U_{\Delta,n}$ in a row $l$, $|l - l_s| \geq 2$, across the strip $P_l$, whose bounding lines are lines of constant time. No strip transverse to the time-zero line is used in this section. This is the parallel case, with a time-integrated partner, which the printed proof reaches only after slicing. The item A8(iv)-T (Assumption A8, item (iv-T)) and Lemma 7.4 concern pairs of sharp-time monomials: the printed proof translates the strip and takes "$A = Q_1$, $B_T = Q_2$ in Theorem 1.1.11" [7, p. 597], where "We assume that $A$ is localized at times $x_{i0} \leqq 0$, and that $B$ is localized at times $x_{i0} \geqq 0$" [7, p. 596], and Theorem 1.1.11 is printed for "monomials of type (1.1.5)" [7, p. 596]. The partner $U_{\Delta,n}$ is of type (1.1.6); the printed proof reaches the pair only after $U_{\Delta,n}$ is sliced in time and the bound is integrated over the slices, the multi-time reading of (1.1.5) that the plan records (Part 2) and GJS do not print. This appendix uses the printed statement of Theorem 1.1.7, which covers the pair (Step 4 of the proof of Lemma D.2), and does not need to decide whether the printed proof reaches it. The strips are parallel because $B$ contains all columns of $\operatorname{supp} g$ in the three rows $l_s - 1, l_s, l_s + 1$: every charged square outside $B$ lies above or below them (Lemma N.38(i)). The price is that $A$ depends on $N_B \geq N_g$; this is why Theorem D.3 is a fixed-$g$ statement and Theorem D.4 needs transverse strips. Remark N.33(3) is used here: squares of the rows of $B$ outside its columns that meet $\operatorname{supp} h_{T,g}$ only in a null set are not charged.

**Remark N.40** (order of quantifiers)**.** For every $g \in \mathcal{C}_{\mathrm{H}}$ and every bounded interval $I$ of positive length the constants $A$ (through $N_B$) and $B'$ (through $N_g$) are fixed; then the bound holds for every $s \in \mathbb{R}$, every finite real linear combination $F$ of cylinder monomials supported in $I$, and every $T > 0$. Neither constant depends on $s$, $F$ or $T$; $A$ depends on $I$ through $N_B$ only, and $B'$ not at all.

**Remark N.41** (boundary cases)**.** $T < 1$. Row $l$ meets $[-T, T]$ in positive measure if and only if $l - \frac{1}{2} < T$ and $l + \frac{1}{2} > -T$. *(a) $T \leq \frac{1}{2}$:* only row 0 does. If moreover $|l_s| \leq 1$, then $\mathcal{D}(h_{T,g}, B) = \varnothing$ by Lemma N.38(i), so $h_{T,g} - h^B_{T,g} = 0$ almost everywhere (Lemma N.5(iii)) and the comparison term vanishes; if $|l_s| \geq 2$, then $h^B_{T,g} = 0$ almost everywhere (everywhere unless $T = \frac{1}{2}$ and $l_s = 2$, when the null set $\{x^0 = \frac{1}{2}\}$ lies in row $1 \subseteq B$), the base term is $\int F(s) d\mu_0$, and $\mathcal{D}$ consists of row-0 squares in columns of $K_g$, at row distance $|l_s| \geq 2$. *(b) $\frac{1}{2} < T < 1$:* exactly the rows $-1, 0, 1$ meet $[-T, T]$ in positive measure. If $l_s = 0$, every charged square lies in rows $-1, 0, 1$ and in a column of $K_g$, hence in $B$, and $\mathcal{D}(h_{T,g}, B) = \varnothing$; otherwise the charged squares outside $B$ lie in rows $l \in \{-1, 0, 1\}$ with $|l - l_s| \geq 2$, and the general case of the proof applies to them. *$T$ not an integer* enters only through the rows met by the slab. *$s$ on a lattice line*, $s \in \frac{1}{2} + \mathbb{Z}$: then $l_s = s + \frac{1}{2}$, the sharp-time pieces lie on the lower edge of row $l_s$, and for $l \leq l_s - 2$ the upper bounding line of $P_l$ is $x^0 = s$; the observable lies on the boundary line, outside the open strip, which is what separation requires. Assigning such an $s$ to the lower row instead (closed squares) moves $B$ down one row; the proof is unchanged. *$I$ meeting one column only*: $K_I = \{k\}$, $R = \{\Delta_{l_s,k}\}$, $N_I^{\square} = 1$. $N_g = 0$ (which implies $g = 0$ a.e.): $h_{T,g} = 0$ a.e., $B' = 0$, and the bound is the base term. Conversely, if $g = 0$ a.e., no square is charged and the comparison term vanishes, whatever $N_g$ is; $B'$ then overestimates it (example: $g = \mathbf{1}_{\mathbb{Q} \cap [0,1]} \in \mathcal{C}_{\mathrm{H}}$ vanishes a.e., while $\operatorname{supp} g = [0, 1]$ gives $N_g = 2$). $N_g = 1$: $B' = 2C_7 c_U e^{-m_1}(1 - e^{-m_1})^{-1}$. *The $n = 0$ term of $\mathscr{P}$* enters only through $Z(h^B)$ (the factor $e^{3\lambda|a_0|N_B}$) and through $\mathcal{U}_{h^B}$ inside $S_2$, and is absent from $c_U$ (Lemma N.18(iii)).

**Remark N.42** (what Theorem D.3 is not, and its inputs)**.** It is not CE1 at slab cutoffs: its bound is $A\|F\|_{L^2} + B'\|F(s)\|_{\mathrm{GJS}}$, with $A$ growing with the observable's box (through

$N_B$ and $S_2(B)$), not $C_8\|F(s)\|_{\mathrm{GJS}}$. The statement "CE1 follows from CE2" is not made; what is proved is a bound of this form for observables of bounded support (trap 8 of the plan). It is uniform in $T$ and $s$ at fixed $g$ and $I$, and it is not uniform in $g$: $A$ and $B'$ grow with $N_B$ and $N_g$. Of GJS only Theorem 1.1.7 is used, in its parallel case; Theorem 1.1.11 and Assumption A8(vi) are not used, and none of GJS's proofs is verified here. The inputs from the rest of this manuscript are: Assumptions A2, A3 and A8(i)–(iv), Remark 2.5 (Lemma N.35), Lemma 3.1(ii) ($h_{T,g} \in \mathcal{C}_{\mathrm{E}}$), Lemma C.2, Theorem C.5, and Lemma C.4, through Lemma D.2 (Step 3 of its proof) and through Proposition N.15(iii) (the independence of $A$ from $s$); with Lemmas D.1 and D.2, all as in Remark N.34 and the status line after Theorem D.3.

## D.e Phase B: uniform in the cutoff, transverse strips

Theorem D.4 proves the moment bound that Theorem 5.1 consumes, for every cutoff $h \in \mathcal{C}_{\mathrm{E}}$ at once, hence uniformly in $T > 0$ and $g \in \mathcal{C}_{\mathrm{H}}$ when $h = h_{T,g}$. Its proof uses GJS Theorem 1.1.7 also across strips transverse to the time-zero line. That case is covered by the printed statement and not reached by its printed proof; it is disclosed by the item A8(iv)-T of Assumption A8, item (iv-T), and every place below where it is used says so.

### D.e.1 Setting

The standing assumption of §D.c.1 holds. Let $h \in \mathcal{C}_{\mathrm{E}}$, $a \in \mathbb{R}$, and let $h'$ be a real measurable function on $\mathbb{R}$ with $\|h'\|_\infty \le 1$ and $\operatorname{supp} h' \subseteq [a, a+2]$.

*The observable.* $V(h') := \lambda \sum_{n=0}^{2p'} a_n W_n(h') \in L^2(d\phi_0)$ is the time-zero observable constructed in Lemma N.20(iv), $W_n(h')$ the limit of Lemma N.20(iii). Its sharp-time version at time 0 is the element

$$V(h')(0) := Y = J_0 V(h') \in L^2(d\mu_0)$$

of Lemma N.21, with $J_0$ the isometry of the proof of its part (i), and $Q_V := V(h')(0)^2 = Y^2$. Formally $V(h')(0) = \lambda \sum_n a_n \int h'(x) :\Phi(0,x)^n: dx$, the right side of (4), before squaring, with $h'$ for its $h$. By Lemma N.22 and Remark N.23(b), $Q_V$ is also, almost everywhere, the sum of the localized monomials of Definition N.26(b) to which Lemma D.2 is applied (Lemma N.43(i) below); so the base term and the comparison term of the proof concern one and the same element of $L^2(d\mu_0)$. Theorem D.4 is a statement about this constructed $V(h')$. The $V(h')$ of §2.1 is the same element of $L^2(d\phi_0)$ by Lemma N.23′(iii); Theorem D.4 does not use that identification, which enters only at Theorem 5.1, Step 1, and at the transfer of Theorem D.4 to Theorem 5.1 (Step 5 there).

*The box and the rings.* We use Definition N.10: $k_a$, the row-0 squares $S_a = \{\Delta_{0,k} : k_a \le k \le k_a + 2\}$, the box $B_{3\times5} = B_{3\times5}(a)$ (rows $-1, 0, 1$, columns $k_a - 1, \dots, k_a + 3$; $|B_{3\times5}| = 15$), and the rings $\mathcal{L}_j$, $j \ge 1$. By Lemma N.11(i) the columns that meet $[a, a+2]$ are exactly $k_a, k_a+1, k_a+2$, so $[a, a+2] \subseteq [k_a - \frac{1}{2}, k_a + \frac{5}{2})$, and the time 0 lies in row 0. Put $R := S_a$.

*The ring constant.* With $x := e^{-m_1} \in (0,1)$ let

$$\overline{\Sigma}_{3\times5} := \sum_{j\ge1} (8j+12)\, e^{-m_1 j} = \frac{4x(5-3x)}{(1-x)^2},$$

the right side of Lemma N.11(vi). That lemma writes $\Sigma_{3\times5}$ for the sum $\sum_{\Delta \not\subseteq B_{3\times5}} e^{-m_1 d(\Delta,R)}$, which depends on $R$, and bounds it by $\overline{\Sigma}_{3\times5}$. Theorem D.4 is stated and proved with $\overline{\Sigma}_{3\times5}$, because its proof applies Theorem 1.1.7 only along the axis-parallel strips of Lemma N.11(iv),

whose orientation is named; using the widest strips instead would give the smaller $\Sigma_{3\times5}$, at the price of tilted (hence transverse) strips also for squares with $|l| = 1 + j$ (Remark N.9).

*The Gaussian constant.* $V_* := \sup_{h'} \|V(h')\|^2_{L^2(d\phi_0)}$, the supremum over all $h'$ as above and all $a$, is finite by Lemma N.20(iv), with $V_* \le \overline{V}_* = \lambda^2 \sum_{n=0}^{2p'} a_n^2\, n!\, I_n$ there. It concerns the constructed $V(h')$ and does not use the identification of Lemma N.23′.

### D.e.2 The observable

**Lemma N.43** (the observable $Q_V$)**.** *For $0 \le n, n' \le 2p'$ and $k, k' \in \{k_a, k_a+1, k_a+2\}$ let $h'_k := h' \mathbf{1}_{[k-\frac12, k+\frac12)}$, and let $Q_{n,n',k,k'}$ be the localized monomial of Definition N.26(b) at time 0 with two factors of degrees $n, n'$ and kernel $\lambda^2 a_n a_{n'} h'_k \otimes h'_{k'}$, formally*

$$Q_{n,n',k,k'} = \lambda^2 a_n a_{n'} \iint h'_k(x)\, h'_{k'}(y)\, {:}\Phi(0,x)^n{:}\, {:}\Phi(0,y)^{n'}{:}\, dx\, dy.$$

(i) *$Q_V = \sum_{n,n'} \sum_{k,k'} Q_{n,n',k,k'}$ $d\mu_0$-almost everywhere.* (ii) *Each $Q_{n,n',k,k'}$ is a monomial of type (1.1.5), in the sense of Definition N.26(b), with $r = 2$ factors at time 0, of individual degrees $n, n' \le 2p' < \bar n$, with kernel $\lambda^2 a_n a_{n'} h'_k \otimes h'_{k'} \in L^2(\mathbb{R}^2)$, localized in $\Delta_{0,k}$ and $\Delta_{0,k'}$, which belong to $S_a$. Its total degree $n + n' \le 4p'$ may exceed $\bar n - 1 = 2p'$; this is admissible because the source bounds only the individual degrees, "We assume a bound on the $n_i$ in (1.1.5), $n_i < \bar n$", and adds "Since there is no restriction on $r$, $Q$ still may have an arbitrarily large degree" [7, p. 593]. This is where that sentence is used. No Wick re-expansion of the product is made: as in Step 1 of the proof of Theorem 5.1, the product of two individually Wick-ordered powers is itself of type (1.1.5).* (iii) *For this decomposition, (1.1.9) and (1.1.8) give*

$$\|Q_V\|_{\mathrm{GJS}} := \sum_{n,n',k,k'} \|Q_{n,n',k,k'}\|_{\mathrm{GJS}} \le 9\, K_1 K_2^{8p'} (4p')!\, \lambda^2 \Big( \sum_{n=0}^{2p'} |a_n| \Big)^2.$$

Part (i) is Remark N.23(b) of §D.a, restated with the present notation; (ii)–(iii) are Steps 1–4 of the proof of Theorem 5.1, transcribed to the columns of $S_a$. That the limits of Definition N.26 are the monomials that GJS print as formal integrals is a reading of the source (Remark N.2(iv), Remark N.27).

*Proof.* (i) Each $h'_k$ is real, measurable, bounded by 1, vanishes outside $[a, a+2]$ and outside the column $[k - \frac12, k + \frac12)$, so it satisfies the hypotheses of Lemma N.22 (bounded, compact support) and of Remark N.23(a) (vanishing outside one column). Since $h'$ vanishes outside $[a, a+2] \subseteq [k_a - \frac12, k_a + \frac52)$, $h' = \sum_{k=k_a}^{k_a+2} h'_k$ pointwise. Remark N.23(b) gives, almost everywhere, $Q_{n,n',k,k'} = \lambda^2 a_n a_{n'}\, J_0 W_n(h'_k)\, J_0 W_{n'}(h'_{k'})$ and $W_n(h') = \sum_k W_n(h'_k)$; hence, $J_0$ being linear,

$$\sum_{n,n',k,k'} Q_{n,n',k,k'} = \Big( J_0\, \lambda \sum_n a_n \sum_k W_n(h'_k) \Big)^2 = \big( J_0 V(h') \big)^2 = Y^2 = Q_V.$$

(ii) The kernel is a product of bounded functions with compact support, hence in $L^2(\mathbb{R}^2)$, and it vanishes unless $x \in [k - \frac12, k + \frac12)$ and $y \in [k' - \frac12, k' + \frac12)$, i.e. unless $(0,x) \in \Delta_{0,k}$ and $(0,y) \in \Delta_{0,k'}$. A factor of degree 0 is ${:}\Phi^0{:} = 1$ and contributes 0 to the local degree. (iii) This is Steps 3–4 of the proof of Theorem 5.1 with the cells $[j, j+1)$ replaced by GJS's columns. For one ordered pair $(k, k')$ and degrees $(n, n')$: if $k = k'$ the local degree of $\Delta_{0,k}$ is $n + n'$ and the weight of (1.1.8) is $K_1 K_2^{2(n+n')} (n+n')!$; if $k \ne k'$ it is $K_1 K_2^{2n} n!\, K_2^{2n'} n'!$. In both cases the weight is at most $K_1 K_2^{8p'} (4p')!$, because $K_2 \ge 1$, $n + n' \le 4p'$ and $n!\, n'! \le (n+n')! \le (4p')!$.

The kernel norm is $\lambda^2|a_n||a_{n'}|\,\|h'_k\|_2\|h'_{k'}\|_2 \le \lambda^2|a_n||a_{n'}|$, since $\|h'_k\|_2^2 = \int_{k-1/2}^{k+1/2} h'^2 \le 1$. There are at most $3^2 = 9$ ordered column pairs, and $\sum_{n,n'}|a_n||a_{n'}| = (\sum_n |a_n|)^2$. (This is (5); its former final equality "$= M^2/C_8$" was deleted on 7 November 2026, item n14 of the plan. Test, $\mathscr{P}(\xi) = \xi^4$: the bound is $9 \cdot 8!\, K_1 K_2^{16} \lambda^2 = 362\,880\, K_1 K_2^{16} \lambda^2$.) □

### D.e.3 The theorem

**Theorem D.4** (Phase B: uniform in the cutoff)**.** *Assume the standing assumption of §D.c.1. For every $h \in \mathcal{C}_{\mathrm{E}}$, every $a \in \mathbb{R}$ and every real measurable $h'$ with $\|h'\|_\infty \le 1$ and* $\operatorname{supp} h' \subseteq [a, a+2]$,

$$\int V(h')(0)^2\, dq_h \le M_B^2,$$

$$M_B^2 := e^{15\lambda|a_0|}\, S_2(B_{3\times 5}) \cdot 9^{p'}\, V_* + C_7\, c_U\, \overline{\Sigma}_{3\times 5} \cdot 9\, K_1 K_2^{8p'} (4p')!\, \lambda^2 \Big(\sum_{n=0}^{2p'} |a_n|\Big)^2,$$

*with $V(h')(0)$, $V_*$ and $\overline{\Sigma}_{3\times 5}$ as in §D.e.1, $c_U$ of Lemma N.18, and $S_2(B_{3\times 5})$ the common value of Definition N.14 at all boxes of 3 rows and 5 columns (Proposition N.15(iii)). In working coordinates (Remark 3.3), $M_B$ depends on $(\lambda, \mathscr{P}, m_0)$ through $C_7$, $m_1$, $K_1$, $K_2$, the unevaluated finite constant $S_2(B_{3\times 5})$ of Theorem C.5, and $V_*$; it does not depend on $h$, $a$ or $h'$. The proof uses GJS Theorem 1.1.7 across strips transverse to the time-zero line, a case its printed proof does not reach (A8(iv)-T, Assumption A8, item (iv-T); §D.e.4).*

*Status of Theorem D.4. Proved*, from Lemmas D.1 and D.2, Lemma C.2, Theorem C.5 with Proposition N.15 (including the common value of $S_2$ at all $3 \times 5$ boxes), and the comparison of Lemma N.21(ii), all as in Remark N.34; from Assumption A8(iv) at the printed statement's scope, including its transverse case A8(iv)-T, which the printed proof does not reach; under the reading of the source of Remark N.2(iv), which it inherits through Lemma D.2 (the pieces $Q_{n,n',k,k'}$ and the $U_{\Delta,n''}$ are taken as GJS's monomials, and the $\mathcal{U}_h$ of Assumption A3 as GJS's $V(h)$); C5 items 1–4, 9(c) and 10–12 are open. It does not use the identification of Lemma N.23′, and Lemma N.22 uses none of the results of Appendix C beyond its regularization (Remark N.23(d)).

*Proof. Step 1: the observable and the box.* By Lemma N.43, $Q_V = V(h')(0)^2$ is, almost everywhere, the finite sum of the localized monomials $Q_{n,n',k,k'}$ of type (1.1.5), with kernels in $L^2$, individual degrees $\le 2p' < \bar{n}$ and localization squares in $R = S_a$, and $\|Q_V\|_{\mathrm{GJS}} \le 9K_1K_2^{8p'}(4p')!\lambda^2(\sum_n |a_n|)^2$ for this decomposition. So Steps 2 and 4 below bound expectations of one and the same element of $L^1(dq_h)$ (Lemma N.32); almost everywhere equal functions have equal integrals under $dq_h$, which has a density with respect to $d\mu_0$. $B := B_{3\times 5}$ is a box containing the squares of $R$. Since $Q_V \ge 0$, $\int Q_V\, dq_h = \langle Q_V\rangle_{h^B} + \big(\langle Q_V\rangle_h - \langle Q_V\rangle_{h^B}\big)$ with $\langle Q_V\rangle_{h^B} \ge 0$.

*Step 2: the base term.* $h^B \in \mathcal{C}_{\mathrm{E}}$ with $\operatorname{supp} h^B \subseteq \overline{B}$, a closed $3 \times 5$ rectangle of Lebesgue measure 15 (Lemma N.5(i)). Lemma C.2 with $\Lambda = \overline{B}$ (hypotheses: $h^B \in \mathcal{C}_{\mathrm{E}}$, $\operatorname{supp} h^B \subseteq \Lambda$, $\Lambda$ compact) gives $Z(h^B) \ge e^{-15\lambda|a_0|}$. Proposition N.15(iv) gives $\|e^{-\mathcal{U}_{h^B}}\|_{L^2(d\mu_0)} \le S_2(B)$, and by its part (iii) $S_2(B) = S_2(B_{3\times 5})$ is one number for all $a$. Lemma N.21(ii), applied to $Y = V(h')(0)$, gives $\|Q_V\|_{L^2(d\mu_0)} = \|Y^2\|_{L^2(d\mu_0)} \le 9^{p'} V_*$; by Lemma N.22 and Remark N.23(b) this $Y^2$ is the $Q_V$ of Step 1. This step concerns the constructed $V(h')$; its identification with the $V(h')$ of §2.1 (Lemma N.23′) is not used here. (Lemma C.2, Theorem C.5 through $S_2$, and the comparison are used as in Remark N.34.) By Cauchy–Schwarz in $L^2(d\mu_0)$,

$$\langle Q_V\rangle_{h^B} \le e^{15\lambda|a_0|}\, \|Q_V\|_{L^2(d\mu_0)}\, \|e^{-\mathcal{U}_{h^B}}\|_{L^2(d\mu_0)} \le e^{15\lambda|a_0|}\, 9^{p'} V_*\, S_2(B_{3\times 5}).$$

*Step 3: the strips.* By Lemma N.11(ii) every square $\Delta = \Delta_{l,k} \not\subseteq B$ lies in exactly one ring $\mathcal{L}_j$, $j \geq 1$. Assign:

(P) if $|l| = 1 + j$: the strip $S_\Delta$ bounded by $x^0 = \frac{1}{2}$ and $x^0 = l - \frac{1}{2}$ (if $l > 0$), resp. by $x^0 = l + \frac{1}{2}$ and $x^0 = -\frac{1}{2}$ (if $l < 0$), which is *parallel to the time-zero line*, has width $j$, and separates $\Delta$ from $\bigcup S_a$ (Lemma N.11(iv), with Lemma N.7(i) for $S = S_a$);

(T) if $|l| \leq j$: then $k = k_a - 1 - j$ or $k = k_a + 3 + j$, and $S_\Delta$ is the strip bounded by $x^1 = k_a + \frac{5}{2}$ and $x^1 = k - \frac{1}{2}$, resp. by $x^1 = k + \frac{1}{2}$ and $x^1 = k_a - \frac{1}{2}$, which is *transverse to the time-zero line*, has width $j$, and separates $\Delta$ from $\bigcup S_a$ (Lemma N.11(iv), with Lemma N.7(ii)).

The four corner squares of each ring ($|l| = 1 + j$ in the two side columns) admit both strips and are assigned to (P). So $w_\Delta = j$ for $\Delta \in \mathcal{L}_j$, and since $|\mathcal{L}_j| = 8j + 12$ (Lemma N.11(ii)) and the terms are positive,

$$\sum_{\Delta \in \mathcal{D}(h,B)} e^{-m_1 w_\Delta} \leq \sum_{j \geq 1} (8j + 12)\, e^{-m_1 j} = \overline{\Sigma}_{3\times 5}.$$

*Step 4: the comparison term.* Apply Lemma D.2(a) with $F = \sum Q_{n,n',k,k'}$, which is $Q_V$ almost everywhere by Lemma N.43(i) (Lemma N.22 and Remark N.23(b)), and its pieces $Q_{n,n',k,k'}$, $R = S_a$, the box $B = B_{3\times 5}$, the cutoff $h \in \mathcal{C}_{\mathrm{E}}$, and the strips of Step 3. The estimates inside that lemma are Source Fact 2.2 with (gate GA1):

- $Q_1 = Q_{n,n',k,k'}$: type (1.1.5), two factors at time 0 of degrees $n, n' \leq 2p' < \bar{n}$, localized in $\Delta_{0,k}, \Delta_{0,k'} \in S_a$, kernel $\lambda^2 a_n a_{n'} h'_k \otimes h'_{k'} \in L^2(\mathbb{R}^2)$;
- $Q_2 = U_{\Delta,n''}$, $1 \leq n'' \leq 2p'$, $\Delta = \Delta_{l,k''} \in \mathcal{D}(h, B)$, $\Delta \in \mathcal{L}_j$: type (1.1.6), localized in $\Delta$ alone, degree $n'' < \bar{n}$, kernel $\lambda a_{n''} h \mathbf{1}_\Delta \in L^2(\mathbb{R}^2)$ (Lemma N.18(i));
- the cutoff $h_\alpha = h^B + \alpha(h - h^B) \in \mathcal{C}_{\mathrm{E}}$, $0 \leq \alpha \leq 1$ (Lemma N.5(ii));
- the strip $S_\Delta$ of Step 3, of width $j$: if $\Delta$ is of class (P), *parallel to the time-zero line*; if $\Delta$ is of class (T), *transverse to the time-zero line. Every application to a square of class (T) is the transverse case of GJS Theorem 1.1.7, used at the printed statement's scope and listed as A8(iv)-T (Assumption A8, item (iv-T)); see §D.e.4.*
- the resulting term, summed over $n''$: $C_7\, c_U\, \|Q_{n,n',k,k'}\|_{\mathrm{GJS}}\, e^{-m_1 j}$.

Lemma D.2(a), Lemma N.43(iii) and Step 3 give

$$\big|\langle Q_V\rangle_h - \langle Q_V\rangle_{h^B}\big| \leq C_7\, c_U\, \|Q_V\|_{\mathrm{GJS}}\, \overline{\Sigma}_{3\times 5} \leq C_7\, c_U\, \overline{\Sigma}_{3\times 5} \cdot 9\, K_1 K_2^{8p'} (4p')!\, \lambda^2 \Big(\sum_{n=0}^{2p'} |a_n|\Big)^2,$$

where the transverse case enters through the squares of class (T).

*Step 5.* Steps 1, 2 and 4 give $\int Q_V\, dq_h \leq M_B^2$. None of the constants depends on $h$, $a$ or $h'$: $S_2(B_{3\times 5})$ by Proposition N.15(iii), $V_*$ by its definition as a supremum, and $c_U$, $\overline{\Sigma}_{3\times 5}$, $C_7$, $m_1$, $K_1$, $K_2$ by construction. □

### D.e.4 Where the transverse case is used

The squares outside $B_{3\times 5}$ fall into the two classes of Step 3:

- *Class (P)*, $|l| = 1+j$ for $\Delta \in \mathcal{L}_j$ (so $|l| \geq 2$): separated from $S_a$ by a strip parallel to the time-zero line, of width $j$, which is the width the bound uses. For the time-integrated partner $U_{\Delta,n''}$ this is the parallel case that the printed proof reaches after slicing $U_{\Delta,n''}$ in time (Remark N.39), as in §D.d.

- *Class (T) in rows* $-1, 0, 1$: separated from $S_a$ only by strips that are not parallel to the time-zero line (Lemma N.11(v)); the bound uses the transverse strip of width $j$.

- *Class (T) in rows* $2 \leq |l| \leq j$: separated also by parallel strips, but only of width $\leq |l| - 1 < j$ (Lemma N.11(iv)); the bound uses the transverse strip of width $j$. Using the parallel strips instead would replace $\overline{\Sigma}_{3\times5}$ by a sum over all columns $k$ with $\int_{\Delta_{l,k}} h > 0$ of $e^{-m_1(|l|-1)}$, which is not bounded uniformly in $h \in \mathcal{C}_{\mathrm{E}}$ (Remark N.12(b)).

So the transverse case is used for every square of class (T), in rows $-1, 0, 1$ and beyond, as Remark N.12(b) also records; the plan's description (Part 1: "Squares in rows $-1$, 0, 1 outside $B_{3\times5}$ are separated from the time-zero observable only by strips transverse to the time-zero line") names the squares for which no parallel strip exists. Each of these applications is Source Fact 2.2 for a sharp-time $Q_1$ (type (1.1.5), at time 0) and a time-integrated $Q_2 = U_{\Delta,n''}$ (type (1.1.6)) across a strip bounded by two lines of constant $x^1$. The printed statement covers it: no orientation is prescribed ("let $d$ be the width of the widest strip bounded by two parallel lines", [7, p. 594]), and Source Fact 2.2 says "(any orientation)". The printed proof does not reach it: it proceeds "After rotating and translating the strip separating $Q_1$ and $Q_2$ in Theorem 1.1.7, and after choosing $A = Q_1$, $B_T = Q_2$ in Theorem 1.1.11" [7, p. 597], and Theorem 1.1.11 takes "monomials of type (1.1.5)" localized at times $\leqq 0$ and $\geqq 0$ [7, p. 596]; a rotation by a right angle turns the sharp-time piece of $Q_V$ into an object with fixed spatial position and integrated times, which is not of type (1.1.5) in the rotated frame. This is the content of the item A8(iv)-T (Assumption A8, item (iv-T)); Theorem D.4 uses the transverse case at the printed statement's scope, the standard under which this manuscript imports A1–A8. Nothing in this appendix proves that case.

### D.e.5 Remarks on Theorem D.4

**Remark N.44** (order of quantifiers)**.** $M_B$ is fixed first, by $(\lambda, \mathscr{P}, m_0)$ through the printed constants $C_7$, $m_1$, $K_1$, $K_2$ and through $S_2(B_{3\times5})$ and $V_*$, in working coordinates (Remark 3.3); then the bound holds for every $h \in \mathcal{C}_{\mathrm{E}}$, every $a \in \mathbb{R}$ and every admissible $h'$. In particular, for $h = h_{T,g}$ it holds for every $g \in \mathcal{C}_{\mathrm{H}}$ and every $T > 0$ with the same $M_B$.

**Remark N.45** (boundary cases)**.** $p' = 1$: $\bar{n} = 3$, all individual degrees are $\leq 2$, and $9^{p'} = 9$. *The position of* $a$: if $a \in \frac{1}{2} + \mathbb{Z}$, $[a, a+2]$ meets the columns $k_a$ and $k_a + 1$ in positive measure and $k_a + 2$ in one point; $B_{3\times5}$ is defined through $k_a$ in every case, so it has 15 squares and contains the columns of $h'$ with a margin of at least one column on each side. *Ring* 1 (Lemma N.11(iii)): 20 squares, of which the 14 in rows $\pm2$ are of class (P) and the 6 in rows $-1, 0, 1$ (columns $k_a - 2$ and $k_a + 4$) of class (T). $\operatorname{supp} h \subseteq \overline{B}_{3\times5}$: then $\mathcal{D}(h, B_{3\times5}) = \varnothing$, the comparison term vanishes, and the bound is the base term. *The coefficient* $a_0$: the measures $dq_h$ do not depend on it (Lemma N.18(iii)), but the bound does, in three places: through $Z(h^B)$ and $\mathcal{U}_{h^B}$ in the base term ($e^{15\lambda|a_0|}$ and $S_2$); through the observable, in $V_*$ (the term $n = 0$ of $V(h')$) and in $\|Q_V\|_{\mathrm{GJS}}$ (pieces with a factor of degree 0, covered by Lemma N.43); in $\mathcal{U}_{h-h^B}$ it is the constant $\lambda a_0 \int (h - h^B)$, which has no covariance. $h' = 0$: both sides are 0 and $\leq M_B^2$.

**Remark N.46** (what Theorem D.4 is not, and what it rests on)**.** It does not prove $U_{\mathrm{GJS}}$, the retired item Assumption A8(vi), which bounds every sharp-time monomial by $C_8\|\cdot\|_{\mathrm{GJS}}$:

it derives the moment bound that Theorem 5.1 consumes from GJS Theorem 1.1.7 as printed (including its transverse case A8(iv)-T) and from Lemma C.2, Lemma C.4 and Theorem C.5; GJS Theorem 1.1.11 is not used, and none of GJS's proofs is verified (traps 1, 4 and 7 of the plan). Its inputs are listed in the status line after the statement. The transfer to Theorem 5.1 (Steps 1 and 5 there) uses in addition the identification of Lemma N.23′(iii). The printed proof of Theorem 1.1.7 takes its constant from Theorem 1.1.8 at the same $h$ [7, p. 597]; that is a fact about the source's proof, which this manuscript does not re-verify (Scope (c) of Assumption A8). $M_B$ is not "independent of everything but $(\lambda, \mathscr{P}, m_0)$" in an explicit way: it depends on them through $C_7$, $m_1$, $K_1$, $K_2$ and the unevaluated constant of Theorem C.5, in working coordinates (Remark 3.3) (trap 10). The Euclidean bound implies the Hamiltonian bound of Theorem 5.1 through Steps 5–6 of its proof; the converse is not claimed (trap 9). The theorem needs the weak-coupling hypothesis through Theorem 1.1.7 (trap 17). Lemma C.2 and Theorem C.5 are load-bearing here (trap 12); their independent audit was round V3, and their repairs are recorded in Appendix B.

## D.f Quantifiers and the dependence of the constants

This section states, for every result of this appendix and for every proof that the note replaces, the order of the quantifiers and, for every constant, what it depends on and what it is uniform in. "Uniform in $X$" means that one value serves for all $X$ in the stated range, the remaining data being fixed. The statements are those of §§D.b–D.e, which agree with Part 1 of `ugjs/PLAN.md`.

### D.f.1 The fixed data

The interaction polynomial $\mathscr{P}(\xi) = \sum_{n=0}^{2p'} a_n \xi^n$, $a_{2p'} > 0$, $p' \geq 1$, is fixed. In §§D.c–D.e and §§4–7 the standing assumption $0 \leq \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$ (§2.6) is in force, and this appendix works in the working coordinates, in which the pair $(m_0, \lambda)$ satisfies the hypotheses printed with Theorem 1.1.7, "Let $m_0$, $K_2$ and $K_1$ be sufficiently large and let $\lambda$ be sufficiently small" [7, p. 594]; Remark 3.3 transports every conclusion back to the original coordinates. §D.b uses neither: Lemma D.1 holds for every $\lambda \geq 0$ and $m_0 > 0$. The constants fixed by these data are:

(F1) $m_1 > 0$ and $C_7$, the constants of Theorem 1.1.7: "There are constants $0 < m_1$ and $O(1)$ independent of $h$, $Q_1$ and $Q_2$" [7, p. 594]; this manuscript reads "independent of $h$" over the class $\mathcal{C}_{\mathrm{E}}$ (Assumption A8, Scope (d)). They are fixed once $(m_0, \lambda, \mathscr{P})$, $K_1$, $K_2$ and $\bar{n}$ are fixed by the choices behind the printed hypotheses; the printed statement makes them independent of $h$, $Q_1$ and $Q_2$, and says nothing more about their dependence.

(F2) $K_1$ and $K_2 \geq 1$, the constants of the printed norms ((1)), and $\bar{n} = 2p' + 1$ (Assumption A8(ii)).

(F3) The derived constants:

- $c_U = \lambda \sum_{n=1}^{2p'} |a_n| K_1 K_2^{2n} n!$ (Lemma N.18), depending on $\lambda, a_1, \dots, a_{2p'}, K_1, K_2$;
- the stability constant $S_2(B)$ (Definition N.14), finite by Theorem C.5 (as in Remark N.34), and not evaluated. It depends on $(\lambda, \mathscr{P}, m_0)$ and on the box $B$ through its shape only (Proposition N.15(iii), whose proof uses Lemma C.4). Independently of that proposition, the bound that the *proof* of Theorem C.5 produces depends on $\Lambda$ only through $|\Lambda|$ and $\operatorname{diam} \Lambda$, monotonically (recorded in

the statement of Theorem C.5); for a box this is a bound depending on the shape only;

– the ring constant $\overline{\Sigma}_{3\times5} := \sum_{j\geq1}(8j+12)e^{-m_1 j} = 4x(5-3x)(1-x)^{-2}$, $x = e^{-m_1}$ (§D.e), depending on $m_1$ only. It bounds the sum $\Sigma_{3\times5} = \sum_{\Delta\not\subseteq B_{3\times5}} e^{-m_1 d(\Delta,R)}$ of Lemma N.11(vi), which depends on $R$, and the sum of $e^{-m_1 w_\Delta}$ over the charged squares outside $B_{3\times5}$ with the widths $w_\Delta$ of the axis-parallel strips assigned in the proof of Theorem D.4;

– $V_*$, depending on $(\lambda, \mathscr{P}, m_0)$, finite (Lemma N.20(iv)); the integrability of $C(x-y)^n$ on $[-1,1]^2$ is Lemma N.25(ii),(iv). $V_*$ is taken for the constructed $V(h')$ of Lemma N.20(iv), which is the $V(h')$ of §2.1 by Lemma N.23′(iii); that identification enters at Theorem 5.1, Steps 1 and 5, and at Corollary 5.2;

– the factor $9^{p'} = (3^{p'})^2$ of the comparison $\|Y\|_4 \leq 3^{p'}\|Y\|_2$ for $Y$ in the chaoses of degree $\leq 2p'$ (proof of Theorem C.5; Lemma N.21; Theorem I.22 of Simon's book, Assumption A4), depending on $p'$ only; it rests on the chaos structure imported in Assumption A3, and it is applied to the constructed $Y = J_0 V(h')$ of Lemma N.21.

None of these depends on a cutoff $h$, on $T$, on $g$, on the position of a box, or on an observable. They are constants of the working coordinates, and Lemma D.2 and Theorems D.3–D.4 say so (the rider of trap 10).

### D.f.2 The order of the quantifiers, statement by statement

*Lemma D.1.* For all $h_1, h_2 \in \mathcal{C}_{\mathrm{E}}$ and all $F \in \mathfrak{F}$ (indeed all $F \in \bigcup_{p_0>1} L^{p_0}(d\mu_0)$, Remark N.29(a)), the map $\alpha \mapsto \langle F\rangle_{h_\alpha}$ is $C^1$ on $[0,1]$ and (41) holds. There is no constant; no hypothesis on $\lambda$ or $m_0$ is used. Status: proved; see the status paragraph after Lemma D.1.

*Lemma D.2*, in the form (a) of §D.c. For every finite family $(F_a)_{a\in\mathfrak{A}}$ of localized monomials of type (1.1.5) or (1.1.6) with kernels in $L^2$ and $n_i < \bar{n}$, every set $R$ of squares containing their localization squares, every box $B \supseteq R$, every $h \in \mathcal{C}_{\mathrm{E}}$, and every assignment $\Delta \mapsto S_\Delta$, $\Delta \in \mathcal{D}(h,B)$, of strips of widths $w_\Delta > 0$ separating $\Delta$ from $R$,

$$\big|\langle F\rangle_h - \langle F\rangle_{h^B}\big| \leq C_7\, c_U \Big(\sum_a \|F_a\|_{\mathrm{GJS}}\Big) \sum_{\Delta\in\mathcal{D}(h,B)} e^{-m_1 w_\Delta}, \qquad F := \sum_a F_a.$$

$C_7$, $m_1$ and $c_U$ are those of (F1)–(F3); none depends on $h$, $B$, $R$, the family or the strips. The right side depends on $h$ only through the finite set $\mathcal{D}(h,B)$ of the squares outside $B$ charged by $h$, and on $F$ through the norm of the chosen decomposition (1.1.9); the bound therefore holds with the infimum over decompositions. Of the cutoff the lemma uses exactly $h \geq 0$, $h \leq 1$ and compact support (Remark N.33(1)): it is a statement for $h \in \mathcal{C}_{\mathrm{E}}$ only, and nothing in this appendix provides it for cutoffs of variable sign, for which $h_\alpha \notin \mathcal{C}_{\mathrm{E}}$ in general and neither Assumption A3 nor Theorem 1.1.7 as imported applies (trap 20 of the plan). The orientation of each application of Theorem 1.1.7 is that of the strip $S_\Delta$, which is chosen before the bound is applied. Lemma D.2 uses Lemma D.1, and Lemma C.4 with Lemma N.23′(iii) in Step 3 of its proof, which decomposes $\mathcal{U}_{h-h^B}$ into the $U_{\Delta,n}$. There, Step 1 of the proof of Lemma D.1 and Lemma N.28(a) give the convergence of the regularizations $\mathcal{U}_w^{(\kappa)}$, $w = h - h^B$, and of their pieces $U_{\Delta,n}^{(\kappa)}$; Lemma N.5(iii) splits $w$ into the squares; and Lemma C.4 identifies the limit of $\mathcal{U}_w^{(\kappa)}$ with $\mathcal{U}_h - \mathcal{U}_{h^B}$.

*Theorem D.3.* Let $g \in \mathcal{C}_{\mathrm{H}}$, let $I \subset \mathbb{R}$ be a bounded interval of positive length and let $s \in \mathbb{R}$; these determine $l_s$, $N_g$, $N_B$ and $B = B_s(g,I)$, with $|B| = 3N_B$, and then

$$A := e^{3\lambda|a_0|N_B} S_2(B), \qquad B' := 2\, C_7\, c_U\, N_g\, \frac{e^{-m_1}}{1-e^{-m_1}}.$$

For every finite linear combination $F$ of cylinder monomials (§6) supported in $I$, at the sharp time $s$, and every $T > 0$, $|\int F\, dq_{h_{T,g}}| \le A\|F\|_{L^2(d\phi_0)} + B'\|F\|_{\mathrm{GJS}}$. The order is: $g$ and $I$; then $A$ and $B'$, which do not depend on $s$ (for $A$ through Proposition N.15(iii), as in Remark N.40); then $s$, $F$ and $T$.

- $A$ depends on $\lambda$ and $a_0$ through Lemma C.2, $Z(h^B) \ge e^{-\lambda|a_0||B|}$, and on $(\lambda, \mathscr{P}, m_0)$ and $N_B$ through $S_2(B)$, which depends on the shape $3 \times N_B$ only. It is uniform in $T > 0$, in $s$, in $F$ and in the position of $B$. It is *not* uniform in $N_B$, which grows with the diameter of the union of $\operatorname{supp} g$ and $I$, in particular with the distance from $I$ to $\operatorname{supp} g$ (item m1 of the R3 verification).

- $B'$ depends on $C_7$, $m_1$, $c_U$ and $N_g$. It is uniform in $T$, in $s$, in $F$ and in the position and the length of $I$; it depends on $g$ only through $N_g$. The term $B'\|F\|_{\mathrm{GJS}}$ comes from Lemma D.2(a), so Theorem D.3 rests on Lemma C.4 through Lemma D.2.

- Both depend on $g$; §7 fixes $g \in \mathcal{C}_{\mathrm{H}}$, so this is all it needs.

- $\|F\|_{\mathrm{GJS}}$ is the norm (1.1.9) of the decomposition of $F$ into its pieces localized in the squares of row $l_s$ on the columns of §D.a (Lemma N.36(v)), bounded with the column count $N_I^{\square}$ of Remark N.3; it does not depend on $s$ (Remark 6.3).

- Theorem D.3 bounds $\int F\, dq_h$ for observables of bounded support, with the constant $A\|F\|_{L^2} + B'\|F\|_{\mathrm{GJS}}$; it is not a form of CE1, since $A$ depends on the support of $F$ through $N_B$ and both constants depend on $g$ (trap 8).

*The consequences of Theorem D.3 in §§4 and 7.*

- Lemma 4.3(ii): at each fixed $T > 0$ and $g \in \mathcal{C}_{\mathrm{H}}$ the square moments are finite by Assumption A3 and Hölder's inequality with the exponents $(2, 2)$; no uniformity in $T$ or $g$ is claimed or used (finding F7 of the plan).

- Lemma 7.1: for every $g \in \mathcal{C}_{\mathrm{H}}$, every bounded interval $I$ of positive length and every cylinder monomial $Q$ supported in $I$, $\sup_{T>0} \|Q\hat{\psi}_T\|^2 \le A\|Q^2\|_{L^2(d\phi_0)} + B'\mathfrak{n}(Q)^2$, with $A, B'$ of Theorem D.3 at $(g, I, 0)$; the same bound holds for $\|Q\Omega_g\|^2$. The bound is uniform in $T$ and depends on $g$ and $I$. Its right side replaces the printed $C_8\,\mathfrak{n}(Q)^2$, so the *statement* of Lemma 7.1 changes.

- Lemmas 7.2–7.3: the square moments at the time $s$ and at the times $\pm t/2$ obey the same bound, with $A$ and $B'$ independent of the time and of $T$.

- Lemma 7.6: for every $g \in \mathcal{C}_{\mathrm{H}}$, every real $f \in C_c^\infty(\mathbb{R})$, $f \ne 0$, supported in an interval $I$, and every $n \ge 1$, $\mathbb{E}_{\mu_g}[X_f^{2n}] \le A_f K_f^{2n}(2n)!$ with $A_f := A + B'K_1$ ($A, B'$ at $(g, I, 0)$), $K_f := \max\{K', \sqrt{2}\,\sigma_f\}$, $K' := K_2^2 (N_I^{\square})^{1/2}\|f\|_2$ and $\sigma_f^2 := \langle f, Cf\rangle$. $A_f$ and $K_f$ do not depend on $n$, which is what Theorem 7.7 uses; $A_f$ depends on $g$ and $I$; $K_f$ depends on $f$ (through $\|f\|_2$, $\sigma_f$ and the column count of $I$), on $K_2$ and, through $C$, on $m_0$, and not on $g$. The radius $1/K_f$ of the exponential moment comes from the route through the GJS norm; nothing in this appendix shows that it is intrinsic to $\mu_g$ (trap 25). The statement of Lemma 7.6 changes ($C_8K_1$ and $K'$ become $A_f$ and $K_f$).

- Theorem 7.7: for given $e_1, \dots, e_k$ the polystrip of its Step 1 can be taken of half-width $s_0 := \min_{j \le k,\ e_j \ne 0} (2kK_{e_j})^{-1}$ ($K_f$ is defined for $f \ne 0$ only); if all $e_j = 0$, then

$F(z) = \mathbb{E}_{\mu_g}[\bar{\zeta}]$ is constant and every $s_0 > 0$ serves, so $s_0 := +\infty$. Indeed, by the Cauchy–Schwarz inequality and Hölder's inequality with $k$ exponents $k$,

$$\mathbb{E}_{\mu_g}\Big[|\zeta|\, e^{\sum_j |\mathrm{Re}\, z_j||X_{e_j}|}\Big] \le \|\zeta\|_{L^2(\mu_g)} \prod_{j=1}^{k} \Big(\mathbb{E}_{\mu_g} e^{2k|\mathrm{Re}\, z_j||X_{e_j}|}\Big)^{1/(2k)},$$

which is finite when $2k|\mathrm{Re}\, z_j| < 1/K_{e_j}$ for every $j$ with $e_j \neq 0$ (a factor with $e_j = 0$ is 1). This $s_0$ depends on $e_1, \dots, e_k$ only, not on $g$. (Part 1 of the plan writes $s_0 = 1/K_f$, the radius of Lemma 7.6 for one $f$.)

- Proposition 7.5 and Theorem 7.8: the gap $m_1$ is the constant of Theorem 1.1.7 (through Lemma 7.4) and is uniform in $g$; $\widehat{C}_Q$ depends on $g$, and only its finiteness is used. Their statements do not change.

*Theorem D.4.* There is a constant $M_B$, determined by the fixed data (F1)–(F3),

$$M_B^2 := e^{15\lambda|a_0|} S_2(B_{3\times 5})\, 9^{p'} V_* + C_7\, c_U\, \overline{\Sigma}_{3\times 5} \cdot 9 K_1 K_2^{8p'} (4p')!\, \lambda^2 \Big(\sum_{n=0}^{2p'} |a_n|\Big)^2,$$

such that for every $h \in \mathcal{C}_{\mathrm{E}}$, every $a \in \mathbb{R}$ and every real measurable $h'$ with $\|h'\|_\infty \le 1$ and $\operatorname{supp} h' \subseteq [a, a+2]$: $\int V(h')(0)^2\, dq_h \le M_B^2$. The order is: $M_B$ first; then $h$, $a$ and $h'$.

- $M_B$ depends on $(\lambda, \mathscr{P}, m_0)$ through $C_7$, $m_1$ (also through $\overline{\Sigma}_{3\times 5}$), $K_1$, $K_2$, the unevaluated constant $S_2(B_{3\times 5})$ of Theorem C.5 and the finite $V_*$, in the working coordinates (Remark 3.3); this is the rider of trap 10, and $M_B$ replaced the former $M$ of Theorem 5.1 on 7 November 2026. Theorem D.4 bounds the constructed observable, $V(h')(0) = J_0 V(h')$ with $V(h')$ the limit of Lemma N.20(iv); its base term and its comparison term concern one element of $L^2(d\mu_0)$, because the two regularizations of a sharp-time Wick power agree (Lemma N.22 and Remark N.23(b)). So $M_B$ and the Euclidean bound do not use the identification of Lemma N.23′, which enters only where $M_B$ is used for the $V(h)$ of §2.1: at Theorem 5.1, Steps 1 and 5, and at Corollary 5.2.

- It is uniform in $h \in \mathcal{C}_{\mathrm{E}}$, hence in $T > 0$ and $g \in \mathcal{C}_{\mathrm{H}}$ when $h = h_{T,g}$; in $a$, because the boxes $B_{3\times 5}(a)$ (Definition N.10) all have the shape $3 \times 5$ and $S_2$ depends on the shape only (Proposition N.15(iii), through Lemma C.4); and in $h'$.

- The factors:
  - $e^{15\lambda|a_0|}$ is Lemma C.2 with $|B_{3\times 5}| = 15$;
  - $S_2(B_{3\times 5})$ bounds $\|e^{-\mathcal{U}_{h^B}}\|_{L^2(d\mu_0)}$ (Proposition N.15(iv));
  - $9^{p'} V_*$ bounds $\|V(h')\|^2_{L^4(d\phi_0)}$, which equals $\|V(h')^2\|_{L^2(d\phi_0)}$, for the constructed $V(h')$ (Lemma N.21);
  - $C_7 c_U \overline{\Sigma}_{3\times 5}$ is Lemma D.2(a) summed over the rings (through Lemma D.2, which uses Lemma C.4), $\overline{\Sigma}_{3\times 5}$ bounding the sum of $e^{-m_1 w_\Delta}$ over the charged squares outside $B_{3\times 5}$;
  - the last factor bounds $\|Q_V\|_{\mathrm{GJS}}$ as in (5), whose "$= M^2/C_8$" no longer applies (plan, item n14).

- *Where the transverse case enters.* Every square of rows $-1, 0, 1$ outside $B_{3\times 5}$ is separated from $R$ only by transverse strips of positive width (Lemma N.11(v)); every side square of the ring $\mathcal{L}_j$ in a row $2 \le |l| \le j$ admits parallel strips only of width $\le |l| - 1 < j$,

and the bound $d(\Delta, R) \geq j$ uses a transverse strip for it too (Remark N.12(b)). The squares of the rows $|l| = j+1$ of $\mathcal{L}_j$ are reached by parallel strips of width $j$. Each application with a transverse strip falls under item A8(iv)-T (Assumption A8, item (iv-T)).

- Theorem D.4 rests on the printed hypotheses of Theorem 1.1.7; Phase B does not remove the weak-coupling restriction (trap 17).

*Theorem 5.1 with $M := M_B$, Corollary 5.2, and §§8–14.*

- Theorem 5.1: for every $g \in \mathcal{C}_{\mathrm{H}}$ and every real measurable $h'$ with $\|h'\|_\infty \leq 1$ and $\operatorname{supp} h'$ in an interval of length 2 (which lies in some $[a, a+2]$), $\Omega_g \in D(V(h'))$ and $\|V(h')\Omega_g\| \leq M_B$. Step 5 applies Theorem D.4 at $h = h_{T,g} \in \mathcal{C}_{\mathrm{E}}$ for every $T > 0$; Step 6 lets $T_k \to \infty$ at fixed $g$. The order "there is $M_B$ such that for all $g$ and $h'$" is the content of $(\mathrm{M}_2)$; the slab bound implies $(\mathrm{M}_2)$, and the converse is not claimed (trap 9). Through Theorem D.4, Theorem 5.1, and with it Corollary 5.2, Lemma 10.2, Theorem 11.1 and items (1) and (5) of Theorem 12.2, rest on the transverse case of Theorem 1.1.7 at the printed statement's scope (item A8(iv)-T, Assumption A8, item (iv-T)). At Step 1 they use the identification of the $V(h')$ of §2.1 with the constructed $V(h')$ of Lemma N.20(iv), to which Theorem D.4 applies: Lemma N.23′(iii); Lemma N.22 and Remark N.23 identify the two constructions of this appendix with each other.

- Corollary 5.2: its number $362\,880\,C_8 K_1 K_2^{16} \lambda^2$ retires with $C_8$. For $\mathscr{P}(\xi) = \xi^4$ ($p' = 2$, $a_4 = 1$, the other $a_n = 0$) the formula above gives

$$M_B^2 = 81\, S_2(B_{3\times 5})\, V_* + 362\,880\, C_7\, c_U\, \overline{\Sigma}_{3\times 5}\, K_1 K_2^{16} \lambda^2,$$

  with $c_U = 24\lambda K_1 K_2^8$ and $V_* \leq 24\lambda^2 \iint_{[-1,1]^2} C(x-y)^4\, dx\, dy$. As a bound for the $V(h)$ of §2.1, Corollary 5.2 inherits from Theorem 5.1 the transverse case A8(iv)-T and the identification of Lemma N.23′ at Step 1.

- §§8–14 consume the local moment only through Lemma 10.2, which applies Theorem 5.1 at the cutoffs $g_s \in \mathcal{C}_{\mathrm{H}}$ of an affine path and at $h_j$ supported in $[j-1, j+1]$; with $M := M_B$, which is uniform in $g$, Lemma 10.2 holds verbatim (cf. trap 11). The constants $A$, $B'$, $A_f$, $K_f$ and $s_0$ do not leave §7 (gate GA6, §§4–7).

- In the original coordinates the conclusions are transported by Remark 3.3; for the local moment this means supports of length $2/s$, or a partition of unity into at most $\lceil s \rceil + 1$ pieces and the constant $(\lceil s \rceil + 1) M_B$, where $s = m_0/M_*$.

### D.f.3 Boundary cases

Squares, rows, columns, $l_s$ and the charged squares $\mathcal{D}(h, B)$ are those of §§D.a–D.c: $\Delta_{l,k} = [l - \frac{1}{2}, l + \frac{1}{2}) \times [k - \frac{1}{2}, k + \frac{1}{2})$, $l_s = \lfloor s + \frac{1}{2} \rfloor$, and $\mathcal{D}(h, B)$ the squares $\Delta \not\subset B$ with $\int_\Delta h > 0$.

*(1) $T < 1$: a slab thinner than one row.* $h_{T,g} = \mathbf{1}_{[-T,T]} \otimes g$ charges at most the rows $l$ with $|[l - \frac{1}{2}, l + \frac{1}{2}) \cap [-T, T]| > 0$, i.e. $|l| < T + \frac{1}{2}$: row 0 alone for $T \leq \frac{1}{2}$, rows $-1, 0, 1$ for $\frac{1}{2} < T < 1$. In Theorem D.3 with $l_s = 0$, the box $B$ (rows $-1, 0, 1$, all columns meeting $\operatorname{supp} g \cup I$ in positive measure) then contains every charged square, $\mathcal{D}(h_{T,g}, B) = \varnothing$, $h^B = h_{T,g}$, and the bound is the base term $A\|F\|_{L^2}$ alone. For other $l_s$ the charged squares outside $B$ lie at row distance $\geq 2$ from $l_s$ and the parallel strips of the general case apply. Theorem D.3 needs no lower bound on $T$. In Theorem D.4 with $T \leq \frac{1}{2}$ the only charged row is 0, and every charged square outside $B_{3\times 5}$ lies in row 0 and is reached only by a transverse strip: the

transverse case is used even for the thinnest slabs. (The fallback Lemma L of the plan needs $T \geq t_* = 1/m_0$; Theorems D.3 and D.4 do not.)

*(2) $T$ not an integer.* The rows at the edge of the slab are charged only in part, so $h_{T,g}\mathbf{1}_\Delta \neq g$ on such squares; $c_U$ uses only $\|h\mathbf{1}_\Delta\|_2 \leq 1$ (Lemma N.18(i)), and no constant depends on $T$. If $T \in \frac{1}{2}+\mathbb{Z}$, the row $l = T + \frac{1}{2}$ meets $[-T, T]$ only in the null set $\{x^0 = T\}$, is not charged, and carries $U_\Delta = 0$; it needs no strip (Remark N.33(3)).

*(3) $s$ on a lattice line.* If $s = l - \frac{1}{2}$ with $l \in \mathbb{Z}$, then $l_s = l$ (Definition N.1): the observable is localized in row $l$, although the line $x^0 = s$ is also the upper edge of row $l-1$. The box $B$ of Theorem D.3 contains rows $l-1$, $l$, $l+1$, hence both rows touching the line. A charged square in row $l'$ with $|l' - l| \geq 2$ is separated from the row-$l$ squares by the parallel strip between $x^0 = \min\{l, l'\} + \frac{1}{2}$ and $x^0 = \max\{l, l'\} - \frac{1}{2}$, of width $|l' - l| - 1$; it separates the closed squares, in particular the line $x^0 = s$ itself. For $l' \leq l - 2$ the line lies on the boundary of that strip; for $l' \geq l + 2$ the true distance is one unit larger than the width used. The constants are unchanged.

*(4) $N_g = 1$, and $N_g = 0$.* If $\operatorname{supp} g$ meets one column in positive measure, the charged squares outside $B$ lie in that column, in rows $l$ with $|l - l_s| \geq 2$, at most two at each strip width $j = |l - l_s| - 1 \geq 1$, and $B' = 2C_7 c_U e^{-m_1}(1 - e^{-m_1})^{-1}$. If $g = 0$ almost everywhere, no square is charged, $dq_{h_{T,g}} = d\mu_0$ (since $\mathcal{U}_{h_{T,g}} = 0$ and $Z = 1$), the comparison term vanishes, and the theorem is the bound $|\int F\, d\mu_0| \leq \|F\|_{L^2} \leq A\|F\|_{L^2}$ ($S_2 \geq 1$). The count must be taken with the charged squares of Lemma D.2(a). With the index set $\{\Delta \not\subset B : \Delta \cap \operatorname{supp} h \neq \varnothing\}$ of its form (b), a column that meets $\operatorname{supp} g$ only in a null set would contribute squares in the rows $l_s - 1$, $l_s$, $l_s + 1$ outside $B$, which only transverse strips separate from $R$; for instance $g := \mathbf{1}_{[0,1/10]} + \mathbf{1}_{\{3\}} \in \mathcal{C}_{\mathrm{H}}$ has $N_g = 1$ while column 3 meets $\operatorname{supp} g = [0, \frac{1}{10}] \cup \{3\}$. Such columns carry $U_\Delta = 0$, and form (a) omits them (Remark N.33(3), which records this example).

*(5) $I$ meeting one column only.* Then $N_I^{\square} = 1$, every cylinder monomial of $F$ has a single localized piece, all its factors in one square $\Delta_{l_s,k}$, and $\|F\|_{\mathrm{GJS}}$ is bounded by (N.a) of §D.a with count 1. If $\operatorname{supp} g$ meets only the same column, $N_B = 1$, $|B| = 3$ and $A = e^{3\lambda|a_0|}S_2(B)$. If $I$ touches a second column in a null set (e.g. $I = [k - \frac{1}{2}, k + \frac{1}{2}]$), the pieces there vanish in $L^2$ and give no localization square. On the former cells $[j, j+1)$ the count can differ (Remark N.3(ii)); this appendix counts on the columns.

*(6) $p' = 1$.* $\bar{n} = 3$; the pieces $U_{\Delta,n}$ have $n \in \{1, 2\}$, the factors of $Q_V$ have degrees $\leq 2 < 3$, and its local degrees are $\leq 4$, which (1.1.7) permits (the bound is on the individual $n_i$). $c_U = \lambda K_1(|a_1|K_2^2 + 2|a_2|K_2^4)$, $9^{p'} = 9$, and $9K_1K_2^{8p'}(4p')! = 216\, K_1K_2^8$. Nothing degenerates.

*(7) Ring $k = 1$.* $\mathcal{L}_1$ has $8 \cdot 1 + 12 = 20$ squares (Lemma N.11(iii)): the fourteen squares of rows $\pm 2$ in the columns $k_a - 2, \ldots, k_a + 4$, reached by parallel strips of width 1; and the six squares of rows $-1, 0, 1$ in the columns $k_a - 2$ and $k_a + 4$, reached by transverse strips of width 1 and by no parallel strip of positive width. The four corners (rows $\pm 2$ in the two side columns) admit both. The ring contributes the term $20e^{-m_1}$ of $\overline{\Sigma}_{3\times 5}$, of which $6e^{-m_1}$ rest on the transverse case alone.

*(8) $a$ such that $[a, a+2]$ meets three columns.* $[a, a+2]$ meets exactly the three columns $k_a, k_a+1, k_a+2$ (Lemma N.11(i)); it meets all three in positive measure unless $a \in \frac{1}{2}+\mathbb{Z}$, when it meets $k_a+2$ only in the point $a+2$. In that case the pieces of $Q_V$ in column $k_a+2$ vanish in $L^2$ and $R$ lies in two columns, while $B_{3\times 5}(a)$ is built on all three, which only enlarges the box: the base term and the ring count are unchanged. Under the "positive measure" reading the box would be $3\times 4$ for $a \in \frac{1}{2}+\mathbb{Z}$; its rings have $(3+2j)(4+2j) - (1+2j)(2+2j) = 8j+10 \leq 8j+12$ squares, each at strip width $\geq j$ from $R$ by Lemma N.7, so that $\overline{\Sigma}_{3\times 5}$ still bounds their sum; its base term has $e^{12\lambda|a_0|} \leq e^{15\lambda|a_0|}$, and its $S_2$ is at most $S_2(B_{3\times 5})$ (Proposition N.15(ii), the $3\times 4$ box lying in a $3 \times 5$ box). So $M_B$ is valid under both readings, and the count in Theorem 5.1, Step 2 ("at most three columns ... in positive measure") holds on the columns, as stated there.

(The ring counts $8j+10$ with strip widths $\geq j$ for $j \leq 12$, and the rows charged by a slab in case (1), are also checked in exact integer and rational arithmetic by `parts/check_d1.py`, parts (4)–(5); *numerical*, run record `ugjs/parts/check_d1.output.txt`, 3 November 2026.)

*(9) The term $n=0$ of $\mathscr{P}$.* $U_{\Delta,0} = \lambda a_0 \int_\Delta h$ is a constant, with zero covariance under every $dq_{h'}$ (Lemma N.18(iii)), and $c_U$ starts at $n=1$. The coefficient $a_0$ does not change $dq_h$. As a coefficient of the interaction it enters $A$ and $M_B$ only through $e^{3\lambda|a_0|N_B}$ and $e^{15\lambda|a_0|}$ (Lemma C.2) and through $S_2$, an artefact of bounding $\|e^{-\mathcal{U}_{h^B}}\|_2$ and $Z(h^B)$ separately; with $\mathscr{P}-a_0$ in place of $\mathscr{P}$ every $dq_h$ is the same and those factors disappear. As a coefficient of the observable it enters $M_B$ through $V_*$ and $\|Q_V\|_{\mathrm{GJS}}$, where the term $n=0$ stays: $V(h')$ contains $\lambda a_0 \int h'$, and the sums $\sum_{n=0}^{2p'}$ include it.

*(10) $\lambda = 0$.* Then $\mathcal{U}_h = 0$, $dq_h = d\mu_0$ for every $h$, both sides of Lemma D.2 vanish, $c_U = V_* = 0$, $S_2 = 1$, and $M_B = 0 = \|V(h')\Omega_g\|$.

*(11) Lemma D.1 at its edges.* For $h_1 = h_2$ both sides of (41) vanish ($\mathcal{U}_{h_2-h_1} = 0$); for constant $F$ the covariance vanishes and $\langle F\rangle_h$ is constant; at $\alpha \in \{0,1\}$ the derivatives are one-sided, and Step 3 of the proof covers them, the dominator not depending on $\alpha$.

## D.g Constants recomputed by hand

This appendix recomputes, line by line, every constant of Theorems D.3 and D.4 and of the consequences of §§4–7: $c_U$, $S_2(B)$, $A$, $B'$, $A_f$, $K_f$, $s_0$, $\overline{\Sigma}_{3\times5}$, $V_*$ and $M_B$, together with the auxiliary quantities they are built from. The statements of `ugjs/PLAN.md`, Part 1, are the targets. The inputs are the printed constants of §D.a.8; Lemma D.1; Source Fact 2.2 (Assumption A8(iv)); Lemma C.2 (the base terms), Lemma C.4 (through Lemmas N.5(iv), N.18(iv) and N.13, i.e. the decomposition of $\mathcal{U}_{h-h^B}$ and the shape invariance of $S_2$), Theorem C.5 (finiteness of $S_2$) and the comparison step in the proof of Theorem C.5 (the factor $9^{p'}$), as in Remark N.34; Remark 2.5, Assumptions A2, A3 and A4 and Lemma 3.1(ii); and the lemmas of §D.a. The recomputation was made independently of the text of §§D.c–D.e and §§4–7; the discrepancies found with the plan and, afterwards, with those sections are listed in Appendix A.5–A.6 of `ugjs/comparison.tex`.

### The constant of the comparison

**Lemma N.48** (the comparison constant is $C_7 c_U$)**.** *Let $h \in \mathcal{C}_{\mathrm{E}}$, let $B$ be a box, and let $F = \sum_a F_a$ be a finite sum of localized monomials of type (1.1.5) or (1.1.6) with $L^2$ kernels and factor degrees $< \bar{n}$, all of whose localization squares lie in a set $R \subseteq B$, with $\|F\|_{\mathrm{GJS}} = \sum_a \|F_a\|_{\mathrm{GJS}}$ by (1.1.9). Assume that $F$ satisfies the hypotheses of Lemma D.1 with $h_1 = h^B$, $h_2 = h$, and that to every $\Delta \in \mathfrak{S}(h)$ with $\Delta \cap B = \varnothing$ a strip $S_\Delta$ of width $w_\Delta > 0$ separating $\Delta$ from $\bigcup R$ is assigned. Then*

$$\big|\langle F\rangle_h - \langle F\rangle_{h^B}\big| \leq C_7\, c_U\, \|F\|_{\mathrm{GJS}} \sum_{\Delta\in\mathfrak{S}(h),\ \Delta\cap B=\varnothing} e^{-m_1 w_\Delta}.$$

*Every estimate behind this bound is Source Fact 2.2 for a pair $(F_a, U_{\Delta,n})$ across $S_\Delta$, so the orientation of $S_\Delta$ is the orientation of that application. Since $w_\Delta$ may be taken arbitrarily close to $d(\Delta,R)$, the bound also holds with $d(\Delta,R)$ in place of $w_\Delta$; the orientations are then not controlled (Remark N.9): where a strip so chosen is transverse to the time-zero line and some $F_a$ is of type (1.1.5), this form uses the transverse case of GJS Theorem 1.1.7 at the printed statement's scope (A8(iv)-T, Assumption A8, item (iv-T)).*

*Proof.* By Lemma D.1 and Lemma N.5(ii), $\langle F\rangle_h - \langle F\rangle_{h^B} = -\int_0^1 \mathrm{Cov}_{h_\alpha}(F, \mathcal{U}_{h-h^B})\, d\alpha$ with $h_\alpha \in \mathcal{C}_{\mathrm{E}}$; by Lemma N.18(iv) the covariance is $\sum_\Delta \sum_{n=1}^{2p'} \sum_a \mathrm{Cov}_{h_\alpha}(F_a, U_{\Delta,n})$, the sum over

$\Delta \in \mathfrak{S}(h)$ with $\Delta \cap B = \varnothing$. Each pair $(F_a, U_{\Delta,n})$ consists of two localized monomials (types (1.1.5)/(1.1.6) and (1.1.6), Lemma N.18(i)) with disjoint localization squares (those of $F_a$ lie in $R \subseteq B$, and $\Delta \cap B = \varnothing$), factor degrees $< \bar{n}$, and the strip $S_\Delta$ separates $\Delta$ from $\bigcup R$, hence from the squares of $F_a$. Source Fact 2.2, at the cutoff $h_\alpha \in \mathcal{C}_{\mathrm{E}}$, in the weakened form of §D.a.3 with the strip $S_\Delta$, gives $|\operatorname{Cov}_{h_\alpha}(F_a, U_{\Delta,n})| \le C_7 e^{-m_1 w_\Delta} \|F_a\|_{\mathrm{GJS}} \|U_{\Delta,n}\|_{\mathrm{GJS}}$. Sum: $\sum_a \|F_a\|_{\mathrm{GJS}} = \|F\|_{\mathrm{GJS}}$, $\sum_n \|U_{\Delta,n}\|_{\mathrm{GJS}} \le c_U$ (Lemma N.18(ii)), and $\int_0^1 d\alpha = 1$. For the last sentence: given $\varepsilon > 0$ choose strips with $w_\Delta > d(\Delta, R) - \varepsilon$; the sum is finite, so letting $\varepsilon \to 0$ gives the bound with $d(\Delta, R)$. □

The orientation of each application is that of the assigned strip; Propositions N.49 and N.50 assign the strips explicitly. The constant of the comparison is $C_7 c_U$ and nothing else; the geometry enters only through the sum.

### The constants of Theorems D.3 and D.4

**Proposition N.49** (the constants of Theorem D.3)**.** *Let $g \in \mathcal{C}_{\mathrm{H}}$, $I \subset \mathbb{R}$ a bounded interval, $s \in \mathbb{R}$, $N_g$ the number of columns meeting* $\operatorname{supp} g$ *in positive measure, and $B = B_s(g, I)$ the rows $l_s - 1, l_s, l_s + 1$ times the smallest interval of $N_B$ columns containing every column that meets* $\operatorname{supp} g$ *or $I$ (in positive measure, or set-theoretically: both readings work below). Let $F$ be a finite linear combination of cylinder monomials at sharp time $s$, supported in $I$, satisfying the hypotheses of Lemma D.1. Then for every $T > 0$*

$$\left| \int F \, dq_{h_{T,g}} \right| \le A \, \|F\|_{L^2(d\phi_0)} + B' \, \|F\|_{\mathrm{GJS}},$$

$$A := e^{3\lambda |a_0| N_B} S_2(B), \qquad B' := \frac{2\, C_7\, c_U\, N_g\, e^{-m_1}}{1 - e^{-m_1}}.$$

*$A$ depends on $(\lambda, \mathscr{P}, m_0)$ and $N_B$ and is uniform in $T > 0$ and $s \in \mathbb{R}$; $B'$ depends on $C_7, m_1, K_1, K_2, \lambda, a_1, \dots, a_{2p'}$ and $N_g$, and is uniform in $T$, $s$ and the position of $I$. All strips used are parallel.*

*Proof. Base term.* $h := h_{T,g} \in \mathcal{C}_{\mathrm{E}}$ (Lemma 3.1(ii)); $h^B \in \mathcal{C}_{\mathrm{E}}$ and $\operatorname{supp} h^B \subseteq \overline{B}$ (Lemma N.5(i)). Cauchy–Schwarz in $L^2(d\mu_0)$ gives

$$|\langle F \rangle_{h^B}| \le Z(h^B)^{-1} \|F\|_{L^2(d\mu_0)} \|e^{-\mathcal{U}_{h^B}}\|_{L^2(d\mu_0)}.$$

Lemma C.2, "For every $h \in \mathcal{C}_{\mathrm{E}}$ with $\operatorname{supp} h \subseteq \Lambda$: $Z(h) \ge e^{-\lambda |a_0|\, |\Lambda|} > 0$", applied with $\Lambda = \overline{B}$ (compact, $|\overline{B}| = |B| = 3N_B$; hypotheses checked above) gives $Z(h^B)^{-1} \le e^{3\lambda |a_0| N_B}$ (as in Remark N.34); Proposition N.15(iv) gives $\|e^{-\mathcal{U}_{h^B}}\|_2 \le S_2(B)$; and $\|F\|_{L^2(d\mu_0)} = \|F\|_{L^2(d\phi_0)}$ because the sharp-time field has the law of $\varphi_0$ (Remark 2.5, as in the proof of Lemma N.21). So $|\langle F \rangle_{h^B}| \le A \|F\|_{L^2(d\phi_0)}$. Uniformity: $A$ contains neither $T$ nor $s$; $B_s(g, I)$ has shape $(3, N_B)$ for every $s$, so $S_2(B)$ is independent of $s$ (Proposition N.15(iii)).

*Comparison term.* The localization squares of $F$ (Remark N.2(ii), pieces $f_i \mathbf{1}_{[k-1/2, k+1/2)}$) lie in $R \subseteq S := \{\Delta_{l_s,k} : k \text{ a column of } B\}$, a rectangle in $B$. If $\Delta_{l,k} \in \mathfrak{S}(h_{T,g})$, then $\int_{\Delta_{l,k}} h_{T,g} = |[-T, T] \cap [l - \frac{1}{2}, l + \frac{1}{2})| \int_{k-1/2}^{k+1/2} g > 0$, so column $k$ meets $\{g > 0\} \subseteq \operatorname{supp} g$ in positive measure: $k$ is one of the $N_g$ columns, all of which are columns of $B$. If moreover $\Delta_{l,k} \cap B = \varnothing$, then $|l - l_s| \ge 2$, so $\rho_t = |l - l_s| \ge 2$ and $\rho_x = 0$ with respect to $S$, and Lemma N.7(i) gives the *parallel* strip $S_\Delta$ bounded by the lines $x^0 = l_s + \frac{1}{2}$ and $x^0 = l - \frac{1}{2}$ (if $l > l_s$), resp. $x^0 = l + \frac{1}{2}$ and $x^0 = l_s - \frac{1}{2}$ (if $l < l_s$), of width $w_\Delta = |l - l_s| - 1 \ge 1$, separating $\Delta$ from $\bigcup S \supseteq \bigcup R$. Hence, with $x = e^{-m_1}$ and $\sum_{j \ge 1} x^j = x/(1 - x)$ (proof of Lemma N.11(vi)),

$$\sum_{\Delta \in \mathfrak{S}(h_{T,g}),\ \Delta \cap B = \varnothing} e^{-m_1 w_\Delta} \le N_g \sum_{|l - l_s| \ge 2} e^{-m_1 (|l - l_s| - 1)} = 2 N_g \sum_{j \ge 1} x^j = \frac{2 N_g e^{-m_1}}{1 - e^{-m_1}}.$$

Lemma N.48 with these strips gives $|\langle F\rangle_{h_{T,g}} - \langle F\rangle_{h^B}| \le B'\|F\|_{\mathrm{GJS}}$. The bound contains neither $T$, nor $s$, nor $I$. For $T < \frac{1}{2}$ only row 0 is charged; the bound is the same. □

**Proposition N.50** (the constants of Theorem D.4)**.** *Let $h \in \mathcal{C}_{\mathrm{E}}$, $a \in \mathbb{R}$, and $h'$ real measurable with $\|h'\|_\infty \le 1$ and $\operatorname{supp} h' \subseteq [a, a+2]$, and assume that $Q_V := V(h')(0)^2$ satisfies the hypotheses of Lemma D.1. Then $\int V(h')(0)^2\, dq_h \le M_B^2$, where*

$$M_B^2 := e^{15\lambda|a_0|}\, S_2(B_{3\times 5})\, 9^{p'} V_* + C_7\, c_U\, \overline{\Sigma}_{3\times 5}\, \mathfrak{Q}, \qquad \mathfrak{Q} := 9K_1 K_2^{8p'}(4p')!\, \lambda^2 \Big(\sum_{n=0}^{2p'} |a_n|\Big)^2,$$

*where $S_2(B_{3\times 5})$ is the common value for all $a$ (Proposition N.15(iii)) and $\overline{\Sigma}_{3\times 5} = \sum_{j\ge 1}(8j+12)e^{-m_1 j} = 4e^{-m_1}(5-3e^{-m_1})(1-e^{-m_1})^{-2}$ is the ring constant of Lemma N.11(vi). Fully explicit up to $S_2$:*

$$M_B^2 \le \overline{M}_B^2 := e^{15\lambda|a_0|}\, S_2(B_{3\times 5})\, 9^{p'}\, \overline{V}_* + C_7\, c_U\, \overline{\Sigma}_{3\times 5}\, \mathfrak{Q},$$

*$\overline{V}_* = \lambda^2 \sum_{n=0}^{2p'} a_n^2 n!\, I_n$ (Lemma N.20). $M_B$ depends on $(\lambda, \mathscr{P}, m_0)$ through $C_7, m_1, K_1, K_2$, $V_*$ and the unevaluated $S_2(B_{3\times 5})$, and is uniform in $h \in \mathcal{C}_{\mathrm{E}}$ (hence in $T > 0$ and $g \in \mathcal{C}_{\mathrm{H}}$), in $a$ and in $h'$. The comparison term uses the strips of Lemma N.11(iv): parallel ones for the squares with $|l| = 1 + j$ (the corner squares included) and transverse ones for the squares with $|l| \le j$ (Remark N.12(b)).*

*Proof. The norm $\mathfrak{Q}$.* By (4) (Theorem 5.1, Step 1),

$$Q_V = \lambda^2 \sum_{n,n'=0}^{2p'} a_n a_{n'} \iint h'(x)h'(y) {:}\Phi(0,x)^n{:}\,{:}\Phi(0,y)^{n'}{:}\, dx\, dy.$$

Split $h' = \sum_k h' \mathbf{1}_{[k-1/2,k+1/2)}$ over the columns meeting $[a, a+2]$ in positive measure, at most three, all among $k_a, k_a+1, k_a+2$ (Lemma N.11(i)). The piece $(n, n', k, k')$ is a localized monomial of type (1.1.5) at time $0 \in [-\frac{1}{2}, \frac{1}{2})$, i.e. in row 0, with kernel $w = \lambda^2 a_n a_{n'} h'\mathbf{1}_k \otimes h'\mathbf{1}_{k'}$, $\|w\|_2 \le \lambda^2 |a_n||a_{n'}|$ (unit columns, $|h'| \le 1$), and factor degrees $n, n' \le 2p' < \bar{n}$. Its weight in (1.1.8) is $K_1 K_2^{2(n+n')}(n+n')!$ if $k = k'$ and $K_1 K_2^{2n} n!\, K_2^{2n'} n'!$ if $k \ne k'$; since $n + n' \le 4p'$, $K_2 \ge 1$ and $n!\, n'! \le (n+n')!$, both are $\le K_1 K_2^{8p'}(4p')!$. By (1.1.9), summing over at most 9 pairs $(k, k')$ and over $(n, n')$: $\|Q_V\|_{\mathrm{GJS}} \le \mathfrak{Q}$. (This is (5), recomputed on the squares of Definition N.1; the count of three columns is Lemma N.11(i).)

*Base term.* $B := B_{3\times 5}(a)$ is a box of 15 squares containing $R$, the localization squares of $Q_V$, which lie in $S_a$. As in Proposition N.49: $|\langle Q_V\rangle_{h^B}| \le e^{15\lambda|a_0|} S_2(B)\|Q_V\|_{L^2(d\mu_0)}$, and $\|Q_V\|_{L^2(d\mu_0)} = \|Y^2\|_{L^2(d\mu_0)} \le 9^{p'} V_*$ by Lemma N.21. The $Q_V$ of the base term and the $Q_V$ whose pieces enter the comparison term below are the same element of $L^2(d\mu_0)$ by Lemma N.22 and Remark N.23(b).

*Comparison term.* Lemma N.48 with $F = Q_V$, $R = S_a$, and, for $\Delta = \Delta_{l,k} \in \mathcal{L}_j$, the strip $S_\Delta$ of Lemma N.11(iv) of width $w_\Delta = j$: parallel if $|l| = 1 + j$ (the corner squares included), transverse if $|l| \le j$. Every application to a square with $|l| \le j$ is the transverse case of GJS Theorem 1.1.7, used at the printed statement's scope (item A8(iv)-T, Assumption A8, item (iv-T)). The terms are positive and the charged squares outside $B$ are among all squares $\Delta \not\subseteq B$, so by Lemma N.11(ii),(vi) the sum is at most $\sum_{\Delta\not\subseteq B} e^{-m_1 w_\Delta} = \overline{\Sigma}_{3\times 5}$, and $|\langle Q_V\rangle_h - \langle Q_V\rangle_{h^B}| \le C_7 c_U\, \mathfrak{Q}\, \overline{\Sigma}_{3\times 5}$. Add the two terms; $\langle Q_V\rangle_h \ge 0$. Uniformity: no term contains $h$, $T$, $g$ or $h'$, and $a$ enters only through $S_2(B_{3\times 5}(a))$, which does not depend on $a$. □

**Remark N.51** (the $\varphi^4$ instance; Corollary 5.2 recomputed)**.** For $\mathscr{P}(\xi) = \xi^4$: $p' = 2$, $a_4 = 1$, all other $a_n = 0$, so $a_0 = 0$ and $\sum_n |a_n| = 1$. Then $c_U = \lambda K_1 K_2^8 \, 4! = 24\,\lambda K_1 K_2^8$; $\mathfrak{Q} = 9 \cdot 8!\, K_1 K_2^{16} \lambda^2 = 362\,880\, K_1 K_2^{16} \lambda^2$; $\overline{V}_* = \lambda^2\, 4!\, I_4 = 24\,\lambda^2 I_4$; $9^{p'} = 81$; and

$$\begin{aligned} M_B^2 &\le 81 \cdot 24\,\lambda^2 I_4\, S_2(B_{3\times 5}) + 24 \cdot 362\,880\, C_7 K_1^2 K_2^{24} \lambda^3\, \overline{\Sigma}_{3\times 5} \\ &= 1944\,\lambda^2 I_4\, S_2(B_{3\times 5}) + 8\,709\,120\, C_7 K_1^2 K_2^{24} \lambda^3\, \overline{\Sigma}_{3\times 5}. \end{aligned}$$

(Arithmetic: $9 \cdot 40\,320 = 362\,880$; $81 \cdot 24 = 1944$; $24 \cdot 362\,880 = 8\,709\,120$.) The former number $362\,880\, C_8 K_1 K_2^{16} \lambda^2$ of Corollary 5.2 was replaced on 7 November 2026, as the plan says; the new value is not a number, since $S_2$ is not evaluated, and $I_4$ depends on $m_0 = M_*$.

### The constants of the consequences

**Lemma N.52** (Gaussian moments of $X_f$)**.** *Let $f \in C_c^\infty(\mathbb{R})$ be real, $X_f := \varphi_0(f)$, $\sigma_f^2 := \langle f, Cf\rangle = \iint f(x) C(x-y) f(y)\, dx\, dy$ (Lemma N.19(ii)). Then for $n \ge 0$*

$$\|X_f^{2n}\|_{L^2(d\phi_0)} = \sigma_f^{2n} \big((4n-1)!!\big)^{1/2} \le (\sqrt{2}\,\sigma_f)^{2n} (2n)!,$$

*with $(-1)!! := 1$.*

*Proof.* $X_f$ is centred Gaussian with variance $\sigma_f^2$, so $\mathbb{E} e^{sX_f} = e^{s^2\sigma_f^2/2}$; comparing coefficients of $s^{2m}$, $\mathbb{E}[X_f^{2m}]/(2m)! = \sigma_f^{2m}/(2^m m!)$, i.e. $\mathbb{E}[X_f^{2m}] = (2m-1)!!\,\sigma_f^{2m}$. With $m = 2n$, $\|X_f^{2n}\|_2 = (\mathbb{E} X_f^{4n})^{1/2} = \sigma_f^{2n}((4n-1)!!)^{1/2}$. Next, $(4n-1)!! = (4n)!/(2^{2n}(2n)!)$ and $(4n)! = \binom{4n}{2n}((2n)!)^2 \le 2^{4n}((2n)!)^2$, so $(4n-1)!! \le 2^{2n}(2n)!$ and $((4n-1)!!)^{1/2} \le 2^n((2n)!)^{1/2} \le 2^n(2n)!$. (Test $n = 1$: $\|X_f^2\|_2 = \sqrt{3}\,\sigma_f^2 \le 4\sigma_f^2$. Numerical check of $(4n-1)!! \le 4^n(2n)!$ for $n \le 40$ in exact integers: `ugjs/parts/cqft2_ringcheck.py`, run record in Remark N.12(c); nothing is deduced from it.) □

**Corollary N.53** (Lemma 7.6 with the constants of Theorem D.3)**.** *Let $g \in \mathcal{C}_{\mathrm{H}}$, let $f \in C_c^\infty(\mathbb{R})$ be real, $f \ne 0$, supported in a bounded interval $I$ of length $L$, and let $A$, $B'$ be the constants of Proposition N.49 for $g$, $I$ and $s = 0$. Put*

$$K'_\square := K_2^2\, (N_I^\square)^{1/2} \|f\|_2, \qquad K_f := \max\{K'_\square, \sqrt{2}\,\sigma_f\}, \qquad A_f := A + B' K_1.$$

*Assume the transfer*

$$\mathbb{E}_{\mu_g}[X_f^{2n}] \le \sup_{T>0} \int X_f(0)^{2n}\, dq_{h_{T,g}}$$

*(the weak-graph passage of Lemma 7.1, as replaced in §§4–7). Then for every $n \ge 1$ and every $0 \le s < 1/K_f$,*

$$\mathbb{E}_{\mu_g}[X_f^{2n}] \le A_f\, K_f^{2n} (2n)!, \qquad \mathbb{E}_{\mu_g} e^{s|X_f|} \le 2 \max\{1, A_f\} \sum_{n\ge 0} (sK_f)^{2n} < \infty.$$

*The constants $A_f$ and $K_f$ do not depend on $n$.*

*Proof.* Proposition N.49 with $F = X_f^{2n}$ (a cylinder monomial at time 0 with $2n$ factors, supported in $I$) gives $\int X_f(0)^{2n} dq_{h_{T,g}} \le A\|X_f^{2n}\|_{L^2(d\phi_0)} + B'\|X_f^{2n}\|_{\mathrm{GJS}}$ for every $T > 0$. By Lemma N.52, $\|X_f^{2n}\|_2 \le (\sqrt{2}\sigma_f)^{2n}(2n)!$; by (N.a) of Remark N.3 with $r = 2n$ and all $f_i = f$, $\|X_f^{2n}\|_{\mathrm{GJS}} \le K_1 K_2^{4n} (2n)!\, (N_I^\square)^n \|f\|_2^{2n} = K_1 (K'_\square)^{2n} (2n)!$. Both are $\le K_f^{2n}(2n)!$ times 1, resp. $K_1$; add. For the exponential bound: $e^{s|x|} \le 2\cosh(sx)$, and by monotone convergence $\mathbb{E}_{\mu_g} \cosh(sX_f) = \sum_{n\ge 0} s^{2n} \mathbb{E}_{\mu_g}[X_f^{2n}]/(2n)! \le 1 + A_f \sum_{n\ge 1} (sK_f)^{2n} \le \max\{1, A_f\} \sum_{n\ge 0} (sK_f)^{2n}$, a geometric series with ratio $(sK_f)^2 < 1$. □

**Remark N.54** (the radius $s_0$ of Theorem 7.7)**.** In the proof of Theorem 7.7, $F(z) = \mathbb{E}_{\mu_g}[\overline{\zeta}\exp(\sum_{j=1}^k z_j X_{e_j})]$ must be defined and analytic in a polystrip $\{|\mathrm{Re}\, z_j| < s_0\}$. By Cauchy–Schwarz and Hölder with $k$ equal exponents,

$$\mathbb{E}_{\mu_g}\Big[|\zeta|\, e^{\sum_j |\mathrm{Re}\, z_j||X_{e_j}|}\Big] \le \|\zeta\|_{L^2(\mu_g)} \prod_{j=1}^{k} \Big(\mathbb{E}_{\mu_g} e^{2k|\mathrm{Re}\, z_j||X_{e_j}|}\Big)^{1/(2k)},$$

finite, locally uniformly, when $2k|\mathrm{Re}\, z_j| < 1/K_{e_j}$ for every $j$ with $e_j \neq 0$ (Corollary N.53, which assumes $f \neq 0$; a factor with $e_j = 0$ is 1). So

$$s_0 := \min_{1\le j\le k,\ e_j\neq 0} \frac{1}{2k\, K_{e_j}}$$

works (any $s_0 > 0$ if all $e_j = 0$); $1/K_f$ is the one-variable radius of Corollary N.53, and any $s_0 > 0$ suffices for Theorem 7.7. $s_0$ depends on $e_1, \dots, e_k$ and $K_2$, $m_0$, not on $g$ (the constants $A_{e_j}$ do, but they do not enter $s_0$).

**Remark N.55** (Lemmas 4.3, 7.1–7.3, Proposition 7.5)**.** (a) Lemma 4.3(ii) needs, at each fixed $T$, only the finiteness of

$$\int F(\Phi(s,\cdot))^2\, dq_{h_{T,g}} \le Z(h_{T,g})^{-1}\, \|F^2\|_{L^2(d\mu_0)}\, \|e^{-\mathcal{U}_{h_{T,g}}}\|_{L^2(d\mu_0)},$$

which holds by Assumption A3 ($e^{-2\mathcal{U}} \in L^1$, $0 < Z$) and Gaussian moments of $F$; no constant is needed. (b) Lemma 7.1 becomes

$$\sup_{T>0} \|Q\hat{\psi}_T\|^2 \le A\|Q^2\|_{L^2(d\phi_0)} + B'\, \mathfrak{n}_\square(Q)^2, \qquad \mathfrak{n}_\square(Q)^2 := K_1 K_2^{4r} (2r)!\, (N_I^\square)^r \prod_i \|f_i\|_2^2,$$

the second bound of (N.a), and Lemmas 7.2–7.3 use the same constants at the times $s$ and $\mp t/2$ (the box $B_s(g, I)$ has the shape $(3, N_B)$ for every $s$). Here

$$\|Q^2\|_{L^2(d\phi_0)} \le \big((4r-1)!!\big)^{1/2} \prod_i \sigma_{f_i}^2,$$

by Hölder with $r$ equal exponents and the moment formula in the proof of Lemma N.52:

$$\mathbb{E}\prod_i X_{f_i}^4 \le \prod_i \big(\mathbb{E} X_{f_i}^{4r}\big)^{1/r} = (4r-1)!! \prod_i \sigma_{f_i}^4.$$

(c) Proposition 7.5 keeps $\widehat{C}_Q = \max\{C_7 e^{2m_1}\|Q\|_{\mathrm{GJS}}^2, e^{3m_1}\|\psi_Q\|^2\}$, with $\|Q\|_{\mathrm{GJS}}$ computed on the squares of Definition N.1; the gap $m_1$ of Theorem 7.8 is unchanged.

### The table

All constants are those of the working coordinates (§2.6 and Remark 3.3). "Cond." marks a constant whose finiteness or bound uses Lemma C.2, Lemma C.4, Theorem C.5 or the comparison step in the proof of Theorem C.5 (as in Remark N.34). "A8(iv)-T" marks a constant whose derivation uses GJS Theorem 1.1.7 across strips transverse to the time-zero line (Assumption A8, item (iv-T)). "Ident." marks a constant used for the $V(h)$ of §2.1, through Lemma N.23′: $M_B$ in Theorem 5.1 and Corollary 5.2. The rows $V_*$, $9^{p'}$ and $M_B$ concern the constructed objects, which Lemma N.22 identifies with the regularized ones of Appendix C.

| constant | formula | depends on | uniform in | derived in |
|---|---|---|---|---|
| $\bar{n}$ | $2p'+1$ | $p'$ | — | §D.a.8; A8(ii) |
| $K_1$, $K_2$ | printed, (1.1.7)–(1.1.8); $K_2 \geq 1$ | fixed with $(m_0, \lambda)$ of §2.6 | $h$, $Q$ | §D.a.8 |
| $C_7$, $m_1$ | printed, (1.1.10) | fixed with $(m_0, \lambda, K_1, K_2)$; not printed | $h$, $Q_1$, $Q_2$ | §D.a.8 |
| $c_U$ | $\lambda \sum_{n=1}^{2p'} \lvert a_n \rvert K_1 K_2^{2n} n!$ | $\lambda$, $a_1, \dots, a_{2p'}$, $K_1$, $K_2$ | $\Delta$, $h \in \mathcal{C}_{\mathrm{E}}$ | L. N.18 |
| $C_7 c_U$ | constant of the comparison; cond. | $C_7$, $c_U$ | $h$, $B$, $F$, $\alpha$ | L. N.48 |
| $S_2(B)$ | $\sup\{\lVert e^{-\mathcal{U}_{h'}} \rVert_{L^2(d\mu_0)} : h' \in \mathcal{C}_{\mathrm{E}}, \operatorname{supp} h' \subseteq \overline{B}\}$; $1 \leq S_2 < \infty$; not evaluated; cond. | $\lambda$, $\mathscr{P}$, $m_0$; shape of $B$ | position of $B$ | P. N.15 |
| $N_g$, $N_B$ | columns meeting $\operatorname{supp} g$ in positive measure; columns of $B_s(g, I)$ | $g$; $g$ and $I$ | $s$, $T$ | P. N.49 |
| $A$ | $e^{3\lambda \lvert a_0 \rvert N_B} S_2(B_s(g, I))$; cond. | $\lambda$, $\mathscr{P}$, $m_0$, $N_B$ | $T > 0$, $s \in \mathbb{R}$ | P. N.49 |
| $B'$ | $2C_7 c_U N_g e^{-m_1} (1 - e^{-m_1})^{-1}$; cond. | $C_7$, $m_1$, $c_U$, $N_g$ | $T$, $s$, position of $I$ | P. N.49 |

| constant | formula | depends on | uniform in | derived in |
|---|---|---|---|---|
| $N_I^{\square}$ | $\#\{k : \lvert I \cap [k - \frac{1}{2}, k + \frac{1}{2}) \rvert > 0\} \leq \lfloor L \rfloor + 2$ | $I$ | — | R. N.3 |
| $\mathfrak{n}_{\square}(Q)^2$ | $K_1 K_2^{4r} (2r)! (N_I^{\square})^r \prod_i \lVert f_i \rVert_2^2$ | $K_1$, $K_2$, $r$, $I$, $f_i$ | $s$ | R. N.3 |
| $\sigma_f$ | $\left( \iint f(x) C(x-y) f(y)\, dx\, dy \right)^{1/2}$ | $f$, $m_0$ | — | L. N.19 |
| $K'_{\square}$ | $K_2^2 (N_I^{\square})^{1/2} \lVert f \rVert_2$ | $K_2$, $I$, $f$ | $n$ | C. N.53 |
| $K_f$ | $\max\{K'_{\square}, \sqrt{2}\, \sigma_f\}$ | $K_2$, $m_0$, $I$, $f$ | $n$, $g$ | C. N.53 |
| $A_f$ | $A + B' K_1$ ($A$, $B'$ for $g$, $I \supseteq \operatorname{supp} f$, $s = 0$); cond. | as $A$, $B'$; $K_1$ | $n$, $T$ | C. N.53 |
| $s_0$ | $\min_{j \leq k,\, e_j \neq 0} (2k K_{e_j})^{-1}$; one-variable radius $1/K_f$ | $e_1, \dots, e_k$, $K_2$, $m_0$ | $g$ | R. N.54 |
| $\mathfrak{Q}$ | $9 K_1 K_2^{8p'} (4p')! \lambda^2 (\sum_{n=0}^{2p'} \lvert a_n \rvert)^2 \geq \lVert Q_V \rVert_{\mathrm{GJS}}$ | $\lambda$, $\mathscr{P}$, $K_1$, $K_2$ | $h'$, $a$ | P. N.50 |
| $\overline{\Sigma}_{3\times 5}$ | $\sum_{j \geq 1} (8j + 12) e^{-m_1 j} = 4e^{-m_1} (5 - 3e^{-m_1}) (1 - e^{-m_1})^{-2} \geq \Sigma_{3\times 5}$ | $m_1$ | $a$, $h$, $R \subseteq S_a$ | L. N.11 |
| $b_0$, $I_n$ | $\log 4 + e^{-1} + m_0^{-2}$; $I_0 = 4$, $I_n \leq (4\pi)^{-n} 2^{n+1} (2b_0^n + 2^n (n! + (\log 2)^n))$ | $m_0$, $n$ | — | L. N.19 |
| $V_*$ | $\sup_{h'} \lVert V(h') \rVert_{L^2(d\phi_0)}^2 \leq \overline{V}_* := \lambda^2 \sum_{n=0}^{2p'} a_n^2 n!\, I_n$, with $V(h') = \lambda \sum_n a_n W_n$ of L. N.20(iv) | $\lambda$, $\mathscr{P}$, $m_0$ | $h'$, $a$ | L. N.20 |
| $9^{p'}$ | $\lVert Y \rVert_{L^4(d\mu_0)}^2 \leq 9^{p'} \lVert Y \rVert_{L^2(d\mu_0)}^2$; cond. | $p'$ | $h'$, $a$ | L. N.21 |
| $M_B^2$ | $e^{15\lambda \lvert a_0 \rvert} S_2(B_{3\times 5})\, 9^{p'} V_* + C_7 c_U \overline{\Sigma}_{3\times 5} \mathfrak{Q}$; cond.; A8(iv)-T; ident. in Thm. 5.1 | $(\lambda, \mathscr{P}, m_0)$ via $C_7$, $m_1$, $K_1$, $K_2$, $V_*$, $S_2(B_{3\times 5})$ | $h \in \mathcal{C}_{\mathrm{E}}$ (so $T$, $g$), $a$, $h'$ | P. N.50 |
| $M_B^2$ ($\varphi^4$) | $\leq 1944 \lambda^2 I_4 S_2(B_{3\times 5}) + 8\,709\,120\, C_7 K_1^2 K_2^{24} \lambda^3 \overline{\Sigma}_{3\times 5}$; cond.; A8(iv)-T; ident. in Cor. 5.2 | $\lambda$, $m_0$, $C_7$, $m_1$, $K_1$, $K_2$ | as $M_B$ | R. N.51 |

In Appendix A the rows that changed on 7 November 2026 are: $K_1, K_2, C_7, C_8, m_1$ ($C_8$ dropped after Phase B); $N_I$ (counted on GJS's columns, bound $\lfloor L \rfloor + 2$ unchanged, Remark N.3(i)); $M^2$ (now $M_B^2$); $M^2$ ($\varphi^4$); $K'$ (same formula, $N_I$ by columns); and the $\Psi_\gamma$ rows (the $M$ in them is $M_B$). Unchanged: $\varepsilon_{\mathscr{P}}$, $\mathfrak{K}(p')$, $\widehat{C}_Q$ (its formula; $\lVert Q \rVert_{\mathrm{GJS}}$ is computed on GJS's squares), $\mathfrak{m}$, $d_0$ and the shell count. New rows: $c_U$, $S_2(B)$, $A$, $B'$, $A_f$, $K_f$ (with $\sigma_f$), $s_0$, $V_*$, $\overline{\Sigma}_{3\times 5}$, $M_B$, and the explicit constants $b_0$, $C_*$, $c_1$, $c_2$, $b_*$ of Appendix C; $S_2(B)$, and through it $A$, $A_f$ and $M_B$, contain the unevaluated finite constant of Theorem C.5.

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